

Principles of

Data Science

Principles of Data Science

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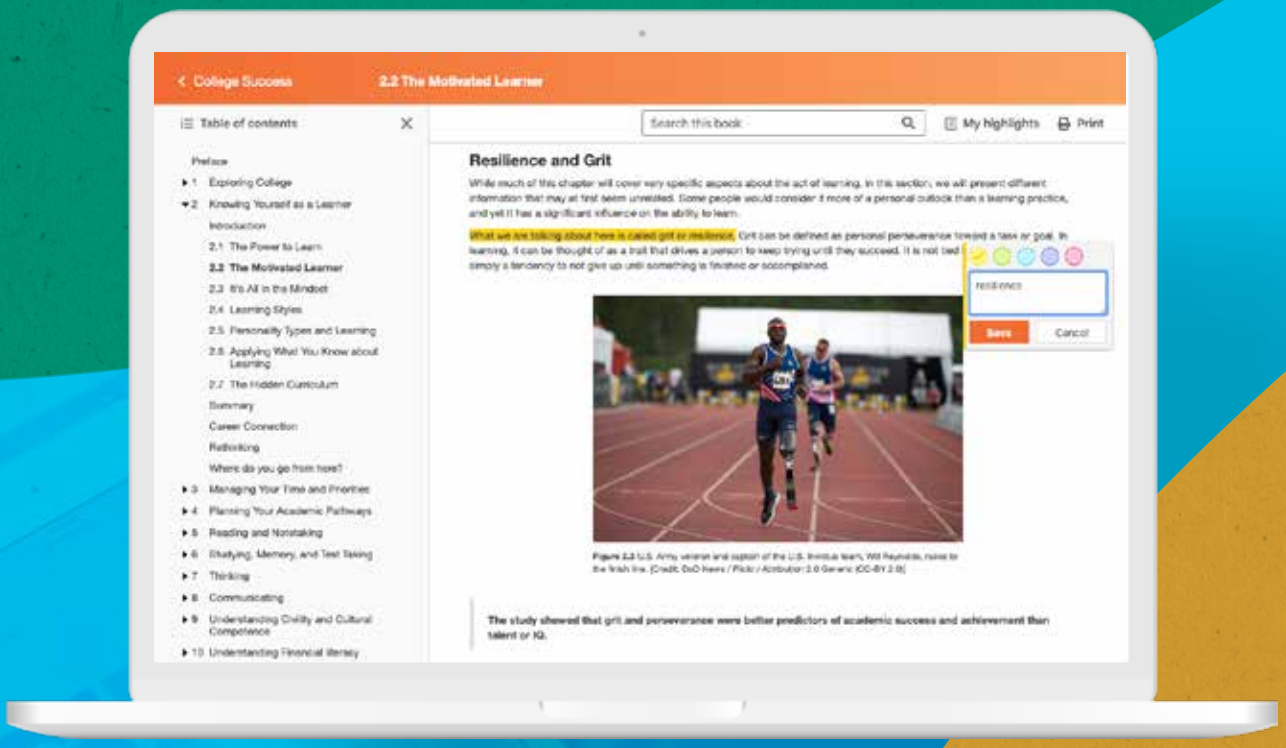
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Preface

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About *Principles of Data Science*

Summary

Principles of Data Science is intended as introductory material for a one- or two-semester course on data science. It is appropriate for undergraduate students interested in the rapidly growing field of data science;

this may include data science majors, data science minors, or students concentrating in business, finance, health care, engineering, the sciences, or a number of other fields where data science has become critically important. The material is designed to prepare students for future coursework and career applications in a data science–related field. It does not assume significant prior coding experience, nor does it assume completion of more than college algebra. The text provides foundational statistics instruction for students who may have a limited statistical background.

Coverage and Scope

Principles of Data Science emphasizes the use of Python code in relevant data science applications. Python provides a versatile programming language with libraries and frameworks for data manipulation, analysis, and machine learning. The book begins with an introduction to Python and presents Python libraries, algorithms, and functions as they are needed throughout. In occasional, focused instances, the authors also use Excel to illustrate the basic manipulation of data using functions, formulas, and tools for calculations, visualization, and financial analysis. R, a programming language used most often for statistical modeling, is briefly described and then summarized and applied to relevant examples in a book appendix. Excel and Python summaries are also provided in appendices at the end of the book.

The table of contents (TOC) is divided into ten chapters, organized in four units, intuitively following the standard data science cycle. The four units are:

Unit 1: *Introducing Data Science and Data Collection*
Unit 2: *Analyzing Data Using Statistics*
Unit 3: *Predicting and Modeling Using Data*
Unit 4: *Maintaining a Professional and Ethical Data Science Practice*

The learning objectives and curriculum of introductory data science courses vary, so this textbook aims to provide broader and more detailed coverage than an average single-semester course. Instructors can choose which chapters or sections they want to include in their particular course.

To enable this flexibility, chapters in this text can be used in a self-contained manner, although most chapters do cross-reference sections and chapters that precede or follow. More importantly, the authors have taken care to build topics gradually, from chapter to chapter, so instructors should bear this in mind when considering alternate sequence coverage.

Unit 1: Introducing Data Science and Data Collection starts off with **Chapter 1**'s explanation of the data science cycle (data collection and preparation, data analysis, and data reporting) and its practical applications in fields such as medicine, engineering, and business. **Chapter 1** also describes various types of datasets and provides the student with basic data summary tools from the Python `pandas` library. **Chapter 2** describes the processes of data collection and cleaning and the challenges of managing large datasets. It previews some of the qualitative ethical considerations that Chapters 7 and 8 later expand on.

Unit 2: Analyzing Data Using Statistics forms a self-contained unit that instructors may assign on a modular, more optional basis, depending on students' prior coursework. **Chapter 3** focuses on measures of center, variation, and position, leading up to probability theory and illustrating how to use Python with binomial and normal distributions. **Chapter 4** goes deeper into statistical analysis, demonstrating how to use Python to calculate confidence intervals, conduct hypothesis tests, and perform correlation and regression analysis.

The three chapters in ***Unit 3: Predicting and Modeling Using Data*** form the core of the book. **Chapter 5** introduces students to the concept and practical applications of time series. Chapter 5 provides focused examples of both Python and Excel techniques useful in forecasting time series, analyzing seasonality, and identifying measures of error. **Chapter 6** starts with distinguishing supervised vs. unsupervised machine learning and then develops some common methods of data classification, including logistic regression, clustering algorithms, and decision trees. Chapter 6 includes Python techniques for more sophisticated

statistical analyses such as regression with bootstrapping and multivariable regression. Finally, Chapter 6 refers back to the topics of data mining and big data introduced in Chapter 2.

Chapter 7 is a pedagogically rich chapter, with a balance of quantitative and qualitative content, covering the role of neural networks in deep learning and applications in large language models. The first four sections discuss the topics of neural networks (standard, recurrent, and convolutional), backpropagation, and deep learning. The real-life application of classifying handwritten numerals is used as an example. The last section dives into the important and rapidly changing technology of natural language processing (NLP), large language models (LLMs), and artificial intelligence (AI). While in-depth coverage of these evolving subjects is beyond the scope of this textbook, the pros/cons, the examples from technical and artistic applications, and the online resources provided in this section all serve as a good starting point for classroom discussion. This topic also naturally segues into the broader professional responsibility discussed in **Chapter 8**.

The final chapters in ***Unit 4: Maintaining a Professional and Ethical Data Science Practice*** help the student apply and adjust the specific techniques learned in the previous chapters to the real-life data analysis, decision-making, and communication situations they will encounter professionally. **Chapter 8** emphasizes the importance of ethical considerations along each part of the cycle: data collection; data preparation, analysis, and modeling; and reporting and visualization. Coverage of the issues in this chapter makes students aware of the subjective and sensitive aspects of privacy and informed consent at every step in the process. At the professional level, students learn more about the evolving standards for the relatively new field of data science, which may differ among industries or between the United States and other countries.

Chapter 9 circles back to some of the statistical concepts introduced in Chapters 3 and 4, with an emphasis on clear visual analysis of data trends. Chapter 9 provides a range of Python techniques to create boxplots and histograms for univariate data; to create line charts and trend curves for time series; to graph binomial, Poisson, and normal distributions; to generate heatmaps from geospatial data; and to create correlation heatmaps from multidimensional data.

Chapter 10 brings the student back to the practical decision-making setting introduced in Chapter 1. Chapter 10 helps the student address how to tailor the data analysis and presentation to the audience and purpose, how to validate the assumptions of a model, and how to write an effective executive summary.

The four **Appendices (A–D)** provide a practical set of references for Excel commands and commands for R statistical software as well as Python commands and algorithms. **Appendix A** uses a baseball dataset from Chapter 1 to illustrate basic Microsoft® Excel® software commands for manipulating, analyzing, summarizing, and graphing data. **Appendix B** provides a brief overview of data analysis with the open-source statistical computing package R (https://openstax.org/r/project), using a stock price example. **Appendix C** lists the approximately 60 Python algorithms used in the textbook, and **Appendix D** lists the code and syntax for the approximately 75 Python functions demonstrated in the textbook. Both **Appendices C and D** are organized in a tabular format, in consecutive chapter order, hyperlinked to the first significant use of each Python algorithm and function. (Instructors may find Appendices C and D especially useful in developing their teaching plan.)

Pedagogical Foundation

Because this is a practical, introductory-level textbook, math equations and code are presented clearly throughout. Particularly in the core chapters, students are introduced to key mathematical concepts, equations, and formulas, often followed by numbered Example Problems that encourage students to apply the concepts just covered in a variety of situations. Technical illustrations and Python code feature boxes build on and supplement the theory. Students are encouraged to try out the Python code from the feature boxes in the Google Colaboratory (https://openstax.org/r/colabresearch) (Colab) platform.

The authors have included a diverse mix of data types and sources for analysis, illustration, and discussion purposes. Some scenarios are fictional and/or internal to standard Python libraries, while other datasets come from external, real-world sources, both corporate and government (such as Federal Reserve Economic Data

(FRED) (https://openstax.org/r/nasdaq1), Statista (https://openstax.org/r/statista1), and Nasdaq (https://openstax.org/r/nasdaq)). Most scenarios are either summarized in an in-line table, or have datasets provided in a downloadable student spreadsheet for import as a .CSV file (Chapter 1 also discusses the .JSON format) and/or with a hyperlink to the external source. Some examples focus on scientific topics (e.g., the "classic" Iris flower dataset, annual temperature changes), while other datasets reflect phenomena with more nuanced socioeconomic issues (gender-based salary differences, cardiac disease markers in patients).

While the book's foundational chapters illustrate practical "techniques and tools," the more process-oriented chapters iteratively build on and emphasize an underlying framework of professional, responsible, and ethical data science practice. Chapter 1 refers the student to several national and international data science organizations that are developing professional standards. Chapter 2 emphasizes avoiding bias in survey and sample design. Chapter 8 discusses relevant privacy legislation. For further class exploration, Chapters 7 and 8 include online resources on mitigating bias and discrimination in machine learning models, including related Python libraries such as HolisticAI (https://openstax.org/r/holistic) and Fairlens (https://openstax.org/r/projectfairlens). Chapter 10 references several executive dashboards that support transparency in government.

The Group Projects at the end of each chapter encourage students to apply the techniques and considerations covered in the book using either datasets already provided or new data sources that they might receive from their instructors or in their own research. For example, project topics include the following: collecting data on animal extinction due to global warming (Chapter 2), predicting future trends in stock market prices (Chapter 5), diagnosing patients for liver disease (Chapter 7), and analyzing the severity of ransomware attacks (Chapter 8).

Key Features

The key in-chapter features, depending on chapter content and topics, may include the following:

- Learning Outcomes (LOs) to guide the student's progress through the chapter
- Example Problems, demonstrating calculations and solutions in-line
- Python code boxes, providing sample input code for and output from Google Colab
- Note boxes providing instructional tips to help with the practical aspects of the math and coding
- Data tables from a variety of social science and industry settings
- Technical charts and heatmaps to visually demonstrate code output and variable relationships
- Exploring Further boxes, with additional resources and online examples to extend learning
- Mathematical formulas and equations
- Links to downloadable spreadsheet containing key datasets referenced in the chapter for easy manipulation of data

End-of-chapter (EOC) elements, depending on chapter content and topics, may include the following:

- Key Terms
- Group Projects
- Chapter Review Questions
- Critical Thinking Questions
- Quantitative Problems

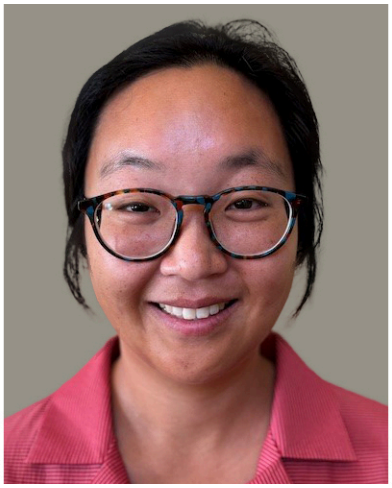
Answers [and Solutions] to Questions in the Book

The student-facing Answer Key at the end of the book provides the correct answer letter and text for Chapter Review questions (multiple-choice). An Instructor Solution Manual (ISM) will be available for verified instructors and downloadable from the restricted OpenStax Instructor Resources web page, with detailed solutions to Quantitative Problems, sample answers for Critical Thinking questions, and a brief explanation of the correct answer for Chapter Review questions. (Sample calculations, tables, code, or figures may be included, as

applicable.) An excerpt of the ISM, consisting of the solutions/sample answers for the odd-numbered questions only, will also be available as a Student Solution Manual (SSM), downloadable from the public OpenStax Student Resources web page. (Answers to the Group Projects are not provided, as they are integrative, exploratory, open-ended assignments.)

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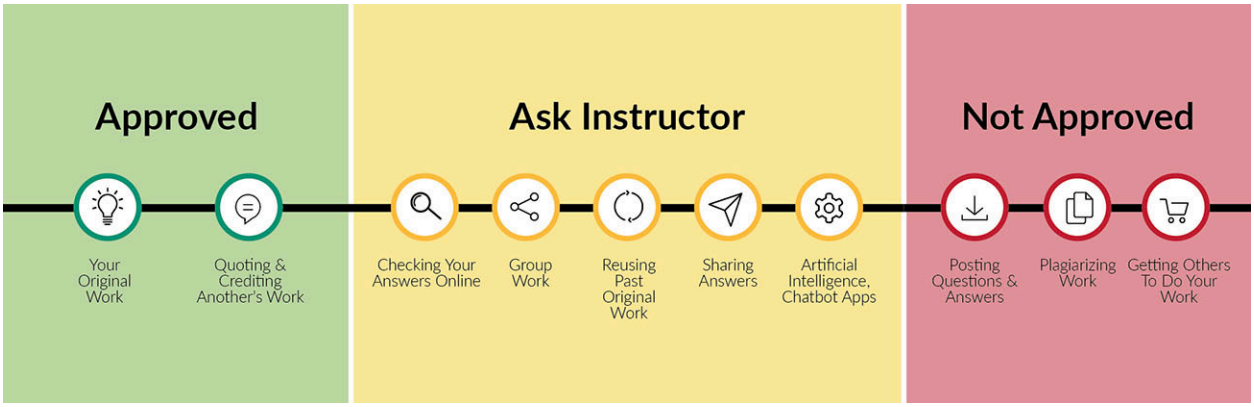
Student and Instructor Resources

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1 What Are Data and Data Science?

Figure 1.1 Petroglyphs are one of the earliest types of data generated by humanity, providing vital information about the daily life of the people who created them. (credit: modification of work "Indian petroglyphs (~100 B.C. to ~1540 A.D.) (Newspaper Rock, southeastern Utah, USA) 24" by James St. John/Flickr, CC BY 2.0)

Chapter Outline

1.1 What Is Data Science?
1.2 Data Science in Practice
1.3 Data and Datasets
1.4 Using Technology for Data Science
1.5 Data Science with Python

Introduction

Many of us use the terms "data" and "data science," but not necessarily with a lot of precision. This chapter will define data science terminology and apply the terms in multiple fields. The chapter will also briefly introduce the types of technology (such as statistical software, spreadsheets, and programming languages) that data scientists use to perform their work and will then take a deeper dive into the use of Python for data analysis. The chapter should help you build a technical foundation so that you can practice the more advanced data science concepts covered in future chapters.

1.1 What Is Data Science?

Learning Outcomes

By the end of this section, you should be able to:

- 1.1.1 Describe the goals of data science.
- 1.1.2 Explain the data science cycle and goals of each step in the cycle.
- 1.1.3 Explain the role of data management in the data science process.

Data science is a field of study that investigates how to collect, manage, and analyze data of all types in order to retrieve meaningful information. Although we will describe data in more detail in Data and Datasets, you can consider *data* to be any pieces of evidence or observations that can be analyzed to provide some insights.

In its earliest days, the work of data science was spread across multiple disciplines, including statistics, mathematics, computer science, and social science. It was commonly believed that the job of data collection, management, and analysis would be carried out by different types of experts, with each job independent of one another. To be more specific, *data collection* was considered to be the province of so-called domain experts (e.g., doctors for medical data, psychologists for psychological data, business analysts for sales, logistic, and marketing data, etc.) as they had a full context of the data; *data management* was for computer scientists/engineers as they knew how to store and process data in computing systems (e.g., a single computer, a server, a data warehouse); and *data analysis* was for statisticians and mathematicians as they knew how to derive some meaningful insights from data. Technological advancement brought about the proliferation of data, muddying the boundaries between these jobs, as shown in Figure 1.2. Now, it is expected that a data scientist or data science team will have some expertise in all three domains.

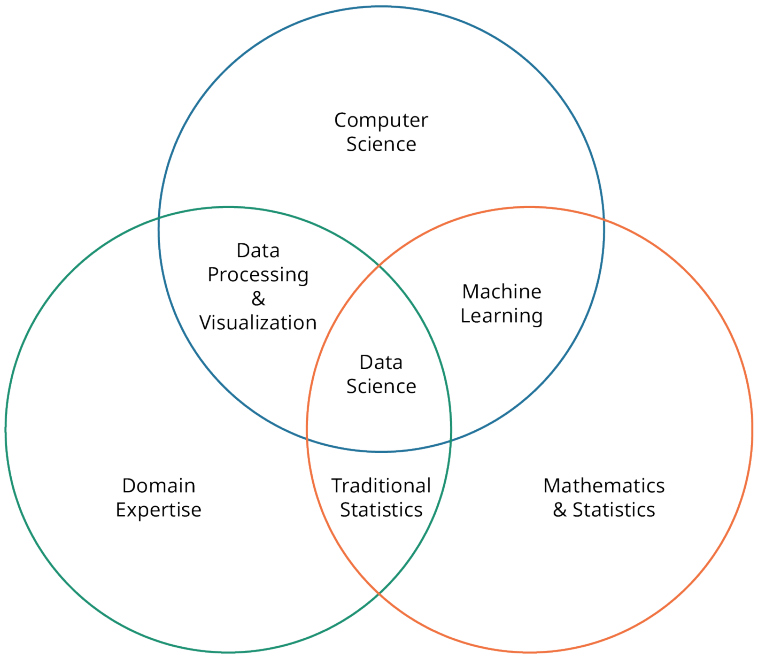


Figure 1.2 The Field of Data Science

One good example of this is the development of personal cell phones. In the past, households typically had only one landline telephone, and the only data that was generated with the telephone was the list of phone numbers called by the household members. Today the majority of consumers own a smartphone, which contains a tremendous amount of data: photos, social media contacts, videos, locations (usually), and perhaps health data (with the consumers' consent), among many other things.

Is the data from a smartphone solely collected by domain experts who are specialized in photos, videos, and such? Probably not. They are automatically logged and collected by the smartphone system itself, which is designed by computer scientists/engineers. For a health care scientist to collect data from many individuals in the "traditional" way, bringing patients into a laboratory and taking vital signs regularly over a period of time takes a lot of time and effort. A smartphone application is a more efficient and productive method, from a data collection perspective.

Data science tasks are often described as a process, and this section provides an overview for each step of that process.

The Data Science Cycle

Data science tasks follow a process, called the **data science cycle**, which includes problem definition, then data collection, preparation, analysis, and reporting, as illustrated in Figure 1.3. See this animation about data science (https://openstax.org/r/youtube), which describes the data science cycle in much the same way.

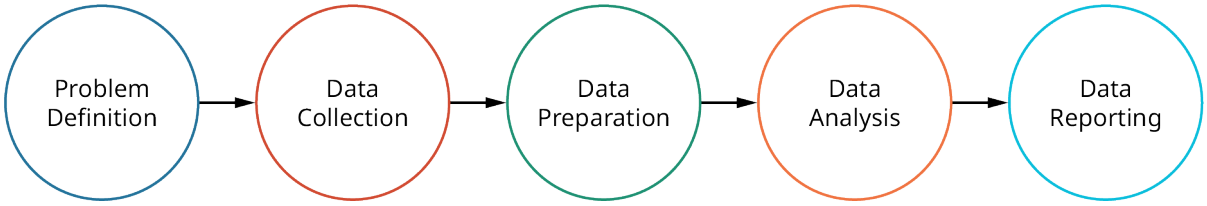


Figure 1.3 The Data Science Cycle

Although data collection and preparation may sound like simple tasks compared to the more important work of analysis, they are actually the most time- and effort-consuming steps in the data science cycle. According to a survey conducted by Anaconda (2020), data scientists spend about half of the entire process in data collection and cleaning, while data analysis and communication take about a quarter to a third of the time each, depending on the job.

Problem Definition, Data Collection, and Data Preparation

The first step in the data science cycle is a precise definition of the problem statement to establish clear objectives for the goal and scope of the data analysis project. Once the problem has been well defined, the data must be generated and collected. **Data collection** is the systematic process of gathering information on variables of interest. Data is often collected purposefully by domain experts to find answers to predefined problems. One example is data on customer responses to a product satisfaction survey. These survey questions will likely be crafted by the product sales and marketing representatives, who likely have a specific plan for how they wish to use the response data once it is collected.

Not all data is generated this purposefully, though. A lot of data around our daily life is simply a by-product of our activity. These by-products are kept as data because they could be used by others to find some helpful insights later. One example is our web search histories. We use a web search engine like Google to search for information about our interests, and such activity leaves a history of the search text we used on the Google server. Google employees can utilize the records of numerous Google users in order to analyze common search patterns, present accurate search results, and potentially, to display relatable advertisements back to the searchers.

Often the collected data is not in an optimal form for analysis. It needs to be processed somehow so that it can be analyzed, in the phase called **data preparation** or *data processing*. Suppose you work for Google and want to know what kind of food people around the globe search about the most during the night. You have users' search history from around the globe, but you probably cannot use the search history data as is. The search keywords will probably be in different languages, and users live all around the Earth, so nighttime will vary by each user's time zone. Even then, some search keywords would have some typos, simply not make sense, or even remain blank if the Google server somehow failed to store that specific search history record. Note that all these scenarios are possible, and therefore data preparation should address these issues so that the actual analysis can draw more accurate results. There are many different ways to manage these issues, which we will discuss more fully in Collecting and Preparing Data.

Data Analysis

Once the data is collected and prepared, it must be analyzed in order to discover meaningful insights, a process called **data analysis**. There are a variety of data analysis methods to choose from, ranging from simple ones like checking minimum and maximum values, to more advanced ones such as modelling a dependent variable. Most of the time data scientists start with simple methods and then move into more advanced ones, based on what they want to investigate further. Descriptive Statistics: Statistical Measurements and Probability Distributions and Inferential Statistics and Regression Analysis discuss when and how to use different analysis methods. Time Series and Forecasting and Decision-Making Using Machine Learning Basics discuss forecasting and decision-making.

Data Reporting

Data reporting involves the presentation of data in a way that will best convey the information learned from data analysis. The importance of data reporting cannot be overemphasized. Without it, data scientists cannot effectively communicate to their audience the insights they discovered in the data. Data scientists work with domain experts from different fields, and it is their responsibility to communicate the results of their analysis in a way that those domain experts can understand. **Data visualization** is a graphical way of presenting and reporting data to point out the patterns, trends, and hidden insights; it involves the use of visual elements such as charts, graphs, and maps to present data in a way that is easy to comprehend and analyze. The goal of data visualization is to communicate information effectively and facilitate better decision-making. Data visualization and basic statistical graphing, including how to create graphical presentations of data using Python, are explored in depth in Visualizing Data. Further details on reporting results are discussed in Reporting Results.

Data Management

In the early days of data analysis (when generated data was mostly structured and not quite so “big”), it was possible to keep data in local storage (e.g., on a single computer or a portable hard drive). With this setup, data processing and analysis was all done locally as well.

When so much more data began to be collected—much of it unstructured as well as structured—cloud-based management systems were developed to store all the data on a designated server, outside a local computer. At the same time, data scientists began to see that most of their time was being spent on data processing rather than analysis itself. To address this concern, modern data management systems not only store the data itself but also perform some basic processing on a cloud. These systems, referred to as **data warehousing**, store and manage large volumes of data from various sources in a central location, enabling efficient retrieval and analysis for business intelligence and decision-making. (Data warehousing is covered in more detail in Handling Large Datasets.)

Today, enterprises simply subscribe to a cloud-warehouse service such as Amazon RedShift (which runs on the Amazon Web Services) or Google BigQuery (which runs on the Google Cloud) instead of buying physical storage and configuring data management/processing systems on their own. These services ensure the data is safely stored and processed on the cloud, all without spending money on purchasing/maintaining physical storage.

1.2 Data Science in Practice

Learning Outcomes

By the end of this section, you should be able to:

- 1.2.1 Explain the interdisciplinary nature of data science in various fields.
- 1.2.2 Identify examples of data science applications in various fields.
- 1.2.3 Identify current issues and challenges in the field of data science

While data science has adopted techniques and theories from fields such as math, statistics, and computer science, its applications concern an expanding number of fields. In this section we introduce some examples of how data science is used in business and finance, public policy, health care and medicine, engineering and sciences, and sports and entertainment.

Data Science in Business

Data science plays a key role in many business operations. A variety of data related to customers, products, and sales can be collected and generated within a business. These include customer names and lists of products they have purchased as well as daily revenue. Business analytics investigate these data to launch new products and to maximize the business revenue/profit.

Retail giant Walmart is known for using business analytics to improve the company's bottom line. Walmart collects multiple *petabytes* (1 petabyte = 1,024 terabytes) of unstructured data every hour from millions of customers (commonly referred to as "big data"); as of 2024, Walmart's customer data included roughly 255 million weekly customer visits (Statista, 2024). Walmart uses this big data to investigate consumer patterns and adjust its inventories. Such analysis helps the company avoid overstocking or understocking and resulted in an estimated online sales increase of between 10% and 15%, translating to an extra $1 billion in revenue (ProjectPro, 2015). One often-reported example includes the predictive technology Walmart used to prepare for Hurricane Frances in 2004. A week before the hurricane's arrival, staff were asked to look back at their data on sales during Hurricane Charley, which hit several weeks earlier, and then come up with some forecasts about product demand ahead of Frances (Hays, 2004). Among other insights, the executives discovered that strawberry Pop-Tart sales increased by about sevenfold during that time. As a result, in the days before Hurricane Frances, Walmart shipped extra supplies of strawberry Pop-Tarts to stores in the storm's path (Hays, 2004). The analysis also provided insights on how many checkout associates to assign at different times of the day, where to place popular products, and many other sales and product details. In addition, the company has launched social media analytics efforts to investigate hot keywords on social media and promptly make related products available (ProjectPro, 2015).

Amazon provides another good example. Ever since it launched its Prime membership service, Amazon has focused on how to minimize delivery time and cost. Like Walmart, it started by analyzing consumer patterns and was able to place products close to customers. To do so, Amazon first divided the United States into eight geographic regions and ensured that most items were warehoused and shipped within the same region; this allowed the company to reduce the shipping time and cost. As of 2023, more than 76% of orders were shipped from within the customer's region, and items in same-day shipping facilities could be made ready to put on a delivery truck in just 11 minutes (Herrington, 2023). Amazon also utilizes machine learning algorithms to predict the demand for items in each region and have the highest-demand items available in advance at the fulfillment center of the corresponding region. This predictive strategy has helped Amazon reduce the delivery time for each product and extend the item selections for two-day shipping (Herrington, 2023).

Data science is utilized extensively in finance as well. Detecting and managing fraudulent transactions is now done by automated machine learning algorithms (IABAC, 2023). Based on the customer data and the patterns of past fraudulent activities, an algorithm can determine whether a transaction is fraudulent in real time. Multiple tech companies, such as IBM and Amazon Web Services, offer their own fraud detection solutions to their corporate clients. (For more information, see this online resource on Fraud Detection through Data Analytics (https://openstax.org/r/iabac).)

Data Science in Engineering and Science

Various fields of engineering and science also benefit from data science. Internet of Things (IoT) is a good example of a new technology paradigm that has benefited from data science. **Internet of Things (IoT)** describes the network of multiple objects interacting with each other through the Internet. Data science plays a crucial role in these interactions since behaviors of the objects in a network are often triggered by data collected by another object in the network. For example, a smart doorbell or camera allows us to see a live stream on our phone and alerts us to any unusual activity.

In addition, weather forecasting has always been a data-driven task. Weather analysts collect different measures of the weather such as temperature and humidity and then make their best estimate for the weather in the future. Data science has made weather forecasting more reliable by adopting more sophisticated prediction methods such as time-series analysis, artificial intelligence (AI), and machine learning (covered in Time Series and Forecasting, Decision-Making Using Machine Learning Basics, and Deep Learning and AI Basics). Such advancement in weather forecasting has also enabled engineers and scientists to predict some natural disasters such as flood or wildfire and has enabled precision farming, with which agricultural engineers can identify an optimal time window to plant, water, and harvest crops. For example, an agronomy

consultant, Ag Automation, has partnered with Hitachi to offer a solution that both automates data collection and remotely monitors and controls irrigation for the best efficiency (Hitachi, 2023).

EXPLORING FURTHER

Using AI for Irrigation Control

See this Ag Automation video (https://openstax.org/r/youtube128) demonstrating the use of data collection for controlling irrigation.

Data Science in Public Policy

Smart cities are among the most representative examples of the use of data science in public policy. Multiple cities around the world, including Masdar City in the United Arab Emirates and Songdo in South Korea, have installed thousands of data-collecting sensors used to optimize their energy consumption. The technology is not perfect, and smart cities may not yet live up to their full potential, but many corporations and companies are pursuing the goal of developing smart cities more broadly (Clewlow, 2024). The notion of a smart city has also been applied on a smaller scale, such as to a parking lot, a building, or a street of lights. For example, the city of San Diego installed thousands of sensors on the city streets to control the streetlights using data and smart technology. The sensors measure traffic, parking occupancy, humidity, and temperature and are able to turn on the lights only when necessary (Van Bocxlaer, 2020). New York City has adopted smart garbage bins that monitor the amount of garbage in a bin, allowing garbage collection companies to route their collection efforts more efficiently (Van Bocxlaer, 2020).

EXPLORING FURTHER

Sensor Networks to Monitor Energy Consumption

See how Songdo (https://openstax.org/r/youtube4) monitors energy consumption and safety with sensor networks.

Data Science in Education

Data science also influences education. Traditional instruction, especially in higher education, has been provided in a one-size-fits-all form, such as many students listening to a single instructor's lecture in a classroom. This makes it difficult for an instructor to keep track of each individual student's learning progress. However, many educational platforms these days are online and can produce an enormous amount of student data, allowing instructors to investigate everyone's learning based on these collected data. For example, online learning management systems such as Canvas (https://openstax.org/r/instructure) compile a grade book in one place, and online textbooks such as zyBooks (https://openstax.org/r/zybooks) collect students' mastery level on each topic through their performance on exercises. All these data can be used to capture each student's progress and offer personalized learning experiences such as intelligent tutoring systems or adaptive learning. ALEKS (https://openstax.org/r/aleks), an online adaptive learning application, offers personalized material for each learner based on their past performance.

Data Science in Health Care and Medicine

The fields of health care and medicine also use data science. Often their goal is to offer more accurate diagnosis and treatment using **predictive analytics**—statistical techniques, algorithms, and machine learning that analyze historical data and make predictions about future events. Medical diagnosis and prescription practices have traditionally been based on a patient's verbal description of symptoms and a doctor's or a

group of doctors' experience and intuition; this new movement allows health care professionals to make decisions that are more data-driven. Data-driven decisions became more feasible thanks to all the personal data collected through personal gadgets—smartphones, smart watches, and smart bands. Such devices collect daily health/activity records, and this in turn helps health care professionals better capture each patient's situation. All this work will enable patients to receive more accurate diagnoses along with more personalized treatment regimens in the long run.

The Precision Medicine Initiative (https://openstax.org/r/obamawhitehouse) is a long-term research endeavor carried out by the National Institutes of Health (NIH) and other research centers, with the goal of better understanding how a person's genetics, environment, and lifestyle can help determine the best approach to prevent or treat disease. The initiative aims to look at as much data as possible to care for a patient's health more proactively. For example, the initiative includes genome sequencing to look for certain mutations that indicate a higher risk of cancer or other diseases.

Another application of data science in health focuses on lowering the cost of health care services. Using historical records of patients' symptoms and prescription, a chatbot that is powered by artificial intelligence can provide automated health care advice. This will reduce the need for patients to see a pharmacist or doctor, which in turn improves health care accessibility for those who are in greater need.

EXPLORING FURTHER

Big Data In Health Care

The 2015 launch of the National Institutes of Health Precision Medicine Initiative was documented in One Woman's Quest to Cure Her Fatal Brain Disease (https://openstax.org/r/youtubevk). "Promise of Precision Medicine" signaled a new approach to health care in the United States—one heavily reliant on big data.

Data Science in Sports and Entertainment

Data science is prevalent in the sports and the entertainment industry as well. Sports naturally produce much data—about the player, positions, teams, seasons, and so on. Therefore, just as there is the concept of business analytics, the analysis of such data in sports is called **sports analytics**. For example, the Oakland Athletics baseball team famously analyzed player recruitment for the 2002 season. The team's management adapted a statistical approach referred to as **sabermetrics** to recruit and position players. With sabermetrics, the team was able to identify critical yet traditionally overlooked metrics such as on-base percentage and slugging percentage. The team, with its small budget compared to other teams, recruited a number of undervalued players with strong scores on these metrics, and in the process, they became one of the most exciting teams in baseball that year, breaking the American League record for 20 wins in a row. Does this story sound familiar? This story was so dramatic that Michael Lewis wrote a book about it, which was also made into a movie: *Moneyball*.

EXPLORING FURTHER

The Sabermetrics YouTube Channel

Sabermetrics is so popular that there is a YouTube channel devoted to it: Simple Sabermetrics (https://openstax.org/r/simplesabermetrics), with baseball animations and tutorials explaining how data impacts the way today's game is played behind the scenes.

In the entertainment industry, data science is commonly used to make data-driven, personalized suggestions that satisfy consumers known as **recommendation systems**. One example of a recommendation system is on

video streaming services such as Netflix. Netflix Research considers subscribers' watch histories, satisfaction with the content, and interaction records (e.g., search history). Their goal is to make perfect personalized recommendations despite some challenges, including the fact that subscribers themselves often not do not know what they want to see.

EXPLORING FURTHER

Careers in Data Science

As you advance in your data science training, consider the many professional options in this evolving field. This helpful graphic from edX (https://openstax.org/r/edx) distinguishes data analyst vs. data science paths. This Coursera article (https://openstax.org/r/coursera) lists typical skill sets by role. Current practitioner discussions are available on forums such as Reddit's r/data science (https://openstax.org/r/reddit).

Trends and Issues in Data Science

Technology has made it possible to collect abundant amounts of data, which has led to challenges in the processing and analyzing of that data. But technology comes to the rescue again! Data scientists now use machine learning to better understand the data, and artificial intelligence can make an automated, data-driven decision on a task. Decision-Making Using Machine Learning Basics and Deep Learning and AI Basics will cover the details of machine learning and artificial intelligence.

With these advances, many people have raised concerns about ethics and privacy. Who is allowed to collect these data, and who has access to them? None of us want someone else to use our personal data (e.g., contact information, health records, location, photos, web search history) without our consent or without knowing the risk of sharing our data. Machine learning algorithms and artificial intelligence are trained to make a decision based on the past data, and when the past data itself inherits some bias, the trained machine learning algorithms and artificial intelligence will make biased decisions as well. Thus, carefully attending to the process of collecting data and evaluating the bias of a trained results is critical. Ethics Throughout the Data Science Cycle will discuss these and other ethical concerns and privacy issues in more depth.

1.3 Data and Datasets

Learning Outcomes

By the end of this section, you should be able to:

- 1.3.1 Define data and dataset.
- 1.3.2 Differentiate among the various data types used in data science.
- 1.3.3 Identify the type of data used in a dataset.
- 1.3.4 Discuss an item and attribute of a dataset.
- 1.3.5 Identify the different data formats and structures used in data science.

What Is Data Science? and Data Science in Practice introduced the many varieties of and uses for data science in today's world. Data science allows us to extract insights and knowledge from data, driving decision-making and innovation in business, health care, entertainment, and so on. As we've seen, the field has roots in math, statistics, and computer science, but it only began to emerge as its own distinct field in the early 2000s with the proliferation of digital data and advances in computing power and technology. It gained significant momentum and recognition around the mid to late 2000s with the rise of big data and the need for sophisticated techniques to analyze and derive insights from large and complex datasets. Its evolution since then has been rapid, and as we can see from the previous discussion, it is quickly becoming a cornerstone of many industries and domains.

Data, however, is not new! Humans have been collecting data and generating datasets from the beginning of

time. This started in the Stone Age when people carved some shapes and pictures, called petroglyphs, on rock. The petroglyphs provide insights on how animals looked and how they carried out their daily life, which is valuable "data" for us. Ancient Egyptians invented a first form of paper—papyrus—in order to journal *their* data. Papyrus also made it easier to store data in bulk, such as listing inventories, noting financial transactions, and recording a story for future generations.

Data

"Data" is the plural of the Latin word "datum," which translates as something that is given or used and is often used to mean a single piece of information or a single point of reference in a dataset. When you hear the word "data," you may think of some sort of "numbers." It is true that numbers are usually considered data, but there are many other forms of data all around us. Anything that we can analyze to compile **information**—high-level insights—is considered **data**.

Suppose you are debating whether to take a certain course next semester. What process do you go through in order to make your decision? First, you might check the course evaluations, as shown in Table 1.1.

Semester	Instructor	Class Size	Rating
Fall 2020	A	100	Not recommended at all
Spring 2021	A	50	Highly recommended
Fall 2021	B	120	Not quite recommended
Spring 2022	B	40	Highly recommended
Fall 2022	A	110	Recommended
Spring 2023	B	50	Highly recommended

Table 1.1 Course Evaluation Records

The evaluation record consists of four kinds of data, and they are grouped as columns of the table: Semester, Instructor, Class Size, and Rating. Within each column there are six different pieces of data, located at each row. For example, there are six pieces of text data under the Semester column: "Fall 2020," "Spring 2021," "Fall 2021," "Spring 2022," "Fall 2022," and "Spring 2023."

The course evaluation ratings themselves do not provide an idea on whether to take the course next semester. The ratings are just a *phrase* (e.g., "Highly recommended" or "Not quite recommended") that *encodes* how recommended the course was in that semester. You need to analyze them to come up with a decision!

Now let's think about how to derive the information from these ratings. You would probably look at all the data, including when in the semester the course was offered, who the instructor is, and class size. These records would allow you to derive information that would help you decide "whether or not to take the course next semester."

EXAMPLE 1.1

Problem

Suppose you want to decide whether or not to put on a jacket today. You research the highest temperatures in the past five days and determine whether you needed a jacket on each day. In this scenario, what data are you using? And what information are you trying to derive?

Solution

Temperature readings and whether you needed a jacket on each of the past five days are two kinds of data you are referring to. Again, they do not indicate anything related to wearing a jacket today. They are simply five pairs of numbers (temperature) and yes/no (whether you needed a jacket) records, with each pair representing a day. Using these data, you are deriving information that you can analyze to help you decide whether to wear a jacket today.

Types of Data

The previous sections talked about how much our daily life is surrounded by data, how much our daily life itself produces new data, and how often we make data-driven decisions without even noticing it. You might have noticed that data comes in various types. Some data are quantitative, which means that they are measured and expressed using numbers. **Quantitative data** deals with quantities and amounts and is usually analyzed using statistical methods. Examples include numerical measurements like height, weight, temperature, heart rate, and sales figures. **Qualitative data** are non-numerical data that generally describe subjective attributes or characteristics and are analyzed using methods such as thematic analysis or content analysis. Examples include descriptions, observations, interviews, and open-ended survey responses (as we'll see in Survey Design and Implementation) that address unquantifiable details (e.g., photos, posts on Reddit). The data types often dictate methods for data analysis, so it is important to be able to identify a type of data. Thus, this section will take a further dive into types of data.

Let's revisit our previous example about deciding whether to take a certain course next semester. In that example, we referred to four pieces of data. They are encoded in different types such as numbers, words, and symbols.

1. The semester the course was offered—Fall 2020, Spring 2021, ..., Fall 2022, Spring 2023
2. The instructor—A and B
3. The class size—100, 50, 120, 40, 110, 50
4. The course rating—"Not recommended at all," ..., "Highly recommended"

There are two primary types of quantitative data—*numeric* and *categorical*—and each of these can be divided into a few subtypes. **Numeric data** is represented in numbers that indicate measurable quantities. It may be followed by some symbols to indicate units. Numeric data is further divided into *continuous data* and *discrete data*. With **continuous data**, the values can be any number. In other words, a value is chosen from an infinite set of numbers. With **discrete data**, the values follow a specific precision, which makes the set of possible values finite.

From the previous example, the class size 100, 150, etc. are numbers with the implied unit "students." Also, they indicate measurable quantities as they are head counts. Therefore, the class size is *numeric data*. It is also *continuous data* since the size numbers seem to be any natural numbers and these numbers are chosen from an infinite set of numbers, the set of natural numbers. Note that whether data is continuous (or discrete) also depends on the context. For example, the same class size data can be discrete if the campus enforces all classes to be 200 seats or less. Such restriction makes the class size values be chosen from a finite set of 200 numbers: 1, 2, 3, ..., 198, 199, 200.

Categorical data is represented in different forms such as words, symbols, and even numbers. A categorical value is chosen from a finite set of values, and the value does not necessarily indicate a measurable quantity. Categorical data can be divided into *nominal data* and *ordinal data*. For **nominal data**, the set of possible values does not include any ordering notion, whereas with **ordinal data**, the set of possible values includes an ordering notion.

The rest—semester, instructor, and ratings—are *categorical data*. They are represented in symbols (e.g., "Fall 2020," "A") or words (e.g., "Highly recommended"), and these values are chosen from the finite set of those symbols and words (e.g., A vs. B). The former two data are nominal since the semester and instructor do not have orders to follow, while the latter is ordinal since there is a notion of degree (Not recommended at all ~ Highly recommended). You may argue that the semester could have *chronological* ordering: Fall 2020 comes before Spring 2021, Fall 2021 follows Fall 2020. If you want to value that notion for your analysis, you could consider the semester data to be ordinal as well—the chronological ordering is indeed critical when you are looking at a time-series dataset. You will learn more about that in Time Series and Forecasting.

EXAMPLE 1.2

Problem

Consider the jacket scenario in Example 1.1. In that example, we referred to two kinds of data:

1. The temperature during past three days—90°F, 85°F, ...
2. On each of those days, whether you needed a jacket—Yes, No, ...

What is the type of each data?

Solution

The temperatures are numbers followed by the unit degrees Fahrenheit (°F). Also, they indicate measurable quantities as they are specific readings from a thermometer. Therefore, the temperature is numeric data. They are also continuous data since they can be any real number, and the set of real numbers is infinite.

The other type of data—whether or not you needed a jacket—is categorical data. Categorical data are represented in symbols (Yes/No), and the values are chosen from the finite set of those symbols. They are also nominal since Yes/No does not have ordering to follow.

Datasets

A **dataset** is a collection of observations or data entities organized for analysis and interpretation, as shown in Table 1.1. Many datasets can be represented as a table where each row indicates a unique data entity and each column defines the structure of the entities.

Notice that the dataset we used in Table 1.1 has six *entities* (also referred to as **items**, *entries*, or *instances*), distinguished by semester. Each entity is defined by a combination of four **attributes** or characteristics (also known as *features* or *variables*)—Semester, Instructor, Class Size, and Rating. A combination of features characterizes an entry of a dataset.

Although the actual values of the attributes are different across entities, note that all entities have values for the same four attributes, which makes them a **structured dataset**. As a structured dataset, these items can be listed as a table where each item is listed along the rows of the table.

By contrast, an **unstructured dataset** is one that lacks a predefined or organized data model. While structured datasets are organized in a tabular format with clearly defined fields and relationships, unstructured data lacks a fixed schema. Unstructured data is often in the form of text, images, videos, audio recordings, or other content where the information doesn't fit neatly into rows and columns.

There are plenty of unstructured datasets. Indeed, some people argue there are more unstructured datasets than structured ones. A few examples include Amazon reviews on a set of products, Twitter posts last year, public images on Instagram, popular short videos on TikTok, etc. These unstructured datasets are often processed into a structured one so that data scientists can analyze the data. We'll discuss different data processing techniques in Collecting and Preparing Data.

EXAMPLE 1.3

Problem

Let's revisit the jacket example: deciding whether to wear a jacket to class. Suppose the dataset looks as provided in Table 1.2:

Date	Temperature	Needed a Jacket?
Oct. 10	80°F	No
Oct. 11	60°F	Yes
Oct. 12	65°F	Yes
Oct. 13	75°F	No

Table 1.2 Jacket Dataset

Is this dataset structured or unstructured?

Solution

It is a structured dataset since 1) every individual item is in the same structure with the same three attributes—Date, Temperature, and Needed a Jacket—and 2) each value strictly fits into a cell of a table.

EXAMPLE 1.4

Problem

How many entries and attributes does the dataset in the previous example have?

Solution

The dataset has four entries, each of which is identified with a specific date (Oct. 10, Oct. 11, Oct. 12, Oct. 13). The dataset has three attributes—Date, Temperature, Needed a Jacket.

EXAMPLE 1.5

Problem

A dataset has a list of keywords that were searched on a web search engine in the past week. Is this dataset structured or unstructured?

Solution

The dataset is an unstructured dataset since each entry in the dataset can be a freeform text: a single word, multiple words, or even multiple sentences.

EXAMPLE 1.6

Problem

The dataset from the previous example is processed so that now each search record is summarized as up to three words, along with the timestamp (i.e., when the search occurred). Is this dataset structured or unstructured?

Solution

It is a structured dataset since every entry of this dataset is in the same structure with two attributes: a short keyword along with the timestamp.

Dataset Formats and Structures (CSV, JSON, XML)

Datasets can be stored in different formats, and it's important to be able to recognize the most commonly used formats. This section covers three of the most often used formats for structured datasets—**comma-separated values (CSV)**, **JavaScript Object Notation (JSON)**, and **Extensible Markup Language (XML)**. While CSV is the most intuitive way of encoding a tabular dataset, much of the data we would collect from the web (e.g., websites, mobile applications) is stored in the JSON or XML format. The reason is that JSON is the most suitable for exchanging data between a user and a server, and XML is the most suitable for complex dataset due to its hierarchy-friendly nature. Since they all store data as plain texts, you can open them using any typical text editors such as Notepad, Visual Studio Code, Sublime Text, or VI editor.

Table 1.3 summarizes the advantages and disadvantages of CSV, JSON, and XML dataset formats. Each of these is described in more detail below.

Dataset Format	Pros	Cons	Typical Use
CSV	• Simple	• Difficult to add metadata • Difficult to parse if there are special characters • Flat structure	• Tabular data
JSON	• Simple • Compatible with many languages • Easy to parse	• Difficult to add metadata • Cannot leave comments	• Data that needs to be exchanged between a user and a server
XML	• Structured (so more readable) • Possible to add metadata	• Verbose • Complex structure with tags	• Hierarchical data structures

Table 1.3 Summary of the CSV, JSON, and XML formats

EXPLORING FURTHER

Popular and Reliable Databases to Search for Public Datasets

Multiple online databases offer public datasets for free. When you want to look for a dataset of interest, the following sources can be your initial go-to.

Government data sources include:

Data.gov (https://openstax.org/r/datagov)

Bureau of Labor Statistics (BLS) (https://openstax.org/r/blsgov1)

National Oceanic and Atmospheric Administration (NOAA) (https://openstax.org/r/noaagov)

World Health Organization (WHO) (https://openstax.org/r/who)

Some reputable nongovernment data sources are:

Kaggle (https://openstax.org/r/kaggle1)

Statista (https://openstax.org/r/statista)

Pew Research Center (https://openstax.org/r/pewresearch)

Comma-Separated Values (CSV)

The CSV stores each item in the dataset in a single line. Variable values for each item are listed all in one line, separated by commas (","). The previous example about signing up for a course can be stored as a CSV file. Figure 1.4 shows how the dataset looks when opened with a *text editor* (e.g., Notepad, TextEdit, MS Word, Google Doc) or programming software in the form of a *code editor* (e.g., Sublime Text, Visual Studio Code, XCode). Notice that commas are used to separate the attribute values within a single line (see Figure 1.4).

```
1  Semester,Instructor,Class Size,Rating
2  Fall 2020,A,100,Not recommended at all
3  Spring 2021,A,50,Highly recommended
4  Fall 2021,B,120,Not quite recommended
5  Spring 2022,B,40,Highly recommended
6  Fall 2022,A,110,Recommended
7  Spring 2023,B,50,Highly recommended
```

Figure 1.4 ch1-courseEvaluations.csv Opened with Visual Studio Code

COMMA FOR NEW LINE?

There is some flexibility on how to end a line with CSV files. It is acceptable to end with or without commas, as some software or programming languages automatically add a comma when generating a CSV dataset.

CSV files can be opened with spreadsheet software such as MS Excel and Google Sheets. The spreadsheet software visualizes CSV files more intuitively in the form of a table (see Figure 1.5). We will cover the basic use of Python for analyzing CSV files in Data Science with Python.

	A	B	C	D
1	Semester	Instructor	Class Size	Rating
2	Fall 2020	A	100	Not recommended at all
3	Spring 2021	A	50	Highly recommended
4	Fall 2021	B	120	Not quite recommended
5	Spring 2022	B	40	Highly recommended
6	Fall 2022	A	110	Recommended
7	Spring 2023	B	50	Highly recommended
8				

Figure 1.5 ch1-courseEvaluations.csv Opened with Microsoft Excel (Used with permission from Microsoft)

HOW TO DOWNLOAD AND OPEN A DATASET FROM THE CH1-DATA SPREADSHEET IN THIS TEXT

A spreadsheet file accompanies each chapter of this textbook. The files include multiple tabs corresponding to a single dataset in the chapter. For example, the spreadsheet file for this chapter (https://openstax.org/r/spreadsheet4) is shown in Figure 1.6. Notice that it includes multiple tabs with the names "ch1-courseEvaluations.csv," "ch1-cancerdoc.csv," and "ch1-riris.csv," which are the names of each dataset.

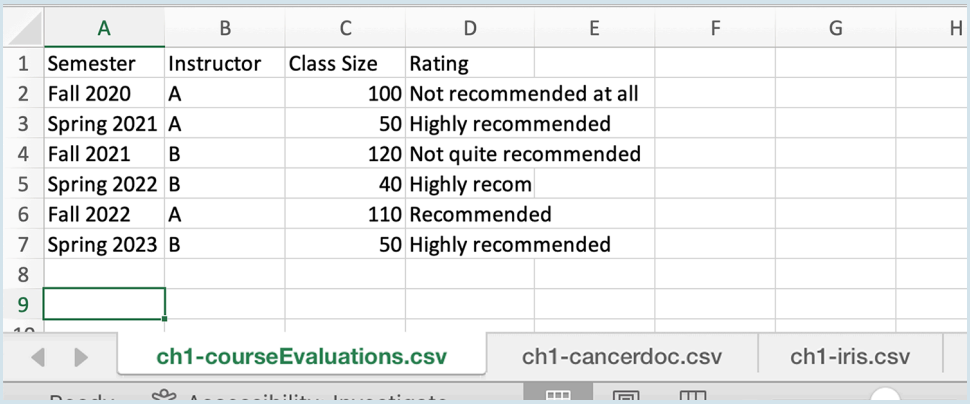


Figure 1.6 The dataset spreadsheet file for Chapter 1 (Used with permission from Microsoft)

To save each dataset as a separate CSV file, choose the tab of your interest and select File > Save As ... > CSV File Format. This will only save the current tab as a CSV file. Make sure the file name is set correctly; it may have used the name of the spreadsheet file—"ch1-data.xlsx" in this case. You should name the generated CSV file as the name of the corresponding tab. For example, if you have generated a CSV file for the first tab of "ch1-data.xlsx," make sure the generated file name is "ch1-courseEvaluations.csv." This will prevent future confusion when following instructions in this textbook.

JavaScript Object Notation (JSON)

JSON uses the syntax of a programming language named JavaScript. Specifically, it follows JavaScript's object syntax. Don't worry, though! You do not need to know JavaScript to understand the JSON format.

Figure 1.7 provides an example of the JSON representation of the same dataset depicted in Figure 1.6.


```
{"Semester":"Fall 2020","Instructor":"A","Class Size":100,"Rating":"Not recommended at all"}
{"Semester":"Spring 2021","Instructor":"A","Class Size":50,"Rating":"Highly recommended"}
{"Semester":"Fall 2021","Instructor":"B","Class Size":120,"Rating":"Not quite recommended"}
{"Semester":"Spring 2022","Instructor":"B","Class Size":40,"Rating":"Highly recommended"}
{"Semester":"Fall 2022","Instructor":"A","Class Size":110,"Rating":"Recommended"}
{"Semester":"Spring 2023","Instructor":"B","Class Size":50,"Rating":"Highly recommended"}
```


Figure 1.7 CourseEvaluations.json Opened with Visual Studio Code

Notice that the JSON format starts and ends with a pair of curly braces ({}). Inside, there are multiple pairs of

two fields that are separated by a colon (:). These two fields that are placed on the left and right of the colon are called a key and value, respectively,—key : value. For example, the dataset in Figure 1.7 has five pairs of key-values with the key "Semester": "Fall 2020", "Semester": "Spring 2021", "Semester": "Fall 2021", "Semester": "Spring 2022", "Semester": "Fall 2022", and "Semester": "Spring 2023".

CourseEvaluations.json (https://openstax.org/r/filed1v) has one key-value pair at the highest level: "Members": [...]. You can see that each item of the dataset is listed in the form of an *array* or list under the key "Members". Inside the array, each item is also bound by curly braces and has a list of key-value pairs, separated by commas. Keys are used to describe attributes in the dataset, and values are used to define the corresponding values. For example, the first item in the JSON dataset above has four keys, each of which maps to each attribute—Semester, Instructor, Class Size, and Rating. Their values are "Fall 2020", "A", 100, and "Not recommended at all".

Extensible Markup Language (XML)

The XML format is like JSON, but it lists each item of the dataset using different symbols named *tags*. An **XML tag** is any block of text that consists of a pair of angle brackets (< >) with some text inside. Let's look at the example XML representation of the same dataset in Figure 1.8. Note that the screenshot of CourseEvaluations.xml below only includes the first three items in the original dataset.

```
<Members>
    <Evaluation>
        <Semester>"Fall 2020"</Semester>
        <Instructor>"A"</Instructor>
        <Classsize>100</Classsize>
        <Rating>"Not recommended at all"</Rating>
    </Evaluation>
    <Evaluation>
        <Semester>"Spring 2021"</Semester>
        <Instructor>"A"</Instructor>
        <Classsize>50</Classsize>
        <Rating>"Highly recommended"</Rating>
    </Evaluation>
    <Evaluation>
        <Semester>"Fall 2021"</Semester>
        <Instructor>"B"</Instructor>
        <Classsize>120</Classsize>
        <Rating>"Not quite recommended"</Rating>
    </Evaluation>
</Members>
```

Figure 1.8 ch1-courseEvaluations.xml with the First Three Entries Only, Opened with Visual Studio Code

CourseEvaluations.xml lists each item of the dataset between a pair of tags, <members> and </members>. Under <members>, each item is defined between <evaluation> and </evaluation>. Since the dataset in Figure 1.8 has three items, we can see three blocks of <evaluation> ... </evaluation>. Each item has four attributes, and they are defined as different XML tags as well—<semester>, <instructor>, <classsize>, and <rating>. They are also followed by closing tags such as </semester>, </instructor>, </classsize>, and </rating>.

PubMed datasets (https://openstax.org/r/pubmed1) provides a list of articles that are published in the National Library of Medicine in XML format. Click Annual Baseline and download/open any .xml file. Note that all the .xml files are so big that they are compressed to .gz files. However, once you download one and attempt to

open it by double-clicking, the file will automatically be decompressed and open. You will see a bunch of XML tags along with information about numerous publications, such as published venue, title, published date, etc.

XML and Image Data

The XML format is also commonly used as an attachment to some image data. It is used to note supplementary information about the image. For example, the Small Traffic Light Dataset (https://openstax.org/r/traffic) in Figure 1.9 comes with a set of traffic light images, placed in one of the three directories: `JPEGImages`, `train_images`, and `valid_images`. Each image directory is accompanied with another directory just for annotations such as `Annotations`, `train_annotations`, and `valid_annotations`.

```
├── Annotations
├── class.txt
├── JPEGImages
├── train_annotations
├── train_images
├── valid_annotations
└── valid_images
```

Figure 1.9 The Directory Structure of the Small Traffic Light Dataset

The annotation directories have a list of XML files, each of which corresponds to an image file with the same filename inside the corresponding image directory (Figure 1.10). Figure 1.11 shows that the first XML file in the Annotations directory includes information about the .jpg file with the same filename.

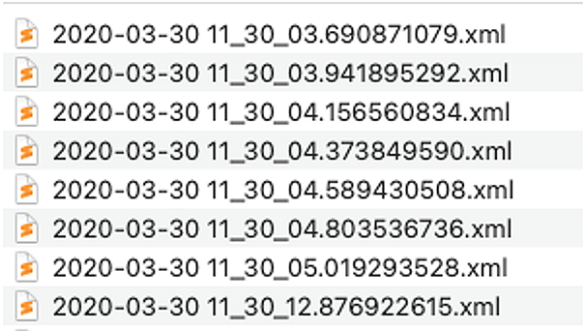


Figure 1.10 List of XML Files under the Annotations Directory in the Small Traffic Light Dataset (source: "Small Traffic Light Dataset," https://www.kaggle.com/datasets/sovitrath/small-traffic-light-dataset-xml-format)

```xml
<annotation>
  <folder/>
  <filename>2020-03-30 11_30_03.690871079.jpg</filename>
  <database/>
  <annotation/>
  <image/>
  <size>
    <height>1080</height>
    <width>1920</width>
    <depth>3</depth>
  </size>
  <segmented/>
  <object>
    <name>green</name>
    <pose/>
    <truncated/>
    <difficult/>
    <bndbox>
      <xmin>616</xmin>
      <ymin>477</ymin>
      <xmax>633</xmax>
      <ymax>521</ymax>
    </bndbox>
  </object>
```

Figure 1.11 2020-03-30 11_30_03.690871079.xml, an Example XML file within the Small Traffic Light Dataset (Source: "Small Traffic Light Dataset," https://www.kaggle.com/datasets/sovitrath/small-traffic-light-dataset-xml-format)

JSON and XML Dataset Descriptions

Both JSON and XML files often include some description(s) of the dataset itself as well (known as *metadata*), and they are included as a separate entry in the file ({} or <>). In Figure 1.12 and Figure 1.13, the actual data entries are listed inside "itemData" and `<data>`, respectively. The rest are used to provide background information on the dataset. For example:

- "creationDateTime": describes when the dataset was created.
- `<name>` is used to write the name of this dataset.
- `<metadata>` is used to describe each column name of the dataset along with its data type.

```
{
    "creationDateTime": "2023-03-22T11:53:27",
    "datasetJSONVersion": "1.0.0",
    "fileOID": "www.sponsor.org.project123.final",
    "sourceSystemVersion": "1.2.3",
    "clinicalData": {
        "items": [
                    {"OID": "ITEMGROUPDATASEQ", "name": "ITEMGROUPDATASEQ", "label": "Record identifier", "type": "integer"},
                    {"OID": "IT.STUDYID", "name": "STUDYID", "label": "Study identifier", "type": "string"},
                    {"OID": "IT.DOMAIN", "name": "DOMAIN", "label": "Domain identifier", "type": "string", "length": 2},
            ],
            "itemData": [
                [1, "MyStudy1", "D1"],
                [2, "MyStudy2", "D2"],
                ...
            ]
        }
    }
}
```

Figure 1.12 An Example JSON File with Metadata

```xml
<?xml version="1.0" encoding="UTF-8"?>
<dataset>
    <name>CEREALS</name>
    <metadata>
        <version>1.2</version>
        <date>04/02/2024</date>
        <description>a list of some popular cereals</description>
        <col name="NAME" type="string"/>
        <col name="MANUFACTURER" type="string"/>
        <col name="CALORIES_PER_SERVING" type="integer"/>
    </metadata>
    <data>
        <row>
            <value>Cheerios</value>
            <value>General Mills</value>
            <value>140</value>
        </row>
        <row>
            <value>Corn Flakes</value>
            <value>Kellogg's</value>
            <value>150</value>
        </row>
    </data>
</dataset>
```

Figure 1.13 An Example XML Dataset with Metadata

The Face Mask Detection (https://openstax.org/r/andrewmvd) dataset has a set of images of human faces with masks on. It follows a similar structure as well. The dataset consists of two directories—annotations and images. The former is in the XML format. The name of each XML file includes any text description about the image with the same filename. For example, "maksssksksss0.xml" includes information on "maksssksksss0.png."

EXAMPLE 1.7

Problem

The Iris Flower dataset (ch1-iris.csv (https://openstax.org/r/iris)) is a classic dataset in the field of data analysis.[1] Download this dataset and open it with a code editor (e.g., Sublime Text, XCode, Visual Studio Code). (We recommend that if you do not have any code editor installed, you install one. All three of these editors are quick and easy to install.) Now, answer these questions:

- How many items are there in the dataset?
- How many attributes are there in the dataset?
- What is the second attribute in the dataset?

Solution

There are 151 rows in the dataset with the header row at the top, totaling 150 items. There are five attributes listed across columns: `sepal_length`, `sepal_width`, `petal_length`, `petal_width`, `species`. The second attribute is `sepal_width`.

1 The Iris Flower dataset was introduced by the British statistician and biologist Ronald Fisher in his 1936 paper "The Use of Multiple Measurements in Taxonomic Problems." This work became a landmark study in the use of multivariate data in classification problems and frequently makes an appearance in data science as a convenient test case for machine learning and neural network algorithms. The Iris Flower dataset is often used as a beginner's dataset to demonstrate various techniques, such as classification of algorithms, formatted in CSV.

EXAMPLE 1.8

Problem

The Jeopardy dataset (ch1-jeopardy.json (https://openstax.org/r/filed15)) is formatted in JSON. Download and open it with a code editor (e.g., Notepad, Sublime Text, Xcode, Visual Studio Code).

- How many items are there in the dataset?
- How many attributes are there in the dataset?
- What is the third item in the dataset?

Solution

There are 409 items in the dataset, and each item has seven attributes: "`category`", "`air_date`", "`question`", "`value`", "`answer`", "`round`", and "`show_number`". The third item is located at index 2 of the first list as shown in Figure 1.14.


```
"category" : string "EVERYBODY TALKS ABOUT IT..."
"air_date" : string "2004-12-31"
"question" :
string "'The city of Yuma in this state has a record average of 4,055 hours of sunshine each year'"
"value" : string "$200"
"answer" : string "Arizona"
"round" : string "Jeopardy!"
"show_number" : string "4680"
```


Figure 1.14 The Third Item in the Jeopardy Dataset

1.4 Using Technology for Data Science

Learning Outcomes

By the end of this section, you should be able to:

- 1.4.1 Explain how statistical software can help with data analysis.
- 1.4.2 Explain the uses of different programs and programming languages for data manipulation, analysis, and visualizations.
- 1.4.3 Explain the uses of various data analysis tools used in data science applications.

Technology empowers data analysts, researchers, and organizations to leverage data, extract actionable insights, and make decisions that optimize processes and improve outcomes in many areas of our lives. Specifically, technology provides the tools, platforms, and algorithms that enable users to efficiently process and analyze data—especially complex datasets. The choice of technology used for a data science project will vary depending on the goals of the project, the size of the datasets, and the kind of analysis required.

Spreadsheet Programs

Spreadsheet programs such as Excel and Google Sheets are software applications consisting of electronic worksheets with rows and columns where data can be entered, manipulated, and calculated. Spreadsheet programs offer a variety of functions for data manipulation and can be used to easily create charts and tables. **Excel** is one of the most widely used spreadsheet programs, and as part of Microsoft Office, it integrates well with other Office products. Excel was first released by Microsoft in 1987, and it has become one of the most popular choices for loading and analyzing tabular data in a spreadsheet format. You are likely to have used Excel in some form or other —perhaps to organize the possible roommate combinations in your dorm room or

to plan a party, or in some instructional context. We refer to the use of Excel to manipulate data in some of the examples of this text because sometimes a spreadsheet is simply the easiest way to work with certain datasets. (See Appendix A: Review of Excel for Data Science for a review of Excel functionality.)

Google Sheets is a cloud-based spreadsheet program provided by Google as part of the Google Workspace. Because it is cloud-based, it is possible to access spreadsheets from any device with an internet connection. This accessibility allows for collaboration and real-time updates among multiple users, enhancing communication within a team and making it ideal for team projects or data sharing among colleagues. Users can leave comments, track changes, and communicate within the spreadsheet itself.

The user interfaces for both Excel and Google Sheets make these programs very user-friendly for many applications. But these programs have some limitations when it comes to large databases or complex analyses. In these instances, data scientists will often turn to a programming language such as Python, R, or SPSS.

Programming Languages

A **programming language** is a formal language that consists of a set of instructions or commands used to communicate with a computer and to instruct it to perform specific tasks that may include data manipulation, computation, and input/output operations. Programming languages allow developers to write algorithms, create software applications, and automate tasks and are better suited than spreadsheet programs to handle complex analyses.

Python and R are two of the most commonly used programming languages today. Both are open-source programming languages. While **Python** started as a general-purpose language that covers various types of tasks (e.g., numerical computation, data analysis, image processing, and web development), **R** is more specifically designed for statistical computing and graphics. Both use simple syntax compared to conventional programming languages such as Java or C/C++. They offer a broad collection of packages for data manipulation, statistical analysis, visualization, machine learning, and complex data modeling tasks.

We have chosen to focus on Python in this text because of its straightforward and intuitive syntax, which makes it especially easy for beginners to learn and apply. Also, Python skills can apply to a wide range of computing work. Python also contains a vast network of libraries and frameworks specifically designed for data analysis, machine learning, and scientific computing. As we'll see, Python libraries such as NumPy (https://openstax.org/r/nump), Pandas (https://openstax.org/r/panda), Matplotlib (https://openstax.org/r/matplot), and Seaborn (https://openstax.org/r/seabo) provide powerful tools for data manipulation, visualization, and machine learning tasks, making Python a versatile choice for handling datasets. You'll find the basics of R and various R code examples in Appendix B: Review of R Studio for Data Science. If you want to learn how to use R, you may want to practice with RStudio (https://openstax.org/r/posit1), a commonly used software application to edit/run R programs.

PYTHON IN EXCEL

Microsoft recently launched a new feature for Excel named "Python in Excel." This feature allows a user to run Python code to analyze data directly in Excel. This textbook does not cover this feature, but instead presents Python separately, as it is also crucial for you to know how to use each tool in its more commonly used environment. If interested, refer to Microsoft's announcement (https://openstax.org/r/youtu).

EXPLORING FURTHER

Specialized Programming Languages

Other programming languages are more specialized for a particular task with data. These include SQL, Scala, and Julia, among others, as briefly described in this article on "The Nine Top Programming Languages for Data Science (https://openstax.org/r/9top)."

Other Data Analysis/Visualization Tools

There are a few other data analysis tools that are strong in data visualization. Tableau (https://openstax.org/r/tableau1) and PowerBI (https://openstax.org/r/micro) are user-friendly applications for data visualization. They are known for offering more sophisticated, interactive visualizations for high-dimensional data. They also offer a relatively easy user interface for compiling an analysis dashboard. Both allow users to run a simple data analysis as well before visualizing the results, similar to Excel.

In general, data visualization aims to make complex data more understandable and usable. The tools and technology described in this section offer a variety of ways to go about creating visualizations that are most accessible. Refer to Visualizing Data for a deeper discussion of the types of visualizations—charts, graphs, boxplots, histograms, etc.—that can be generated to help find the meaning in data.

EXPLORING FURTHER

Evolving Professional Standards

Data science is a field that is changing daily; the introduction of artificial intelligence (AI) has increased this pace. Technological, social, and ethical challenges with AI are discussed in Natural Language Processing, and ethical issues associated with the whole data science process, including the use of machine learning and artificial intelligence, are covered in Ethics Throughout the Data Science Cycle. A variety of data science professional organizations are working to define and update process and ethical standards on an ongoing basis. Useful references may include the following:

- Initiative for Analytics and Data Science Standards (IADSS) (https://openstax.org/r/iadss1)
- Data Science Association (DSA) (https://openstax.org/r/datascienceassn1)
- Association of Data Scientists (ADaSci) (https://openstax.org/r/adasci1)

1.5 Data Science with Python

Learning Outcomes

By the end of this section, you should be able to

- 1.5.1 Load data to Python.
- 1.5.2 Perform basic data analysis using Python.
- 1.5.3 Use visualization principles to graphically plot data using Python.

Multiple tools are available for writing and executing Python programs. Jupyter Notebook is one convenient and user-friendly tool. The next section explains how to set up the Jupyter Notebook environment using Google Colaboratory (Colab) and then provides the basics of two open-source Python libraries named `Pandas` and `Matplotlib`. These libraries are specialized for data analysis and data visualization, respectively.

EXPLORING FURTHER

Python Programming

In the discussion below, we assume you are familiar with basic Python syntax and know how to write a simple program using Python. If you need a refresher on the basics, please refer to Das, U., Lawson, A., Mayfield, C., & Norouzi, N. (2024). *Introduction to Python Programming*. OpenStax. https://openstax.org/books/introduction-python-programming/pages/1-introduction (https://openstax.org/r/page1).

Jupyter Notebook on Google Colaboratory

Jupyter Notebook is a web-based environment that allows you to run a Python program more interactively, using programming code, math equations, visualizations, and plain texts. There are multiple web applications or software you could use to edit a Jupyter Notebook, but in this textbook we will use Google's free application named **Google Colaboratory (Colab)** (https://openstax.org/r/colab1), often abbreviated as Colab. It is a cloud-based platform, which means that you can open, edit, run, and save a Jupyter Notebook on your Google Drive.

Setting up Colab is simple. On your Google Drive, click New > More. If your Google Drive has already installed Colab before, you will see Colaboratory under More. If not, click "Connect more apps" and install Colab by searching "Colaboratory" on the app store (Figure 1.15). For further information, see the Google Colaboratory Ecosystem (https://openstax.org/r/1pp) animation.

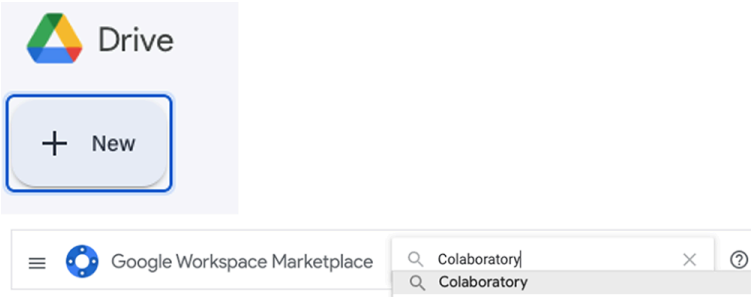


Figure 1.15 Install Google Colaboratory (Colab)

Now click New > More > Google Laboratory. A new, empty Jupyter Notebook will show up as in Figure 1.16.

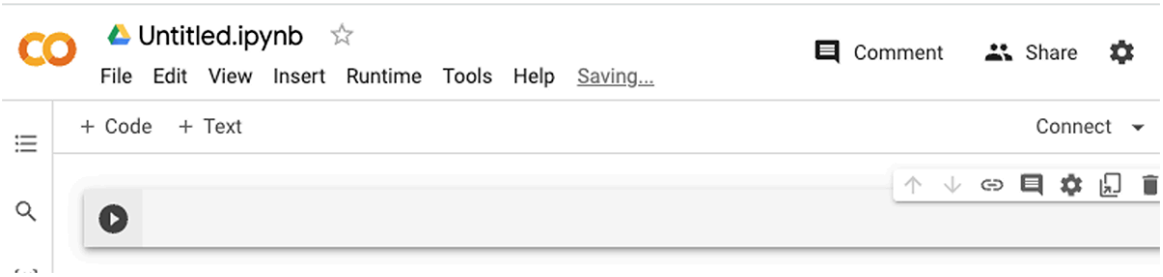


Figure 1.16 Google Colaboratory Notebook

The gray area with the play button is called a cell. A **cell** is a block where you can type either code or plain text. Notice that there are two buttons on top of the first cell—"+ Code" and "+ Text." These two buttons add a code or text cell, respectively. A code cell is for the code you want to run; a text cell is to add any text description or note.

Let's run a Python program on Colab. Type the following code in a code cell.

PYTHON CODE

```
print ("hello world!")
```

The resulting output will look like this:

```
hello world!
```

You can write a Python program across multiple cells and put text cells in between. Colab would treat all the code cells as part of a single program, running from the top to bottom of the current Jupyter Notebook. For example, the two code cells below run as if it is a single program.

When running one cell at a time from the top, we see the following outputs under each cell.

PYTHON CODE

```
a = 1
print ("The a value in the first cell:", a)
```

The resulting output will look like this:

```
The a value in the first cell: 1
```

PYTHON CODE

```
b = 3
print ("a in the second cell:", a)
print ("b in the second cell:", b)
a + b
```

The resulting output will look like this:

```
a in the second cell: 1
b in the second cell: 3
4
```

CONVENTIONAL PYTHON VERSUS JUPYTER NOTEBOOK SYNTAX

While conventional Python syntax requires `print()` syntax to print something to the program console,

Jupyter Notebook does not require `print()`. On Jupyter Notebook, the line `a+b` instead of `print(a+b)` also prints the value of `a+b` as an output. But keep in mind that if there are multiple lines of code that trigger printing some values, *only* the output from the last line will show.

You can also run multiple cells in bulk. Click Runtime on the menu, and you will see there are multiple ways of running multiple cells at once (Figure 1.17). The two commonly used ones are "Run all" and "Run before." "Run all" runs all the cells in order from the top; "Run before" runs all the cells before the currently selected one.

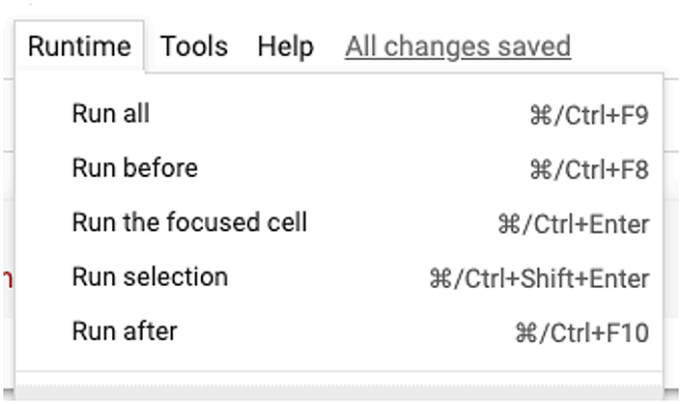


Figure 1.17 Multiple Ways of Running Cells on Colab

One thing to keep in mind is that being able to split a long program into multiple blocks and run one block at a time raises chances of user error. Let's look at a modified code from the previous example.

PYTHON CODE

```
a = 1
print ("the value in the first cell:", a)
```

The resulting output will look like this:

```
the value in the first cell: 1
```

PYTHON CODE

```
b = 3
print ("a in the second cell:", a)
print ("b in the second cell:", b)
a + b
```

The resulting output will look like this:

```
a in the second cell: 1
```

```
b in the second cell: 3
4
```

PYTHON CODE

```
a = 2
a + b
```

The resulting output will look like this:

```
5
```

The modified code has an additional cell at the end, updating `a` from 1 to 2. Notice that now `a+b` returns 5 as `a` has been changed to 2. Now suppose you need to run the second cell for some reason, so you run the second cell again.

PYTHON CODE

```
a = 1
print ("the a value in the first cell:", a)
```

The resulting output will look like this:

```
the a value in the first cell: 1
```

PYTHON CODE

```
b = 3
print ("a in the second cell:", a)
print ("b in the second cell:", b)
a + b
```

The resulting output will look like this:

```
a in the second cell: 2
b in the second cell: 3
5
```

PYTHON CODE

```
a = 2
a + b
```

The resulting output will look like this:

```
5
```

The value of `a` has changed to 2. This implies that the execution order of each cell matters! If you have run the third cell before the second cell, the value of `a` will have the value from the third one even though the third cell is located below the second cell. Therefore, it is recommended to use "Run all" or "Run before" after you make changes across multiple cells of code. This way your code is guaranteed to run sequentially from the top.

Python Pandas

One of the strengths of Python is that it includes a variety of free, open-source libraries. Libraries are a set of already-implemented methods that a programmer can refer to, allowing a programmer to avoid building common functions from scratch.

`Pandas` is a Python library specialized for data manipulation and analysis, and it is very commonly used among data scientists. It offers a variety of methods, which allows data scientists to quickly use them for data analysis. You will learn how to analyze data using `Pandas` throughout this textbook.

Colab already has `Pandas` installed, so you just need to import `Pandas` and you are set to use all the methods in `Pandas`. Note that it is convention to abbreviate pandas to pd so that when you call a method from `Pandas`, you can do so by using `pd` instead of having to type out `Pandas` every time. It offers a bit of convenience for a programmer!

PYTHON CODE

```
# import Pandas and assign an abbreviated identifier "pd"
import pandas as pd
```

EXPLORING FURTHER

Installing Pandas on Your Computer

If you wish to install Pandas on your own computer, refer to the installation page of the Pandas website (https://openstax.org/r/pyd).

Load Data Using Python Pandas

The first step for data analysis is to load the data of your interest to your Notebook. Let's create a folder on Google Drive where you can keep a CSV file for the dataset and a Notebook for data analysis. Download a

public dataset, ch1-movieprofit.csv (https://openstax.org/r/movieprofit), and store it in a Google Drive folder. Then open a new Notebook in that folder by entering that folder and clicking New > More > Google Colaboratory.

Open the Notebook and allow it to access files in your Google Drive by following these steps:

First, click the Files icon on the side tab (Figure 1.18).



Figure 1.18 Side Tab of Colab

Then click the Mount Drive icon (Figure 1.19) and select "Connect to Google Drive" on the pop-up window.

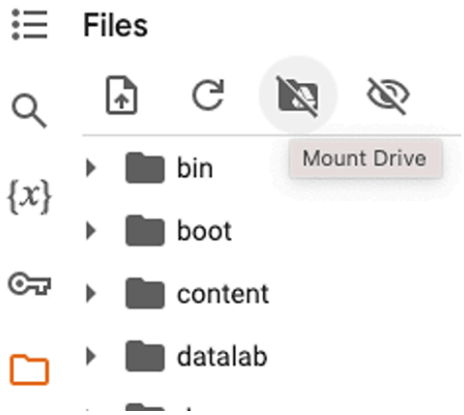


Figure 1.19 Features under Files on Colab

Notice that a new cell has been inserted on the Notebook as a result (Figure 1.20).

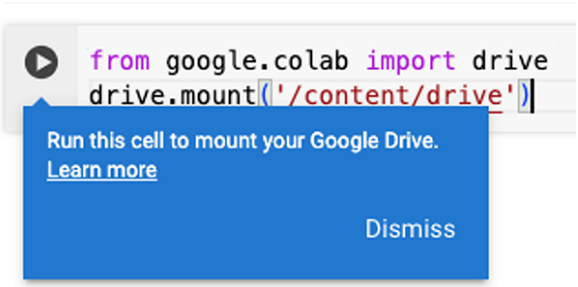


Figure 1.20 An Inserted Cell to Mount Your Google Drive

Connect your Google Drive by running the cell, and now your Notebook file can access all the files under content/drive. Navigate folders under drive to find your Notebook and ch1-movieprofit.csv (https://openstax.org/r/movieprofit) files. Then click "..." > Copy Path (Figure 1.21).

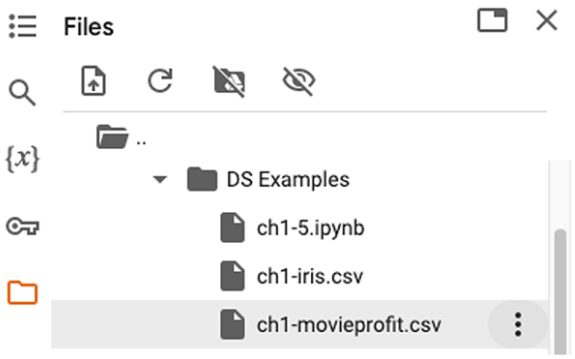


Figure 1.21 Copying the Path of a CSV File Located in a Google Drive Folder

Now replace [Path] with the copied path in the below code. Run the code and you will see the dataset has been loaded as a table and stored as a Python variable data.

PYTHON CODE

```
# import Pandas and assign an abbreviated identifier "pd"
import pandas as pd

data = pd.read_csv("[Path]")
data
```

The resulting output will look like this:

	Unnamed: 0	Title	Year	Genre	Rating	Duration	US_Gross_Million	Worldwide_Gross_Million	Votes
0	1	Avatar	2009	Action	7.8	162	760.51	2847.40	1,236,962
1	2	Avengers: Endgame	2019	Action	8.4	181	858.37	2797.50	1,108,641
2	3	Titanic	1997	Drama	7.9	194	659.33	2201.65	1,162,142
3	4	Star Wars: Episode VII - The Force Awakens	2015	Action	7.8	138	936.66	2069.52	925,551
4	5	Avengers: Infinity War	2018	Action	8.4	149	678.82	2048.36	1,062,517
...	...	...	...	...	...	...	...	...	...
961	962	The A-Team	2010	Action	6.7	117	77.22	177.24	259,316
962	963	Tootsie	1982	Comedy	7.4	116	177.20	177.20	107,311
963	964	In the Line of Fire	1993	Action	7.2	128	102.31	177.00	104,598
964	965	Analyze This	1999	Comedy	6.7	103	106.89	176.89	154,726
965	966	The Hitman's Bodyguard	2017	Action	6.9	118	75.47	176.60	230,821

The `read_csv()` method in `Pandas` loads a CSV file and stores it as a DataFrame. A **DataFrame** is a data type that `Pandas` uses to store multi-column tabular data. Therefore, the variable data holds the table in ch1-movieprofit.csv (https://openstax.org/r/movieprofit) in the form of a `Pandas` DataFrame.

DATAFRAME VERSUS SERIES

`Pandas` defines two data types for tabular data—DataFrame and Series. While DataFrame is used for multi-column tabular data, Series is used for single-column data. Many methods in `Pandas` support both DataFrame and Series, but some are only for one or the other. It is always good to check if the method you

are using works as you expect. For more information, refer to the Pandas documentation (https://openstax.org/r/docs) or Das, U., Lawson, A., Mayfield, C., & Norouzi, N. (2024). *Introduction to Python Programming*. OpenStax. https://openstax.org/books/introduction-python-programming/pages/1-introduction (https://openstax.org/r/page1).

EXAMPLE 1.9

Problem

Remember the Iris dataset we used in Data and Datasets? Load the dataset ch1-iris.csv (https://openstax.org/r/iris) to a Python program using `Pandas`.

Solution

The following code loads the ch1-iris.csv (https://openstax.org/r/iris) that is stored in a Google Drive. Make sure to replace the path with the actual path to ch1-iris.csv (https://openstax.org/r/iris) on your Google Drive.

PYTHON CODE

```
import pandas as pd

data = pd.read_csv("[Path to ch1-iris.csv]") # Replace the path
data
```

The resulting output will look like this:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

EXPLORING FURTHER

Can I load a file that is uploaded to someone else's Google Drive and shared with me?

Yes! This is useful especially when your Google Drive runs out of space. Simply add the shortcut of the shared file to your own drive. Right-click > Organize > Add Shortcut will let you select where to store the shortcut. Once done, you can call `pd.read_csv()` using the path of the shortcut.

Summarize Data Using Python Pandas

You can compute basic statistics for data quite quickly by using the `DataFrame.describe()` method. Add and run the following code in a new cell. It calls the `describe()` method upon `data`, the DataFrame we defined earlier with ch1-movieprofit.csv (https://openstax.org/r/movieprofit).

PYTHON CODE



```
data = pd.read_csv("[Path to ch1-movieprofit.csv]")
data.describe()
```

like this:

	Unnamed: 0	Rating	Duration	US_Gross_Million	Worldwide_Gross_Million
count	966.00000	966.000000	966.000000	966.000000	966.000000
mean	483.50000	6.814286	117.506211	156.158975	410.140600
std	279.00448	0.894383	21.615612	110.629617	294.758791
min	1.00000	3.300000	69.000000	0.010000	176.600000
25%	242.25000	6.200000	101.250000	90.832500	223.277500
50%	483.50000	6.800000	116.000000	129.245000	309.345000
75%	724.75000	7.400000	130.000000	187.090000	472.645000
max	966.00000	9.200000	238.000000	936.660000	2847.400000

`describe()` returns a table whose columns are a subset of the columns in the entire dataset and whose rows are different statistics. The statistics include the number of unique values in a column (`count`), mean (`mean`), standard deviation (`std`), minimum and maximum values (`min/max`), and different quartiles (`25%/50%/75%`), which you will learn about in Measures of Variation. Using this representation, you can compute such statistics of different columns easily.

EXAMPLE 1.10

Problem

Summarize the IRIS dataset using `describe()` of ch1-iris.csv (https://openstax.org/r/iris) you loaded in the

previous example.

Solution

The following code in a new cell returns the summary of the dataset.

PYTHON CODE

```
data = pd.read_csv("[Path to ch1-iriscsv]")
data.describe()
```

The resulting output will look like this:

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

Select Data Using Python Pandas

The `Pandas` DataFrame allows a programmer to use the column name itself when selecting a column. For example, the following code prints all the values in the "US_Gross_Million" column in the form of a Series (remember the data from a single column is stored in the Series type in `Pandas`).

PYTHON CODE

```
data = pd.read_csv("[Path to ch1-movieprofit.csv]")

data["US_Gross_Million"]
```

like this:

```
0   760.51
1   858.37
2   659.33
3   936.66
```

```
4   678.82
    ...
961   77.22
962  177.20
963  102.31
964  106.89
965   75.47
Name: US_Gross_Million, Length: 966, dtype: float64
```

`DataFrame.iloc[]` enables a more powerful selection—it lets a programmer select by both column and row, using column and row indices. Let's look at some code examples below.

PYTHON CODE



```
data.iloc[:, 2] # select all values in the second column
```

The resulting output will look like this:

```
0   2009
1   2019
2   1997
3   2015
4   2018
    ...
961  2010
962  1982
963  1993
964  1999
965  2017
Name: Year, Length: 966, dtype: object
```

PYTHON CODE



```
data.iloc[2,:] # select all values in the third row
```

The resulting output will look like this:

```
Unnamed: 0             3
Title            Titanic
Year              1997
Genre             Drama
Rating              7.9
Duration             194
```

```
US_Gross_Million       659.33
Worldwide_Gross_Million   2201.65
Votes                1,162,142
Name: 2, dtype: object
```

To pinpoint a specific value within the “US_Gross_Million” column, you can use an index number.

PYTHON CODE

```
print (data["US_Gross_Million"][0]) # index 0 refers to the top row
print (data["US_Gross_Million"][2]) # index 2 refers to the third row
```

The resulting output will look like this:

```
760.51
659.33
```

You can also use `DataFrame.iloc[]` to select a specific group of cells on the table. The example code below shows different ways of using `iloc[]`. There are multiple ways of using `iloc[]`, but this chapter introduces a couple of common ones. You will learn more techniques for working with data throughout this textbook.

PYTHON CODE

```
data.iloc[:, 1] # select all values in the second column (index 1)
```

The resulting output will look like this:

```
0                     Avatar
1              Avengers: Endgame
2                     Titanic
3   Star Wars: Episode VII - The Force Awakens
4            Avengers: Infinity War
            ...
961                  The A-Team
962                    Tootsie
963               In the Line of Fire
964                  Analyze This
965             The Hitman's Bodyguard
Name: Title, Length: 966, dtype: object
```

PYTHON CODE

```
data.iloc[[1, 3], [2, 3]]
# select the rows at index 1 and 3, the columns at index 2 and 3
```

The resulting output will look like this:

	Year	Genre
1	2019	Action
3	2015	Action

EXAMPLE 1.11

Problem

Select a “sepal_width” column of the IRIS dataset using the column name.

Solution

The following code in a new cell returns the “sepal_width” column.

PYTHON CODE

```
data = pd.read_csv("[Path to ch1-iris.csv]")

data["sepal_width"]
```

The resulting output will look like this:

```
0   3.5
1   3.0
2   3.2
3   3.1
4   3.6
   ...
145  3.0
146  2.5
147  3.0
148  3.4
149  3.0
Name: sepal_width, Length: 150, dtype: float64
```

EXAMPLE 1.12

Problem

Select a "petal_length" column of the IRIS dataset using `iloc[]`.

Solution

The following code in a new cell returns the "petal_length" column.

PYTHON CODE

```
data.iloc[:, 2]
```

The resulting output will look like this:

```
0   1.4
1   1.4
2   1.3
3   1.5
4   1.4
   ...
145  5.2
146  5.0
147  5.2
148  5.4
149  5.1
Name: petal_length, Length: 150, dtype: float64
```

Search Data Using Python Pandas

To search for some data entries that fulfill specific criteria (i.e., filter), you can use `DataFrame.loc[]` of `Pandas`. When you indicate the filtering criteria inside the brackets, [], the output returns the filtered rows within the DataFrame. For example, the code below filters out the rows whose genre is comedy. Notice that the output only has 307 out of the full 3,400 rows. You can check the output on your own, and you will see their Genre values are all "Comedy."

PYTHON CODE

```
data = pd.read_csv("[Path to ch1-movieprofit.csv]")

data.loc[data['Genre'] == 'Comedy']
```

The resulting output will look like this:

	Unnamed: 0	Title	Year	Genre	Rating	Duration	US_Gross_Million	Worldwide_Gross_Million	Votes
161	162	Mamma Mia!	2008	Comedy	6.5	108	144.13	611.26	245,380
170	171	The Hangover Part II	2011	Comedy	6.4	102	254.46	586.76	499,126
183	184	Mei ren yu	2016	Comedy	6.2	94	3.23	553.81	9,374
186	187	Ted	2012	Comedy	6.9	106	218.82	549.37	612,897
208	209	Meet the Fockers	2004	Comedy	6.3	115	279.26	522.66	271,402
...	...	...	...	...	...	...	...	...	...
954	955	How to Lose a Guy in 10 Days	2003	Comedy	6.4	116	105.81	177.50	238,604
957	958	The 40 Year Old Virgin	2005	Comedy	7.1	116	109.45	177.38	436,221
960	961	The Descendants	2011	Comedy	7.3	115	82.58	177.24	242,388
962	963	Tootsie	1982	Comedy	7.4	116	177.20	177.20	107,311
964	965	Analyze This	1999	Comedy	6.7	103	106.89	176.89	154,726

EXAMPLE 1.13

Problem

Using `DataFrame.loc[]`, search for all the items of *Iris-virginica* species in the IRIS dataset.

Solution

The following code returns a filtered DataFrame whose species are *Iris-virginica*. All such rows show up as an output.

PYTHON CODE



```
data = pd.read_csv("[Path to ch1-iris.csv]")

data.loc[data['species'] == 'Iris-virginica']
```

The resulting figure will look like this:

	sepal_length	sepal_width	petal_length	petal_width	species
100	6.3	3.3	6.0	2.5	Iris-virginica
101	5.8	2.7	5.1	1.9	Iris-virginica
102	7.1	3.0	5.9	2.1	Iris-virginica
103	6.3	2.9	5.6	1.8	Iris-virginica
104	6.5	3.0	5.8	2.2	Iris-virginica
105	7.6	3.0	6.6	2.1	Iris-virginica
106	4.9	2.5	4.5	1.7	Iris-virginica
107	7.3	2.9	6.3	1.8	Iris-virginica
108	6.7	2.5	5.8	1.8	Iris-virginica

```
(Rows 109 through 149 not shown.)
```

EXAMPLE 1.14

Problem

This time, search for all the items whose species is *Iris-virginica* and whose sepal width is wider than 3.2.

Solution

You can use a Boolean expression—in other words, an expression that evaluates as either True or False—inside `data.loc[]`.

PYTHON CODE



```
data.loc[(data['species'] == 'Iris-virginica') & (data['sepal_width'] > 3.2)]
```

The resulting output will look like this:

	sepal_length	sepal_width	petal_length	petal_width	species
100	6.3	3.3	6.0	2.5	Iris-virginica
109	7.2	3.6	6.1	2.5	Iris-virginica
117	7.7	3.8	6.7	2.2	Iris-virginica
124	6.7	3.3	5.7	2.1	Iris-virginica
131	7.9	3.8	6.4	2.0	Iris-virginica
136	6.3	3.4	5.6	2.4	Iris-virginica
144	6.7	3.3	5.7	2.5	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica

Visualize Data Using Python Matplotlib

There are multiple ways to draw plots of data in Python. The most common and straightforward way is to import another library, `Matplotlib`, which is specialized for data visualization. `Matplotlib` is a huge library, and to draw the plots you only need to import a submodule named `pyplot`.

Type the following import statement in a new cell. Note it is convention to denote `matplotlib.pyplot` with `plt`, similarly to denoting `Pandas` with `pd`.

PYTHON CODE



```
import matplotlib.pyplot as plt
```

`Matplotlib` offers a method for each type of plot, and you will learn the `Matplotlib` methods for all of the commonly used types throughout this textbook. In this chapter, however, let's briefly look at how to draw a plot using `Matplotlib` in general.

Suppose you want to draw a scatterplot between "US_Gross_Million" and "Worldwide_Gross_Million" of the movie profit dataset (ch1-movieprofit.csv (https://openstax.org/r/movieprofit)). You will investigate scatterplots in more detail in Correlation and Linear Regression Analysis. The example code below draws such a scatterplot using the method `scatter()`. `scatter()` takes the two columns of your interest—data["US_Gross_Million"] and data["Worldwide_Gross_Million"]—as the inputs and assigns them for the *x*- and *y*-axes, respectively.

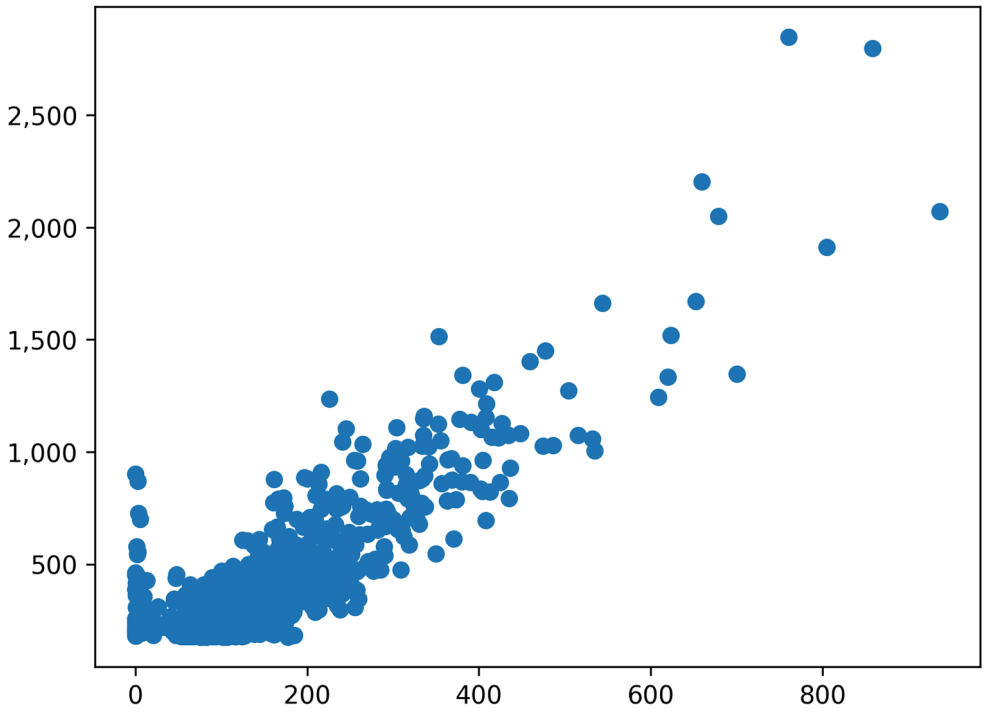
PYTHON CODE



```
data = pd.read_csv("[Path to ch1-movieprofit.csv]")
```

```
# draw a scatterplot using matplotlib's scatter()
plt.scatter(data["US_Gross_Million"], data["Worldwide_Gross_Million"])
```

The resulting output will look like this:



Notice that it simply has a set of dots on a white plane. The plot itself does not show what each axis represents, what this plot is about, etc. Without them, it is difficult to capture what the plot shows. You can set these with the following code. The resulting plot below indicates that there is a positive correlation between domestic gross and worldwide gross.

PYTHON CODE

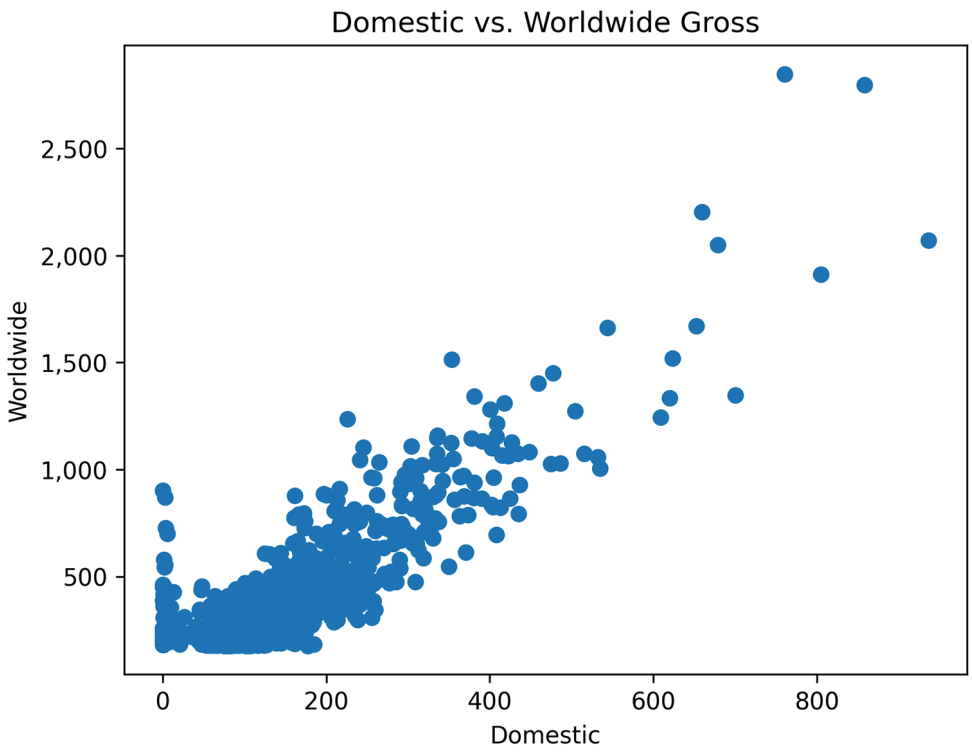
```
# draw a scatterplot
plt.scatter(data["US_Gross_Million"], data["Worldwide_Gross_Million"])

# set the title
plt.title("Domestic vs. Worldwide Gross")

# set the x-axis label
plt.xlabel("Domestic")

# set the y-axis label
plt.ylabel("Worldwide")
```

The resulting output will look like this:



You can also change the range of numbers along the *x*- and *y*-axes with `plt.xlim()` and `plt.ylim()`. Add the following two lines of code to the cell in the previous Python code example, which plots the scatterplot.

PYTHON CODE

```
# draw a scatterplot
plt.scatter(data["US_Gross_Million"], data["Worldwide_Gross_Million"])

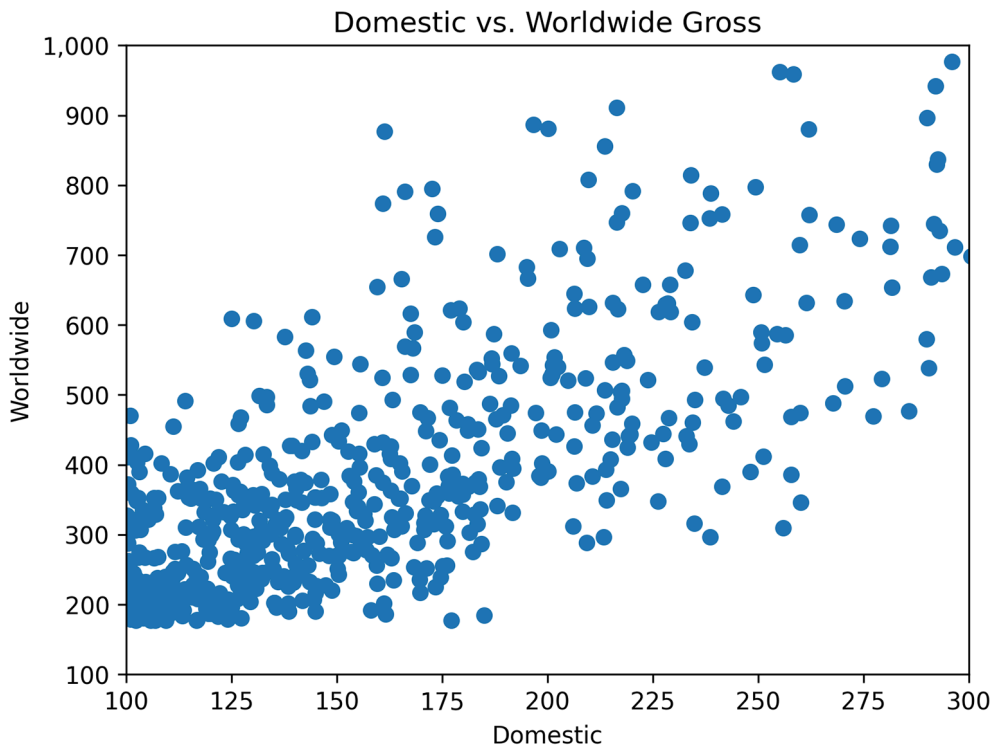
# set the title
plt.title("Domestic vs. Worldwide Gross")

# set the x-axis label
plt.xlabel("Domestic")

# set the y-axis label
plt.ylabel("Worldwide")

# set the range of values of the x- and y-axes
plt.xlim(1*10**2, 3*10**2) # x axis: 100 to 300
plt.ylim(1*10**2, 1*10**3) # y axis: 100 to 1,000
```

The resulting output will look like this:



The resulting plot with the additional lines of code has a narrower range of values along the *x*- and *y*-axes.

EXAMPLE 1.15

Problem

Using the iris dataset, draw a scatterplot between petal length and height of *Setosa Iris*. Set the title, *x*-axis label, and *y*-axis label properly as well.

Solution

PYTHON CODE

```
import matplotlib.pyplot as plt

data = pd.read_csv("[Path to ch1-iris.csv]")

# select the rows whose species are Setosa Iris
setosa = data.loc[(data['species'] == 'Iris-setosa')]

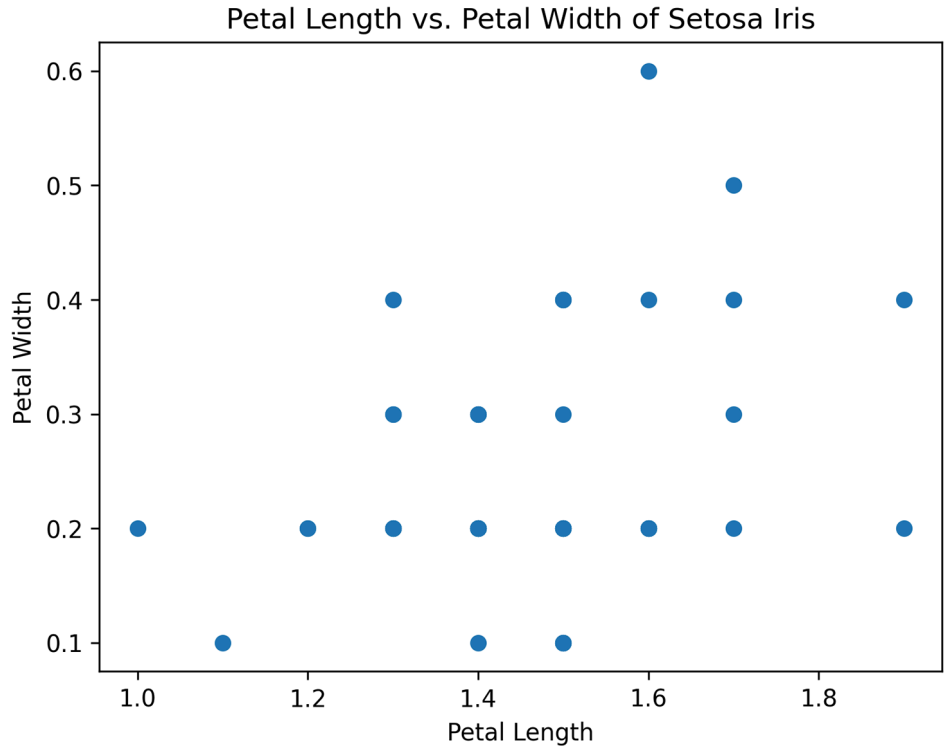
# draw a scatterplot
plt.scatter(setosa["petal_length"], setosa["petal_width"])
```

```
# set the title
plt.title("Petal Length vs. Petal Width of Setosa Iris")

# set the x-axis label
plt.xlabel("Petal Length")

# set the y-axis label
plt.ylabel("Petal Width")
```

The resulting output will look like this:



Datasets

Note: The primary datasets referenced in the chapter code may also be [downloaded here (https://openstax.org/r/spreadsheetsd1)].

Key Terms

attribute characteristic or feature that defines an item in a dataset
categorical data data that is represented in different forms and do not indicate measurable quantities
cell a block or rectangle on a table that is specified with a combination of a row number and a column number
comma-separated values (CSV) format of a dataset in which each item takes up a single line and its values are separated by commas (“,”)
continuous data data whose value is chosen from an infinite set of numbers
data anything that we can analyze to compile some high-level insights
data analysis the process of examining and interpreting raw data to uncover patterns, discover meaningful insights, and make informed decisions
data collection the systematic process of gathering information on variables of interest
data preparation (data processing) the second step within the data science cycle; converts the collected data into an optimal form for analysis
data reporting the presentation of data in a way that will best convey the information learned from data analysis
data science a field of study that investigates how to collect, manage, and analyze data in order to retrieve meaningful information from some seemingly arbitrary data
data science cycle a process used when investigating data
data visualization the graphical representation of data to point out the patterns and trends involving the use of visual elements such as charts, graphs, and maps
data warehousing the process of storing and managing large volumes of data from various sources in a central location for easier access and analysis by businesses
DataFrame a data type that `Pandas` uses to store a multi-column tabular data
dataset a collection of related and organized information or data points grouped together for reference or analysis
discrete data data that follows a specific precision
Excel a spreadsheet program with a graphical user interface developed by Microsoft to help with the manipulation and analysis of data
Extensible Markup Language (XML) format of a dataset with which uses tags
Google Colaboratory (Colab) software for editing and running Jupyter Notebook files
Google Sheets a spreadsheet program with a graphical user interface developed by Google to help with the manipulation and analysis of data
information some high-level insights that are compiled from data
Internet of Things (IoT) the network of multiple objects interacting with each other through the Internet
item an element that makes up a dataset; also referred to as an *entry* and an *instance*
JavaScript Object Notation (JSON) format of a dataset that follows the syntax of the JavaScript programming language
Jupyter Notebook a web-based document that helps users run Python programs more interactively
nominal data data whose values do not include any ordering notion
numeric data data that are represented in numbers and indicate measurable quantities
ordinal data data whose values include an ordering notion
Pandas a Python library specialized for data manipulation and analysis
predictive analytics statistical techniques, algorithms, and machine learning that analyze historical data and make predictions about future events, an approach often used in medicine to offer more accurate diagnosis and treatment
programming language a formal language that consists of a set of instructions or commands used to communicate with a computer and instruct it to perform specific tasks
Python a programming language that has extensive libraries and is commonly used for data analysis

qualitative data non-numerical data that generally describe subjective attributes or characteristics and are analyzed using methods such as thematic analysis or content analysis
quantitative data data that can be measured by specific quantities and amounts and are often analyzed using statistical methods
R an open-source programming language that is specifically designed for statistical computing and graphics
recommendation system a system that makes data-driven, personalized suggestions for users
sabermetrics a statistical approach to sports team management
sports analytics use of data and business analytics in sports
spreadsheet program a software application consisting of electronic worksheets with rows and columns where data can be entered, manipulated, and calculated
structured data dataset whose individual items have the same structure
unstructured data dataset whose individual items have different structures
XML tag any block of text that consists of a pair of angle brackets (< >) with some text inside



Group Project

Project A: Data Source Quality

As a student of, or a new professional working in, data science, you will not always be collecting new primary data. It's just as important to be able to locate, critically evaluate, and properly clean existing sources of secondary data. (Collecting and Preparing Data will cover the topic of data collection and cleaning in more detail.)

Some reputable government data sources are:
Data.gov (https://openstax.org/r/datagov)

Bureau of Labor Statistics (BLS) (https://openstax.org/r/blsgov1)
National Oceanic and Atmospheric Administration (NOAA) (https://openstax.org/r/noaagov)

Some reputable nongovernment data sources are:
Kaggle (https://openstax.org/r/kaggle1)
Statista (https://openstax.org/r/statista)
Pew Research Center (https://openstax.org/r/pewresearch)

Using the suggested sources or similar-quality sources that you research on the Internet, find two to three datasets about the field or industry in which you intend to work. (You might also try to determine whether similar data sets are available at the national, state/province, and local/city levels.) In a group, formulate a specific, typical policy issue or business decision that managers in these organizations might make. For the datasets you found, compare and contrast their size, collection methods, types of data, update frequency and recency, and relevance to the decision question you have identified.

Project B: Data Visualization

Using one of the data sources mentioned in the previous project, find a dataset that interests you. Download it as a CSV file. Use Python to read in the CSV file as a `Pandas` DataFrame. As a group, think of a specific question that might be addressed using this dataset, discuss which features of the data seem most important to answer your question, and then use the Python libraries `Pandas` and `Matplotlib` to select the features and make graphs that might help to answer your question about the data. Note, you will learn many sophisticated techniques for doing data analysis in later chapters, but for this project, you should stick to simply isolating some data and visualizing it using the tools present in this chapter. Write a brief report on your findings.

Project C: Privacy, Ethics, and Bias

Identify at least one example from recent current events or news articles that is related to each of the

following themes (starting references given in parentheses):

a. Privacy concerns related to data collection (See the Protecting Personal Privacy (https://openstax.org/r/gaog) website of the U.S. Government Accountability Office.)
b. Ethics concerns related to data collection, including fair use of copyrighted materials (See the U.S. Copyright Office guidelines (https://openstax.org/r/fairuse).)
c. Bias concerns related to data collection (See the National Cancer Institute (NCI) article (https://openstax.org/r/bias) on data bias.)

Suppose that you are part of a data science team working for an organization on data collection for a major project or product. Discuss as a team how the issues of privacy, ethics, and equity (avoiding bias) could be addressed, depending on your position in the organization and the type of project or product.

Chapter Review

1. Select the incorrect step and goal pair of the data science cycle.
   a. Data collection: collect the data so that you have something for analysis.
   b. Data preparation: have the collected data stored in a server as is so that you can start the analysis.
   c. Data analysis: analyze the prepared data to retrieve some meaningful insights.
   d. Data reporting: present the data in an effective way so that you can highlight the insights found from the analysis.

2. Which of the following best describes the evolution of data management in the data science process?
   a. Initially, data was stored locally on individual computers, but with the advent of cloud-based systems, data is now stored on designated servers outside of local storage.
   b. Data management has remained static over time, with most data scientists continuing to store and process data locally on individual computers.
   c. The need for data management arose as a result of structured data becoming unmanageable, leading to the development of cloud-based systems for data storage.
   d. Data management systems have primarily focused on analysis rather than processing, resulting in the development of modern data warehousing solutions.

3. Which of the following best exemplifies the interdisciplinary nature of data science in various fields?
   a. A historian traveling to Italy to study ancient manuscripts to uncover historical insights about the Roman Empire
   b. A mathematician solving complex equations to model physical phenomena
   c. A biologist analyzing a large dataset of genetic sequences to gain insights about the genetic basis of diseases
   d. A chemist synthesizing new compounds in a laboratory

Critical Thinking

1. For each dataset (https://openstax.org/r/spreadsheetsd1), list the attributes.
   a. Spotify dataset
   b. CancerDoc dataset

2. For each dataset (https://openstax.org/r/spreadsheetsd1), define the type of the data based on following criteria and explain why:
   - Numeric vs. categorical
   - If it is numeric, continuous vs. discrete; if it is categorical, nominal vs. ordinal

   a. “artist_count” attribute of Spotify dataset
   b. “mode” attribute of Spotify dataset
   c. “key” attribute of Spotify dataset

d. the second column in CancerDoc dataset

3. For each dataset (https://openstax.org/r/spreadsheetsd1), identify the type of the dataset—structured vs. unstructured. Explain why.
   a. Spotify dataset
   b. CancerDoc dataset

4. For each dataset (https://openstax.org/r/spreadsheetsd1), list the first data entry.
   a. Spotify dataset
   b. CancerDoc dataset

5. Open the WikiHow dataset (ch1-wikiHow.json (https://openstax.org/r/filed)) and list the attributes of the dataset.

6. Draw scatterplot between bpm (*x*-axis) and danceability (*y*-axis) of the Spotify dataset (https://openstax.org/r/filed) using:
   a. Python `Matplotlib`
   b. A spreadsheet program such as MS Excel or Google Sheets (Hint: Search "Scatterplot" on Help.)

7. Regenerate the scatterplot of the Spotify dataset (https://openstax.org/r/filed), but with a custom title and *x*-/*y*-axis label. The title should be "BPM vs. Danceability." The *x*-axis label should be titled "bpm" and range from the minimum to the maximum bpm value. The *y*-axis label should be titled "danceability" and range from the minimum to the maximum Danceability value.
   a. Python `Matplotlib` (Hint: `DataFrame.min()` and `DataFrame.max()` methods return min and max values of the DataFrame. You can call these methods upon a specific column of a DataFrame as well. For example, if a DataFrame is named `df` and has a column named "col1", `df["col1"].min()` will return the minimum value of the "col1" column of `df`.)
   b. A spreadsheet program such as MS Excel or Google Sheets (Hint: Calculate the minimum and maximum value of each column somewhere else first, then simply use the value when editing the scatterplot.)

8. Based on the Spotify dataset (https://openstax.org/r/spreadsheet4), filter the following using Python `Pandas:`
   a. Tracks whose artist is Taylor Swift
   b. Tracks that were sung by Taylor Swift and released earlier than 2020

Quantitative Problems

1. Based on the Spotify dataset (https://openstax.org/r/spreadsheet4), calculate the average bpm of the songs released in 2023 using:
   a. Python `Pandas`
   b. A spreadsheet program such as MS Excel or Google Sheets (Hint: The formula AVERAGE() computes the average across the cells specified in the parentheses. For example, within Excel, typing in the command "=AVERAGE(A1:A10)" in any empty cell will calculate the numeric average for the contents of cells A1 through A10. Search "AVERAGE function" on Help as well.)

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2

Collecting and Preparing Data

Figure 2.1 Periodic population surveys such as censuses help governments plan resources to supply public services. (credit: modification of work "Census 2010 @ La Fuente" by Jenn Turner/Flickr, CC BY 2.0)

Chapter Outline

2.1 Overview of Data Collection Methods
2.2 Survey Design and Implementation
2.3 Web Scraping and Social Media Data Collection
2.4 Data Cleaning and Preprocessing
2.5 Handling Large Datasets

Introduction

Data collection and preparation are the first steps in the data science cycle. They involve systematically gathering the necessary data to meet a project's objectives and ensuring its readiness for further analysis. Well-executed data collection and preparation serve as a solid foundation for effective, data-driven decision-making and aid in detecting patterns, trends, and insights that can drive business growth and efficiency.

With today's ever-increasing volume of data, a *robust* approach to data collection is crucial for ensuring accurate and meaningful results. This process requires following a comprehensive and systematic methodology designed to ensure the quality, reliability, and validity of data gathered for analysis. It involves identifying and sourcing relevant data from diverse sources, including internal databases, external repositories, websites, and user-generated information. And it requires meticulous planning and execution to guarantee the accuracy, comprehensiveness, and reliability of the collected data.

Preparing, or "wrangling," the collected data adequately prior to analysis is equally important. Preparation involves scrubbing, organizing, and transforming the data into a format suitable for analysis. Data preparation plays a pivotal role in detecting and resolving any inconsistencies or errors present in the data, thereby enabling accurate analysis. The rapidly advancing technology and widespread use of the internet have added complexity to the data collection and preparation processes. As a result, data analysts and organizations face many challenges, such as identifying relevant data sources, managing large data volumes, identifying outliers or erroneous data, and handling unstructured data. By mastering the art and science of collecting and preparing data, organizations can leverage valuable insights to drive informed decision-making and achieve

business success.

2.1 Overview of Data Collection Methods

Learning Outcomes

By the end of this section, you should be able to:

- 2.1.1 Define data collection and its role in data science.
- 2.1.2 Describe different data collection methods commonly used in data science, such as surveys and experiments.
- 2.1.3 Recognize scenarios where specific data collection methods are most appropriate.

Data collection refers to the systematic and well-organized process of gathering and accurately conveying important information and aspects related to a specific phenomenon or event. This involves using statistical tools and techniques to collect data, identify its attributes, and capture relevant contextual information. The gathered data is crucial for making sound interpretations and gaining meaningful insights. Additionally, it is important to take note of the environment and geographic location from where the data was obtained, as it can significantly influence the decision-making process and overall conclusions drawn from the data.

Data collection can be carried out through various methods, depending on the nature of the research or project and the type of data being collected. Some common methods for data collection include experiments, surveys, observation, focus groups, interviews, and document analysis.

This chapter will focus on the use of surveys and experiments to collect data. Social scientists, marketing specialists, and political analysts regularly use surveys to gather data on topics such as public opinion, customer satisfaction, and demographic information. Pharmaceutical companies heavily rely on experimental data from clinical trials to test the safety and efficacy of new drugs. This data is then used by their legal teams to gain regulatory approval and bring drugs to market.

Before collecting data, it is essential for a data scientist to have a clear understanding of the project's objectives, which involves identifying the research question or problem and defining the target population or sample. If a survey or experiment is used, the design of the survey/experiment is also a critical step, requiring careful consideration of the type of questions, response options, and overall structure. A survey may be conducted online, via phone, or in person, while experimental research requires a controlled environment to ensure data validity and reliability.

Types of Data

Observational and transactional data play important roles in data analysis and related decision-making, each offering unique insights into different aspects of real-world phenomena and business operations. **Observational data**, often used in qualitative research, is collected by systematically observing and recording behavior without the active participation of the researcher. **Transactional data** refers to any type of information related to transactions or interactions between individuals, businesses, or systems, and it is more often used in quantitative research.

Many fields of study use observational data for their research. Table 2.1 summarizes some examples of fields that rely on observational data, the type of data they collect, and the purpose of their data collection.

Field	Data Collected By	Purpose
Education	Teachers	To monitor and assess student behavior and learning progress in the classroom
Psychology	Therapists and psychologists	To gather information about their clients' behavior, thoughts, and emotions

Table 2.1 Fields Where Observation Methods Are Used

Field	Data Collected By	Purpose
Health care	Medical professionals	To diagnose and monitor patients' conditions and progress
Market research	Businesses	To gather information about consumer behavior and preferences to improve their products, services, and marketing strategies
Environmental science	Scientists	To gather data about the natural environment and track changes over time
Criminal investigations	Law enforcement officers	To gather evidence and information about criminal activity
Animal behavior	Zoologists	To study and understand the behavior of various animal species
Transportation planning	Urban planners and engineers	To collect data on traffic patterns and transportation usage to make informed decisions about infrastructure and transit systems

Table 2.1 Fields Where Observation Methods Are Used

Transactional data is collected by directly recording transactions that occur in a particular setting, such as a retail store or an online platform that allows for accurate and detailed information on actual consumer behavior. It can include financial data, but it also includes data related to customer purchases, website clicks, user interactions, or any other type of activity that is recorded and tracked.

Transactional data can be used to understand patterns and trends, make predictions and recommendations, and identify potential opportunities or areas for improvement. For example, the health care industry may focus on transactional data related to patient interactions with health care providers and facilities, such as appointments, treatments, and medications prescribed. The retail industry may use transactional data on customer purchases and product returns, while the transportation industry may analyze data related to ticket sales and passenger traffic.

While observational data provides detailed descriptions of behavior, transactional data provides numerical data for statistical analysis. There are strengths and limitations with each of these, and the examples in this chapter will make use of both types.

EXAMPLE 2.1

Problem

Ashley loves setting up a bird feeder in her backyard and watching the different types of birds that come to feed. She has always been curious about the typical number of birds that visit her feeder each day and has estimated the number based on the amount of food consumed. However, she has to visit her grandmother's house for three days and is worried about leaving the birds without enough food. In order to prepare the right amount of bird food for her absence, Ashley has decided to measure the total amount of feed eaten each day to determine the total amount of food needed for her three-day absence. Which method of data collection is best suited for Ashley's research on determining the total amount of food required for her three-day absence—observational or transactional? Provide a step-by-step explanation of the chosen method.

Solution

Ashley wants to ensure that there is enough food for her local birds while she is away for three days. To do this, she will carefully observe the feeder daily for two consecutive weeks. She will record the total amount of feed eaten each day and make sure to refill the feeder each morning before the observation. This will provide a consistent amount of food available for the birds. After two weeks, Ashley will use the total amount of food consumed and divide it by the number of days observed to estimate the required daily food. Then, she will multiply the daily food by three to determine the total amount of bird food needed for her three-day absence. By directly observing and recording the bird food, as well as collecting data for two weeks, Ashley will gather accurate and reliable information. This will help her confidently prepare the necessary amount of bird food for her feathered friends while she is away, thus ensuring that the birds are well-fed and taken care of during her absence.

EXAMPLE 2.2

Problem

A group of data scientists working for a large hospital have been tasked with analyzing their transactional data to identify areas for improvement. In the past year, the hospital has seen an increase in patient complaints about long wait times for appointments and difficulties scheduling follow-up visits. Samantha is one of the data scientists tasked to collect data in order to analyze these issues.

a. What methodology should be employed by Samantha to collect pertinent data for analyzing the recent surge in patient complaints regarding extended appointment wait times and difficulties in scheduling follow-up visits at the hospital?
b. What strategies could be used to analyze the data?

Solution

a. Explore the stored information as transactional data
b. Collecting transactional data for analysis can be achieved by utilizing various sources within the hospital setting. These sources include:

1. Electronic Health Records (EHRs): Samantha can gather data from the hospital's electronic health records system. This data may include patients' appointment schedules, visit durations, and wait times. This information can help identify patterns and trends in appointment scheduling and wait times.
2. Appointment Booking System: Samantha can gather data from the hospital's appointment booking system. This data can include appointment wait times, appointment types (e.g., primary care, specialist), and scheduling difficulties (e.g., appointment availability, cancellations). This information can help identify areas where the booking system may be causing delays or challenges for patients.
3. Hospital Call Center: Samantha can gather data from the hospital's call center, which is responsible for booking appointments over the phone. This data can include call wait times, call duration, and reasons for call escalations. This information can help identify areas for improvement in the call center's processes and procedures.
4. Historical Data: Samantha can analyze historical data, such as appointment wait times and scheduling patterns, to identify any changes that may have contributed to the recent increase in complaints. This data can also be compared to current data to track progress and improvements in wait times and scheduling.

Collecting Data Through Experiments

Collecting data through scientific experiments requires a well-designed experimental scheme, describing the research objectives, variables, and procedures. The establishment of a control specimen is crucial, and data is obtained through systematic properties, measurements, or characteristics. It is crucial to follow ethical guidelines for the proper documentation and ethical utilization of the collected data (see Ethics in Data Collection).

Consider this example: Scientist Sally aimed to investigate the impact of sunlight on plant growth. The research inquiry was to determine whether increased exposure to sunlight enhances the growth of plants. Sally experimented with two groups of plants wherein one group received eight hours of sunlight per day, while the other only received four hours. The height of each plant was measured and documented every week for four consecutive weeks. The main research objective was to determine the growth rate of plants exposed to eight hours of sunlight compared to those with only four hours. A total of 20 identical potted plants were used, with one group allocated to the "sunlight" condition and the other to the "limited sunlight" condition. Both groups were maintained under identical environmental conditions, including temperature, humidity, and soil moisture. Adequate watering was provided to ensure equal hydration of all plants. The measurements of plant height were obtained and accurately recorded every week. This approach allowed for the collection of precise and reliable data on the impact of sunlight on plant growth, which can serve as a valuable resource for further research and understanding of this relationship.

2.2 Survey Design and Implementation

Learning Outcomes

By the end of this section, you should be able to:

- 2.2.1 Describe the elements of survey design and identify the steps data scientists take to ensure the reliability of survey results.
- 2.2.2 Describe methods for avoiding bias in survey questions.
- 2.2.3 Describe various sampling techniques and the advantages of each.

Surveys are a common strategy for gathering data in a wide range of domains, including market research, social sciences, and education. Surveys collect information from a *sample* of individuals and often use questionnaires to collect data. **Sampling** is the process of selecting a subset of a larger population to represent and analyze information about that population.

Designing the Survey

The process of data collection through surveys is a crucial aspect of research—and one that requires careful planning and execution to gather accurate and reliable data. The first step, as stated earlier, is to clearly define the research objectives and determine the appropriate target population. This will help you structure the survey and identify the specific questions that need to be included.

Constructing good surveys is hard. A survey should begin with simple and easy-to-answer questions and progress to more complex or sensitive ones. This can help build a rapport with the respondents and increase their willingness to answer more difficult questions. Additionally, the researcher may consider mixing up the response options for multiple-choice questions to avoid response bias. To ensure the quality of the data collected, the survey questionnaire should undergo a pilot test with a small group of individuals from the target population. This allows the researcher to identify any potential issues or confusion with the questions and make necessary adjustments before administering the survey to the larger population.

Open-Ended Versus Closed-Ended Questions

Surveys should generally contain a mix of closed-ended and open-ended questions to gather both quantitative and qualitative data.

Open-ended questions allow for more in-depth responses and provide the opportunity for unexpected insights. They also allow respondents to elaborate on their thoughts and provide detailed and personal responses. **Closed-ended questions** have predetermined answer choices and are effective in gathering quantitative data. They are quick and easy to answer, and their clear and structured format allows for quantifiable results.

Avoiding Bias in Survey Questions

Unbiased sampling and unbiased survey methodology are essential for ensuring accurate and reliable results. One well-known real-life instance of sampling bias leading to inaccurate findings is the 1936 *Literary Digest* poll. This survey aimed to forecast the results of the US presidential election and utilized a mailing list of telephone and automobile owners. This approach was considered biased toward affluent individuals and therefore favored Republican voters. As a consequence, the poll predicted a victory for Republican nominee Alf Landon. However, the actual outcome was a landslide win for Franklin D. Roosevelt (Lusinchi, 2012). This discrepancy can be attributed to the biased sampling method as well as the use of primarily closed-ended questions, which may not have accurately captured the opinions of all voters.

An example of a biased survey question in a survey conducted by a shampoo company might be "Do you prefer our brand of shampoo over cheaper alternatives?" This question is biased because it assumes that the respondent prefers the company's brand over others. A more unbiased and accurate question would be "What factors do you consider when choosing a shampoo brand?" This allows for a more detailed and accurate response. The biased question could have led to inflated results in favor of the company's brand.

Sampling

The next step in the data collection process is to choose a participant *sample* to ideally represent the restaurant's customer base. Sampling could be achieved by randomly selecting customers, using customer databases, or targeting specific demographics, such as age or location.

Sampling is necessary in a wide range of data science projects to make data collection more manageable and cost-effective while still drawing meaningful conclusions. A variety of techniques can be employed to determine a subset of data from a larger population to perform research or construct hypotheses about the entire population. The choice of a sampling technique depends upon the nature and features of the population being studied as well as the objectives of the research. When using a survey, researchers must also consider the tool(s) that will be used for distributing the survey, such as through email, social media, or physically distributing questionnaires at the restaurant. It's crucial to make the survey easily accessible to the chosen sample to achieve a higher response rate.

A number of sampling techniques and their advantages are described below. The most frequently used among these are *simple random selection*, *stratified sampling*, *cluster sampling*, and *convenience sampling*.

1. **Simple random selection.** Simple random selection is a statistical technique used to pick a representative sample from a larger population. This process involves randomly choosing individuals or items from the population, ensuring that each selected member of the population has an identical chance of being contained in the sample. The main step in simple random selection is to define the population of interest and assign a unique identification number to each member. This could be done using a random number generator, a computer program designed to generate a sequence of random numbers, or a random number table, which lists numbers in a random sequence. The primary benefit of this technique is its ability to minimize bias and deliver a fair representation of the population.

   In the health care field, simple random sampling is utilized to select patients for medical trials or surveys, allowing for a diverse and unbiased sample (Elfil & Negida, 2017). Similarly, in finance, simple random sampling can be applied to gather data on consumer behavior and guide decision-making in financial institutions. In engineering, this technique is used to select random samples of materials or components

for quality control testing. In the political arena, simple random sampling is commonly used to select randomly registered voters for polls or surveys, ensuring equal representation and minimizing bias in the data collected.

2. **Stratified sampling.** Stratified sampling involves splitting the population into subgroups based on specified factors, such as age, area, income, or education level, and taking a random sample from each stratum in proportion to its size in the population. Stratified sampling allows for a more accurate representation of the population as it ensures that all subgroups are adequately represented in the sample. This can be especially useful when the variables being studied vary significantly between the stratified groups.
3. **Cluster sampling.** With cluster sampling, the population is divided into natural groups or clusters, such as schools, communities, or cities, with a random sample of these clusters picked and all members within the chosen clusters included in the sample. Cluster sampling is helpful to represent the entire population even if it is difficult or time-consuming due to challenges such as identifying clusters, sourcing a list of clusters, traveling to different clusters, and communicating with them. Additionally, data analysis and sample size calculation may be more complex, and there is a risk of bias in the sample. However, cluster sampling can be more cost-effective.

   An example of cluster sampling would be a study on the effectiveness of a new educational program in a state. The state is divided into clusters based on school districts. The researcher uses a random selection process to choose a sample of school districts and then collects data from all the schools within those districts. This method allows the researcher to obtain a representative sample of the state's student population without having to visit each individual school, saving time and resources.
4. **Convenience sampling.** Convenience sampling applies to selecting people or items for the sample based on their availability and convenience to the data science research. For example, a researcher may choose to survey students in their classroom or manipulate data from social media users. Convenience sampling is effortless to achieve, and it is useful for exploratory studies. However, it may not provide a representative sample as it is prone to selection bias in that individuals who are more readily available or willing to participate may be overrepresented.

   An example of convenience sampling would be conducting a survey about a new grocery store in a busy shopping mall. A researcher stands in front of the store and approaches people who are coming out of the store to ask them about their shopping experience. The researcher only includes responses from those who agreed to participate, resulting in a sample that is convenient but may not be representative of the entire population of shoppers in the mall.
5. **Systematic sampling.** Systematic sampling is based on starting at a random location in the dataset and then selecting every nth member from a population to be contained in the sample. This process is straightforward to implement, and it provides a representative sample when the population is randomly distributed. However, if there is a pattern in the **sampling frame** (the organizing structure that represents the population from which a sample is drawn), it may lead to a biased sample.

   Suppose a researcher wants to study the dietary habits of students in a high school. The researcher has a list of all the students enrolled in the school, which is approximately 1,000 students. Instead of randomly selecting a sample of students, the researcher decides to use systematic sampling. The researcher first assigns a number to each student, going from 1 to 1,000. Then, the researcher randomly selects a number from 1 to 10—let's say they select 4. This number will be the starting point for selecting the sample of students. The researcher will then select every 10th student from the list, which means every student with a number ending in 4 (14, 24, 34, etc.) will be included in the sample. This way, the researcher will have a representative sample of 100 students from the high school, which is 10% of the population. The sample will consist of students from different grades, genders, and backgrounds, making it a diverse and representative sample.

6. **Purposive sampling.** With purposive sampling, one or more specific criteria are used to select participants who are likely to provide the most relevant and useful information for the research study. This can involve selecting participants based on their expertise, characteristics, experiences, or behaviors that are relevant to the research question.

   For example, if a researcher is conducting a study on the effects of exercise on mental health, they may use purposive sampling to select participants who have a strong interest or experience in physical fitness and have a history of mental health issues. This sampling technique allows the researcher to target a specific population that is most relevant to the research question, making the findings more applicable and generalizable to that particular group. The main advantage of purposive sampling is that it can save time and resources by focusing on individuals who are most likely to provide valuable insights and information. However, researchers need to be transparent about their sampling strategy and potential biases that may arise from purposely selecting certain individuals.
7. **Snowball sampling.** Snowball sampling is typically used in situations where it is difficult to access a particular population; it relies on the assumption that people with similar characteristics or experiences tend to associate with each other and can provide valuable referrals. This type of sampling can be useful in studying hard-to-reach or sensitive populations, but it may also be biased and limit the generalizability of findings.
8. **Quota sampling.** Quota sampling is a non-probability sampling technique in which experimenters select participants based on predetermined quotas to guarantee that a certain number or percentage of the population of interest is represented in the sample. These quotas are based on specific demographic characteristics, such as age, gender, ethnicity, and occupation, which are believed to have a direct or indirect relationship with the research topic. Quota sampling is generally used in market research and opinion polls, as it allows for a fast and cost-effective way to gather data from a diverse range of individuals. However, it is important to note that the results of quota sampling may not accurately represent the entire population, as the sample is not randomly selected and may be biased toward certain characteristics. Therefore, the findings from studies using quota sampling should be interpreted with caution.
9. **Volunteer sampling.** Volunteer sampling refers to the fact that the participants are not picked at random by the researcher, but instead volunteer themselves to be a part of the study. This type of sampling is commonly used in studies that involve recruiting participants from a specific population, such as a specific community or organization. It is also often used in studies where convenience and accessibility are important factors, as participants may be more likely to volunteer if the study is easily accessible to them. Volunteer sampling is not considered a random or representative sampling technique, as the participants may not accurately represent the larger population. Therefore, the results obtained from volunteer sampling may not be generalizable to the entire population.

Sampling Error

Sampling error is the difference between the results obtained from a sample and the true value of the population parameter it is intended to represent. It is caused by chance and is inherent in any sampling method. The goal of researchers is to minimize sampling errors and increase the accuracy of the results. To avoid sampling error, researchers can increase sample size, use probability sampling methods, control for extraneous variables, use multiple modes of data collection, and pay careful attention to question formulation.

Sampling Bias

Sampling bias occurs when the sample used in a study isn't representative of the population it intends to generalize to, leading to skewed or inaccurate conclusions. This bias can take many forms, such as *selection bias*, where certain groups are systematically over- or underrepresented, or *volunteer bias*, where only a specific subset of the population participates. Researchers use the sampling techniques summarized earlier to avoid sampling bias and ensure that each member of the population has an equal chance of being included in

the sample. Additionally, careful consideration of the sampling frame should ideally encompass all members of the target population and provide a clear and accessible way to identify and select individuals or units for inclusion in the sample. Sampling bias can occur at various stages of the sampling process, and it can greatly impact the accuracy and validity of research findings.

Measurement Error

Measurement errors are inaccuracies or discrepancies that surface during the process of collecting, recording, or analyzing data. They may occur due to human error, environmental factors, or inherent inconsistencies in the phenomena being studied. *Random error*, which arises unpredictably, can affect the precision of measurements, and *systematic error* may consistently bias measurements in a particular direction. In data analysis, addressing measurement error is crucial for ensuring the reliability and validity of results. Techniques for mitigating measurement error include improving data collection methods, calibrating instruments, conducting validation studies, and employing statistical methods like error modeling or sensitivity analysis to account for and minimize the impact of measurement inaccuracies on the analysis outcomes.

A Sampling Case Study

Consider a research study that wants to randomly select a group of college students from a larger population to examine the effects of exercise on their mental health outcomes. Using student ID numbers generated by a computer program, 100 participants from the larger population were randomly selected to participate in the study to achieve the desired accuracy. This process ensured that every student in the university had an equal chance of being selected to participate. The participants were then randomly assigned to either the exercise group or the control group. This method of random sampling ensures that the sample is representative of the larger population, providing a more accurate representation of the relationship between exercise and mental health outcomes for college students.

Types of sampling error that could occur in this study include the following:

1. **Sampling bias.** One potential source of bias in this study is *self-selection bias*. As the participants are all college students, they may not be representative of the larger population, as college students tend to have more access and motivation to exercise compared to the general population. This could limit the generalizability of the study's findings. In addition, if the researchers only recruit participants from one university, there may be *under-coverage bias*. This means that certain groups of individuals, such as nonstudents or students from other universities, may be excluded from the study, potentially leading to biased results.
2. **Measurement error.** Measurement errors could occur, particularly if the researchers are measuring the participants' exercise and mental health outcomes through self-report measures. Participants may not accurately report their exercise habits or mental health symptoms, leading to inaccurate data.
3. **Non-response bias.** Some participants in the study may choose not to participate or may drop out before the study is completed. This could introduce non-response bias, as those who choose not to participate or drop out may differ from those who remain in the study in terms of their exercise habits or mental health outcomes.
4. **Sampling variability.** The sample of 100 participants is a relatively small subset of the larger population. As a result, there may be sampling variability, meaning that the characteristics and outcomes of the participants may differ from those of the larger population simply due to chance.
5. **Sampling error in random assignment.** In this study, the researchers randomly assign participants to either the exercise group or the control group. However, there is always a possibility of sampling error in the random assignment process, meaning that the groups may not be perfectly balanced in terms of their exercise habits or other characteristics.

   These types of sampling errors can affect the accuracy and generalizability of the study's findings.

Researchers need to be aware of these potential errors and take steps to minimize them when designing and conducting their studies.

EXAMPLE 2.3

Problem

Mark is a data scientist who works for a marketing research company. He has been tasked to lead a study to understand consumer behavior toward a new product that is about to be launched in the market. As data scientists, they know the importance of using the right sampling technique to collect accurate and reliable data. Mark divided the population into different groups based on factors such as age, education, and income. This ensures that he gets a representative sample from each group, providing a more accurate understanding of consumer behavior. What is the name of the sampling technique used by Mark to ensure a representative sample from different groups of consumers for his study on consumer behavior toward a new product?

Solution

The sampling technique used by Mark is called stratified sampling. This involves dividing the population into subgroups or strata based on certain characteristics and then randomly selecting participants from each subgroup. This ensures that each subgroup is represented in the sample, providing a more accurate representation of the entire population. This type of sampling is often used in market research studies to get a more comprehensive understanding of consumer behavior and preferences. By using stratified sampling, Mark can make more reliable conclusions and recommendations for the new product launch based on the data he collects.

2.3 Web Scraping and Social Media Data Collection

Learning Outcomes:

By the end of this section, you should be able to:

- 2.3.1 Discuss the uses of web scraping for collecting and preparing data for analysis.
- 2.3.2 Apply regular expressions for data manipulation and pattern matching.
- 2.3.3 Write Python code to scrape data from the web.
- 2.3.4 Apply various methods for parsing, extracting, processing, and storing data.

Web scraping and social media data collection are two approaches used to gather data from the internet. **Web scraping** involves pulling information and data from websites using a web data extraction tool, often known as a web scraper. One example would be a travel company looking to gather information about hotel prices and availability from different booking websites. Web scraping can be used to automatically gather this data from the various websites and create a comprehensive list for the company to use in its business strategy without the need for manual work.

Social media data collection involves gathering information from various platforms like Twitter and Instagram using application programming interface or monitoring tools. An **application programming interface (API)** is a set of protocols, tools, and definitions for building software applications allowing different software systems to communicate and interact with each other and enabling developers to access data and services from other applications, operating systems, or platforms. Both web scraping and social media data collection require determining the data to be collected and analyzing it for accuracy and relevance.

Web Scraping

There are several techniques and approaches for scraping data from websites. See Table 2.2 for some of the

common techniques used. (Note: The techniques used for web scraping will vary depending on the website and the type of data being collected. It may require a combination of different techniques to effectively scrape data from a website.)

Web Scraping Technique	Details
Web Crawling	• Follows links on a web page to navigate to other pages and collect data from them • Useful for scraping data from multiple pages of a website
XPath	• Powerful query language • Navigates through the elements in an HTML document • Often used in combination with HTML parsing to select specific elements to scrape
Regular Expressions	• Search for and extract specific patterns of text from a web page • Useful for scraping data that follows a particular format, such as dates, phone numbers, or email addresses
HTML Parsing	• Analyzes the HTML (HyperText Markup Language) structure of a web page and identifies the specific tags and elements that contain the desired data • Often used for simple scraping tasks
Application Programming Interfaces (APIs)	• Authorize developers to access and retrieve data instantly without the need for web scraping • Often a more efficient and reliable method for data collection
(XML) API Subset	• XML (Extensible Markup Language) is another markup language used exchanging data • This method works similarly to using the HTML API subset by making HTTP requests to the website's API endpoints and then parsing the data received in XML format
(JSON) API Subset	• JSON (JavaScript Object Notation) is a lightweight data interchange format that is commonly used for sending and receiving data between servers and web applications • Many websites provide APIs in the form of JSON, making it another efficient method for scraping data

Table 2.2 Techniques and Approaches for Scraping Data from Websites

Social Media Data Collection

Social media data collection can be carried out through various methods such as API integration, social listening, social media surveys, network analysis, and image and video analysis. APIs provided by social media platforms allow data scientists to collect structured data on user interactions and content. **Social listening** involves monitoring online conversations for insights on customer behavior and trends. Surveys conducted on social media can provide information on customer preferences and opinions. **Network analysis**, or the examination of relationships and connections between users, data, or entities within a network, can reveal influential users and communities. It involves identifying and analyzing influential individuals or groups as well as understanding patterns and trends within the network. Image and video analysis can provide insights into visual trends and user behavior.

An example of social media data collection is conducting a Twitter survey on customer satisfaction for a food delivery company. Data scientists can use Twitter's API to collect tweets containing specific hashtags related to the company and analyze them to understand customers' opinions and preferences. They can also use social listening to monitor conversations and identify trends in customer behavior. Additionally, creating a social media survey on Twitter can provide more targeted insights into customer satisfaction and preferences. This data can then be analyzed using data science techniques to identify key areas for improvement and drive informed business decisions.

Using Python to Scrape Data from the Web

As noted previously, web scraping is a strategy of gathering data from the internet using automated mechanisms or programs. Python is one of the popular programming languages used for web scraping due to its various libraries and frameworks that make it easy to pull and process data from websites.

To scrape data such as a table from a website using Python, we follow these steps:

1. **Import the `pandas` library.** The first step is to import the pandas library (https://openstax.org/r/pandas), which is a popular Python library for data analysis and manipulation.

```
import pandas as pd
```

2. **Use the `read_html()` function.** This function is used to read HTML tables from a web page and convert them into a list of DataFrame objects. Recall from What Are Data and Data Science? that a **DataFrame** is a data type that `pandas` uses to store multi-column tabular data.

```
df = pd.read_html("https://.....
```

3. **Access the desired data.** If the data on the web page is divided into different tables, we need to specify which table we want to extract. We have used indexing to access the desired table (for example: index 4) from the list of DataFrame objects returned by the `read_html()` function. The index here represents the table order in the web page.
4. **Store the data in a DataFrame.** The result of the `read_html()` function is a list of DataFrame objects, and each DataFrame represents a table from the web page. We can store the desired data in a DataFrame variable for further analysis and manipulation.
5. **Display the DataFrame.** By accessing the DataFrame variable, we can see the extracted data in a tabular format.
6. **Convert strings to numbers.** As noted in Chapter 1, a **string** is a data type used to represent a sequence of characters, such as letters, numbers, and symbols that are enclosed by matching single (') or double (") quotes. If the data in the table is in string format and we want to perform any numerical operations on it, we need to convert the data to numerical format. We can use the `to_numeric()` function from `pandas` to convert strings to numbers and then store the result in a new column in the DataFrame.

```
df['column_name'] = pd.to_numeric(df['column_name'])
```

This will create a new column in the DataFrame with the converted numerical values, which can then be used for further analysis or visualization.

In computer programming, indexing usually starts from 0. This is because most programming languages use 0 as the initial index for arrays, matrices, or other data structures. This convention has been adopted to simplify the implementation of some algorithms and to make it easier for programmers to access and manipulate data. Additionally, it aligns with the way computers store and access data in memory. In the context of parsing tables from HTML pages, using 0 as the initial index allows programmers to easily access and manipulate data from different tables on the same web page. This enables efficient data processing and analysis, making the

task more manageable and streamlined.

EXAMPLE 2.4

Problem

Extract data table "Current Population Survey: Household Data: (Table A-13). Employed and unemployed persons by occupation, Not seasonally adjusted" from the FRED (Federal Reserve Economic Data) (https://openstax.org/r/fred) website in the link (https://fred.stlouisfed.org/release/tables?rid=50&eid=3149#snid=4498 (https://openstax.org/r/stlouisfed)) using Python code. The data in this table provides a representation of the overall employment and unemployment situation in the United States. The table is organized into two main sections: employed persons and unemployed persons.

Solution

PYTHON CODE



```
# Import pandas
import pandas as pd

# Read data from the URL
df_list = pd.read_html(
"https://fred.stlouisfed.org/release/tables?rid=50&eid=3149#snid=4498")

# Since pd.read_html() returns a list of DataFrames, select the first DataFrame
df = df_list[0]

# Print the first 5 rows of the DataFrame
print(df.head(5))
```

The resulting output will look like this:

```
   Unnamed: 0                                                Name  \
0         NaN                                 Monthly, Employed
1         NaN                          Total, 16 years and over
2         NaN  Management, professional, and related occupations
3         NaN  Management, business, and financial operations...
4         NaN              Professional and related occupations

                      May 2024                      Apr 2024  \
0                          NaN                           NaN
1  161,341  Thousands of Persons  161,590  Thousands of Persons
2   70,897  Thousands of Persons   70,548  Thousands of Persons
3   30,910  Thousands of Persons   30,172  Thousands of Persons
4   39,987  Thousands of Persons   40,376  Thousands of Persons

                      May 2023             Units
0                          NaN               NaN
1  161,002  Thousands of Persons  Thous. of Persons
2   70,388  Thousands of Persons  Thous. of Persons
3   30,830  Thousands of Persons  Thous. of Persons
4   39,557  Thousands of Persons  Thous. of Persons
```

In Python, there are several libraries and methods that can be used for parsing and extracting data from text. These include the following:

1. Regular expressions (regex or RE) (https://openstax.org/r/docpython). This is a built-in library in Python that allows for pattern matching and extraction of data from strings. It uses a specific syntax to define patterns and rules for data extraction.
2. Beautiful Soup (https://openstax.org/r/pypi). This is an external library that is mostly used for scraping and parsing HTML and XML code. It can be utilized to extract specific data from web pages or documents.
3. Natural Language Toolkit (NLTK) (https://openstax.org/r/nltk). This is a powerful library for natural language processing in Python. It provides various tools for tokenizing, parsing, and extracting data from text data. (**Tokenizing** is the process of breaking down a piece of text or string of characters into smaller units called tokens, which can be words, phrases, symbols, or individual characters.)
4. TextBlob (https://openstax.org/r/textblob). This library provides a simple interface for most natural language processing assignments, such as argument and part-of-speech tagging. It can also be utilized for parsing and extracting data from text.
5. SpaCy (https://openstax.org/r/spacy). This is a popular open-source library for natural language processing. It provides efficient methods for tokenizing, parsing, and extracting data from text data.

Overall, the library or method used for parsing and extracting data will depend on the specific task and type of data being analyzed. It is important to research and determine the best approach for a given project.

Regular Expressions in Python

Regular expressions, also known as **regex**, are a set of symbols used to define a search pattern in text data. In Python, these expressions are supported by the `re` module (function), and their syntax is similar to other programming languages. The use of regular expressions offers researchers a robust method for identifying and manipulating patterns in text. With this powerful tool, specific words, characters, or patterns of characters

can be searched and matched. Typical applications include data parsing, input validation, and extracting targeted information from larger text sources. Common use cases in Python involve recognizing various types of data, such as dates, email addresses, phone numbers, and URLs, within extensive text files. Moreover, regular expressions are valuable for tasks like data cleaning and text processing. Despite their versatility, regular expressions can be elaborate, allowing for advanced search patterns utilizing meta-characters like `*`, `?`, and `+`. However, working with these expressions can present challenges, as they require a thorough understanding and careful debugging to ensure successful implementation.

USING META-CHARACTERS IN REGULAR EXPRESSIONS

- The `*` character is known as the “star” or “asterisk” and is used to match zero or more occurrences of the preceding character or group in a regular expression. For example, the regular expression `"a*"` would match an `"a"` followed by any number (including zero) of additional `"a"`s, such as `"a"`, `"aa"`, `"aaa"`, etc.
- The `?` character is known as the "question mark" and is used to indicate that the preceding character or group is optional. It matches either zero or one occurrences of the preceding character or group. For example, the regular expression `"a?b"` would match either `"ab"` or `"b"`.
- The `+` character is known as the "plus sign" and is used to match one or more occurrences of the preceding character or group. For example, the regular expression `"a+b"` would match one or more `"a"`s followed by a `"b"`, such as `"ab"`, `"aab"`, `"aaab"`, etc. If there are no `"a"`s, the match will fail. This is different from the `*` character, which would match zero or more `"a"`s followed by a `"b"`, allowing for a possible match without any `"a"`s.

EXAMPLE 2.5

Problem

Write Python code using regular expressions to search for a selected word “Python” in a given string and print the number of times it appears.

Solution

PYTHON CODE



```
## import the regular expression module
import re
# create a string with story problem
story = "Samantha is a sixth-grade student who uses the popular coding language
Python to collect and analyze weather data for science project. She creates a
program that collects data from an online weather API and stores it in a CSV file
for 10 days. With the help of her teacher, she uses Python to visualize the data
and discovers patterns in temperature, humidity, and precipitation."
# create a regex pattern to match a selected word
pattern = "Python"
```

```
# use the "findall" to search for matching patterns in the data string
words = re.findall(pattern, story)

# print the number of repeated python
print ("The word 'python' is repeated", len(words), "times in the story.")

# print the matched patterns
print(words) # output: ['Python']
```

The resulting output will look like this:

```
The word 'python' is repeated 2 times in the story.
['Python', 'Python']
```

In Python, "`len`" is short for "length" and is used to determine the number of items in a collection. It returns the total count of items in the collection, including spaces and punctuation marks in a string.

Parsing and Extracting Data

Splitting and slicing are two methods used to manipulate text strings in programming. **Splitting a string** means dividing a text string into smaller parts or substrings based on a specified separator. The separator can be a character, string, or regular expression. This can be useful for separating words, phrases, or data values within a larger string. For example, the string `"Data Science"` can be split into two substrings `"Data"` and `"Science"` by using a space as the separator.

Slicing a string refers to extracting a portion or section of a string based on a specified range of indices. An index refers to the position of a character in a string, starting from 0 for the first character. The range specifies the start and end indices for the slice, and the resulting substring includes all characters within that range. For example, the string `"Data Science"` can be sliced to extract `"Data"` by specifying the range from index 0 to 4, which includes the first four characters. Slicing can also be used to manipulate strings by replacing, deleting, or inserting new content into specific positions within the string.

Parsing and extracting data involves the analysis of a given dataset or string to extract specific pieces of information. This is accomplished using various techniques and functions, such as splitting and slicing strings, which allow for the structured retrieval of data. This process is particularly valuable when working with large and complex datasets, as it provides a more efficient means of locating desired data compared to traditional search methods. Note that parsing and extracting data differs from the use of regular expressions, as regular expressions serve as a specialized tool for pattern matching and text manipulation. In contrast, parsing and data extraction offers a comprehensive approach to identifying and extracting specific data within a dataset.

Parsing and extracting data using Python involves using the programming language to locate and extract specific information from a given text. This is achieved by utilizing the `re` library, which enables the use of regular expressions to identify and retrieve data based on defined patterns. This process can be demonstrated through an example of extracting data related to a person purchasing an iPhone at an Apple store.

The code in the following Python feature box uses regular expressions (regex) to match and extract specific data from a string. The string is a paragraph containing information about a person purchasing a new phone

from the Apple store. The objective is to extract the product name, model, and price of the phone. First, the code starts by importing the necessary library for using regular expressions. Then, the string data is defined as a variable. Next, regex is used to search for specific patterns in the string. The first pattern searches for the words `"product: "` and captures anything that comes after it until it reaches a comma. The result is then stored in a variable named `"product"`. Similarly, the second pattern looks for the words `"model: "` and captures anything that comes after it until it reaches a comma. The result is saved in a variable named `"model"`. Finally, the third pattern searches for the words `"price: "` and captures any sequence of numbers or symbols that follows it until the end of the string. The result is saved in a variable named `"price"`. After all the data is extracted, it is printed out to the screen, using concatenation to add appropriate labels before each variable.

This application of Python code demonstrates the effective use of regex and the `re` library to parse and extract specific data from a given data. By using this method, the desired information can be easily located and retrieved for further analysis or use.

PYTHON CODE



```
## import necessary library
import re
# Define data to be parsed
data = "Samantha went to the Apple store to purchase a new phone. She was
specifically looking for the latest and most expensive model available. As she
looked at the different options, she came across the product: iPhone 12, the
product name caught her attention, as it was the newest version on the market. She
then noticed the model: A2172, which confirmed that this was indeed the latest and
most expensive model she was looking for. The price made her hesitate for a moment,
but she decided that it was worth it price: $799. She purchased the iPhone 12 and
was excited to show off her new phone to her friends."

# Use regex to match and extract data based on specific pattern
product = re.search(r"product: (.+?),", data).group(1)
model = re.search(r"model: (.+?),", data).group(1)
price = re.search(r"price: (.+?.+?.+?.+?)", data).group(1)

# Print the extracted data
print("product: " + product)
print("model: " + model)
print("price: " + price)
```

The resulting output will look like this:

```
product: iPhone 12 model: A2172 price: $799
```

Processing and Storing Data

Once the data is collected, it should be processed and stored in a suitable format for further analysis. This is where data processing and storage come into play. **Data processing** manages the raw data first by cleaning it through the removal of irrelevant information and then by transforming it into a structured format. The

cleaning process includes identifying and correcting any errors, inconsistencies, and missing values in a dataset and is essential for ensuring that the data is accurate, reliable, and usable for analysis or other purposes. Python is often utilized for data processing due to its flexibility and ease of use, and it offers a wide range of tools and libraries specifically designed for data processing. Once the data is processed, it needs to be stored for future use. (We will cover data storage in Data Cleaning and Preprocessing.) Python has several libraries that allow for efficient storage and manipulation of data in the form of DataFrames.

One method of storing data using Python is using the `pandas` library to create a DataFrame and then using the `to_csv()` function to save the DataFrame as a CSV (comma-separated values) file. This file can then be easily opened and accessed for future analysis or visualization. For example, the code in the following Python sidebar is a Python script that creates a *dictionary* with data about the presidents of the United States (https://openstax.org/r/wikipedia), including their ordered number and state of birth. A **data dictionary** is a data structure that stores data in key-value pairs, allowing for efficient retrieval of data using its key. It then uses the built-in CSV library (https://openstax.org/r/librarycsv) to create a CSV file and write the data to it. This code is used to store the US presidents' data in a structured format for future use, analysis, or display.

PYTHON CODE



```
import csv

# Create a dictionary to store the data
presidents = {
    "1": ["George Washington", "Virginia"],
    "2": ["John Adams", "Massachusetts"],
    "3": ["Thomas Jefferson", "Virginia"],
    "4": ["James Madison", "Virginia"],
    "5": ["James Monroe", "Virginia"],
    "6": ["John Quincy Adams", "Massachusetts"],
    "7": ["Andrew Jackson", "South Carolina"],
    "8": ["Martin Van Buren", "New York"],
    "9": ["William Henry Harrison", "Virginia"],
    "10": ["John Tyler", "Virginia"],
    "11": ["James K. Polk", "North Carolina"],
    "12": ["Zachary Taylor", "Virginia"],
    "13": ["Millard Fillmore", "New York"],
    "14": ["Franklin Pierce", "New Hampshire"],
    "15": ["James Buchanan", "Pennsylvania"],
    "16": ["Abraham Lincoln", "Kentucky"],
    "17": ["Andrew Johnson", "North Carolina"],
    "18": ["Ulysses S. Grant", "Ohio"],
    "19": ["Rutherford B. Hayes", "Ohio"],
    "20": ["James A. Garfield", "Ohio"],
    "21": ["Chester A. Arthur", "Vermont"],
    "22": ["Grover Cleveland", "New Jersey"],
    "23": ["Benjamin Harrison", "Ohio"],
    "24": ["Grover Cleveland", "New Jersey"],
    "25": ["William McKinley", "Ohio"],
```

```
    "26": ["Theodore Roosevelt", "New York"],
    "27": ["William Howard Taft", "Ohio"],
    "28": ["Woodrow Wilson", "Virginia"],
    "29": ["Warren G. Harding", "Ohio"],
    "30": ["Calvin Coolidge", "Vermont"],
    "31": ["Herbert Hoover", "Iowa"],
    "32": ["Franklin D. Roosevelt", "New York"],
    "33": ["Harry S. Truman", "Missouri"],
    "34": ["Dwight D. Eisenhower", "Texas"],
    "35": ["John F. Kennedy", "Massachusetts"],
    "36": ["Lyndon B. Johnson", "Texas"],
    "37": ["Richard Nixon", "California"],
    "38": ["Gerald Ford", "Nebraska"],
    "39": ["Jimmy Carter", "Georgia"],
    "40": ["Ronald Reagan", "Illinois"],
    "41": ["George H. W. Bush", "Massachusetts"],
    "42": ["Bill Clinton", "Arkansas"],
    "43": ["George W. Bush", "Connecticut"],
    "44": ["Barack Obama", "Hawaii"],
    "45": ["Donald Trump", "New York"],
    "46": ["Joe Biden", "Pennsylvania"]
}

# Open a new CSV file in write mode
with open("presidents.csv", "w", newline='') as csv_file:
    # Specify the fieldnames for the columns
    fieldnames = ["Number", "Name", "State of Birth"]
    # Create a writer object
    writer = csv.DictWriter(csv_file, fieldnames=fieldnames)
    # Write the header row
    writer.writeheader()
    # Loop through the presidents dictionary
    for key, value in presidents.items():
        # Write the data for each president to the CSV file
        writer.writerow({
            "Number": key,
            "Name": value[0],
            "State of Birth": value[1]
            })
# Print a success message
print("Data successfully stored in CSV file.")
```

The resulting output will look like this:

```
Data successfully stored in CSV file.
```

2.4 Data Cleaning and Preprocessing

Learning Outcomes

By the end of this section, you should be able to:

- 2.4.1 Apply methods to deal with missing data and outliers.
- 2.4.2 Explain data standardization techniques, such as normalization, transformation, and aggregation.
- 2.4.3 Identify sources of noise in data and apply various data preprocessing methods to reduce noise.

Data cleaning and preprocessing is an important stage in any data science task. It refers to the technique of organizing and converting raw data into usable structures for further analysis. It involves extracting irrelevant or duplicate data, handling missing values, and correcting errors or inconsistencies. This ensures that the data is accurate, comprehensive, and ready for analysis. Data cleaning and preprocessing typically involve the following steps:

1. **Data integration. Data integration** refers to merging data from multiple sources into a single dataset.
2. **Data cleaning.** In this step, data is assessed for any errors or inconsistencies, and appropriate actions are taken to correct them. This may include removing duplicate values, handling missing data, and correcting formatting misconceptions.
3. **Data transformation.** This step prepares the data for the next step by transforming the data into a format that is suitable for further analysis. This may involve converting data types, **scaling** or normalizing numerical data, or encoding categorical variables.
4. **Data reduction.** If the dataset contains a large number of columns or features, data reduction techniques may be used to select only the most appropriate ones for analysis.
5. **Data discretization. Data discretization** involves grouping continuous data into categories or ranges, which can help facilitate analysis.
6. **Data sampling.** In some cases, the data may be too large to analyze in its entirety. In such cases, a sample of the data can be taken for analysis while still maintaining the overall characteristics of the original dataset.

The goal of data cleaning and preprocessing is to guarantee that the data used for analysis is accurate, consistent, and relevant. It helps to improve the quality of the results and increase the efficiency of the analysis process. A well-prepared dataset can lead to more accurate insights and better decision-making.

Handling Missing Data and Outliers

Missing data refers to any data points or values that are not present in a dataset. This could be due to data collection errors, data corruption, or nonresponse from participants in a study. Missing data can impact the accuracy and validity of an analysis, as it reduces the sample size and potentially introduces bias.

Some specific examples of missing data include the following:

1. A survey participant forgetting to answer a question
2. A malfunction in data collection equipment resulting in missing values
3. A participant choosing not to answer a question due to sensitivity or discomfort

An **outlier** is a data point that differs significantly from other data points in a given dataset. This can be due to human error, measurement error, or a true outlier value in the data. Outliers can skew statistical analysis and bias results, which is why it is important to identify and handle them properly before analysis.

Missing data and outliers are common problems that can affect the accuracy and reliability of results. It is important to identify and handle these issues properly to ensure the integrity of the data and the validity of the analysis. You will find more details about outliers in Measures of Center, but here we summarize the

measures typically used to handle missing data and outliers in a data science project:

1. **Identify the missing data and outliers.** The first stage is to identify which data points are missing or appear to be outliers. This can be done through visualization techniques, such as scatterplots, box plots, IQR (interquartile range), or histograms, or through statistical methods, such as calculating the mean, median, and standard deviation (see Measures of Center and Measures of Variation as well as Encoding Univariate Data).

   It is important to distinguish between different types of missing data. **MCAR (missing completely at random) data** is missing data not related to any other variables, with no underlying cause for its absence. Consider data collection with a survey asking about driving habits. One of the demographic questions asks for income level. Some respondents accidentally skip this question, and so there is missing data for income, but this is not related to the variables being collected related to driving habits.

   MAR (missing at random) data is missing data related to other variables but not to any unknown or unmeasured variables. As an example, during data collection, a survey is sent to respondents and the survey asks about loneliness. One of the questions asks about memory retention. Some older respondents might skip this question since they may be unwilling to share this type of information. The likelihood of missing data for loneliness factors is related to age (older respondents). Thus, the missing data is related to an observed variable of age but not directly related to loneliness measurements.

   MNAR (missing not at random) data refers to a situation in which the absence of data depends on observed data but not on unobserved data. For example, during data collection, a survey is sent to respondents and the survey asks about debt levels. One of the questions asks about outstanding debt that the customers have such as credit card debt. Some respondents with high credit card debt are less likely to respond to certain questions. Here the missing credit card information is related to the unobserved debt levels.
2. **Determine the reasons behind missing data and outliers.** It is helpful to understand the reasons behind the missing data and outliers. Some expected reasons for missing data include measurement errors, human error, or data not being collected for a particular variable. Similarly, outliers can be caused by incorrect data entry, measurement errors, or genuine extreme values in the data.
3. **Determine how to solve missing data issues.** Several approaches can be utilized to handle missing data. One option is to withdraw the missing data altogether, but this can lead to a loss of important information. Other methods include **imputation**, where the absent values are replaced with estimated values based on the remaining data or using predictive models to fill in the missing values.
4. **Consider the influence of outliers.** Outliers can greatly affect the results of the analysis, so it is important to carefully consider their impact. One approach is to delete the outliers from the dataset, but this can also lead to a loss of valuable information. Another option is to deal with the outliers as an individual group and analyze them separately from the rest of the data.
5. **Use robust statistical methods.** When dealing with outliers, it is important to use statistical methods that are not affected by extreme values. This includes using median instead of mean and using nonparametric tests instead of parametric tests, as explained in Statistical Inference and Confidence Intervals.
6. **Validate the results.** After handling the missing data and outliers, it is important to validate the results to ensure that they are robust and accurate. This can be done through various methods, such as cross-validation or comparing the results to external data sources.

Handling missing data and outliers in a data science task requires careful consideration and appropriate methods. It is important to understand the reasons behind these issues and to carefully document the process to ensure the validity of the results.

EXAMPLE 2.6

Problem

Starting in 1939, the United States Bureau of Labor Statistics tracked employment on a monthly basis. The number of employers in the construction field between 1939 and 2019 is presented in Figure 2.2.

a. Determine if there is any outlier in the dataset that deviates significantly from the overall trend.
b. In the event that the outlier is not a reflection of real employment numbers, how would you handle the outliers?

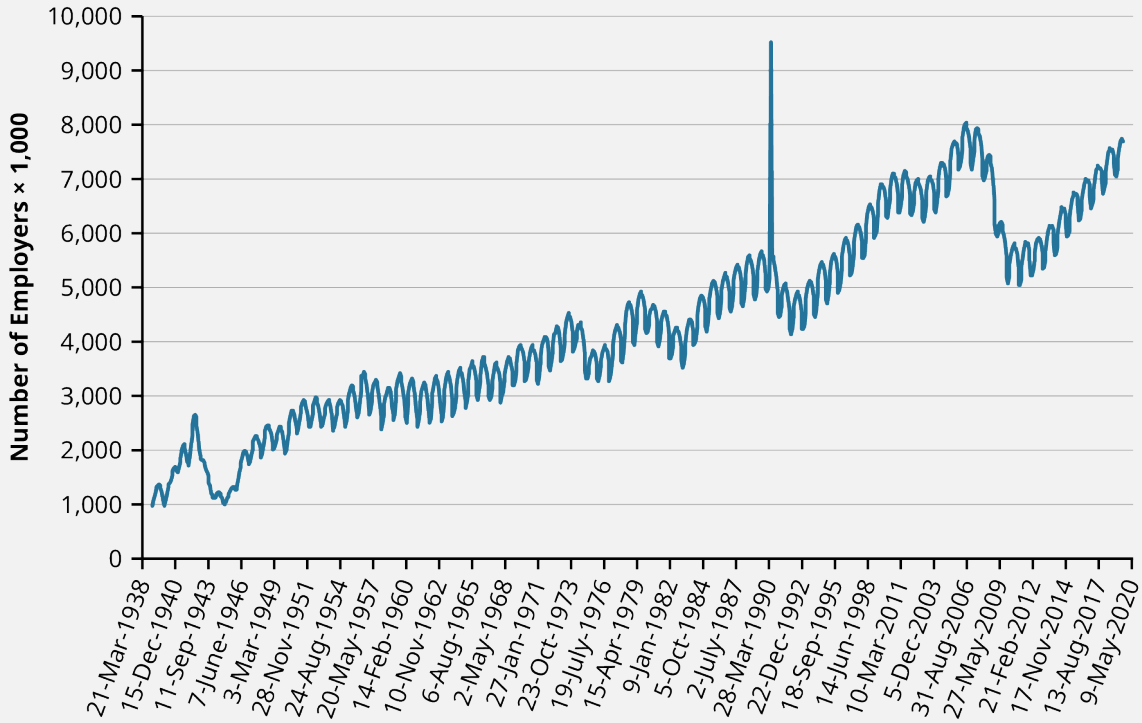


Figure 2.2 US Retail Employment with Outliers
(Data source: Bureau of Labor Statistics; credit: modification of work by Hyndman, R.J., & Athanasopoulos, G. (2021) Forecasting: principles and practice, 3rd edition, OTexts: Melbourne, Australia. OTexts.com/fpp3. Accessed on April 23, 2024)

Solution

a. Based on the visual evidence, it appears that an outlier is present in the employment data for March 28, 1990. The employment level during this month shows a significant jump from approximately 5,400 to 9,500, exceeding the overall maximum employment level recorded between 1939 and 2019.
b. A possible approach to addressing this outlier is to replace it with median, calculated by taking the mean of the points before and after the outlier. This method can help to improve the smoothness and realism of the employment curve as well as mitigate any potential impacts the outlier may have on statistical analysis or modeling processes. (See Table 2.3.)

Date	Number of Employers × 1000
1/1/1990	4974
2/1/1990	4933
3/1/1990	4989

Date	Number of Employers × 1000
4/1/1990	5174
5/1/1990	5370
6/1/1990	9523
7/1/1990	5580
8/1/1990	5581
9/1/1990	5487
10/1/1990	5379
11/1/1990	5208
12/1/1990	4963

Table 2.3 Monthly Employment over 1990

The median employment level from May 1, 1990, to July 1, 1990, is 5,289, as a replacement value for the outlier on May 28, 1990.

The adjusted data is presented in Figure 2.3.

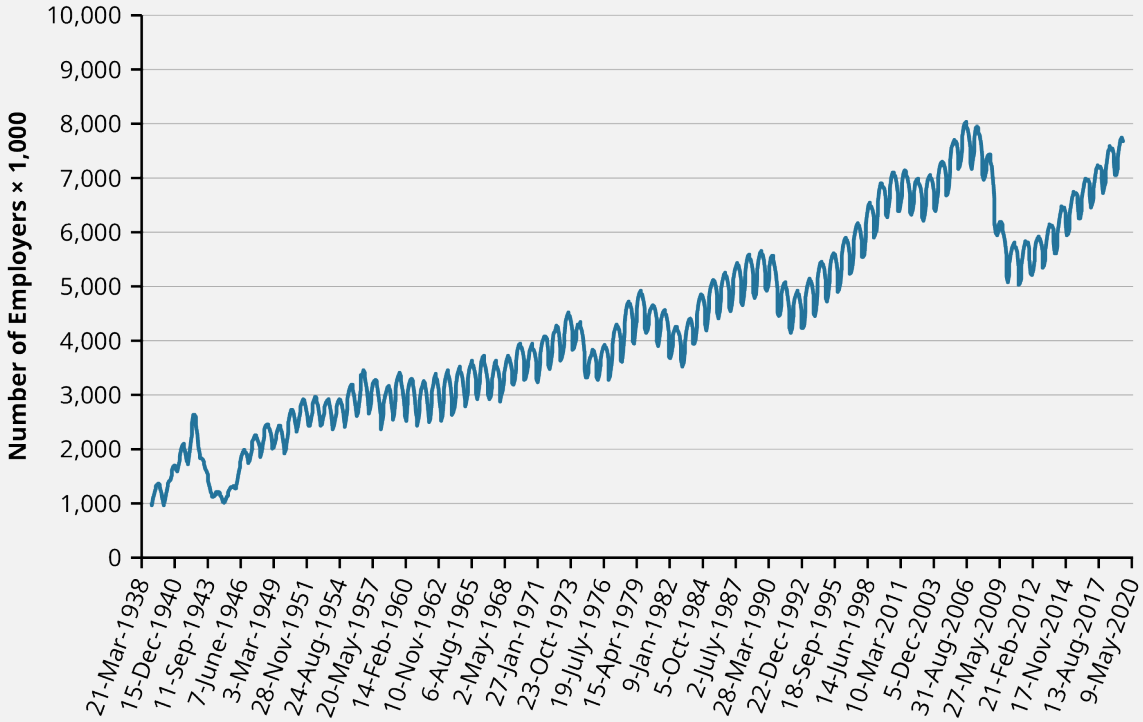


Figure 2.3 US Retail Employment Without Outliers
(Data source: Bureau of Labor Statistics; credit: modification of work by Hyndman, R.J., & Athanasopoulos, G. (2021) Forecasting: principles and practice, 3rd edition, OTexts: Melbourne, Australia. OTexts.com/fpp3. Accessed on April 23, 2024)

Data Standardization, Transformation, and Validation

Data standardization, transformation, and validation are critical steps in the data analysis preprocessing pipeline. **Data standardization** is the process of systematically transforming collected information into a consistent and manageable format. This procedure involves the elimination of inconsistencies, errors, and

duplicates as well as converting data from various sources into a unified format, often termed a *normal form* (defined in the next section). **Data transformation** involves modifying the data to make it more suitable for the analysis that is planned. **Data validation** ensures that the data is accurate and consistent and meets certain criteria or standards. Table 2.4 summarizes these processes, which are explored in more depth later in the chapter.

	Purpose	Techniques	Example
Normalization	To eliminate redundant data and ensure data dependencies make sense	Involve breaking down a larger database into smaller, more manageable tables and establishing relationships between them.	A customer database table with redundant data can be normalized by splitting it into two tables for customer information and orders and establishing the relationship between them.
Transformation	To make the data more consistent, coherent, and meaningful for analysis.	Include data cleaning, data merging, data splitting, data conversion, and data aggregation.	A dataset with different date formats, such as "MM/DD/YYYY" and "YYYY/MM/DD." Data transformation can be used to convert all dates to a single format
Validation	To improve the quality of data used in analysis. It ensures that the data is accurate, relevant, and consistent.	Include data profiling, data audits, and data cleansing.	In a survey, the respondent's age is recorded as 150 years. Through data validation, this value can be identified as erroneous.

Table 2.4 Comparison of Data Standardization, Transformation, and Validation Processes

Data Normalization

The first step in **standardizing data** is to establish guidelines and rules for formatting and structuring the data. This may include setting conventions for naming, data types, and formatting A **normal form (NF)** is a guideline or set of rules used in database design to ensure that a database is well-structured, organized, and free from certain types of data irregularities, such as redundancy and inconsistency. The most commonly used normal forms are *1NF, 2NF, 3NF* (First, Second, and Third Normal Form), and ***BCNF*** **(Boyce-Codd Normal Form)**.

Normalization is the process of applying these rules to a database. The data must be organized and cleaned, which involves removing duplicates and erroneous data, filling in missing values, and logically arranging the data. To uphold data standardization, regular quality control measures should be implemented, including periodic data audits to ascertain the accuracy and consistency of the data. It is also important to document the standardization process, including the guidelines and procedures followed. Periodic review and updates of data standards are necessary to ensure the ongoing reliability and relevance of the data.

Data normalization ensures that data is maintainable regardless of its source. Consider a marketing team collecting data on their customers' purchasing behavior so they can make some decisions about product placement. The data is collected from numerous sources, such as online sales transactions, in-store purchases, and customer feedback surveys. In its raw form, this data could be disorganized and unreliable, making it

difficult to analyze. It is hard to draw meaningful insights from badly organized data.

To normalize this data, the marketing team would go through multiple steps. First, they would identify the key data elements, such as customer name, product purchased, and transaction date. Then, they would ensure that these elements are consistently formatted in all data sources. For instance, they might employ the same date format across all data sources or standardize customer names to the first name and the last name fields. Subsequently, they would eliminate any redundant or irrelevant data elements. In this case, if the data is collected from both online and in-store purchases, they might choose one or the other to avoid duplication. The marketing team would ensure that the data is properly structured and organized. This could involve creating a data table with domains for each data element, such as a customer ID, product code, and purchase amount. By normalizing the data, the marketing team can efficiently follow and investigate the customers' purchasing behavior, identify patterns and trends, and make data-driven judgments to enhance their marketing systems.

The normalization formula is a statistical formula utilized to scale down a dataset, typically to between one and zero. The largest data would have a normalized value of one, and the smallest data point would be zero. Note that the presence of outliers can significantly impact the calculated values of minimum/maximum. As such, it is important to first remove any outliers from the dataset before performing normalization. This ensures more accurate and representative results.

The normalization formula:

$$x_{\text{norm}} = \frac{x - \min}{\max - \min}$$

EXAMPLE 2.7

Problem

A retail company with eight branches across the country wants to analyze its product sales to identify its top-selling items. The company collects data from each branch and stores it in Table 2.5, listing the sales and profits for each product. From previous reports, it discovered that its top-selling products are jewelry, TV accessories, beauty products, DVDs, kids' toys, video games, women's boutique apparel, and designer and fashion sunglasses. However, the company wants to arrange these products in order from highest to lowest based on best sales and profits. Determine which product is the top-selling product by normalizing the data in Table 2.5.

Branch	Product	Sales ($)	Profits ($)
Branch 1	Jewelry	50000	20000
Branch 2	TV Accessories	25000	12500
Branch 3	Beauty Products	30000	15000
Branch 4	DVDs	15000	7500
Branch 5	Kids’ Toys	45000	22500
Branch 6	Video Games	35000	17500
Branch 7	Women’s Boutique Apparel	40000	20000
Branch 8	Designer & Fashion Sunglasses	55000	27500

Table 2.5 Retail Company Sales and Profits

Solution

Using the normalization formula, the maximum sale is $55,000, and the minimum sale is $15,000, as shown in Table 2.6.

Branch	Product	Sales ($)	Profits ($)	Normalization Scale
Branch 1	Jewelry	50000	20000	0.88
Branch 2	TV Accessories	25000	12500	0.25
Branch 3	Beauty Products	30000	15000	0.38
Branch 4	DVDs	15000	7500	0.00
Branch 5	Kids' Toys	45000	22500	0.75
Branch 6	Video Games	35000	17500	0.50
Branch 7	Women's Boutique Apparel	40000	20000	0.63
Branch 8	Designer & Fashion Sunglasses	55000	27500	1.00

Table 2.6 Data of Retail Company Sales on Normalization Scale

Overall, the retail company's top-selling products generate the most profits for the company, with "Designer & Fashion Sunglasses" being the highest in the normalization scale. The company can use this information to focus on promoting and restocking these items at each branch to continue driving sales and profits.

Data Transformation

Data transformation is a statistical technique used to modify the original structure of the data in order to make it more suitable for analysis. Data transformation can involve various mathematical operations such as logarithmic, square root, or exponential transformations. One of the main reasons for transforming data is to address issues related to statistical assumptions. For example, some statistical models assume that the data is normally distributed. If the data is not distributed normally, this can lead to incorrect results and interpretations. In such cases, transforming the data can help to make it closer to a normal distribution and improve the accuracy of the analysis.

One commonly used data transformation technique is the **log transformation**, which requires taking the logarithm of the data values. Log transformation is often used when the data is highly skewed, meaning most of the data points fall toward one end of the distribution. This can cause problems in data analysis as the data may not follow a normal distribution. By taking the logarithm of the values, the distribution can be shifted toward a more symmetrical shape, making it easier to analyze. Another common transformation technique is the **square root transformation**, which involves taking the square root of the data values. Similar to log transformation, square root transformation is often used to address issues with skewness and make the data more normally distributed. Square root transformation is also useful when the data contains values close to zero, as taking the square root of these values can bring them closer to the rest of the data and reduce the impact of extreme values. **Exponential transformations** involve taking the exponent of the data values. Whichever operation is used, data transformation can be a useful tool for data analysts to address issues with data distribution and improve the accuracy of their analyses.

Dealing with Noisy Data

Noisy data refers to data that retains errors, outliers, or irrelevant information that can conceal true patterns and relationships within the dataset. The presence of noisy data in the dataset causes difficulty in drawing

accurate conclusions and making predictions from the data. Most noisy data is caused by human errors in data entry, technical errors in data collection or transmission, or natural variability in the data itself. Noisy data is removed and cleaned by identifying and correcting errors, removing outliers, and filtering out irrelevant information. Noisy data can negatively impact data analysis and modeling, and it may indicate that there are issues with the model's structure or assumptions. Noisy data is unwanted information that can be removed.

Strategies to reduce the noisy data are summarized in Table 2.7.

Strategy	Example
Data cleaning	Removing duplicate or irrelevant data from a dataset, such as deleting repeated rows in a spreadsheet or filtering out incomplete or error-prone data entries.
Data smoothing	A technique used in data analysis to remove outliers or noise from a dataset in order to reveal underlying patterns or trends. One example is smoothing a dataset of daily stock market index values over the course of a month. The values may fluctuate greatly on a day-to-day basis, making it difficult to see any overall trend. By calculating a seven-day average, we can smooth out these fluctuations and see the overall trend of the index over the course of the month.
Imputation	An example of imputation is in a hospital setting where a patient's medical history is incomplete due to missing information. The hospital staff can use imputation to estimate the missing data based on the patient's known medical conditions and past treatments.
Binning	A researcher is studying the age demographics of a population in a city. Instead of looking at individual ages, the researcher decides to bin the data into age groups of 10 years (e.g., 0–10, 10–20, 20–30, etc.). This allows for a more comprehensive and easier analysis of the data.
Data transformation	Consider the following dataset that shows the number of COVID-19 cases recorded in a country at different points in time—01/01/2020, 02/02/2020, 03/01/2020—are 1000, 10000, 100000, respectively. To transform this data using log, we can take the log base 10 of the number of cases column. This would result in a transformed data as follows: 01/01/2020, 02/02/2020, 03/01/2020 are 3, 4, 5, respectively.
Dimensionality reduction	The original dataset would have high dimensionality due to the large number of variables (100 companies) and time points (5 years). By applying principal component analysis, we can reduce the dimensionality of the dataset to a few principal components that represent the overall trends and patterns in the stock market.
Ensemble methods	An example of an ensemble method is the random forest algorithm. It combines multiple decision trees, each trained on a random subset of the data, to make a more accurate prediction. This helps reduce overfitting and increase the overall performance of the model. The final prediction is made by aggregating the predictions of each individual tree.

Table 2.7 Strategies to Reduce Noisy Data

Data Validation

Data validation is the process of ensuring the accuracy and quality of data examined against defined rules and standards. This approach involves identifying and correcting any errors or inconsistencies in the collected data as well as ensuring that the data is relevant and reliable for analysis. Data validation can be performed

through a variety of techniques, including manual checks, automated scripts, and statistical analysis. Some common inspections in data validation include checking for duplicates, checking for mislaid values, and verifying data against external sources or references. Before collecting data, it is important to determine the conditions or criteria that the data needs to meet to be considered valid. This can include factors such as precision, completeness, consistency, and timeliness. For example, a company may set up a data validation process to ensure that all customer information entered into its database follows a specific format. This would involve checking for correct spellings and proper formatting of phone numbers and addresses and validating the correctness of customer names and account numbers. The data would also be checked against external sources, such as official government records, to verify the accuracy of the information. Any discrepancies or errors would be flagged for correction before the data is used for analysis or decision-making purposes. Through this data validation process, the company can ensure that its customer data is accurate, reliable, and compliant with industry standards.

Another method to assess the data is to cross-reference it with reliable sources to identify any discrepancies or errors in the collected data. A number of tools and techniques can be used to validate data. These can include statistical analysis, data sampling, data profiling, and data auditing. It is important to identify and remove outliers before validating the data. Reasonability checks involve using common sense to check if the data is logical and makes sense—for example, checking if a person's age is within a reasonable range or if a company's revenue is within a reasonable range for its industry. If possible, data should be verified with the source to ensure its accuracy. This can involve contacting the person or organization who provided the data or checking against official records. It is always a good idea to involve multiple team members or experts in the validation process to catch any errors or inconsistencies that may have been overlooked by a single person. Documentation of the validation process, including the steps taken and any issues identified, is important in future data audits or for reference purposes. Data validation is a continuous process, and data should be monitored and updated to ensure its accuracy and validity.

Consider a marketing company conducting a survey on customer satisfaction for a new product launch. The company collected data from 1,000 respondents, but when the company started analyzing the data, it noticed several inconsistencies and missing values. The company's data analyst realized that the data standardization and validation processes were not adequately performed before the survey results were recorded. To correct this issue, the data analyst first identified and removed all duplicate entries, reducing the total number of responses to 900. Then, they used automated scripts to identify and fill in missing values, which accounted for 95 responses. The remaining 805 responses were then checked for data accuracy using statistical analysis. After the data standardization and validation process, the company had a clean and reliable dataset of 805 responses. The results showed that the product had a satisfaction rate of 85%, which was significantly higher than the initial analysis of 78%. As a result of this correction, the marketing team was able to confidently report the actual satisfaction rate and make better-informed decisions for future product development.

Data Aggregation

Data aggregation is the process by which information from multiple origins is gathered and merged into a single set that provides insights and meaningful conclusions. It involves gathering, managing, and delivering data from different sources in a structured manner to facilitate analysis and decision-making. Data aggregation can be performed manually or by using automated tools and techniques. The data aggregation process is utilized to identify patterns and trends between different data points, which extracts valuable insights. Some standard types of data aggregation are spatial aggregation, statistical aggregation, attribute aggregation, and temporal aggregation. This methodology is commonly operated in marketing, finance, health care, and research to analyze large sets of data. Data aggregation is used in various industries to combine and analyze large sets of data. Examples include calculating total sales for a company from different departments, determining average temperature for a region including multiple cities, and analyzing website traffic by country. It is also used in fields such as stock market indices, population growth, customer satisfaction scores, credit scores, and airline flight delays. Governments and utility companies also utilize data

aggregation to study energy consumption patterns.

Text Preprocessing

Text preprocessing is a technique of preparing data in text format for further analysis and natural language processing tasks. It involves transforming unstructured text data into a more structured format to be interpreted by algorithms and models. Some common techniques used in text preprocessing are reviewed in Table 2.8.

Preprocessing Technique	Explanation	Example
Tokenization	Breaking text data into individual words or phrases (tokens)	Original Text: “Tokenization is the process of breaking down a sentence, paragraph, or entire text into smaller parts called tokens.” Tokenized Text: “Tokenization”, “is”, “the”, “process”, “of”, “breaking”, “down”, “a”, “sentence”, “,”, “paragraph”, “,”, “or”, “entire”, “text”, “into”, “smaller”, “parts”, “called”, “tokens”, “.”
Lowercasing	Converting all text to lowercase to avoid multiple representations of the identical word	Consider the following sentence: “John likes to eat pizza.” After lowercasing it, the sentence becomes “john likes to eat pizza.”
Stopwords removal	Filtering out commonly occurring words that do not add meaning or context	Consider: “The sun was shining bright in the sky and the birds were chirping. It was a lovely day in the park and people were enjoying the beautiful weather.” After removing stopwords, the paragraph would be transformed into: “Sun shining bright sky birds chirping. Lovely day park people enjoying beautiful weather.”
Lemmatization and stemming	Reducing words to their root forms to reduce complexity and improve model performance	For example, the words “running,” “runs,” and “ran” would all be lemmatized to the base form ”run.”

Table 2.8 Summary of Text Preprocessing Techniques

Preprocessing Technique	Explanation	Example
Part-of-speech tagging	Identifying the grammatical components of each word where each word in a sentence is assigned a specific part-of-speech tag (e.g., noun, verb, or adjective)	In the sentence "I went to the market to buy some fruits," the words "went" and "buy" would be tagged as verbs, "market" and "fruits" as a noun, and "some" as adjectives.
Named entity recognition	Recognizing and categorizing named entities such as people, places, and organizations	Consider the following text: "John went to Paris last summer with his colleagues at Microsoft." By using named entity recognition, we can tag the named entities in this sentence as follows: "John (person) went to Paris (location) last summer with his colleagues at Microsoft (organization)."
Sentiment analysis	Identifying and categorizing the emotions expressed in text	Let's say a customer has left a review for a new laptop they purchased. The review reads: "I am extremely satisfied with my purchase. The laptop has exceeded all of my expectations and has greatly improved my work efficiency. Thanks for an amazing product!" To perform sentiment analysis, the text will first undergo preprocessing, which involves cleaning and preparing the text data for analysis. This may include removing punctuation, converting all letters to lowercase, and removing stopwords (commonly used words that do not add much meaning to a sentence, such as "the" or "and"). After preprocessing, sentiment analysis techniques will be applied to analyze the emotions and opinions expressed in the review. The analysis may identify key positive words such as "satisfied" and "amazing" and measure their overall sentiment score. It may also take into account the context and tone of the review to accurately determine the sentiment.

Table 2.8 Summary of Text Preprocessing Techniques

Preprocessing Technique	Explanation	Example
Spell-checking and correction	Correcting spelling errors to improve accuracy	Suppose we have a text like: “The writting was very inpprecise and had many trgical mistakes.” With spell-checking and correction, this text can be processed and corrected to: “The writing was very imprecise and had many tragic mistakes.” This involves identifying and correcting misspelled words. In this case, “writting” was corrected to “writing,” “inpprecise” to “imprecise,” and “trgical” to “tragic.” This not only improves the readability and accuracy of the text but also helps in better understanding and analysis of the text.
Encoding	Converting text into a numerical representation for machine learning algorithms to process	Let’s say we have a dataset of customer reviews for a restaurant. Each review is a string of text, such as: “I had a great experience at this restaurant, the food was delicious and the service was top-notch.” To encode this text data, we can use techniques such as one-hot encoding or word embedding. One-hot encoding involves creating a binary vector for each word in the review, where the size of the vector is equal to the size of the vocabulary. For example, if the review contains the words “great,” “experience,” “restaurant,” “food,” “delicious,” “service,” and “top-notch,” the one-hot encoded vectors for these words would be: great: [1, 0, 0, 0, 0, 0, 0]; experience: [0, 1, 0, 0, 0, 0, 0]; restaurant: [0, 0, 1, 0, 0, 0, 0]; food: [0, 0, 0, 1, 0, 0, 0]; delicious: [0, 0, 0, 0, 1, 0, 0]; service: [0, 0, 0, 0, 0, 1, 0]; top-notch: [0, 0, 0, 0, 0, 0, 1]. These one-hot encoded vectors can now be used as input features for machine learning models.
Removing special characters and punctuation	Simplifying the text for analysis	Consider the input: “Hello, this is a text with @special# characters!*” and the output: “Hello this is a text with special characters”

Table 2.8 Summary of Text Preprocessing Techniques

Text preprocessing is crucial to effectively use text data for tasks such as text classification and information extraction. By transforming the raw text data, the accuracy and performance of machine learning models can greatly improve and provide meaningful insights and predictions.

Text preprocessing is especially important in artificial intelligence, as it can lay the foundation for effective text analysis, classification, information retrieval, sentiment analysis, and many other *natural language processing* tasks (see Natural Language Processing). A well-designed preprocessing pipeline can lead to better model performance, improved efficiency, and more accurate insights from text data.

An example of text preprocessing in artificial intelligence involves the conversion of text data into a more standardized and consistent format. This can include tasks such as removing accents and diacritics, expanding contractions, and converting numbers into their written representations (e.g., “10” to “ten”). Normalizing text data helps to reduce the number of variations and therefore improves the efficiency and accuracy of natural

language processing tasks. It also makes the data more easily interpretable for machine learning algorithms. For example, when analyzing customer reviews, it may be beneficial to normalize text data so that variations of the same word (e.g., “colour” and “color”) are treated as the same, providing a more accurate understanding of sentiment toward a product or service.

2.5 Handling Large Datasets

Learning Outcomes

By the end of this section, you should be able to:

- 2.5.1 Recognize the challenges associated with large data, including storage, processing, and analysis limitations.
- 2.5.2 Implement techniques for efficient storage and retrieval of large datasets, including compression, indexing, and chunking.
- 2.5.3 Discuss database management systems and cloud computing and their key characteristics with regard to large datasets.

Large datasets, also known as **big data**, are extremely large and complex sets of data that traditional data processing methods and tools are unable to handle. These datasets typically include sizeable volumes, variety, and velocity of data, making it challenging to process, manage, and analyze them using traditional methods. Large datasets can be generated by a variety of sources, including social media, sensors, and financial transactions. They generally possess a high degree of complexity and may have structured, unstructured, or semi-structured data. Large datasets are covered in more depth in Other Machine Learning Techniques.

We have also already discussed a number of the techniques and strategies used to gain meaningful insights from big data. Survey Design and Implementation discussed sampling techniques that allow those working with data to analyze and examine large datasets by select a representative subset, or *representative random sample*. Preprocessing techniques covered in Data Cleaning and Preprocessing are also used to clean, normalize, and transform data to make sure it is consistent before it can be analyzed. This includes handling missing values, removing outliers, and standardizing data formats.

In this section we will consider several other aspects of data management that are especially useful with big data, including data compression, data storage, data indexing, and chunking. In addition, we’ll discuss database management systems—software that allows for the organization, manipulation, and retrieval of data that is stored in a structured format—and cloud computing.

Data Compression

Data compression is a method of reducing file size while retaining essential information; it can be applied to many types of databases. Data compression is classified into two categories, lossy and lossless:

- **Lossy compression** reduces the size of data by permanently extracting particular data that is considered irrelevant or redundant. This method can significantly decrease file sizes, but it also results in a loss of some data. Lossy compression is often utilized for multimedia files and images, where slight modifications in quality may not be noticeable to the human eye. Examples of lossy compression include MP3 and JPEG.
- **Lossless compression** aims to reduce file size without removing any data. This method achieves compression by finding patterns and redundancies in the data and representing them more efficiently. This allows for the reconstruction of the original data without any loss in quality. Lossless compression is commonly used for text and numerical data, where every piece of information is crucial. Examples of lossless compression include ZIP, RAR, and PNG.

  There are several methods of data lossless compression, including Huffman coding. **Huffman coding** works by assigning shorter binary codes to the most frequently used characters or symbols in a given dataset and longer codes to less frequently used characters, as shown in Figure 2.4. This results in a more

efficient use of binary digits and reduces the overall size of the data without losing any information during compression. Huffman coding is applied to data that require fast and efficient compression, such as video compression, image compression, and data transmission and storage.

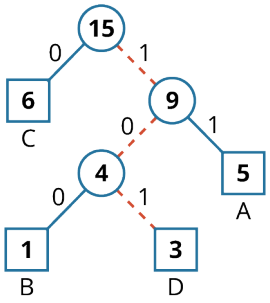


Figure 2.4 Huffman Tree Diagrams

Data Storage

Big data requires storage solutions that can handle large volumes of diverse data types, offer high performance for data access and processing, guarantee scalability for growing datasets, and are thoroughly reliable. The choice of storage solution will depend on data volume, variety, velocity, performance requirements, scalability needs, budget constraints, and existing infrastructure. Organizations often deploy a combination of storage technologies to address different uses and requirements within their big data environments. Some common types of storage solutions used for big data include:

1. **Relational databases**: **Relational databases**. organize data in tables and uses structured query language (SQL) for data retrieval and management. They are commonly used for traditional, structured data such as financial data.
2. **NoSQL databases**. These databases are designed to handle unstructured data, such as social media content or data from sensors, and use non-relational data models.
3. **Data warehouses.** A **data warehouse** is a centralized repository of data that combines data from multiple sources and allows for complex queries and analysis. It is commonly used for business intelligence and reporting intentions.
4. **Cloud storage**. **Cloud storage** involves storing data in remote servers accessed over the internet. It offers scalability, cost-effectiveness, and remote accessibility.
5. **Object storage**. With **object storage**, data are stored as objects that consist of both data and metadata. This method is often used for storing large volumes of unstructured data, such as images and videos.

Data Indexing

Data indexing is an important aspect of data management and makes it easier to retrieve specific data quickly and efficiently. It is a crucial strategy for optimizing the performance of databases and other data storage systems. **Data indexing** refers to the process of managing and saving the collected and generated data in a database or other data storage system in a way that allows for efficient and fast return of specific data.

Indexing techniques vary in how they organize and store data, but they all aim to improve data retrieval performance. *B-tree indexing* is illustrated in Figure 2.5. It involves organizing data in a tree-like structure with a root node and branches that contain pointers to other nodes. Each node contains a range of data values and pointers to child nodes, allowing for efficient searching within a specific range of values.

Hashes indexing involves using a hash function to map data to a specific index in a table. This allows for direct access to the data based on its hashed value, making retrieval faster than traditional sequential searching. *Bitmap indexing* is a technique that involves creating a bitmap for each different value in a dataset. The bitmaps are then combined to quickly identify records that match a specific set of values, allowing efficient data retrieval.

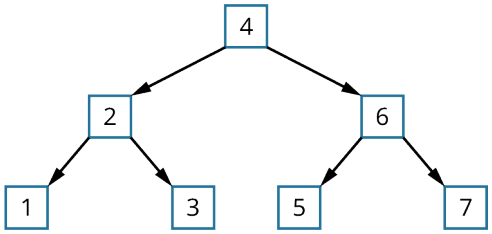


Figure 2.5 B-Tree Indexing

Data Chunking

Data chunking, also known as *data segmentation* or *data partitioning*, is a technique used to break down large datasets into smaller, more manageable chunks and make them easier to manage, process, analyze, and store. Chunking is particularly useful when datasets are too large to be processed or analyzed as a single unit. By dividing the data into smaller chunks, various processing tasks can be distributed across multiple computing nodes or processing units.

Data chunking is used in data storage systems and data transmission over networks, and it is especially useful when working with large datasets that exceed the capacity of a single machine or when transferring data over a network with limited bandwidth. The divided data in data chunking is known as a *chunk* or *block*. The size of each chunk can vary depending on the requirements and range from a few kilobytes to several gigabytes. These chunks are typically organized sequentially, with each chunk containing a set of data from the larger dataset. The process of data chunking also involves adding **metadata** (data that provides information about other data), such as the chunk number and the total number of chunks, to each chunk. This metadata allows the chunks to be reassembled into the original dataset after being transmitted or stored separately. Data chunking has several advantages, including the following:

1. **Increased speed.** By dividing a large dataset into smaller chunks, data processing and transmission can be performed more quickly, reducing the overall processing time.
2. **Better utilization of resources.** Data chunking enables data to be distributed and processed across multiple machines, making better use of available computing resources.
3. **Increased fault tolerance.** In case of data corruption or loss, data chunking allows for the retrieval of only the affected chunk rather than the entire dataset.
4. **Flexibility.** Data chunking allows for the transfer and processing of only the required chunks rather than the entire dataset, providing flexibility in managing large datasets.

Database Management Systems

Database management is a crucial aspect of data science projects as it involves organizing, storing, retrieving, and managing large volumes of data. In a data science project, a database management system (DBMS) is used to ensure the efficient storage and retrieval of data. Database management systems are software tools used for managing data in a structured format. Their functions are summarized in Table 2.9.

DBMS Function	Description	Benefit
Data storage	Provides a centralized warehouse for storing different types of data in a structured format	Makes data easy to retrieve and analyze
Data retrieval	Allows for efficient and fast retrieval of data from the database using queries and filters	Makes data more accessible to data scientists

Table 2.9 Functions of Database Management Systems (DBMS)

DBMS Function	Description	Benefit
Data organization	Helps to manage data in a structured format	Makes data more manageable for performing analysis and identifying patterns or relationships between different data points
Data security	Provides strong security measures to protect sensitive data from unauthorized access	Protects sensitive data such as personal information or financial data from unauthorized access
Data integration	Permits the integration of data from multiple sources	Makes it possible to combine and analyze data from different datasets

Table 2.9 Functions of Database Management Systems (DBMS)

Implementation of database management techniques has become important for hospitals in achieving better patient outcomes and reducing costs. This strategy involves the collection and analysis of patient data from different sources, including electronic health records, medical imaging, and lab results. For example, hospitals are utilizing this approach to improve treatment for patients with chronic conditions such as diabetes and heart disease. By leveraging data-driven insights and identifying patterns, health care experts can develop personalized remedy plans for each patient, guiding them to enhanced health and wellness as well as cost savings for the hospital. This showcases the significance of incorporating advanced data management techniques in health care systems. Through accurate and efficient management and analysis of patient data, hospitals and health care providers are able to make informed decisions, eventually resulting in a more efficient and effective health care system overall.

Cloud Computing

Cloud computing delivers a cost-effective solution for storing vast amounts of data, enabling seamless collaboration and data transfer among remote groups. This technology comprises remote-access tools for storage, processing, and analytics, facilitating multiple users' access regardless of their physical location. Moreover, cloud computing boasts a diverse range of data collection tools, including machine learning, and data warehouses, streamlining the data assembly operation and improving overall efficiency. Cloud computing equips data scientists with the necessary resources and flexibility to effectively collect, manage, and analyze data for their projects. Some examples of cloud storage are Amazon AWS, Microsoft Azure, and Google Cloud.

EXAMPLE 2.8

Problem

The CEO of a large insurance company faced the challenge of increasing digital processes and documents, leading to a need for more storage capacity and rising costs for maintaining servers and hardware. What are all the available options for the insurance company facing the need for additional storage capacity at a higher cost, and which solution would be the most effective for decreasing costs and increasing capacity while ensuring data security?

Solution

1. Migrating to a cloud storage system: This would allow the company to offload the burden of maintaining physical servers and hardware while also providing virtually unlimited storage capacity. The cost of cloud storage is also generally more flexible and scalable, allowing the company to only pay for the storage it needs. Additionally, with the use of cloud-based document management systems, the company can streamline and automate its processes, reducing the need for physical documentation

and increasing efficiency. However, this decision would require careful consideration of security measures and potentially training for employees to adapt to the new system.
2. Implementing a data archiving strategy: Instead of immediately migrating to the cloud or investing in new technology, the company could consider implementing a data archiving strategy. This involves identifying and storing infrequently used data in a separate, low-cost storage system, freeing up space on servers and reducing costs.
3. Outsourcing data storage and management: The company could consider outsourcing its data storage and management to a third-party provider. This would eliminate the need for maintaining physical servers and hardware, and the provider may also offer advanced security measures and data backup options.
4. Consolidating data and processes: The company could assess its current processes and systems to identify areas where data and processes can be consolidated to reduce the need for multiple storage systems and streamline workflows.
5. Implementing a virtual desktop infrastructure: A virtual desktop infrastructure allows employees to access their desktop and applications from a central server, reducing the need for individual storage space on devices. This can also improve security and data backup capabilities.
6. Upgrading or redesigning its current storage system: The company could invest in upgrading or redesigning its current storage system to improve efficiency and increase storage capacity.
7. Utilizing hybrid cloud storage: Instead of fully migrating to the cloud, the company could consider a hybrid approach where certain sensitive data is stored on-premises while less critical data is stored in the cloud. This approach can offer the benefits of both on-premises and cloud storage.
8. Conducting regular audits of data usage and storage: The company could conduct regular audits of its data usage and storage to identify areas of redundancy or inefficiency and adjust accordingly. This can help optimize storage and reduce costs over time.

EXAMPLE 2.9

Problem

What is the descending order of storage capacity in Example 2.8, starting from highest to lowest, while maintaining the same cost and the same level of security?

Solution

The available storage options in Example 2.8, ranked from highest to lowest capacity at the same cost and security level, are:

1. Cloud storage system: This option would provide virtually unlimited storage capacity, as the company can scale up as needed at a relatively low cost. However, the overall capacity and cost would depend on the specific cloud storage plan chosen.
2. Data archiving strategy: By identifying and storing infrequently used data, the company can free up space on its current servers and reduce costs. However, the storage capacity will be limited to the existing servers.
3. Outsourcing data storage and management: By outsourcing to a third-party provider, the company can potentially access higher storage capacity than its current setup, as the provider may have more advanced storage options. However, this would also depend on the specific plan and cost negotiated.
4. Implementing a virtual desktop infrastructure: This option may provide slightly lower storage capacity compared to outsourcing, as it still relies on a central server. However, it can still be an improvement compared to the company's current setup.

5. Upgrading or redesigning its current storage system: This option may also result in lower storage capacity compared to outsourcing, as it only involves improving the existing server setup rather than using an external provider. However, it can still provide a significant increase in capacity depending on the specific upgrades implemented.
6. Hybrid cloud storage: This option can provide a mix of higher and lower storage capacity depending on the specific data stored in the cloud and on-premises. Sensitive data may have lower capacity on the cloud, while less critical data can be stored at higher capacity.
7. Consolidating data and processes: By streamlining processes and eliminating redundancies, the company can potentially reduce the need for excessive storage capacity. However, this would also depend on the company's current setup and level of optimization.
8. Regular audits of data usage and storage: This option may not directly impact storage capacity, but by identifying and eliminating redundancies, the company can optimize its existing storage capacity and potentially reduce costs in the long run.



Datasets

Note: The primary datasets referenced in the chapter code may also be [downloaded here (https://openstax.org/r/drive)].)

Key Terms

application programming interface (API) set of protocols, tools, and definitions for building software applications that allow different software systems to communicate and interact with each other and enable developers to access data and services from other applications, operating systems, or platforms

***BCNF* (Boyce-Codd Normal Form)** a normal form that is similar to 3NF but stricter in its requirements for data independence; it ensures that all attributes in a table are functionally dependent on the primary key and nothing else

big data complex, high-volume and varied data that are challenging to process with traditional methods

closed-ended questions clear, structured questions with predetermined answer choices that are effective in gathering quantitative data

cloud computing the delivery of computing services, such as storage, processing power, and software applications, over the internet

cloud storage type of online storage that allows users and organizations to store, access, and manage data over the internet with data. This data is stored on remote servers managed by a cloud storage service provider.

data aggregation the process of collecting and combining information from multiple sources into a single database or dataset

data chunking a technique used to break down large datasets into smaller, more manageable chunks and make them easier to manage, process, analyze, and store; also known as data segmentation

data compression the process of reducing the size of a data file or transmission to make it easier and faster to store, transmit, or manipulate

data dictionary a data structure that stores data in key-value pairs, allowing for efficient retrieval of data using its key

data discretization the process of converting continuous data into discrete categories or intervals to allow for easier analysis and processing of the data

data indexing the process of organizing and storing data in a way that makes it easier to search and retrieve information

data integration the process of combining data from multiple sources, such as databases, applications, and files, into a unified view

data processing the process of collecting, organizing, and manipulating data to produce meaningful information involving the transformation of raw data into a more useful and understandable format

data standardization the process of transforming data into a common scale or range to help remove any unit differences or variations in magnitude

data transformation the process of modifying data to make it more suitable for the planned analysis

data validation procedure to ensure that the data is reliable and trustworthy for use in the data science project, done by performing checks and tests to identify and correct any errors in the data

data warehouse a large, organized repository of integrated data from various sources used to support decision-making processes within a company or organization

DataFrame a two-dimensional data structure in Python that is used for data manipulation and analysis

exponential transformation a data transformation operation that involves taking the exponent of the data values

Huffman coding reduces file size by assigning shorter binary codes to the most frequently used characters or symbols in a given dataset and longer codes to less frequently used characters

imputation the process of estimating missing values in a dataset by substituting them with a statistically reasonable value

log transformation data transformation technique that requires taking the logarithm of the data values and is often used when the data is highly skewed

lossless compression aims to reduce file size without removing any data, achieving compression by finding patterns and redundancies in the data and representing them more efficiently

lossy compression reduces the size of data by permanently extracting particular data that is considered irrelevant or redundant

measurement error inaccuracies or discrepancies that surface during the process of collecting, recording, or analyzing data caused by human error, environmental factors, or inconsistencies in the data

metadata data that provides information about other data

missing at random (MAR) data missing data related to other variables but not to any unknown or unmeasured variables; the missing data can be accounted for and included in the analysis through statistical techniques

missing completely at random (MCAR) data type of missing data that is not related to any other variables, with no underlying cause for its absence

missing not at random (MNAR) data type of data missing in a way that depends on unobserved data

network analysis the examination of relationships and connections between users, data, or entities within a network to identify and analyze influential individuals or groups and understand patterns and trends within the network

noisy data data that contains errors, inconsistencies, or random fluctuations that can negatively impact the accuracy and reliability of data analysis and interpretation

normal form a guideline or set of rules used in database design to ensure that a database is well-structured, organized, and free from certain types of data irregularities, such as redundancy and inconsistency

normalization the process of transforming numerical data into a standard or common range, usually between 0 and 1, to eliminate differences in scale and magnitude that may exist between different variables in a dataset

NoSQL databases databases designed to handle unstructured data, such as social media content or data from sensors, using non-relational data models

object storage method of storage where data are stored as objects that consist of both data and metadata and often used for storing large volumes of unstructured data, such as images and videos

observational data data that is collected by observing and recording natural events or behaviors in their normal setting without any manipulation or experimentation; typically collected through direct observation or through the use of instruments such as cameras or sensors

open-ended questions survey questions that allow for more in-depth responses and provide the opportunity for unexpected insights

outlier a data point that differs significantly from other data points in a given dataset

regular expressions (regex) a sequence of characters that define a search pattern, used for matching and manipulating strings of text; commonly used for cleaning, processing, and manipulating data in a structured or unstructured format

relational databases databases that organize data in tables and use structured query language (SQL) for data retrieval and management; often used with financial data

sampling the process of selecting a subset of a larger population to represent and gather information about that population

sampling bias occurs when the sample used in a study isn't representative of the population it intends to generalize; can lead to skewed or inaccurate conclusions

sampling error the difference between the results obtained from a sample and the true value of the population parameter it is intended to represent

sampling frame serves as a guide for selecting a representative sample of the population and determining the coverage of the survey and ultimately affects the validity and accuracy of the results

scaling one of the steps involved in data normalization; refers to the process of transforming the numerical values of a feature to a particular range

slicing a string extracting a portion or section of a string based on a specified range of indices

social listening the process of monitoring and analyzing conversations and discussions happening online, specifically on social media platforms, to track mentions, keywords, and hashtags related to a specific topic or business to understand the sentiment, trends, and overall public perception

splitting a string dividing a text string into smaller parts or substrings based on a specified separator
square root transformation data transformation technique that requires taking the square root of the data values; is useful when the data contains values close to zero
standardizing data the process of transforming data into a common scale or range to help remove any unit differences or variations in magnitude
string a data type used to represent a sequence of characters, such as letters, numbers, and symbols, in a computer program
text preprocessing the process of cleaning, formatting, and transforming raw text data into a form that is more suitable and easier for natural language processing to understand and analyze
tokenizing the process of breaking down a piece of text or string of characters into smaller units called tokens, which can be words, phrases, symbols, or individual characters
transactional data a type of information that records financial transactions, including purchases, sales, and payments. It includes details such as the date, time, and parties involved in the transaction.
web scraping a method of collecting data from websites using software or scripts. It involves extracting data from the HTML code of a web page and converting it into a usable format.



Group Project

Project A: Collecting Data on Extinct Animals Due to the Impact of Global Warming

The main objective of this project is to gather data before and after 1900 on the various species that have gone extinct in order to analyze the impact of climate change on these animals and determine the most affected species. To collect data on extinct animals, you need to utilize various sources such as biodiversity databases, academic papers, government reports, and conservation organizations' data.

As a group, complete the following steps:

1. **Selection of data collection method.** The students will determine the most appropriate data collection method based on the research topic and available resources. This could include observational techniques, transactional data, surveys, experimental data, or web scraping. The method chosen should allow for the collection of reliable and relevant data on extinct animals.
2. **Independent research on reliable sources of data.** Each student will conduct independent research to locate credible sources of data on animal extinctions.
3. **Data organization into a table format.** Using the collected information, students will create a table to present data on the distribution and causes of animal extinctions per continent. The table will include columns for the animal's common name, scientific name, date of extinction, continent, and primary reason for extinction. Table 2.10 provides an example.

Common Name	Scientific Name	Date of Extinction	Continent	Primary Reason for Extinction
Passenger pigeon	*Ectopistes migratorius*	1914	North America	Overhunting
Quagga	*Equus quagga*	1883	Africa	Habitat loss

Table 2.10 Project A Required Data and Example

4. **Collaborative data cleaning.** Students will work together to clean the collected data, identifying and addressing any missing or duplicate information. They will also ensure that the data is standardized, allowing for accurate analysis and comparison.
5. **Data storage determination.** In this step, the students will determine the storage size of the collected

data, taking into consideration the amount of information gathered from various sources. They will also decide on the most efficient and accessible way to store the data, ensuring all group members have access to it for further analysis and collaboration.

6. **Source citation.** As part of the data collection process, students will document and list all the sources used to gather information on extinct animals. This will ensure proper credit is given to the original sources and allow for further verification of the data.

Project B: Data Collection Using Experimental Technique

Your group project will involve recording temperature changes at a specific location over the course of 24 hours. This project will be completed in subgroups, with each group member responsible for recording data during a specific time segment of the day. Before beginning your research, your group will select a location to observe. (This could be a park, a city street, a backyard, or any other outdoor location. It is important that the location is accessible and allows for consistent monitoring throughout the day.) Each subgroup will be responsible for recording temperature data for a six-hour time period. The first subgroup will record data from midnight to 6 a.m., the second subgroup from 6 a.m. to 12 p.m., the third subgroup from 12 p.m. to 6 p.m., and the fourth subgroup from 6 p.m. to midnight.

It is crucial that each subgroup records data at the exact same time each day in order to maintain consistency. During the data collection process, it is important for each subgroup to maintain the same surrounding conditions. This includes using the same tools, such as a thermometer, and staying in the same location throughout the time period. Any uncontrolled changes or errors should be noted and recorded to ensure accuracy in the data.

Repeat this process for seven days to gather a sufficient amount of data, and once all the data has been collected, calculate the average temperature for each time segment of the day by adding up all the recorded temperatures and dividing by the number of data points.

Once all data has been collected, the group will come together to analyze and graph the temperature changes over the course of 24 hours. This will allow for a visual representation of the temperature patterns throughout the day.

Discuss potential sources of error and ways to improve the data collection process in the future. Then summarize the collected data process, present the project to the rest of the group, and discuss any insights or lessons learned from the data collection process.



Critical Thinking

1. Megan is interested in opening a gaming center that utilizes virtual reality (VR) technology. She wants to gauge the level of interest in VR gaming among the residents of her city. She plans to conduct a survey using a random sampling method and then use this data to determine the potential demand for a VR gaming center in her city. What is the nature of the collected data, experimental or observational?

2. A high school student is interested in determining the relationship between the weight of an object and the distance it can travel when launched from a catapult. To collect data for analysis, the student plans to launch objects of varying weight and then measure the corresponding distance traveled. Using this data, the student plans to plot a graph with weight on the x-axis and distance on the y-axis.
   a. Will this data collection result in observational data or experimental data?
   b. How can the student gather the necessary data to establish this relationship?

3. A high school is conducting a survey to understand the opinions of the school's students toward the food options available in the cafeteria. They have divided the student population into four groups based on grade level (9th, 10th, 11th, 12th). A random sample of 60 students is selected from each group and asked to rate the cafeteria food on a scale of 1–10. The results are shown in the following table:

Grade Level	Number of Students	Average Rating
9th	60	8.1
10th	60	7.4
11th	60	6.3
12th	60	6.1

Table 2.11 High School Survey

What sampling technique was used to collect this data?

4. A research study is being conducted to understand the opinions of people on the new city budget. The population consists of residents from different neighborhoods in the city with varying income levels. The following table provides the data of 130 participants randomly selected for the study, categorized by their neighborhood and income level.

Neighborhood	Income Level	Number of Participants
Northside	Low Income	35
Southside	Middle Income	50
Eastside	High Income	45

Table 2.12 Neighborhood and Income level Category

What sampling technique was used to collect this data?

5. John is looking to gather data on the popularity of 3D printing among local college students to determine the potential demand for a 3D printing lab on campus. He has decided on using surveys as his method of data collection. What sampling methods would be most effective for obtaining accurate results?

6. A group of researchers were tasked with gathering data on the price of electronics from various online retailers for a project. Suggest an optimal approach for collecting data on electronics prices from different online stores for a research project.

7. As the pandemic continues to surge, the US Centers for Disease Control and Prevention (CDC) releases daily data on COVID-19 cases and deaths in the United States, but the CDC does not report on Saturdays and Sundays, due to delays in receiving reports from states and counties. Then the Saturday and Sunday cases are added to the Monday cases. This results in a sudden increase in reported numbers of Monday cases, causing confusion and concern among the public. (See the table provided.) The CDC explains that the spike is due to the inclusion of delayed or improperly recorded data from the previous week. This causes speculation and panic among the public, but the CDC reassures them that there is no hidden outbreak happening.

Date	Day	New Cases
10/18/2021	Monday	3115
10/19/2021	Tuesday	4849
10/20/2021	Wednesday	3940
10/21/2021	Thursday	4821
10/22/2021	Friday	4357
10/23/2021	Saturday	0
10/24/2021	Sunday	0
10/25/2021	Monday	8572
10/26/2021	Tuesday	4463
10/27/2021	Wednesday	5323
10/28/2021	Thursday	5012
10/29/2021	Friday	4710
10/30/2021	Saturday	0
10/31/2021	Sunday	0
11/1/2021	Monday	10415
11/2/2021	Tuesday	5096
11/3/2021	Wednesday	6882
11/4/2021	Thursday	5400
11/5/2021	Friday	6759
11/6/2021	Saturday	0
11/7/2021	Sunday	0

Table 2.13 Sample of COVID-19 Data Cases Within 23 Days (source: https://data.cdc.gov/Case-Surveillance)

11/8/2021	Monday	10069
11/9/2021	Tuesday	5297

Table 2.13 Sample of COVID-19 Data Cases Within 23 Days (source: https://data.cdc.gov/Case-Surveillance)

a. What simple strategies can the data scientists utilize to address the issue of missing data in their analysis of COVID-19 data from the CDC?
b. How can data scientists incorporate advanced methods to tackle the issue of missing data in their examination of COVID-19 cases and deaths reported by the CDC?

8. What distinguishes data standardization from validation when gathering data for a data science project?

9. Paul was trying to analyze the results of an experiment on the effects of music on plant growth, but he kept facing unexpected data. He was using automated instrumentation to collect growth data on the plants at periodic time intervals. He found that some of the data points were much higher or lower than anticipated, and he suspected that there was noise (error) present in the data. He wanted to investigate the main source of the error in the data so that he could take corrective actions and thus minimize the impact of the error(s) on the experiment results. What is the most likely source of error that Paul might encounter in this experiment: human error, instrumentation error, or sampling error? Consider the various sources of noise (error).

10. The government is choosing the best hospital in the United States based on data such as how many patients they have, their mortality rates, and overall quality of care. They want to collect this data accurately and fairly to improve the country's health care.
a. What is the best way to collect this data? Can we use web scraping to gather data from hospital websites? Can we also use social media data collection to gather patient reviews and feedback about the hospitals?
b. How can we ensure the data is unbiased and accurate? Can we use a mix of these methods to create a comprehensive analysis?

11. Sarah, a data scientist, was in charge of hiring a new sales representative for a big company. She had received a large number of applications. To make the hiring process more efficient, she decided to streamline the process by using text processing to analyze the candidates' applications to identify their sentiment, tone, and relevance. This would help Sarah gain a better understanding of their communication skills, problem-solving abilities, and overall fit for the sales representative role. She compiled two lists of keywords, one for successful candidates and one for rejected candidates. After generating summaries of the keywords from each application, Sarah was ready to make informed and efficient hiring decisions. Identify the best candidate based on the list of keywords associated with each candidate, as reviewed in the figure shown. Your answer should be supported with strong reasoning and evidence.

Candidates	Customer-Centric	Team Player	Collaborative	Sales Experience	Leadership	Multitasking	Goal-Oriented	Team-Oriented	Effective	Inflexible	Uncooperative	Indecisive	Negative	Lacking	Limited
John	✓	✓	✓	✓	✓										
Sarah		✓	✓	✓			✓		✓						
Lin			✓	✓										✓	✓
Aysha	✓				✓					✓	✓	✓	✓	✓	
Alex		✓		✓			✓		✓	✓					
Peter	✓	✓	✓	✓						✓	✓		✓	✓	
Jamal	✓	✓		✓		✓	✓	✓							
Brian	✓	✓	✓	✓	✓	✓	✓					✓		✓	
Samantha	✓	✓	✓	✓	✓		✓	✓	✓						
Miguel					✓	✓	✓	✓	✓	✓	✓	✓			

Figure 2.6 Data Collection Based on Text Processing

12. How can a data analyst for a large airline company efficiently collect and organize a significant amount of data pertaining to flight routes, passenger demographics, and operational metrics in a manner that ensures reliable and up-to-date records, complies with industry standards, and adheres to ethical guidelines and privacy regulations? The data will be utilized to identify potential areas for improvement, such as flight delays and high passenger traffic, so that the company can implement effective strategies to address these issues.

13. A chain of restaurants that serve fast food needs to improve their menu by providing healthier options to their customers. They want to track the nutrition information of their dishes, including burgers, fries, and milkshakes, to make it easier for the managers to compare their calorie, fat, and sugar content. What data management tools can restaurants utilize to track nutritional data to make healthier options readily available and enhance their menu?

14. The NFL team's talent acquisition specialist is on a mission to find the most talented high school football player to join their team. The specialist is responsible for gathering extensive information on potential players, such as their stats, performance history, and overall potential. The objective is to pinpoint the standout athlete who will bring a significant advantage to the NFL team. To achieve this, the specialist utilizes cloud computing to sift through vast amounts of data and identify the ideal high school player who can make a valuable contribution to the team's success in the upcoming season. Why is cloud computing the perfect tool for collecting and analyzing data on high school football players?

References

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3

Descriptive Statistics: Statistical Measurements and Probability Distributions

Figure 3.1 Statistics and data science play significant roles in stock market analysis, offering insights into market trends and risk assessment for better financial decision-making. (credit: modification of work “That was supposed to be going up, wasn't it?” by Rafael Matsunaga/Flickr, CC BY 2.0)

Chapter Outline

3.1 Measures of Center
3.2 Measures of Variation
3.3 Measures of Position
3.4 Probability Theory
3.5 Discrete and Continuous Probability Distributions

Introduction

Statistical analysis is the science of collecting, organizing, and interpreting data to make decisions. Statistical analysis lies at the core of data science, with applications ranging from consumer analysis (e.g., credit scores, retirement planning, and insurance) to government and business concerns (e.g., predicting inflation rates) to medical and engineering analysis.

Statistical analysis is an essential aspect of data science, involving the systematic collection, organization, and interpretation of data for decision-making. It serves as the foundation for various applications in consumer analysis such as credit scoring, retirement planning, and insurance as well as in government and business decision-making processes such as inflation rate prediction and marketing strategies. As a consumer, statistical analysis plays a significant role in various decision-making processes. For instance, when considering a large financial decision such as purchasing a house, the probability of interest rate fluctuations and their impact on mortgage financing must be taken into account.

Part of statistical analysis involves **descriptive statistics**, which refers to the collection, organization, summarization, and presentation of data using various graphs and displays. Once data is collected and summarized using descriptive statistics methods, the next step is to analyze the data using various probability tools and probability distributions in order to come to conclusions about the dataset and formulate predictions that will be useful for planning and estimation purposes.

Descriptive statistics includes measures of the center and dispersion of data, measures of position, and generation of various graphical displays. For example, a human resources administrator might be interested in generating some statistical measurements for the distribution of salaries at a certain company. This might involve calculating the mean salary, the median salary, and standard deviation, among others. The administrator might want to present the data to other employees, so generating various graphical displays such as histograms, box plots, and scatter plots might also be appropriate.

The human resources administrator might also want to derive estimates and predictions about the average salary in the company five years into the future, taking into account inflation effects, or they might want to create a model to predict an individual employee's salary based on the employee's years of experience in the field.

Such analyses are based on the statistical methods and techniques we will discuss in this chapter, building on the introduction to statistical analysis presented in What Are Data and Data Science? and Collecting and Preparing Data. Statistical analysis utilizes a variety of technological tools to automate statistical calculations and generate graphical displays. This chapter also will demonstrate the use of Python to generate results and graphs. The supplementary material at the end of this book will show the same using Excel (Appendix A) and R (Appendix B).

In this chapter, you will also study probability concepts and probability distributions. Probability theory helps us deal with quantifying uncertainty, which is always inherent in real-world data. We will see that in real-world datasets we often have to deal with noise and randomness that will be analyzed using statistical analyses.

Probability analysis provides the tools to model, understand, and quantify uncertainties, allowing data scientists to make informed decisions from data. Probability theory also forms the basis for different types of analyses, such as confidence intervals and hypothesis testing, which will be discussed further in Inferential Statistics and Regression Analysis. Such methods rely on probability models to make predictions and detect patterns. In addition, machine learning (discussed in Time Series and Forecasting) uses probabilistic models to help represent uncertainty and make predictions based on the collected data. Probability analysis helps data scientists to select data models, interpret the results, and assess the reliability of these conclusions. For a more detailed review of statistical concepts, please refer to Introductory Statistics 2e (https://openstax.org/books/introductory-statistics-2e/pages/1-introduction).

3.1 Measures of Center

Learning Outcomes

By the end of this section, you should be able to:

- 3.1.1 Define and calculate mean, trimmed mean, median, and mode for a dataset.
- 3.1.2 Determine the effect of outliers on the mean and the median.
- 3.1.3 Use Python to calculate measures of center for a dataset.

Measures of center are statistical measurements that provide a central, or typical, representation of a dataset. These measures can help indicate where the bulk of the data is concentrated and are often called the data's **central tendency**. The most widely used measures of the center of a dataset are the *mean* (average), the *median*, and the *mode*.

Mean and Trimmed Mean

The **mean**, or average, (sometimes referred to as the *arithmetic mean*) is the most commonly used measure of the center of a dataset. Sometimes the mean can be skewed by the presence of **outliers**, or data values that are significantly different as compared to the remainder of the dataset. In these instances, the *trimmed mean* is often used to provide a more representative measure of the center of the dataset, as we will discuss in the following section.

Mean

To calculate the mean, add the values of all the items in a dataset and divide by the number of items. For example, if the scores on your last three exams were 87, 92, and 73, then the mean score would be $\frac{87+92+73}{3} = 84$. If you had a large number of data values, you would proceed in the same way. For example, to calculate the mean value of 50 exam scores, add the 50 scores together and divide by 50. If the 50 scores add up to 4,050, for example, the mean score is $\frac{4050}{50}$, or 81.

In data science applications, you will encounter two types of datasets: sample data and population data. **Population data** represents all the outcomes or measurements that are of interest. **Sample data** represents outcomes or measurements collected from a subset, or part, of the population of interest. Of course in many applications, collecting data from an entire population is not practical or feasible, and so we often rely on sample data.

The notation $\bar{x}$ is used to indicate the **sample mean**, where the mean is calculated based on data taken from a sample. The notation $\sum x$ is used to denote the sum of the data values, and n is used to indicate the number of data values in the sample, also known as the **sample size**.

The sample mean can be calculated using the following formula:

$$\bar{x} = \frac{\sum x}{n}$$

The notation μ is used to indicate the **population mean**, where the mean is calculated based on data taken from the entire population, and N is used to indicate the number of data values in the population, also known as the **population size**. The population mean can be calculated using the following formula:

$$\mu = \frac{\sum x}{N}$$

The mean can also be determined by its frequency distribution. For every unique data value in the dataset, the **frequency distribution** gives the number of times, or **frequency**, that this unique value appears in the dataset. In this type of situation, the mean can be calculated by multiplying each distinct value by its frequency, summing these values, and then dividing this sum by the total number of data values. Here is the corresponding formula for the sample mean using the frequency distribution:

$$\bar{x} = \frac{\sum x \cdot f}{n}$$

When all the values in the dataset are unique, this reduces to the previous formula given for the sample mean.

EXAMPLE 3.1

Problem

During a clinical trial, a sample is taken of 10 patients and pulse rates are measured in beats per minute:

68, 92, 76, 51, 65, 83, 94, 72, 88, 59

Calculate the mean pulse rate for this sample.

Solution

Add the 10 data values, and the sum is 748. Divide this sum by the number of data values, which is 10. The result is:

$$\bar{x} = \frac{\sum x}{n} = \frac{748}{10} = 74.8$$

EXAMPLE 3.2

Problem

A college professor records the ages of 25 students in a data science class as shown:

Student Age	Number of Students (Frequency)
19	3
20	4
21	8
22	6
23	2
27	1
31	1
Total	**25**

Calculate the mean age for this sample of students.

Solution

Substitute the values from the table in the following formula:

$$\bar{x} = \frac{\sum x \cdot f}{n} = \frac{19 \cdot 3 + 20 \cdot 4 + 21 \cdot 8 + 22 \cdot 6 + 23 \cdot 2 + 27 \cdot 1 + 31 \cdot 1}{25} = \frac{541}{25} = 21.64$$

Trimmed Mean

A **trimmed mean** helps mitigate the effects of outliers, which are data values that are significantly different from most of the other data values in the dataset. In the dataset given in Example 3.1, a pulse rate of 35 or 120 would be considered outlier data values since these pulse rates are significantly different as compared to the rest of the values in Example 3.1. We will see that there is a formal method for determining outliers, and in fact there are several methods to identify outliers in a dataset.

The presence of outlier data values tends to disproportionally skew the mean and produce a potentially misleading result for the mean.

To calculate the trimmed mean for a dataset, first sort the data in ascending order (from smallest to largest). Then decide on a certain percentage of data values to be deleted from the lower and upper ends of the dataset. This might represent the extent of outliers in the dataset; trimmed mean percentages of 10% and 20% are common. Then delete the specified percentage of data values from both the lower end and upper end of the dataset. Then find the mean for the remaining undeleted data values.

As an example, to calculate a 10% trimmed mean, first sort the data values from smallest to largest. Then delete the lower 10% of the data values and delete the upper 10% of the data values. Then calculate the mean for the resulting dataset. Any outliers would tend to be deleted as part of the trimmed mean calculation, and thus the trimmed mean would then be a more representative measure of the center of the data for datasets containing outliers.

EXAMPLE 3.3

Problem

A real estate agent collects data on a sample of recently sold homes in a certain neighborhood, and the data are shown in the following dataset:

397900, 452600, 507400, 488300, 623400, 573200, 1689300, 403890, 612300, 599000, 2345800, 499000, 525000, 675000, 385000

1. Calculate the mean of the dataset.
2. Calculate a 20% trimmed mean rate for the dataset.

Solution

1. For the mean, add the 15 data values, and the sum is 10,777,090. Divide this sum by the number of data values, which is 15. The result is:

$$\bar{x} = \frac{\sum x}{n} = \frac{10{,}777{,}090}{15} = 718{,}472.70$$

2. For the trimmed mean, first order the data from smallest to largest. The sorted dataset is:

   385000, 397900, 403890, 452600, 488300, 499000, 507400, 525000, 573200, 599000, 612300, 623400, 675000, 1689300, 2345800

   Twenty percent of 15 data values is 3, and this indicates that 3 data values are to be deleted from each of the lower end and upper end of the dataset. The resulting 9 undeleted data values are:

   452600, 488300, 499000, 507400, 525000, 573200, 599000, 612300, 623400

   Then find the mean for the remaining data values. The sum of these 9 data values is 4,880,200. Divide this sum by the number of data values (9). The result is:

$$\bar{x} = \frac{\sum x}{n} = \frac{4{,}880{,}200}{9} = 542{,}244.40$$

Notice how the mean calculated in Part (a) is significantly larger as compared to the trimmed mean calculated in Part (b). The reason is the presence of several large outlier home prices. Once these outlier data values are removed by the trimmed mean calculation, the resulting trimmed mean is more representative of the typical home price in this neighborhood as compared to the mean.

Median

The median provides another measure of central tendency for a dataset. The **median** is generally a better measure of the central tendency when there are outliers (extreme values) in the dataset. Since the median focuses on the middle value of the ordered dataset, the median is preferred when outliers are present because the median is not affected by the numerical values of the outliers.

To determine the median of a dataset, first order the data from smallest to largest, and then find the middle value in the ordered dataset. For example, to find the median value of 50 exam scores, find the score that splits the data into two equal parts. The exam scores for 25 students will be below the median, and 25 students will have exam scores above the median.

If there is an odd number of data values in the dataset, then there will be one data value that represents the middle value, and this is the median. If there is an even number of data values in the dataset, then to find the median, add the two middle values together and divide by 2 (this is essentially finding the mean of the two middle values in the dataset).

EXAMPLE 3.4

Problem

The same dataset of pulse rates from Example 3.1 is:

68, 92, 76, 51, 65, 83, 94, 72, 88, 59

Calculate the median pulse rate for this sample.

Solution

First, order the 10 data values from smallest to largest. Divide this sum by the number of data values, which is 10. The result is:

51, 59, 65, 68, 72, 76, 83, 88, 92, 94

Since there is an even number of data values, add the two middle values together and divide by 2.
The two middle values are 72 and 76.

$$\text{Median} = \frac{72 + 76}{2} = \frac{148}{2} = 74$$

You can also quickly find the sample median of a dataset as follows.

Let n represent the number of data values in the sample.

- If n is odd, then the median is the data value in position $\frac{n+1}{2}$.
- If n is even, the median is the mean of the observations in position $\frac{n}{2}$ and position $\frac{n}{2} + 1$.

For example, let's say a dataset has 25 data values. Since n is odd, to identify the position of the median, calculate $\frac{n+1}{2}$, which is $\frac{25+1}{2}$, or 13. This indicates that the median is located in the 13th data position.

As another example, let's say a dataset has 100 data values. Since n is even, to identify the position of the median, calculate $\frac{n}{2}$, which is $\frac{100}{2}$, which is 50, and also calculate $\frac{n}{2} + 1$, which is $50 + 1$, which is 51. This indicates that the median is calculated as the mean of the 50th and 51st data values.

Mode

Another measure of center is the **mode**. The mode is the data value that occurs with the greatest frequency. If there are no repeating data values in a dataset, then there is no mode. If two data values occur with same greatest frequency, then there are two modes, and we say the data is bimodal. For example, assume that the weekly closing stock price for a technology stock, in dollars, is recorded for 20 consecutive weeks as follows:

50, 53, 59, 59, 63, 63, 72, 72, 72, 72, 72, 76, 78, 81, 83, 84, 84, 84, 90, 93

To find the mode, determine the most frequent score, which is 72, which occurs five times. Thus, the mode of this dataset is 72.

The mode can also be applied to non-numeric (qualitative) data, whereas the mean and the median can only be applied for numeric (quantitative) data. For example, a restaurant manager might want to determine the mode for responses to customer surveys on the quality of the service of a restaurant, as shown in Table 3.1.

Customer Service Rating	Number of Respondents
Excellent	267
Very Good	410
Good	392
Fair	107
Poor	18

Table 3.1 Customer Survey Results for Customer Survey Rating

Based on the survey responses, the mode is the Customer Service Rating of "Very Good," since this is the data value with the greatest frequency.

Influence of Outliers on Measures of Center

As mentioned earlier, when outliers are present in a dataset, the mean may not represent the center of the dataset, and the median will provide a better measure of center. The reason is that the median focuses on the middle value of the ordered dataset. Thus, any outliers at the lower end of the dataset or any outliers at the upper end of the dataset will not affect the median. Note: A formal method for identifying outliers is presented in Measures of Position when measures of position are discussed. The following example illustrates the point that the median is a better measure of central tendency when potential outliers are present.

EXAMPLE 3.5

Problem

Suppose that in a small company of 40 employees, one person earns a salary of $3 million per year, and the other 39 individuals each earn $40,000. Which is the better measure of center: the mean or the median?

Solution

The mean, in dollars, would be arrived at mathematically as follows:

$$\bar{x} = \frac{3{,}000{,}000 + 39(40{,}000)}{40} = 114{,}000$$

However, the median would be $40,000 since this is the middle data value in the ordered dataset. There are 39 people who earn $40,000 and one person who earns $3,000,000.

Notice that the mean is not representative of the typical value in the dataset since $114,000 is not reflective of the average salary for most employees (who are earning $40,000). The median is a much better measure of the "average" than the mean in this case because 39 of the values are $40,000 and one is $3,000,000. The data value of $3,000,000 is an outlier. The median result of $40,000 gives us a better sense of the center of the dataset.

Using Python for Measures of Center

We learned in What Are Data and Data Science? how the `DataFrame.describe()` method is used to

summarize data. Recall that the method `describe()` is defined for a DataFrame type object so should be called upon a DataFrame type variable (e.g. given a DataFrame `d`, use `d.describe()`).

Figure 3.2 shows the output of `DataFrame.describe()` on the "Movie Profit" dataset we used in What Are Data and Data Science?, movie_profit.csv. The mean and 50% quartile show the average and median of each column. For example, the average worldwide gross earnings are $410.14 million, and the median earnings are $309.35 million. Note that average and/or median of some columns are not as meaningful as the others. The first column—Unnamed: 0—was simply used as an identifier of each item in the dataset, so the average and mean of this column is not quite useful. `DataFrame.describe()` still computes the values because it can (and it does *not* care which column is meaningful to do so or not).

	Unnamed: 0	Rating	Duration	US_Gross_Million	Worldwide_Gross_Million
count	966.00000	966.000000	966.000000	966.000000	966.000000
mean	483.50000	6.814286	117.506211	156.158975	410.140600
std	279.00448	0.894383	21.615612	110.629617	294.758791
min	1.00000	3.300000	69.000000	0.010000	176.600000
25%	242.25000	6.200000	101.250000	90.832500	223.277500
50%	483.50000	6.800000	116.000000	129.245000	309.345000
75%	724.75000	7.400000	130.000000	187.090000	472.645000
max	966.00000	9.200000	238.000000	936.660000	2847.400000

Figure 3.2 The Output of `DataFrame.describe()` with the Movie Profit Dataset

EXPLORING FURTHER

Working with Python

See the Python website (https://openstax.org/r/python) for more details on using, installing, and working with Python. See this additional documentation, for more specific information on the statistics module (https://openstax.org/r/docspython).

3.2 Measures of Variation

Learning Outcomes

By the end of this section, you should be able to:

- 3.2.1 Define and calculate the range, the variance, and the standard deviation for a dataset.
- 3.2.2 Use Python to calculate measures of variation for a dataset.

Providing some measure of the spread, or *variation*, in a dataset is crucial to a comprehensive summary of the dataset. Two datasets may have the same mean but can exhibit very different spread, and so a measure of dispersion for a dataset is very important. While measures of central tendency (like mean, median, and mode) describe the center or average value of a distribution, measures of dispersion give insights into how much individual data points deviate from this central value.

The following two datasets are the exam scores for a group of three students in a biology course and in a statistics course.

Dataset A: Exam scores for students in a biology course: 40, 70, 100
Dataset B: Exam scores for students in a statistics course: 69, 70, 71

Notice that the mean score for both Dataset A and Dataset B is 70.

However, the datasets are significantly different from one another:

Dataset A has larger variability where one student scored 30 points below the mean and another student scored 30 points above the mean.
Dataset B has smaller variability where the exam scores are much more tightly clustered around the mean of 70.

This example illustrates that publishing the mean of a dataset is often inadequate to fully communicate the characteristics of the dataset. Instead, data scientists will typically include a measure of variation as well.

The three primary measures of variability are range, variance, and standard deviation, and these are described next.

Range

Range is a measure of dispersion for a dataset that is calculated by subtracting the minimum from the maximum of the dataset:

$$\text{Range} = \text{Max} - \text{Min}$$

Range is a straightforward calculation but makes use of only two of the data values in a dataset. The range can also be affected by outliers.

EXAMPLE 3.6

Problem

Calculate the range for Dataset A and Dataset B:

Dataset A: Exam scores for students in a biology course: 40, 70, 100
Dataset B: Exam scores for students in a statistics course: 69, 70, 71

Solution

For Dataset A, the maximum data value is 100 and the minimum data value is 40.
The range is then calculated as:

$$\begin{aligned} \text{Range} &= \text{Max} - \text{Min} \\ \text{Range} &= 100 - 40 \\ \text{Range} &= 60 \end{aligned}$$

For Dataset B, the maximum data value is 71 and the minimum data value is 69.
The range is then calculated as:

$$\begin{aligned} \text{Range} &= \text{Max} - \text{Min} \\ \text{Range} &= 71 - 69 \\ \text{Range} &= 2 \end{aligned}$$

The range clearly indicates that there is much less spread in Dataset B as compared to Dataset A.

One drawback to the use of the range is that it doesn't take into account every data value. The range only uses two data values from the dataset: the minimum (min) and the maximum (max). Also the range is influenced by outliers since an outlier might appear as a minimum or maximum data value and thus skew the results. For these reasons, we typically use other measures of variation, such as variance or standard deviation.

Variance

The **variance** provides a measure of the spread of data values by using the squared deviations from the mean. The more the individual data values differ from the mean, the larger the variance.

A financial advisor might use variance to determine the volatility of an investment and therefore help guide financial decisions. For example, a more cautious investor might opt for investments with low volatility.

The formula used to calculate variance also depends on whether the data is collected from a sample or a population. The notation s^2 is used to represent the *sample variance*, and the notation σ^2 is used to represent the *population variance*.

Formula for the sample variance:

$$s^2 = \frac{\sum(x - \bar{x})^2}{n - 1}$$

Formula for the population variance:

$$\sigma^2 = \frac{\sum(x - \mu)^2}{N}$$

In these formulas:
x represents the individual data values
$\bar{x}$ represents the sample mean
n represents the sample size
μ represents the population mean
N represents the population size

ALTERNATE FORMULA FOR VARIANCE

An alternate formula for the variance is available. It is sometimes used for more efficient computations:

$$\sigma^2 = \frac{\sum x^2}{N} - \mu^2$$

In the formulas for sample variance and population variance, notice the denominator for the sample variance is $n - 1$, whereas the denominator for the population variance is N. The use of $n - 1$ in the denominator of the sample variance is used to provide the best estimate for the population variance, in the sense that if repeated samples of size n are taken and the sample mean computed each time, then the average of those sample means will tend to the population mean as the number of repeated samples increase.

It is important to note that in many data science applications, population data is unavailable, and so we typically calculate the sample variance. For example, if a researcher wanted to estimate the percentage of smokers for all adults in the United States, it would be impractical to collect data from every adult in the United States.

Notice that the sample variance is a sum of squares. Its units of measurement are squares of the units of measurement of the original data. Since these square units are different than the units in the original data, this can be confusing. By contrast, standard deviation is measured in the same units as the original dataset, and thus the standard deviation is more commonly used to measure the spread of a dataset.

Standard Deviation

The **standard deviation** of a dataset provides a numerical measure of the overall amount of variation *in a dataset in the same units as the data*; it can be used to determine whether a particular data value is close to or

far from the mean, relative to the typical distance from the mean.

The standard deviation is always positive or zero. It is small when the data values are all concentrated close to the mean, exhibiting little variation, or spread. It is larger when the data values are spread out more from the mean, exhibiting more variation. A smaller standard deviation implies less variability in a dataset, and a larger standard deviation implies more variability in a dataset.

Suppose that we are studying the variability of two companies (A and B) with respect to employee salaries. The average salary for both companies is $60,000. For Company A, the standard deviation of salaries is $8,000, whereas the standard deviation for salaries for Company B is $19,000. Because Company B has a higher standard deviation, we know that there is more variation in the employee salaries for Company B as compared to Company A.

There are two different formulas for calculating standard deviation. Which formula to use depends on whether the data represents a sample or a population. The notation s is used to represent the sample standard deviation, and the notation σ is used to represent the population standard deviation. In the formulas shown, $\bar{x}$ is the sample mean, μ is the population mean, n is the sample size, and N is the population size.

Formula for the sample standard deviation:

$$s = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$$

Formula for the population standard deviation:

$$\sigma = \sqrt{\frac{\sum(x - \mu)^2}{N}}$$

Notice that the sample standard deviation is calculated as the square root of the variance. This means that once the sample variance has been calculated, the sample standard deviation can then be easily calculated as the square root of the sample variance, as in Example 3.7.

EXAMPLE 3.7

Problem

A biologist calculates that the sample variance for the amount of plant growth for a sample of plants is 8.7 cm^2. Calculate the sample standard deviation.

Solution

The sample standard deviation (s) is calculated as the square root of the variance.

$$s = \sqrt{s^2} = \sqrt{8.7} = 2.9 \text{ cm}$$

EXAMPLE 3.8

Problem

Assume the sample variance (s^2) for a dataset is calculated as 42.2. Based on this, calculate the sample standard deviation.

Solution

The sample standard deviation (s) is calculated as the square root of the variance.

$$s = \sqrt{s^2} = \sqrt{42.2} = 6.5 \text{ years}$$

This result indicates that the standard deviation is about 6.5 years.

Notice that the sample variance is the square of the sample standard deviation, so if the sample standard deviation is known, the sample variance can easily be calculated.

USE OF TECHNOLOGY FOR CALCULATING MEASURES OF VARIABILITY

Due to the complexity of calculating variance and standard deviation, technology is typically utilized to calculate these measures of variability. For example, refer to the examples shown in Coefficient of Variation on using Python for measures of variation.

Coefficient of Variation

A data scientist might be interested in comparing variation with different units of measurement of different means, and in these scenarios the **coefficient of variation (CV)** can be used. The coefficient of variation measures the variation of a dataset by calculating the standard deviation as a percentage of the mean. Note: coefficient of variation is typically expressed in a percentage format.

$$\begin{aligned} \text{CV} &= \frac{\sigma}{\mu} \times 100\% \\ \text{Sample CV} &= \frac{s}{\bar{x}} \times 100\% \end{aligned}$$

EXAMPLE 3.9

Problem

Compare the relative variability for Company A versus Company B using the coefficient of variation, based on the following sample data:

Company A: $\text{Sample Mean} = \$68{,}000,\ \text{Sample Standard Deviation} = \$9{,}200$

Company B: $\text{Sample Mean} = \$71{,}000,\ \text{Sample Standard Deviation} = \$6{,}400$

Solution

Calculate the coefficient of variation for each company:

$$\text{CV for Company A} = \frac{s}{\bar{x}} \times 100\% = \frac{9{,}200}{68{,}000} \times 100\% = 13.5\%$$

$$\text{CV for Company B} = \frac{s}{\bar{x}} \times 100\% = \frac{6{,}400}{71{,}000} \times 100\% = 9.0\%$$

Company A exhibits more variability relative to the mean as compared to Company B.

Using Python for Measures of Variation

`DataFrame.describe()` computes standard deviation as well on each column of a dataset. The **std** lists the standard deviation of each column (See Figure 3.3).

	Unnamed: 0	Rating	Duration	US_Gross_Million	Worldwide_Gross_Million
count	966.00000	966.000000	966.000000	966.000000	966.000000
mean	483.50000	6.814286	117.506211	156.158975	410.140600
std	279.00448	0.894383	21.615612	110.629617	294.758791
min	1.00000	3.300000	69.000000	0.010000	176.600000
25%	242.25000	6.200000	101.250000	90.832500	223.277500
50%	483.50000	6.800000	116.000000	129.245000	309.345000
75%	724.75000	7.400000	130.000000	187.090000	472.645000
max	966.00000	9.200000	238.000000	936.660000	2847.400000

Figure 3.3 The Output of `DataFrame.describe()` with the Movie Profit Dataset

3.3 Measures of Position

Learning Outcomes

By the end of this section, you should be able to:

- 3.3.1 Define and calculate percentiles, quartiles, and z-scores for a dataset.
- 3.3.2 Use Python to calculate measures of position for a dataset.

Common measures of position include percentiles and quartiles as well as z-scores, all of which are used to indicate the relative location of a particular datapoint.

Percentiles

If a student scores 47 on a biology exam, it is difficult to know if the student did well or poorly compared to the population of all other students taking the exam. **Percentiles** provide a way to assess and compare the distribution of values and the position of a specific data point in relation to the entire dataset by indicating the percentage of data points that fall below it. Specifically, a percentile is a value on a scale of one hundred that indicates the percentage of a distribution that is equal to or below it. Let's say the student learns they scored in the 90th percentile on the biology exam. This percentile indicates that the student has an exam score higher than 90% of all other students taking the test. This is the same as saying that the student's score places the student in the top 10% of all students taking the biology test. Thus, this student scoring in the 90th percentile did very well on the exam, even if the actual score was 47.

To calculate percentiles, the data must be ordered from smallest to largest and then the ordered data divided into hundredths. If you score in the 80th percentile on an aptitude test, that does not necessarily mean that you scored 80% on the test. It means that 80% of the test scores are the same as or less than your score and the remaining 20% of the scores are the same as or greater than your score.

Percentiles are useful for comparing many types of values. For example, a stock market mutual fund might report that the performance for the fund over the past year was in the 80th percentile of all mutual funds in the peer group. This indicates that the fund performed better than 80% of all other funds in the peer group. This also indicates that 20% of the funds performed better than this particular fund.

To calculate percentiles for a *specific data value* in a dataset, first order the dataset from smallest to largest and count the number of data values in the dataset. Locate the measurement of interest and count how many data values fall below the measurement. Then the percentile for the measurement is calculated as follows:

$$\text{Percentile} = \frac{\text{number of data values below the measurement}}{\text{total number of data values}} \times 100\% = \frac{n}{N} \times 100\%$$

EXAMPLE 3.10

Problem

The following ordered dataset represents the scores of 15 employees on an aptitude test:

51, 63, 65, 68, 71, 75, 75, 77, 79, 82, 88, 89, 89, 92, 95

Determine the percentile for the employee who scored 88 on the aptitude test.

Solution

There are 15 data values in total, and there are 10 data values below 88.

$$\text{Percentile} = \frac{\text{number of data values below the measurement}}{\text{total number of data values}} \times 100\% = \frac{10}{15} \times 100\% = 67\text{th percentile}$$

Quartiles

While percentiles separate data into 100 equal parts, **quartiles** separate data into quarters, or four equal parts. To find the quartiles, first find the median, or second quartile. The first quartile, Q_1, is the middle value, or median, of the lower half of the data, and the third quartile, Q_3, is the middle value of the upper half of the data.

Note the following correspondence between quartiles and percentiles:

- The first quartile corresponds to the 25th percentile.
- The second quartile (which is the median) corresponds to the 50th percentile.
- The third quartile corresponds to the 75th percentile.

EXAMPLE 3.11

Problem

Consider the following ordered dataset, which represents the time in seconds for an athlete to complete a 40-yard run:

5.4, 6.0, 6.3, 6.8, 7.1, 7.2, 7.4, 7.5, 7.9, 8.2, 8.7

Solution

The median, or second quartile, is the middle value in this dataset, which is 7.2. Notice that 50% of the data values are below the median, and 50% of the data values are above the median. The lower half of the data values are 5.4, 6.0, 6.3, 6.8, 7.1. Note that these are the data values below the median. The upper half of the data values are 7.4, 7.5, 7.9, 8.2, 8.7, which are the data values above the median.

To find the first quartile, Q_1, locate the middle value of the lower half of the data (5.4, 6.0, 6.3, 6.8, 7.1). The middle value of the lower half of the dataset is 6.3. Notice that one-fourth, or 25%, of the data values are below this first quartile, and 75% of the data values are above this first quartile.

To find the third quartile, Q_3, locate the middle value of the upper half of the data (7.4, 7.5, 7.9, 8.2, 8.7). The middle value of the upper half of the dataset is 7.9. Notice that one-fourth, or 25%, of the data values are above this third quartile, and 75% of the data values are below this third quartile.

Thus, the quartiles Q_1, Q_2, Q_3 for this dataset are 6.3, 7.2, 7.9, respectively.

The **interquartile range (IQR)** is a number that indicates the spread of the middle half, or the middle 50%, of the data. It is the difference between the third quartile, Q_3, and the first quartile, Q_1.

$$\text{IQR} = Q_3 - Q_1$$

Note that the IQR provides a measure of variability that excludes outliers.

In Example 3.11, the IQR can be calculated as:

$$\text{IQR} = Q_3 - Q_1 = 7.9 - 6.3 = 1.6$$

Quartiles and the IQR can be used to flag possible outliers in a dataset. For example, if most employees at a company earn about $50,000 and the CEO of the company earns $2.5 million, then we consider the CEO's salary to be an outlier data value because this salary is significantly different from all the other salaries in the dataset. An outlier data value can also be a value much lower than the other data values in a dataset, so if one employee only makes $15,000, then this employee's low salary might also be considered an outlier.

To detect outliers, you can use the quartiles and the IQR to calculate a lower and an upper bound for outliers. Then any data values below the lower bound or above the upper bound will be flagged as outliers. These data values should be further investigated to determine the nature of the outlier condition and whether the data values are valid or not.

To calculate the lower and upper bounds for outliers, use the following formulas:

$$\begin{aligned} \text{Lower Bound for Outliers} &= Q_1 - (1.5 \cdot \text{IQR}) \\ \text{Upper Bound for Outliers} &= Q_3 + (1.5 \cdot \text{IQR}) \end{aligned}$$

These formulas typically use 1.5 as a cutoff value to identify outliers in a dataset.

EXAMPLE 3.12

Problem

Calculate the IQR for the following 13 home prices and determine if any of the home prices values are potential outliers. Data values are in US dollars.

389950, 230500, 158000, 479000, 639000, 114950, 5500000, 387000, 659000, 529000, 575000, 488800, 1095000

Solution

Order the data from smallest to largest.

114950, 158000, 230500, 387000, 389950, 479000, 488800, 529000, 575000, 639000, 659000, 1095000, 5500000

First, determine the median of the dataset. There are 13 data values, so the median is the middle data value, which is 488,800.

Next, calculate the Q_1 and Q_3.

For the first quartile, look at the data values below the median. The two middle data values in this lower half of the data are 230,500 and 387,000. To determine the first quartile, find the mean of these two data values.

For the third quartile, look at the data values above the median. The two middle data values in this upper half of the data are 639,000 and 659,000. To determine the third quartile, find the mean of these two data values.

$$Q_1 = \frac{230{,}500+387{,}000}{2} = 308{,}750$$
$$Q_3 = \frac{639{,}000+659{,}000}{2} = 649{,}000$$

Now, calculate the interquartile range (IQR):

$$\text{IQR} = 649{,}000 - 308{,}750 = 340{,}250$$

Calculate the value of 1.5 interquartile range (IQR):

$$(1.5)(\text{IQR}) = (1.5)(340{,}250) = 510{,}375$$

Calculate the lower and upper bound for outliers:

$$\text{Lower Bound} = Q_1 - (1.5)(\text{IQR}) = 308{,}750 - 510{,}375 = -201{,}625$$
$$\text{Upper Bound} = Q_3 + (1.5)(\text{IQR}) = 649{,}000 + 510{,}375 = 1{,}159{,}375$$

The lower bound for outliers is −201,625. Of course, no home price is less than −201,625, so no outliers are present for the lower end of the dataset.

The upper bound for outliers is 1,159,375. The data value of 5,500,000 is greater than the upper bound of 1,159,375. Therefore, the home price of $5,500,000 is a potential outlier. This is important because the presence of outliers could potentially indicate data errors or some other anomalies in the dataset that should be investigated. For example, there may have been a data entry error and a home price of $550,000 was erroneously entered as $5,500,000.

z-scores

The **z-score** is a measure of the position of an entry in a dataset that makes use of the mean and standard deviation of the data. It represents the number of standard deviations by which a data value differs from the mean. For example, suppose that in a certain neighborhood, the mean selling price of a home is $350,000 and the standard deviation is $40,000. A particular home sells for $270,000. Based on the selling price of this home, we can calculate the relative standing of this home compared to other home sales in the same neighborhood.

The corresponding z-score of a measurement considers the given measurement in relation to the mean and standard deviation for the entire population. The formula for a z-score calculation is as follows:

$$z = \frac{x - \mu}{\sigma}$$

Where:
x is the measurement
μ is the mean
σ is the standard deviation

Notice that when a measurement is below the mean, the corresponding z-score will be a negative value. If the measurement is exactly equal to the mean, the corresponding z-score will be zero. If the measurement is above the mean, the corresponding z-score will be a positive value.

z-scores can also be used to identify outliers. Since z-scores measure the number of standard deviations from the mean for a data value, a z-score of 3 would indicate a data value that is 3 standard deviations above the mean. This would represent a data value that is significantly displaced from the mean, and typically, a z-score less than −3 or a z-score greater than +3 can be used to flag outliers.

EXAMPLE 3.13

Problem

For the home example in Example 3.12, the x value is the home price of $270,000, the mean μ is $350,000, and the standard deviation σ is $40,000. Calculate the z-score.

Solution

The z-score can be calculated as follows:

$$z = \frac{x - \mu}{\sigma} = \frac{270{,}000 - 350{,}000}{40{,}000} = \frac{-80{,}000}{40{,}000} = -2$$

This z-score of −2 indicates that the selling price for this home is 2 standard deviations below the mean, which represents a data value that is significantly below the mean.

Using Python to Calculate Measures of Position for a Dataset

`DataFrame.describe()` computes different measures of position as well on each column of a dataset. See min, 25%, 50%, 75%, and max in Figure 3.4.

	Unnamed: 0	Rating	Duration	US_Gross_Million	Worldwide_Gross_Million
count	966.00000	966.000000	966.000000	966.000000	966.000000
mean	483.50000	6.814286	117.506211	156.158975	410.140600
std	279.00448	0.894383	21.615612	110.629617	294.758791
min	1.00000	3.300000	69.000000	0.010000	176.600000
25%	242.25000	6.200000	101.250000	90.832500	223.277500
50%	483.50000	6.800000	116.000000	129.245000	309.345000
75%	724.75000	7.400000	130.000000	187.090000	472.645000
max	966.00000	9.200000	238.000000	936.660000	2847.400000

Figure 3.4 The Output of `DataFrame.describe()` with the Movie Profit Dataset

3.4 Probability Theory

Learning Outcomes

By the end of this section, you should be able to:

- 3.4.1 Describe the basic concepts of probability and apply these concepts to real-world applications in data science.
- 3.4.2 Apply conditional probability and Bayes' Theorem.

Probability is a numerical measure that assesses the likelihood of occurrence of an event. Probability applications are ubiquitous in data science since many decisions in business, science, and engineering are based on probability considerations. We all use probability calculations every day as we decide, for instance, whether to take an umbrella to work, the optimal route for a morning commute, or the choice of a college major.

Basic Concepts of Probability

We have all used probability in one way or another on a day-to-day basis. Before leaving the house, you might

want to know the probability of rain. The probability of obtaining heads on one flip of a coin is one-half, or 0.5.

A data scientist is in interested in expressing probability as a number between 0 and 1 (inclusive), where 0 indicates impossibility (the event will not occur) and 1 indicates certainty (the event will occur). The probability of an event falling between 0 and 1 reflects the degree of uncertainty associated with the event.

Here is some terminology we will be using in probability-related analysis:

- An **outcome** is the result of a single trial in a probability experiment.
- The **sample space** is the set of all possible outcomes in a probability experiment.
- An **event** is some subset of the sample space. For example, an event could be rolling an even number on a six-sided die. This event corresponds to three outcomes, namely rolling a 2, 4, or 6 on the die.

To calculate probabilities, we can use several approaches, including relative frequency probability, which is based on actual data, and theoretical probability, which is based on theoretical conditions.

Relative Frequency Probability

Relative frequency probability is a method of determining the likelihood of an event occurring based on the observed frequency of its occurrence in a given sample or population. A data scientist conducts or observes a procedure and determines the number of times a certain Event A occurs. The probability of Event A, denoted as $P(A)$, is then calculated based on data that has been collected from the experiment, as follows:

$$P(A) = \frac{\text{number of times Event } A \text{ has occured}}{\text{number of times the procedure was repeated}}$$

EXAMPLE 3.14

Problem

A polling organization asks a sample of 400 people if they are in favor of increased funding for local schools; 312 of the respondents indicate they are in favor of increased funding. Calculate the probability that a randomly selected person will be in favor of increased funding for local schools.

Solution

Using the data collected from this polling, a total of 400 people were asked the question, and 312 people were in favor of increased school funding. The probability for a randomly selected person being in favor of increased funding can then be calculated as follows (notice that Event A in this example corresponds to the event that a person is in favor of the increased funding):

$$\begin{aligned} P(A) &= \text{Probability of In Favor} \\ &= \frac{\text{number of times Event } A \text{ has occurred}}{\text{number of times the procedure was repeated}} = \frac{312}{400} = 0.78 = 78\% \end{aligned}$$

EXAMPLE 3.15

Problem

A medical patient is told they need knee surgery, and they ask the doctor for an estimate of the probability of success for the surgical procedure. The doctor reviews data from the past two years and determines there were 200 such knee surgeries performed and 188 of them were successful. Based on this past data, the doctor calculates the probability of success for the knee surgery (notice that Event A in this example corresponds to the event that a patient has a successful knee surgery result).

Solution

Using the data collected from the past two years, there were 200 surgeries performed, with 188 successes. The probability can then be calculated as:

$$P(A) = \text{Probability of Success} = \frac{\text{number of times Event } A \text{ has occured}}{\text{number of times the procedure was repeated}} = \frac{188}{200} = 0.94$$

The doctor informs the patient that there is a 94% chance of success for the pending knee surgery.

Theoretical Probability

Theoretical probability is the method used when the outcomes in a probability experiment are equally likely—that is, under theoretical conditions.

The formula used for theoretical probability is similar to the formula used for **empirical probability**. Theoretical probability considers all the possible outcomes for an experiment that are known ahead of time so that past data is not needed in the calculation for theoretical probability.

$$\text{Theoretical Probability} = \frac{\text{number of outcomes for the event of interest}}{\text{total number of outcomes in the sample space}}$$

For example, the theoretical probability of rolling an even number when rolling a six-sided die is $\frac{3}{6}$ (which is $\frac{1}{2}$, or 0.5). There are 3 outcomes corresponding to rolling an even number, and there are 6 outcomes total in the sample space. Notice this calculation can be done without conducting any experiments since the outcomes are equally likely.

EXAMPLE 3.16

Problem

A student is working on a multiple-choice question that has 5 possible answers. The student does not have any idea about the correct answer, so the student randomly guesses. What is the probability that the student selects the correct answer?

Solution

Since the student is guessing, each answer choice is equally likely to be selected. There is 1 correct answer out of 5 possible choices. The probability of selecting the correct answer can be calculated as:

$$\begin{aligned}\text{Probability of Correct Answer} &= \frac{\text{number of outcomes for the event of interest}}{\text{total number of outcomes in the sample space}} \\ &= \frac{1}{5} = 0.20 = 20\%\end{aligned}$$

Notice in Example 3.16 that probabilities can be written as fractions, decimals, or percentages.

Also note that any probability must be between 0 and 1 inclusive. An event with a probability of zero will never occur, and an event with a probability of 1 is certain to occur. A probability greater than 1 is not possible, and a negative probability is not possible.

Complement of an Event

The **complement of an event** is the set of all outcomes in the sample space that are *not* included in the event. The complement of Event A is usually denoted by A' (A prime). To find the probability of the complement of Event A, subtract the probability of Event A from 1.

$$P(A') = 1 - P(A)$$

EXAMPLE 3.17

Problem

A company estimates that the probability that an employee will provide confidential information to a hacker is 0.1%. Determine the probability that an employee will not provide any confidential information during a hacking attempt.

Solution

Let Event A be the event that the employee will provide confidential information to a hacker. Then the complement of this Event A' is the event that an employee will not provide any confidential information during a hacking attempt.

$$\begin{aligned} P(A') &= 1 - P(A) \\ P(A') &= 1 - 0.001 = 0.999 \end{aligned}$$

There is a 99.9% probability that an employee will not provide any confidential information during a hacking attempt.

Conditional Probability and Bayes' Theorem

Data scientists are often interested in determining conditional probabilities, or the occurrence of one event that is conditional or dependent on another event. For example, a medical researcher might be interested to know if an asthma diagnosis for a patient is dependent on the patient's exposure to air pollutants. In addition, when calculating conditional probabilities, we can sometimes revise a probability estimate based on additional information that is obtained. As we'll see in the following section, Bayes' Theorem allows new information to be used to refine a probability estimate.

Conditional Probability

A **conditional probability** is the probability of an event given that another event has already occurred. The notation for conditional probability is $P(A|B)$, which denotes the probability of Event A, given that Event B has occurred. The vertical line between A and B denotes the "given" condition. (In this notation, the vertical line does not denote division).

For example, we might want to know the probability of a person getting a parking ticket given that a person did not put any money in a parking meter. Or a medical researcher might be interested in the probability of a patient developing heart disease given that the patient is a smoker.

If the occurrence of one event affects the probability of occurrence for another event, we say that the events are *dependent*; otherwise, the events are *independent*. **Dependent events** are events where the occurrence of one event affects the probability of occurrence of another event. **Independent events** are events where the probability of occurrence of one event is *not* affected by the occurrent of another event. The dependence of events has important implications in many fields such as marketing, engineering, psychology, and medicine.

EXAMPLE 3.18

Problem

Determine if the two events are dependent or independent:

1. Rolling a 3 on one roll of a die, rolling a 4 on a second roll of a die

2. Obtaining heads on one flip of a coin and obtaining tails on a second flip of a coin
3. Selecting five basketball players from a professional basketball team and a player's height is greater than 6 feet
4. Selecting an Ace from a deck of 52 cards, returning the card back to the original stack, and then selecting a King
5. Selecting an Ace from a deck of 52 cards, not returning the card back to the original stack, and then selecting a King

Solution

1. The result of one roll does not affect the result for the next roll, so these events are independent.
2. The results of one flip of the coin do not affect the results for any other flip of the coin, so these events are independent.
3. Typically, basketball players are tall individuals, and so they are more likely to have heights greater than 6 feet as opposed to the general public, so these events are dependent.
4. By selecting an Ace from a deck of 52 cards and then replacing the card, this restores the deck of cards to its original state, so the probability of selecting a King is not affected by the selection of the Ace. So these events are independent.
5. By selecting an Ace from a deck of 52 cards and then not replacing the card, this will result in only 51 cards remaining in the deck. Thus, the probability of selecting a King is affected by the selection of the Ace, so these events are dependent.

There are several ways to use conditional probabilities in data science applications.

Conditional probability can be defined as follows:

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}, \text{ where } P(B) \neq 0$$

When assessing the conditional probability of $P(A|B)$, if the two events are independent, this indicates that Event A is not affected by the occurrence of Event B, so we can write that $P(A|B) = P(A)$ for independent events.

If we determine that the $P(A|B)$ is not equal to $P(A)$, this indicates that the events are dependent.

$P(A|B) = P(A)$ implies independent events, where $P(B) \neq 0$.

$P(A|B) \neq P(A)$ implies dependent events.

EXAMPLE 3.19

Problem

Table 3.2 shows the number of nursing degrees and non-nursing degrees at a university for a specific year, and the data is broken out by age groups. Calculate the probability that a randomly chosen graduate obtained a nursing degree, given that the graduate is in the age group of 23 and older.

Age Group	Nursing Degrees	Non-Nursing Degrees	Total
22 and under	1036	1287	2323
23 and older	986	932	1918
Total	2022	2219	4241

Table 3.2 Number of Nursing and Non-Nursing Degrees at a University by Age Group

Solution

Since we are given that the group of interest are those graduates in the age group of 23 and older, focus only on the second row in the table.

Looking only at the second row in the table, we are interested in the probability that a randomly chosen graduate obtained a nursing degree. The reduced sample space consists of 1,918 graduates, and 986 of them received nursing degrees. So the probability can be calculated as:

$$P(\text{nursing degree} \mid \text{age group of 23 and older}) = \frac{986}{1{,}918} = 0.514$$

Another method to analyze this example is to rewrite the conditional probability using the equation for $P(A \text{ and } B)$, as follows:

$$\begin{aligned} P(A \text{ and } B) &= P(A) \cdot P(B|A) \\ \text{rewrite as: } P(B|A) &= \frac{P(A \text{ and } B)}{P(A)} \end{aligned}$$

We can now use this equation to calculate the probability that a randomly chosen graduate obtained a nursing degree, given that the graduate is in the age group of 23 and older. The probability of A and B is the probability that a graduate received a nursing degree and is also in the age group of 23 and older. From the table, there are 986 graduates who earned a nursing degree and are also in the age group of 23 and older. Since this number of graduates is out of the total sample size of 4,241, we can write the probability of Events A and B as:

$$P(A \text{ and } B) = \frac{986}{4{,}241}$$

We can also calculate the probability that a graduate is in the age group of 23 and older. From the table, there are 1,918 graduates in this age group out of the total sample size of 4,241, so we can write the probability for Event A as:

$$P(A) = \frac{1{,}918}{4{,}241}$$

Next, we can substitute these probabilities into the formula for $P(B|A)$, as follows:

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)} = \frac{\frac{986}{4{,}241}}{\frac{1{,}918}{4{,}241}} = \frac{986}{4{,}241} \cdot \frac{4{,}241}{1{,}918} = \frac{986}{1{,}918} = 0.514$$

Probability of *At Least One*

The probability of *at least one occurrence* of an event is often of interest in many data science applications. For example, a doctor might be interested to know the probability that at least one surgery to be performed this week will involve an infection of some type.

The phrase "at least one" implies the condition of one or more successes. From a sample space perspective, one or more successes is the complement of "no successes." Using the complement rule discussed earlier, we can write the following probability formula:

$$P(\text{at least one success}) = 1 - P(\text{no successes})$$

As an example, we can find the probability of rolling a die 3 times and obtaining at least one four on any of the rolls. This can be calculated by first finding the probability of not observing a four on any of the rolls and then subtracting this probability from 1. The probability of not observing a four on a roll of the die is 5/6. Thus, the probability of rolling a die 3 times and obtaining at least one four on any of the rolls is $1 - \left(\frac{5}{6}\right)^3 = 0.421$.

EXAMPLE 3.20

Problem

From past data, hospital administrators determine the probability that a knee surgery will be successful is 0.89.

1. During a certain day, the hospital schedules four knee surgeries to be performed. Calculate the probability that all four of these surgeries will be successful.
2. Calculate the probability that none of these knee surgeries will be successful.
3. Calculate the probability that at least one of the knee surgeries will be successful.

Solution

1. For all four surgeries to be successful, we can interpret that as the first surgery will be successful, and the second surgery will be successful, and the third surgery will be successful, and the fourth surgery will be successful. Since the probability of success for one knee surgery does not affect the probability of success for another knee surgery, we can assume these events are independent. Based on this, the probability that all four surgeries will be successful can be calculated using the probability formula for $P(A \text{ and } B)$ by multiplying the probabilities together:

$$\begin{aligned} P(A \text{ and } B \text{ and } C \text{ and } D) &= P(A) \cdot P(B) \cdot P(C) \cdot P(D) \\ &= 0.89 \cdot 0.89 \cdot 0.89 \cdot 0.89 = 0.627 \end{aligned}$$

   There is about a 63% chance that all four knee surgeries will be successful.

2. The probability that a knee surgery will be unsuccessful can be calculated using the complement rule. If the probability of a successful surgery is 0.89, then the probability that the surgery will be unsuccessful is 0.11:

$$P(A') = 1 - P(A) = 1 - 0.89 = 0.11$$

   Based on this, the probability that all four surgeries will be unsuccessful can be calculated using the probability formula for $P(A \text{ and } B)$ by multiplying the probabilities together:

$$\begin{aligned}(A' \text{ and } B' \text{ and } C' \text{ and } D') &= P(A') \cdot P(B') \cdot P(C') \cdot P(D') \\ &= 0.11 \cdot 0.11 \cdot 0.11 \cdot 0.11 = 0.00015\end{aligned}$$

Since this is a very small probability, it is very unlikely that none of the surgeries will be successful.

3. To calculate the probability that at least one of the knee surgeries will be successful, use the probability formula for "at least one," which is calculated as the complement of the event "none are successful."

$$\begin{aligned}P(\text{at least one success}) &= 1 - P(\text{no successes}) \\ P(\text{at least one success}) &= 1 - 0.00015 = 0.99985\end{aligned}$$

This indicates there is a very high probability that at least one of the knee surgeries will be successful.

Bayes' Theorem

Bayes' Theorem is a statistical technique that allows for the revision of probability estimates based on new information or evidence that allows for more accurate and efficient decision-making in uncertain situations. Bayes' Theorem is often used to help assess probabilities associated with medical diagnoses such as the probability a patient will develop cancer based on test screening results. This can be important in medical analysis to help assess the impact of a *false positive*, which is the scenario where the patient does not have the ailment but the screening test gives a false indication that the patient does have the ailment.

Bayes' Theorem allows the calculation of the conditional probability $P(A|B)$. There are several forms of Bayes' Theorem, as shown:

$$\begin{aligned}P(A|B) &= \frac{P(A) \cdot P(B|A)}{P(B)} \\ P(A|B) &= \frac{P(A) \cdot P(B|A)}{P(A) \cdot P(B|A) + P(A') \cdot P(B|A')}\end{aligned}$$

EXAMPLE 3.21

Problem

Assume that a certain type of cancer affects 3% of the population. Call the event that a person has cancer "Event A," so:

$$P(A) = 0.03$$

A patient can undergo a screening test for this type of cancer. Assume the probability of a true positive from the screening test is 75%, which indicates that probability that a person has a positive test result given that they actually have cancer is 0.75. Also assume the probability of a false positive from the screening test is 15%, which indicates that probability that a person has a positive test result given that they do not have cancer is 0.15.

A medical researcher is interested in calculating the probability that a patient actually has cancer given that the screening test shows a positive result.

The researcher is interested in calculating $P(A|B)$, where Event A is the person actually has cancer and Event B is the event that the person shows a positive result in the screening test. Use Bayes' Theorem to calculate this conditional probability.

Solution

From the example, the following probabilities are known:

$$\begin{aligned} P(A) &= 0.03 \\ P(A') &= 0.97 \end{aligned}$$

The conditional probabilities can be interpreted as follows:

$$\begin{aligned} P(B|A) &= P(\text{positive test result} \mid \text{patient has cancer}) = 0.75 \\ P(B|A') &= P(\text{positive test result} \mid \text{patient does not have cancer}) = 0.15 \end{aligned}$$

Substituting these probabilities into the formula for Bayes' Theorem results in the following:

$$P(A|B) = \frac{P(A) \cdot P(B|A)}{P(A) \cdot P(B|A) + P(A') \cdot P(B|A')}$$

$$P(\text{ patient has cancer} \mid \text{positive test result}) = \frac{0.03 \cdot 0.75}{0.03 \cdot 0.75 + 0.97 \cdot 0.15} = 0.134$$

This result from Bayes' Theorem indicates that even if a patient receives a positive test result from the screening test, this does not imply a high likelihood that the patient has cancer. There is only a 13% chance that the patient has cancer given a positive test result from the screening test.

3.5 Discrete and Continuous Probability Distributions

Learning Outcomes

By the end of this section, you should be able to:

- 3.5.1 Describe fundamental aspects of probability distributions.
- 3.5.2 Apply discrete probability distributions including binomial and Poisson distributions.
- 3.5.3 Apply continuous probability distributions including exponential and normal distributions.
- 3.5.4 Use Python to apply various probability distributions for probability applications.

Probability distributions are used to model various scenarios to help with probability analysis and predictions, and they are used extensively to help formulate probability-based decisions. For example, if a doctor knows that the weights of newborn infants follow a normal (bell-shaped) distribution, the doctor can use this information to help identify potentially underweight newborn infants, which might indicate a medical condition warranting further investigation. Using a normal distribution, the doctor can calculate that only a small percentage of babies have weights below a certain threshold, which might prompt the doctor to further investigate the cause of the low weight. Or a medical researcher might be interested in the probability that a person will have high blood pressure or the probability that a person will have type O blood.

Overview of Probability Distributions

To begin our discussion of probability distributions, some terminology will be helpful:

- **Random variable**—a variable where a single numerical value is assigned to a specific outcome from an experiment. Typically the letter x is used to denote a random variable. For example, assign the numerical values 1, 2, 3, … 13 to the cards selected from a standard 52-card deck of Ace, 2, 3, … 10, Jack, Queen, King. Notice we cannot use "Jack" as the value of the random variable since by definition a random variable must be a numerical value.
- **Discrete random variable**—a random variable is considered discrete if there is a finite or countable number of values that the random variable can take on. (If there are infinitely many values, the number of values is countable if it possible to count them individually.) Typically, a discrete random variable is the result of a count of some kind. For example, if the random variable x represents the number of cars in a

parking lot, then the values that x can take on can only be whole numbers since it would not make sense to have $x = 15.37$ cars in the parking lot.

- **Continuous random variable**—a random variable is considered continuous if the value of the random variable can take on any value within an interval. Typically, a continuous random variable is the result of a measurement of some kind. For example, if the random variable x represents the weight of a bag of apples, then x can take on any value such as $x = 2.45734$ pounds of apples.

To summarize, the difference between discrete and continuous probability distributions has to do with the nature of the random variables they represent. **Discrete probability distributions** are associated with variables that take on a finite or countably infinite number of distinct values. **Continuous probability distributions** deal with random variables that can take on any value within a given range or interval. It is important to identify and distinguish between discrete and continuous random variables since different statistical methods are used to analyze each type.

EXAMPLE 3.22

Problem

A coin is flipped three times. Determine a possible random variable that can be assigned to represent the number of heads observed in this experiment.

Solution

One possible random variable assignment could be to let x count the number of heads observed in each possible outcome in the sample space. When flipping a coin three times, there are eight possible outcomes, and x will be the numerical count corresponding to the number of heads observed for each outcome. Notice that the possible values for the random variable x are 0, 1, 2 and 3, as shown in Table 3.3.

Result for Flip #1	Result for Flip #2	Result for Flip #3	Value of Random Variable x
Heads	Heads	Heads	3
Heads	Heads	Tails	2
Heads	Tails	Heads	2
Heads	Tails	Tails	1
Tails	Heads	Heads	2
Tails	Heads	Tails	1
Tails	Tails	Heads	1
Tails	Tails	Tails	0

Table 3.3 Result of Three Random Coin Flips

EXAMPLE 3.23

Problem

Identify the following random variables as either discrete or continuous random variables:

1. The amount of gas, in gallons, used to fill a gas tank
2. Number of children per household in a certain neighborhood

3. Number of text messages sent by a certain student during a particular day
4. Number of hurricanes affecting Florida in a given year
5. The amount of rain, in inches, in Detroit, Michigan, in a certain month

Solution

1. The number of gallons of gas used to fill a gas tank can take on any value, such as 12.3489, so this represents a continuous random variable.
2. The number of children per household in a certain neighborhood can only take on certain discrete values such as 0, 1, 2, 3, etc., so this represents a discrete random variable.
3. The number of text messages sent by a certain student during a particular day can only take on certain discrete values such as 26, 10, 17, etc., so this represents a discrete random variable.
4. The number of hurricanes affecting Florida in a given year can only take on certain values such as 0, 1, 2, 3, etc., so this represents a discrete random variable.
5. The number of inches of rain in Detroit, Michigan, in a certain month can take on any value, such as 2.0563, so this represents a continuous random variable.

Discrete Probability Distributions: Binomial and Poisson

Discrete random variables are of interest in many data science applications, and there are several probability distributions that apply to discrete random variables. In this chapter, we present the binomial distribution and the Poisson distribution, which are two commonly used probability distributions used to model discrete random variables for different types of events.

Binomial Distribution

The **binomial distribution** is used in applications where there are two possible outcomes for each trial in an experiment and the two possible outcomes can be considered as success or failure. For example, when a baseball player is at-bat, the player either gets a hit or does not get a hit. There are many applications of binomial experiments that occur in medicine, psychology, engineering, science, marketing, and other fields.

There are many statistical experiments where the results of each trial can be considered as either a success or a failure. For example, when flipping a coin, the two outcomes are heads or tails. When rolling a die, the two outcomes can be considered to be an even number appears on the face of the die or an odd number appears on the face of the die. When conducting a marketing study, a customer can be asked if they like or dislike a certain product. Note that the word “success” here does not necessarily imply a good outcome. For example, if a survey was conducted of adults and each adult was asked if they smoke, we can consider the answer “yes” to be a success and the answer “no” to be a failure. This means that the researcher can define success and failure in any way; however, the binomial distribution is applicable when there are only two outcomes in each trial of an experiment.

The requirements to identify a binomial experiment and apply the binomial distribution include:

- The experiment of interest is repeated for a fixed number of trials, and each trial is independent of other trials. For example, a market researcher might select a sample of 20 people to be surveyed where each respondent will reply with a “yes” or “no” answer. This experiment consists of 20 trials, and each person’s response to the survey question can be considered as independent of another person’s response.
- There are only two possible outcomes for each trial, which can be labeled as “success” or “failure.”
- The probability of success remains the same for each trial of the experiment. For example, from past data we know that 35% of people prefer vanilla as their favorite ice cream flavor. If a group of 15 individuals are surveyed to ask if vanilla is their favorite ice cream flavor, the probability of success for each trial will be 0.35.
- The random variable x will count the number of successes in the experiment. Notice that since x will count

the number of successes, this implies that x will be a discrete random variable. For example, if the researcher is counting the number of people in the group of 15 that respond to say vanilla is their favorite ice cream flavor, then x can take on values such as 3 or 7 or 12, but x could not equal 5.28 since x is counting the number of people.

When working with a binomial experiment, it is useful to identify two specific parameters in a binomial experiment:

1. The number of trials in the experiment. Label this as n.
2. The probability of success for each trial (which is a constant value). Label this as p.

We then count the number of successes of interest as the value of the discrete random variable. Label this as x.

EXAMPLE 3.24

Problem

A medical researcher is conducting a study related to a certain type of shoulder surgery. A sample of 20 patients who have recently undergone the surgery is selected, and the researcher wants to determine the probability that 18 of the 20 patients had a successful result from the surgery. From past data, the researcher knows that the probability of success for this type of surgery is 92%.

1. Does this experiment meet the requirements for a binomial experiment?
2. If so, identify the values of n, p, and x in the experiment.

Solution

1. This experiment does meet the requirements for a binomial experiment since the experiment will be repeated for 20 trials, and each response from a patient will be independent of other responses. Each reply from a patient will be one of two responses—the surgery was successful or the surgery was not successful. The probability of success remains the same for each trial at 92%. The random variable x can be used to count the number of patients who respond that the surgery was successful.
2. The number of trials is 20 since 20 patients are being surveyed, so $n = 20$.
   The probability of success for each surgery is 92%, so $p = 0.92$.
   The number of successes of interest is 18 since the researcher wants to determine the probability that 18 of the 20 patients had a successful result from the surgery, so $x = 18$.

When calculating the probability for x successes in a binomial experiment, a binomial probability formula can be used, but in many cases technology is used instead to streamline the calculations.

The **probability mass function (PMF)** for the binomial distribution describes the probability of getting exactly x successes in n independent Bernoulli trials, each with a probability p of success. The PMF is given by the formula:

$$P(X) = x = \binom{n}{k} p^x (1-p)^{n-x}$$

Where:
$P(X = x)$ is the probability that the random variable X takes on the value of exactly x successes
n is the number of trials in the experiment
p is the probability of success
x is the number of successes in the experiment

$\binom{n}{k}$ refers to the number of ways to choose x successes from $\binom{n}{k} = \frac{n!}{(n-x)!x!}$

Note: The notation $n!$ is read as n factorial and is a mathematical notation used to express the multiplication of $n\,(n-1)\,(n-2)\ldots(3)(2)(1)$. For example, $5! = (5)(4)(3)(2)(1) = 120$.

EXAMPLE 3.25

Problem

For the binomial experiment discussed in Example 3.24, calculate the probability that 18 out of the 20 patients will respond to indicate that the surgery was successful. Also, show a graph of the binomial distribution to show the probability distribution for all values of the random variable x.

Solution

In Example 3.24, the parameters of the binomial experiment are:
$n = 20$
$p = 0.92$
$x = 18$

Substituting these values into the binomial probability formula, the probability for 18 successes can be calculated as follows:

$$\begin{aligned} P(x) &= \frac{n!}{(n-x)!x!}p^x(1-p)^{n-x} \\ P(18) &= \frac{20!}{(20-18)!18!}0.92^{18}(1-0.92)^{20-18} \\ P(18) &= \frac{20!}{2!18!}0.92^{18}(0.18)^2 \\ P(18) &= 190(0.223)(0.032) \\ P(18) &= 0.271 \end{aligned}$$

Based on this result, the probability that 18 out of the 20 patients will respond to indicate that the surgery was successful is 0.271, or approximately 27%.

Figure 3.5 illustrates this binomial distribution, where the horizontal axis shows the values of the random variable x, and the vertical axis shows the binomial probability for each value of x. Note that values of x less than 14 are not shown on the graph since these corresponding probabilities are very close to zero.

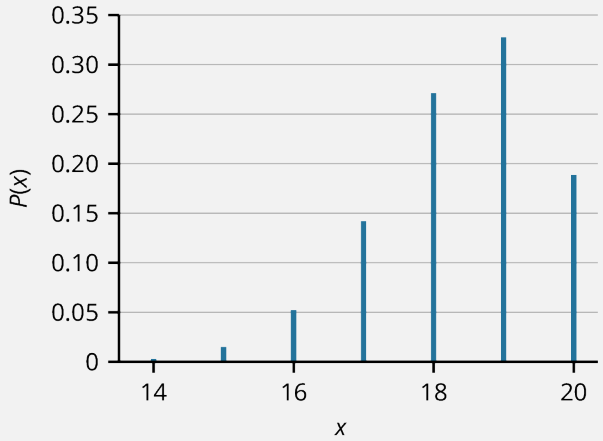


Figure 3.5 Graph of the Binomial Distribution for $n = 20$ and $p = 0.92$

Since these computations tend to be complicated and time-consuming, most data scientists will use technology (such as Python, R, Excel, or others) to calculate binomial probabilities.

Poisson Distribution

The goal of a binomial experiment is to calculate the probability of a certain number of successes in a specific number of trials. However, there are certain scenarios where a data scientist might be interested to know the probability of a certain number of occurrences for a random variable in a specific interval, such as an interval of time.

For example, a website developer might be interested in knowing the probability that a certain number of users visit a website per minute. Or a traffic engineer might be interested in calculating the probability of a certain number of accidents per month at a busy intersection.

The **Poisson distribution** is applied when counting the number of occurrences in a certain interval. The random variable then counts the number of occurrences in the interval.

A common application for the Poisson distribution is to model arrivals of customers for a queue, such as when there might be 6 customers per minute arriving at a checkout lane in the grocery store and the store manager wants to ensure that customers are serviced within a certain amount of time.

The Poisson distribution is a discrete probability distribution used in these types of situations where the interest is in a specific certain number of occurrences for a random variable in a certain interval such as time or area.

The Poisson distribution is used where the following conditions are met:

- The experiment is based on counting the number of occurrences in a specific interval where the interval could represent time, area, volume, etc.
- The number of occurrences in one specific interval is independent of the number of occurrences in a different interval.

Notice that when we count the number of occurrences that a random variable x occurs in a specific interval, this will represent a discrete random variable. For example, the count of the number of customers that arrive per hour to a queue for a bank teller might be 21 or 15, but the count could not be 13.32 since we are counting the number of customers and hence the random variable will be discrete.

To calculate the probability of x successes, the Poisson probability formula can be used, as follows:

$$P(x) = \frac{\mu^x e^{-\mu}}{x!}, \text{ where } x = 0, 1, 2, \ldots$$

Where:
μ is the average or mean number of occurrences per interval
e is the constant 2.71828...

EXAMPLE 3.26

Problem

From past data, a traffic engineer determines the mean number of vehicles entering a parking garage is 7 per 10-minute period. Calculate the probability that the number of vehicles entering the garage is 9 in a certain 10-minute period. Also, show a graph of the Poisson distribution to show the probability distribution for various values of the random variable x.

Solution

This example represents a Poisson distribution in that the random variable x is based on the number of vehicles entering a parking garage per time interval (in this example, the time interval of interest is 10 minutes). Since the average is 7 vehicles per 10-minute interval, we label the mean μ as 7. Since the

engineer want to know the probability that 9 vehicles enter the garage in the same time period, the value of the random variable x is 9.

Thus, in this example, the parameters of the Poisson distribution are:

$$\mu = 7$$
$$x = 9$$

Substituting these values into the Poisson probability formula, the probability for 9 vehicles entering the garage in a 10-minute interval can be calculated as follows:

$$P(x) = \frac{\mu^x e^{-\mu}}{x!}$$
$$P(9) = \frac{7^9 e^{-7}}{9!} = 0.101$$

Thus, there is about a 10% probability of 9 vehicles entering the garage in a 10-minute interval.

Figure 3.6 illustrates this Poisson distribution, where the horizontal axis shows the values of the random variable x and the vertical axis shows the Poisson probability for each value of x.

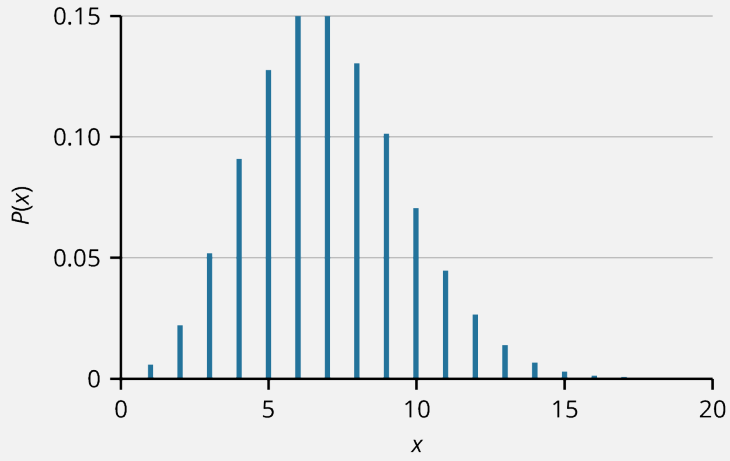


Figure 3.6 Poisson Distribution for $\mu = 7$

As with calculations involving the binomial distribution, data scientists will typically use technology to solve problems involving the Poisson distribution.

Normal Continuous Probability Distributions

Recall that a random variable is considered continuous if the value of the random variable can take on any of infinitely many values. We used the example about that if the random variable x represents the weight of a bag of apples, then x can take on any value such as $x = 2.45734$ pounds of apples.

Many probability distributions apply to continuous random variables. These distributions rely on determining the probability that the random variable falls within a distinct range of values, which can be calculated using a probability density function (PDF). The **probability density function (PDF)** calculates the corresponding area under the probability density curve to determine the probability that the random variable will fall within this specific range of values. For example, to determine the probability that a salary falls between $50,000 and $70,000, we can calculate the area under the probability density function between these two salaries.

Note that the total area under the probability density function will always equal 1. The probability that a continuous random variable takes on a specific value x is 0, so we will always calculate the probability for a random variable falling within some interval of values.

In this section, we will examine an important continuous probability distribution that relies on the probability density function, namely the *normal distribution*. Many variables, such as heights, weights, salaries, and blood pressure measurements, follow a normal distribution, making it especially important in statistical analysis. In addition, the normal distribution forms the basis for more advanced statistical analysis such as confidence intervals and hypothesis testing, which are discussed in Inferential Statistics and Regression Analysis.

The **normal distribution** is a continuous probability distribution that is symmetrical and bell-shaped. It is used when the frequency of data values decreases with data values above and below the mean. The normal distribution has applications in many fields including engineering, science, finance, medicine, marketing, and psychology.

The normal distribution has two parameters: the mean, μ, and the standard deviation, σ. The mean represents the center of the distribution, and the standard deviation measures the spread, or dispersion, of the distribution. The variable x represents the realization, or observed value, of the random variable X that follows a normal distribution.

The typical notation used to indicate that a random variable follows a normal distribution is as follows: $X \sim N(\mu, \sigma)$ (see Figure 3.7). For example, the notation $X \sim N(5.2, 3.7)$ indicates that the random variable follows a normal distribution with mean of 5.2 and standard deviation of 3.7.

A normal distribution with mean of 0 and standard deviation of 1 is called the **standard normal distribution** and can be notated as $X \sim N(0, 1)$. Any normal distribution can be standardized by converting its values to z-scores. Recall that a z-score tells you how many standard deviations from the mean there are for a given measurement.

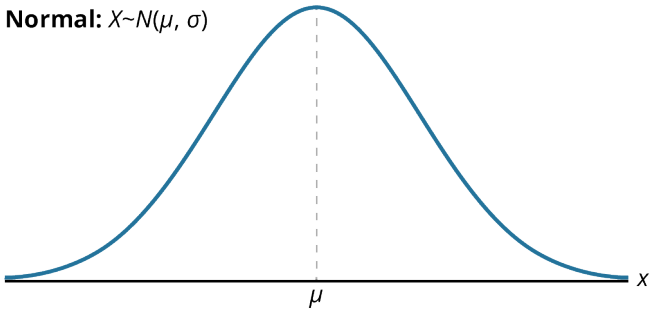


Figure 3.7 Graph of the Normal (Bell-Shaped) Distribution

The curve in Figure 3.7 is symmetric on either side of a vertical line drawn through the mean, μ. The mean is the same as the median, which is the same as the mode, because the graph is symmetric about μ. As the notation indicates, the normal distribution depends only on the mean and the standard deviation. Because the area under the curve must equal 1, a change in the standard deviation, σ, causes a change in the shape of the normal curve; the curve becomes fatter and wider or skinnier and taller depending on σ. A change in μ causes the graph to shift to the left or right. This means there are an infinite number of normal probability distributions.

To determine probabilities associated with the normal distribution, we find specific areas under the normal curve. There are several methods for finding this area under the normal curve, and we typically use some form of technology. Python, Excel, and R all provide built-in functions for calculating areas under the normal curve.

EXAMPLE 3.27

Problem

Suppose that at a software company, the mean employee salary is \$60,000 with a standard deviation of \$7,500. Assume salaries at this company follow a normal distribution. Use Python to calculate the

probability that a random employee earns more than $68,000.

Solution

A normal curve can be drawn to represent this scenario, in which the mean of $60,000 would be plotted on the horizontal axis, corresponding to the peak of the curve. Then, to find the probability that an employee earns more than $68,000, calculate the area under the normal curve to the right of the data value $68,000.

Figure 3.8 illustrates the area under the normal curve to the right of a salary of $68,000 as the shaded-in region.

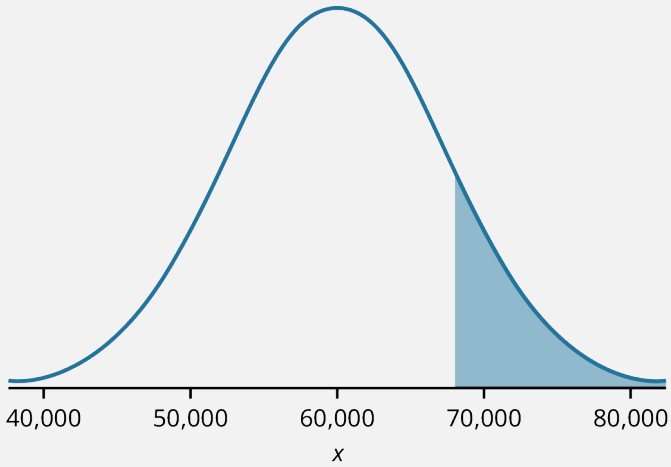


Figure 3.8 Bell-Shaped Distribution for Example 3.27. The shaded region under the normal curve corresponds to the probability that an employee earns more than $68,000.

To find the actual area under the curve, a Python command can be used to find the area under the normal probability density curve to the right of the data value of $68,000. See Using Python with Probability Distributions for the specific Python program and results. The resulting probability is calculated as 0.143.

Thus, there is a probability of about 14% that a random employee has a salary greater than $75,000.

The **empirical rule** is a method for determining approximate areas under the normal curve for measurements that fall within one, two, and three standard deviations from the mean for the normal (bell-shaped) distribution. (See Figure 3.9).

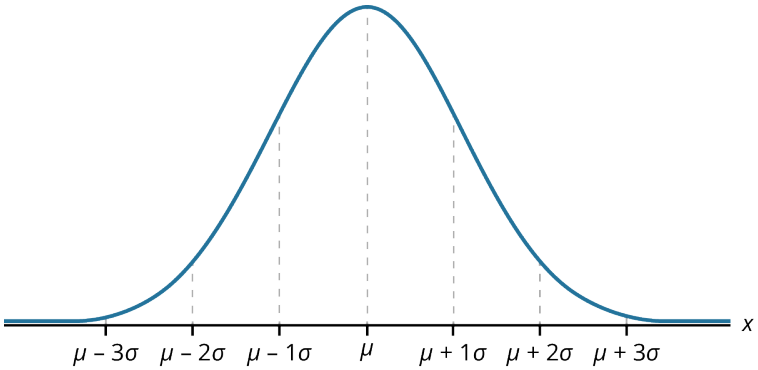


Figure 3.9 Normal Distribution Showing Mean and Increments of Standard Deviation

If x is a continuous random variable and has a normal distribution with mean μ and standard deviation σ, then the empirical rule states that:

- About 68% of the x-values lie between -1σ and $+1\sigma$ units from the mean μ (within one standard deviation of the mean).

- About 95% of the x-values lie between -2σ and $+2\sigma$ units from the mean μ (within two standard deviations of the mean).
- About 99.7% of the x-values lie between -3σ and $+3\sigma$ units from the mean μ (within three standard deviations of the mean). *Notice that almost all the x-values lie within three standard deviations of the mean.*
- The z-scores for $+1\sigma$ and -1σ are $+1$ and -1, respectively.
- The z-scores for $+2\sigma$ and -2σ are $+2$ and -2, respectively.
- The z-scores for $+3\sigma$ and -3σ are $+3$ and -3, respectively.

EXAMPLE 3.28

Problem

An automotive designer is interested in designing automotive seats to accommodate the heights for about 95% of customers. Assume the heights of adults follow a normal distribution with mean of 68 inches and standard deviation of 3 inches. For what range of heights should the designer model the car seats to accommodate 95% of drivers?

Solution

According to the empirical rule, the area under the normal curve within two standard deviations of the mean is 95%. Thus, the designer should design the seats to accommodate heights that are two standard deviations away from the mean. The lower bound of heights would be $68 - 2(3)$ inches, and the upper bound of heights would be $68 + 2(3)$ inches. Thus, the car seats should be designed to accommodate driver heights between 62 and 74 inches.

EXPLORING FURTHER

Statistical Applets to Explore Statistical Concepts

Applets are very useful tools to help visualize statistical concepts in action. Many applets can simulate statistical concepts such as probabilities for the normal distribution, use of the empirical rule, creating box plots, etc.

Visit the Utah State University applet website (https://openstax.org/r/usu) and experiment with various statistical tools.

Using Python with Probability Distributions

Python provides a number of built-in functions for calculating probabilities associated with both discrete and continuous probability distributions such as binomial distribution and the normal distribution. These functions are part of a library called scipy.stats (https://openstax.org/r/scipy).

Here are a few of these probability density functions available within Python:

- `binom()`—calculate probabilities associated with the binomial distribution
- `poisson()`—calculate probabilities associated with the Poisson distribution
- `expon()`—calculate probabilities associated with the exponential distribution
- `norm()`—calculate probabilities associated with the normal distribution

To import these probability density functions within Python, use the import command. For example, to import the `binom()` function use the following command:

```
from scipy.stats import binom
```

Using Python with the Binomial Distribution

The `binom()` function in Python allows calculations of binomial probabilities. The probability mass function for the binomial distribution within Python is referred to as `binom.pmf()`.

The syntax for using this function is `binom.pmf(x, n, p)`

Where:
n is the number of trials in the experiment
p is the probability of success
x is the number of successes in the experiment

Consider the previous Example 3.24 worked out using the Python `binom.pmf()` function. A medical researcher is conducting a study related to a certain type of shoulder surgery. A sample of 20 patients who have recently undergone the surgery is selected, and the researcher wants to determine the probability that 18 of the 20 patients had a successful result from the surgery. From past data, the researcher knows that the probability of success for this type of surgery is 92%. Round your answer to 3 decimal places.

In this example:
n is the number of trials in the experiment $= 20$
p is the probability of success $= 0.92$
x is the number of successes in the experiment $= 18$

The corresponding function in Python is written as:

```
binom.pmf (18, 20, 0.92)
```

The `round()` function is then used to round the probability result to 3 decimal places.

Here is the input and output of this Python program:

PYTHON CODE



```
# import the binom function from the scipy.stats library
from scipy.stats import binom

# define parameters x, n, and p:
x = 18
n = 20
p = 0.92

# use binom.pmf() function to calculate binomial probability
# use round() function to round answer to 3 decimal places
round (binom.pmf(x, n, p), 3)
```

The resulting output will look like this:

```
0.271
```

Using Python with the Normal Distribution

The `norm()` function in Python allows calculations of normal probabilities. The probability density function is sometimes called the cumulative density function, and so this is referred to as `norm.cdf()` within Python. The `norm.cdf()` function returns the area under the normal probability density function to the left of a specified measurement.

The syntax for using this function is
`norm.cdf(x, mean, standard_deviation)`

Where:
`x` is the measurement of interest
`mean` is the mean of the normal distribution
`standard_deviation` is the standard deviation of the normal distribution

Let's work out the previous Example 3.27 using the Python `norm.cdf()` function.

Suppose that at a software company, the mean employee salary is $60,000 with a standard deviation of $7,500. Use Python to calculate the probability that a random employee earns more than $68,000.

In this example:
x is the measurement of interest $= 68{,}000$
mean is the mean of the normal distribution $= 60{,}000$
standard deviation is the standard deviation of the normal distribution $= 7{,}500$

The corresponding function in Python is written as:
`norm.cdf(68000, 60000, 7500)`

The `round()` function is then used to round the probability result to 3 decimal places.

Notice that since this example asks to find the area to the right of a salary of $68,000, we can first find the area to the left using the `norm.cdf()` function and subtract this area from 1 to then calculate the desired area to the right.

Here is the input and output of the Python program:

PYTHON CODE

```
# import the norm function from the scipy.stats library
from scipy.stats import norm
# define parameters x, mean and standard_deviation:
x = 68000
mean = 60000
standard_deviation = 7500
# use norm.cdf() function to calculate normal probability - note this is
# the area to the left
# subtract this result from 1 to obtain area to the right of the x-value
# use round() function to round answer to 3 decimal places
round (1 - norm.cdf(x, mean, standard_deviation), 3)
```

The resulting output will look like this:

```
0.143
```

Key Terms

Bayes' Theorem a method used to calculate a conditional probability when additional information is obtained to refine a probability estimate

binomial distribution a probability distribution for a discrete random variable that is the number of successes in n independent trials. Each of these trials has two possible outcomes, success or failure, with the probability of success in each trial the same.

central tendency the statistical measure that represents the center of a dataset

coefficient of variation (CV) measures the variation of a dataset by calculating the standard deviation as a percentage of the mean

complement of an event the set of all outcomes in the sample space that are not included in the event

conditional probability the probability of an event given that another event has already occurred

continuous probability distribution probability distribution that deals with random variables that can take on any value within a given range or interval

continuous random variable a random variable where there is an infinite number of values that the variable can take on

dependent events events where the probability of occurrence of one event is affected by the occurrence of another event

descriptive statistics the organization, summarization, and graphical display of data

discrete probability distribution probability distribution associated with variables that take on a finite or countably infinite number of distinct values

discrete random variable a random variable where there is only a finite or countable infinite number of values that the variable can take on

empirical probability a probability that is calculated based on data that has been collected from an experiment

empirical rule a rule that provides the percentages of data values falling within one, two, and three standard deviations from the mean for a bell-shaped (normal) distribution

event a subset of the sample space

frequency a count of the number of times that an event or observation occurs in an experiment or study

frequency distribution a method of organizing and summarizing a dataset that provides the frequency with which each value in the dataset occurs

independent events events where the probability of occurrence of one event is not affected by the occurrence of another event

interquartile range (IQR) a number that indicates the spread of the middle half, or middle 50%, of the data; the difference between the third quartile (Q_3) and the first quartile (Q_1)

mean (also called arithmetic mean) a measure of center of a dataset, calculated by adding up the data values and dividing the sum by the number of data values; average

median the middle value in an ordered dataset

mode the most frequently occurring data value in a dataset

mutually exclusive events events that cannot occur at the same time

normal distribution a bell-shaped distribution curve that is used to model many measurements, including IQ scores, salaries, heights, weights, blood pressures, etc.

outcome the result of a single trial in a probability experiment

outliers data values that are significantly different from the other data values in a dataset

percentiles numbers that divide an ordered dataset into hundredths; used to describe the relative standing of a particular value within a dataset by indicating the percentage of data points that fall below it

Poisson distribution a probability distribution for discrete random variables used to calculate probabilities for a certain number of occurrences in a specific interval

population data data representing all the outcomes or measurements that are of interest

population mean the average for all measurements of interest corresponding to the entire group under

study

population size the number of measurements for the entire group under study

probability a numerical measure that assesses the likelihood of occurrence of an event

probability analysis provides the tools to model, understand, and quantify uncertainties, allowing data scientists to make informed decisions from data

probability density function (PDF) a function that is used to describe the probability distribution of a continuous random variable

probability distribution a mathematical function that assigns probabilities to various outcomes

probability mass function (PMF) a function that is used to define the probability distribution of a discrete random variable

quartiles numbers that divide an ordered dataset into quarters; the second quartile is the same as the median

random variable a variable where a single numerical value is assigned to a specific outcome from an experiment

range a measure of dispersion for a dataset calculated by subtracting the minimum from the maximum of the dataset

relative frequency probability a method of determining the likelihood of an event occurring based on the observed frequency of its occurrence in a given sample or population

sample data data representing outcomes or measurements collected from a subset or part of a population

sample mean the average for a subset of the measurements of interest

sample size the number of measurements for the subset taken from the overall population

sample space the set of all possible outcomes in a probability experiment

standard deviation a measure of the spread of a dataset, given in the same units as the data, that indicates how far a typical data value is from the mean

standard normal distribution a normal distribution with mean of 0 and standard deviation of 1

statistical analysis the science of collecting, organizing, and interpreting data to make decisions

theoretical probability a probability that is calculated based on an assessment of equally likely outcomes

trimmed mean a calculation for the average or mean of a dataset where some percentage of data values are removed from the lower and upper end of the dataset; typically used to mitigate the effects of outliers on the mean

variance the measure of the spread of data values in a dataset based on the squared deviations from the mean, which is the average of the squared deviations of the observations from the mean

z-score a measure of the position of a data value in the dataset, calculated by subtracting the mean from the data value and then dividing the difference by the standard deviation



Group Project

Analysis of Salary Data by Gender

There are many sources of salary data where median salaries for a certain year are published by gender. One source for data on earnings by demographics is the Bureau of Labor Statistics (https://openstax.org/r/bls).

As a group:

- Research salary data by gender for a certain year.
- Compile descriptive statistics where available for statistical measurements including the mean, median, quartiles, and standard deviation by gender.
- Create graphical presentations for this data using histograms, side-by-side box plots, and time series graphs.
- Discuss the following:
  a. From your analysis, does there appear to be a wage gap for men's vs. women's salaries?

b. If so, is the wage gap improving over time or worsening over time?

Quantitative Problems

1.

Dataset A Description	Data
Number of cars entering a parking garage in a 1-hour time period	Sample of 10 counts: 15, 18, 4, 15, 8, 11, 13, 7, 16, 24

a. For Dataset A, calculate the mean and a 10% trimmed mean (round answers to 1 decimal place). Use technology as appropriate.
b. Is there any benefit in using the trimmed mean versus the mean for this dataset?

2.

Dataset B Description	Data
Weights of packages on a delivery truck (weights in pounds)	Sample of 12 packages: 3.2, 9.1, 4.3, 1.9, 2.6, 4.7, 0.7, 1.3, 3.9, 6.4, 5.0, 25.4

a. For Dataset B, calculate the mean and median (round answers to 1 decimal place). Use technology as appropriate.
b. Is there any preference in using the mean versus the median for this dataset?
c. If so, would the mean or the median be preferred as a measure of central tendency for this dataset?

3.

Dataset C Description	Data
Amount of time for a pain reliever to provide relief from migraine headache (time in minutes)	Sample of 10 patients: 12.1, 13.8, 9.4, 15.9, 11.5, 14.6, 18.1, 12.7, 11.0, 14.2

a. For Dataset C, calculate the range, standard deviation, and variance (round answers to 1 decimal place). Use technology as appropriate and include appropriate units as part of your answers.
b. What does the standard deviation of this dataset communicate regarding the amount of time to provide relief from a migraine headache?
c. Would a medical researcher prefer to use the standard deviation or the variation for this dataset, and why?

4.

Dataset D Description	Data
Credit card balances for consumers (in dollars)	Sample of 9 consumers: $878.32, $671.54, $990.35, $801.85, $4303.76, $1285.03, $289.44, $78.61, $952.83

a. For Dataset D, calculate the quartiles of the dataset.
b. For this same dataset, calculate the interquartile range (IQR).
c. Determine if there are any potential outliers in this dataset.

5.

Dataset E Description	Data
Salaries for employees at a technology company (in dollars)	Sample of 10 employees: $72,500; $84,300; $58,900; $55,100; $82,000; $94,600; $82,200; $104,600; $63,800; $77,500

For Dataset E, determine the percentile for the employee with a salary of $82,000. Round your answer to

the nearest whole number percentile.

6. At a website development company, the average number of sick days taken by employees per year is 9 days with a standard deviation of 2.3 days.
   a. A certain employee has taken 2 sick days for the year. Calculate the z-score for this employee (round your answer to 2 decimal places).
   b. What does the z-score communicate about this employee's number of sick days as compared to other employees in the company?
   c. A different employee has taken 17 sick days for the year. Calculate the z-score for this employee (round your answer to 2 decimal places).
   d. What does the z-score communicate about this employee's number of sick days as compared to other employees in the company?

7. Two cards are selected at random from a standard 52-card deck. Find the probability that the second card is an Ace, given that the first card selected was an Ace (assume the first card is not replaced before selecting the second card). Round your answer to 4 decimal places.

8. From past data, it's known that the probability that an individual owns a dog is 0.42. Three random people are asked if they own a dog. (Round answers to 3 decimal places).
   a. Find the probability that all three people own a dog.
   b. Find the probability that none of the three people own a dog.
   c. Find the probability that at least one of the people owns a dog.

9. In a group of 100 people, 35 people wear glasses, 40 people wear contacts, 12 wear both glasses and contacts, and 25 wear neither glasses nor contacts.
   a. Are the events "person wears glasses" and "person wears contacts" mutually exclusive?
   b. Calculate the probability that a randomly chosen person wears glasses or contacts. Round your answer to 3 decimal places.

10. A medical researcher collects statistical data as follows:
   - 23% of senior citizens (65 or older) get the flu each year.
   - 32% of people under 65 get the flu each year.
   - In the population, there are 15% senior citizens.

   a. Are the events "being a senior" and "getting the flu" dependent or independent?
   b. Find the probability that a person selected at random is a senior and will get the flu.
   c. Find the probability that a person selected at random is under 65 years old and will get the flu.

11. A certain type of COVID-related virus affects 5% of the population. A screening test is available to detect the virus. Assume the probability of a true positive from the screening test is 85%, meaning that the probability that a person has a positive test result, given that they actually have the virus, is 0.85. Also assume the probability of a false positive from the screening test is 12%, which indicates the probability that a person has a positive test result, given that they do not have the virus, is 0.12. Calculate the probability that a person actually has the virus given that the screening test shows a positive result. Round your answer to 3 decimal places.

12. You are taking a multiple-choice test with 10 questions where each question has 5 possible answers: (a), (b), (c), (d), and (e). You did not study, so you will guess at each question. What is the probability of getting exactly 5 questions correct?

13. The average arrival rate of trucks at a truck loading station is 15 per day. Calculate the probability that 17 trucks arrive at the loading station on a particular day.

14. In a certain city, cell phone bills are known to follow a normal distribution with a mean of $92 and standard deviation of $14.

a. Find the probability that a randomly selected cell phone customer has a monthly bill less than $75.
b. Find the probability that a randomly selected cell phone customer has a monthly bill more than $85.



4

Inferential Statistics and Regression Analysis

Figure 4.1 Inferential statistics is used extensively in data science to draw conclusions about a larger population—and drive decision-making. (credit: modification of work "Compiling Health Statistics" by Ernie Branson/Wikimedia Commons, Public Domain)

Chapter Outline

4.1 Statistical Inference and Confidence Intervals
4.2 Hypothesis Testing
4.3 Correlation and Linear Regression Analysis
4.4 Analysis of Variance (ANOVA)

Introduction

Inferential statistics plays a key role in data science applications, as its techniques allow researchers to infer or generalize observations from samples to the larger population from which they were selected. If the researcher had access to a full set of population data, then these methods would not be needed. But in most real-world scenarios, population data cannot be obtained or is impractical to obtain, making inferential analysis essential. This chapter will explore the techniques of inferential statistics and their applications in data science.

Confidence intervals and hypothesis testing allow a data scientist to formulate conclusions regarding population parameters based on sample data.

One technique, **correlation analysis**, allows the determination of a statistical relationship between two numeric quantities, often referred to as variables. A **variable** is a characteristic or attribute that can be measured or observed. A **correlation** between two variables is said to exist where there is an association between them. Finance professionals often use correlation analysis to predict future trends and mitigate risk in a stock portfolio. For example, if two investments are strongly correlated, an investor might not want to have both investments in a certain portfolio since the two investments would tend to move in the same direction as market prices rose or fell. To diversify a portfolio, an investor might seek investments that are not strongly correlated with one another.

Regression analysis takes correlation analysis one step further by *modeling* the relationship between the two numeric quantities or variables when a correlation exists. In statistics, modeling refers specifically to the

process of creating a mathematical representation that describes the relationship between different variables in a dataset. The model is then used to understand, explain, and predict the behavior of the data.

This chapter focuses on **linear regression**, which is analysis of the relationship between one *dependent variable* and one *independent variable*, where the relationship can be modeled using a linear equation. The foundations of regression analysis have many applications in data science, including in machine learning models where a mathematical model is created to determine a relationship between input and output variables of a dataset. Several such applications of regression analysis in machine learning are further explored in Decision-Making Using Machine Learning Basics. In Time Series and Forecasting, we will use time series models to analyze and predict data points for data collected at different points in time.

4.1 Statistical Inference and Confidence Intervals

Learning Outcomes

By the end of this section, you should be able to:

- 4.1.1 Estimate parameters, create confidence intervals, and calculate sample size requirements.
- 4.1.2 Apply bootstrapping methods for parameter estimation.
- 4.1.3 Use Python to calculate confidence intervals and conduct hypothesis tests.

Data scientists interested in inferring the value of a population truth or parameter such as a population mean or a population proportion turn to inferential statistics. A data scientist is often interested in making generalizations about a population based on the characteristics derived from a sample; inferential statistics allows a data scientist to draw conclusions about a population based on sample data. In addition, inferential statistics is used by data scientists to assess model performance and compare different algorithms in machine learning application. Inferential statistics provides methods for generating predictive forecasting models, and this allows data scientists to generate predictions and trends to assist in effective and accurate decision-making. In this section, we explore the use of confidence intervals, which is used extensively in inferential statistical analysis.

We begin by introducing confidence intervals, which are used to estimate the range within which a population parameter is likely to fall. We discuss estimation of parameters for the mean both when the standard deviation is known and when it is not known. We discuss sample size determination, building on the sampling techniques presented in Collecting and Preparing Data. And we discuss bootstrapping, a method used to construct a confidence interval based on repeated sampling. Hypothesis Testing will move on to hypothesis testing, which is used to make inferences about unknown parameters.

Estimating Parameters with Confidence Intervals

A **point estimate** is a single value that is used to estimate a population parameter. For example, a **sample mean** is a *point estimate* that is representative of the true population mean in that the sample mean is used as an estimate for the unknown population mean. When researchers collect data from a sample to make inferences about a population, they calculate a point estimate based on the observed sample data. (See Survey Design and Implementation for coverage of sampling techniques.) The point estimate serves as the best guess or approximation for the parameter's actual value.

A **confidence interval** estimates the *range* within which a population parameter, such as a mean or a proportion, is likely to fall. The confidence interval provides a level of uncertainty associated with the estimate and is expressed as a range of values. A confidence interval will provide both a lower and an upper bound for the population parameter, where the point estimate is centered within the interval.

Table 4.1 describes the point estimates and corresponding population parameters for the mean and proportion.

	Population Parameter	Point Estimate
Mean	Population mean is denoted as μ.	Point estimate is the sample mean $\bar{x}$.
Proportion	Population proportion is denoted as p.	Point estimate is the sample proportion $\hat{p}$.

Table 4.1 Population Parameters and Point Estimates for Mean and Proportion

Let's say a researcher is interested in estimating the mean income for all residents of California. Since it is not feasible to collect data from every resident of California, the researcher selects a random sample of 1,000 residents and calculates the sample mean income for these 1,000 people. This average income estimate from the sample then provides an estimate for the population mean income of all California residents. The sample mean is chosen as the point estimate for the population mean in that the sample mean provides the most *unbiased estimate* for the population mean. In the same way, the **sample proportion** provides the most unbiased estimate for the population proportion. An **unbiased estimator** is a statistic that provides an accurate estimate for the corresponding population parameter without overestimating or underestimating the parameter.

In order to calculate a confidence interval, two quantities are needed:

- The point estimate
- The margin of error

As noted, the point estimate is a single number that is used to estimate the population parameter. The **margin of error** (usually denoted by *E*) provides an indication of the maximum error of the estimate. The margin of error can be viewed as the maximum distance around the point estimate where the population parameter exists based on a specified confidence.

Once the point estimate and margin of error are determined, the confidence interval is calculated as follows:

$$\begin{aligned} \text{Lower Bound of the Confidence Interval} &= \text{Point Estimate} - \text{Margin of Error} \\ \text{Upper Bound of the Confidence Interval} &= \text{Point Estimate} + \text{Margin of Error} \end{aligned}$$

The margin of error reflects the desired *confidence level* of the researcher. The **confidence level** is the probability that the interval estimate will contain the population parameter, given that the estimation process on the parameter is repeated over and over. Confidence levels typically range from 80% to 95% confidence.

In addition to the confidence level, both the variability of the sample and the sample size will affect the margin of error.

Once the confidence interval has been calculated, a concluding statement is made that reflects both the confidence level and the unknown population parameter. The concluding statement indicates that there is a certain level of confidence that the population parameter is contained within the bounds of the confidence interval.

As an example, consider a data scientist who is interested in forecasting the median income for all residents of California. Income data is collected from a sample of 1,000 residents, and the median income level is $68,500. Also assume the margin of error for a 95% confidence interval is $4,500 (more details on the calculation of margin of error are provided next). Now the data scientist can construct a 95% confidence interval to forecast income levels as follows:

$$\begin{aligned}\text{Lower Bound of the Confidence Interval} &= \text{Point Estimate} - \text{Margin of Error}\\ &= \$68500 - \$4500\\ \text{Upper Bound of the Confidence Interval} &= \text{Point Estimate} + \text{Margin of Error}\\ &= \$68500 + \$4500 = \$73000\end{aligned}$$

The data scientist can then state the following conclusion: There is 95% confidence that the forecast for median income for all residents of California is between $64,000 and $73,000.

EXAMPLE 4.1

Problem

A medical researcher is interested in estimating the population mean age for patients suffering from arthritis. A sample of 100 patients is taken, and the sample mean is determined to be 64 years old. Assume that the corresponding margin of error for a 95% confidence interval is calculated to be 4 years.

Calculate the confidence interval and provide a conclusion regarding the confidence interval.

Solution

The point estimate is the sample mean, which is 64, and the margin of error is given as 4 years.

The 95% confidence interval can be calculated as follows:

$$\begin{aligned}\text{Lower Bound of the Confidence Interval} &= \text{Point Estimate} - \text{Margin of Error}\\ &= 64 - 4 = 60\\ \text{Upper Bound of the Confidence Interval} &= \text{Point Estimate} + \text{Margin of Error}\\ &= 64 + 4 = 68\end{aligned}$$

Concluding statement:
The researcher is 95% confident that the mean population age for patients suffering from arthritis is contained in the interval from 60 to 68 years of age.

Sampling Distribution for the Mean

A researcher takes repeated samples of size 1,000 from the residents of New York to collect data on mean income of residents of New York.

For each sample of size 1,000, we can calculate a sample mean, $\bar{x}$. If the researcher were to take 50 such samples (each of sample size 1,000), a series of sample means can be calculated:

$$\bar{x}_1, \bar{x}_2, \bar{x}_3, \ldots, \bar{x}_{50}$$

A probability distribution based on all possible random samples of a certain size from a population—or **sampling (or sample) distribution**—can then be analyzed. For example, we can calculate the mean of these sample means, and we can calculate the standard deviation of these sample means.

There are two important properties of the sample distribution of these sample means:

1. The mean of the sample means (notated as $\mu_{\bar{x}}$) is equal to the population mean μ.
   a. Written mathematically: $\mu_{\bar{x}} = \mu$
2. The standard deviation of the sample means (notated as $\sigma_{\bar{x}}$) is equal to the population standard deviation σ divided by the square root of the sample size n.
   a. Written mathematically: $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$

The **central limit theorem** describes the relationship between the sample distribution of sample means and

the underlying population. This theorem is an important tool that allows data scientists and researchers to use sample data to generate inferences for population parameters.

Conditions of the central limit theorem:

- If random samples are taken from any population with mean μ and standard deviation, where the sample size is at least 30, then the distribution of the sample means approximates a normal (bell-shaped) distribution.
- If random samples are taken from a population that is normally distributed with mean μ and standard deviation σ, then the distribution of the sample means approximates a normal (bell-shaped) distribution for any sample size

EXAMPLE 4.2

Problem

An economist takes random samples of size 50 to estimate the mean salary of chemical engineers. Assume that the population mean salary is \$85,000 and the population standard deviation is \$9,000. It is unknown if the distribution of salaries follows a normal distribution.

Calculate the mean and standard deviation of the sampling distribution of the sample means and comment on the shape of the distribution for the sample means.

Solution

The mean of the sample means is equal to the population mean μ.

$$\mu_{\overline{x}} = \mu = \$85{,}000$$

The standard deviation of the sample means is equal to the population standard deviation σ divided by the square root of the sample size n.

$$\sigma_{\overline{x}} = \frac{\sigma}{\sqrt{n}} = \frac{\$9{,}000}{\sqrt{50}} = \$1272.8$$

Since the sample size of 50 is greater than 30, the distribution of the sample means approximates a normal (bell-shaped) distribution.

Confidence Interval for the Mean When the Population Standard Deviation Is Known

Although in many situations the population standard deviation is unknown, in some cases a reasonable estimate for the population standard deviation can be obtained from past studies or historical data.

Here is what is needed to calculate the confidence interval for the mean when the population standard deviation is known:

a. A random sample is selected from the population.
b. The sample size is at least 30, or the underlying population is known to follow a normal distribution.
c. The population standard deviation (σ) is known.

Once these conditions are met, the margin of error E is calculated according to the following formula:

$$E = z_c \frac{\sigma}{\sqrt{n}}$$

Where:

z_c is called the critical value of the normal distribution and is calculated as the z-score, which includes the area

corresponding to the confidence level between $-z_c$ and $+z_c$. For example, for a 95% confidence interval, the corresponding critical value is $z_c = 1.96$ since an area of 0.95 under the normal curve is contained between the z-scores of -1.96 and $+1.96$.

σ is the population standard deviation.

n is the sample size.

Note: $\frac{\sigma}{\sqrt{n}}$ is called the **standard error of the mean**.

Typical values of z_c for various confidence levels are shown in Table 4.2.

Confidence Level	Value of z_c
80% confidence	1.280
90% confidence	1.645
95% confidence	1.960
99% confidence	2.575

Table 4.2 Typical Values of z_c for Various Confidence Levels

Graphically, the critical values can be marked on the normal distribution curve, as shown in Figure 4.2. (See Discrete and Continuous Probability Distributions for a review of the normal distribution curve.) Figure 4.2 is an example for a 95% confidence interval where the area of 0.95 is centered as the area under the standard normal curve showing the values of $-z_c$ and $+z_c$. Since the standard normal curve is symmetric, and since the total area under the curve is known to be 1, the area in each of the two tails can be calculated to be 0.025. Thus, the **critical value** is that z-score which cuts off an area of 0.025 in the upper tail of the standard normal curve.

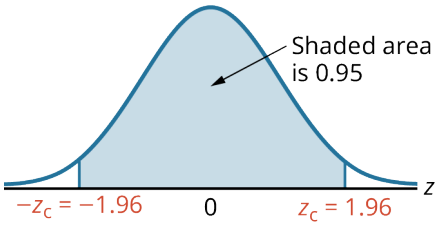


Figure 4.2 Critical Values of z_c Marked on Standard Normal Curve for 95% Confidence Interval

Once the margin of error has been calculated, the confidence interval can be calculated as follows:

$$\text{Sample Mean} \pm \text{Margin of Error}$$

Note that this is the general format for constructing any confidence interval, namely the margin of error is added and subtracted to the sample statistic to generate the upper and lower bounds of the confidence interval, respectively. A **sample statistic** describes some aspect of the sample, such as a sample mean or sample proportion.

General Form of a Confidence Interval: $\text{Sample Statistic} \pm \text{Margin of Error}$

EXAMPLE 4.3

Problem

A college professor collects data on the amount of time spent on homework assignments per week for a

sample of 50 students in a statistics course. From the sample of 50 students, the mean amount of time spent on homework assignments per week is 12.5 hours. The population standard deviation is known to be 6.3 hours.

The professor would like to forecast the amount of time spent on homework in future semesters. Create a forecasted confidence interval using both a 90% and 95% confidence interval and provide a conclusion regarding the confidence interval. Also compare the widths of the two confidence intervals. Which confidence interval is wider?

Solution

Calculation for 90% Confidence Interval

For a 90% confidence interval, the corresponding critical value is $z_c = 1.645$.

The margin of error is calculated as follows:

$$E = z_c \frac{\sigma}{\sqrt{n}} = 1.645 \cdot \frac{6.3}{\sqrt{50}} = 1.47$$

The 90% confidence interval is calculated as follows:

$$\begin{aligned} \text{Lower Bound of the Confidence Interval} &= \text{Point Estimate} - \text{Margin of Error} \\ &= 12.5 - 1.47 = 11.03 \\ \text{Upper Bound of the Confidence Interval} &= \text{Point Estimate} + \text{Margin of Error} \\ &= 12.5 + 1.47 = 13.97 \end{aligned}$$

Concluding statement:

The college professor can forecast with 90% confident that the mean amount of time spent on homework assignments in a future semester is the interval from 11.03 to 13.97 hours per week.

Calculation for 95% Confidence Interval

For a 95% confidence interval, the corresponding critical value is $z_c = 1.960$.

The margin of error is calculated as follows:

$$E = z_c \frac{\sigma}{\sqrt{n}} = 1.960 \cdot \frac{6.3}{\sqrt{50}} = 1.75$$

The 95% confidence interval is calculated as follows:

$$\begin{aligned} \text{Lower Bound of the Confidence Interval} &= \text{Point Estimate} - \text{Margin of Error} \\ &= 12.5 - 1.75 = 10.75 \\ \text{Upper Bound of the Confidence Interval} &= \text{Point Estimate} + \text{Margin of Error} \\ &= 12.5 + 1.75 = 14.25 \end{aligned}$$

Concluding statement:

The college professor can forecast with 95% confident that the mean amount of time spent on homework assignments in a future semester is the interval from 10.75 to 14.25 hours per week.

Comparison of 90% and 95% confidence intervals:

The 90% confidence interval extends from 11.03 to 13.97.
The 95% confidence interval extends from 10.75 to 14.25.

Notice that the 95% confidence interval is wider. If the confidence level is increased, with all other

parameters held constant, we should expect that the confidence interval will become wider. Another way to consider this: the wider the confidence interval, the more likely the interval is to contain the true population mean. This makes intuitive sense in that if you want to be more certain that the true value of the parameter is within an interval, then the interval needs to be wider to account for a larger range of potential values. As an analogy, consider a person trying to catch a fish. The wider the net used, the higher the probability of catching a fish.

You might notice that the sample size n is located in the denominator of the formula for margin of error. This indicates that as the sample size increases, the margin of error will decrease, which will then result in a narrower confidence interval. On the other hand, as the sample size decreases, the margin of error increases and the confidence interval becomes wider.

Confidence Interval for the Mean When the Population Standard Deviation Is Unknown

A confidence interval can still be determined when the population standard deviation is unknown by calculating a sample standard deviation based on the sample data. This is actually the more common application of confidence intervals.

Here are the conditions required to use this procedure:

a. A random sample is selected from the population.
b. The sample size is at least 30, or the underlying population is known to follow a normal distribution.
c. The population standard deviation (σ) is unknown; the sample standard deviation (s) can be calculated.

Once these requirements are met, the margin of error (E) is calculated according to the following formula:

$$E = t_c \frac{s}{\sqrt{n}}$$

t_c is called the *critical value of the t-distribution*. The **t-distribution** is a bell-shaped, symmetric distribution similar to the normal distribution, though the t-distribution has "thicker tails" as compared to the normal distribution. The comparison of the normal and t-distribution curves is shown in Figure 4.3.

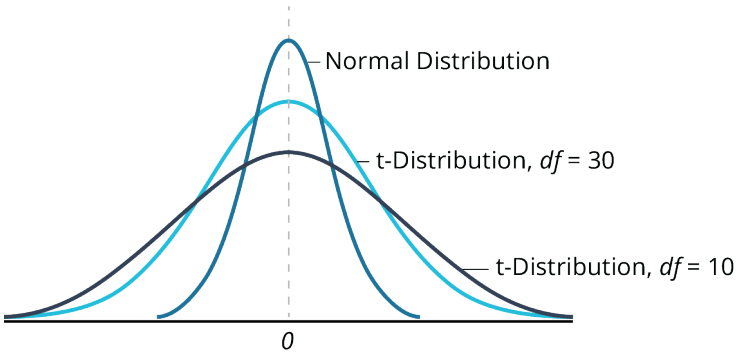


Figure 4.3 Comparison of t-Distribution and Normal Distribution Curves

The t-distribution is actually a family of curves, determined by a parameter called *degrees of freedom (df)*, where *df* is equal to $n - 1$. The critical value t_c is similar to a *z*-score and specifies the area under the t-distribution curve corresponding to the confidence level between $-t_c$ and $+t_c$. Values of t_c can be obtained using either a look-up table or technology.

For example, for a 95% confidence interval and sample size of 30, the corresponding critical value is $t_c = 2.045$ since an area of 0.95 under the t-distribution curve is contained between the t-scores of -2.045 and $+2.045$.

s is the sample standard deviation.

n is the sample size.

df is degrees of freedom, where $df = n - 1$.

Typical values of t_c for various confidence levels and degrees of freedom are shown in Table 4.3.

	Confidence Level		
Degrees of Freedom (*df*)	90% Confidence	95% Confidence	99% Confidence
1	6.314	12.706	63.657
2	2.920	4.203	9.925
3	2.353	3.182	5.841
4	2.132	2.776	4.604
5	2.015	2.571	4.032
10	1.812	2.228	3.169
15	1.753	2.131	2.947
20	1.725	2.086	2.845
25	1.708	2.060	2.787
30	1.697	2.042	2.750

Table 4.3 Typical Values of t_c for Various Confidence Levels and Degrees of Freedom

Note that Python can be used to calculate these critical values. Python provides a function called `t.ppf()` that generates the value of the t-distribution corresponding to a specified area under the t-distribution curve and specified degrees of freedom. This function is part of the scipy.stats library (https://openstax.org/r/stats).

The syntax for the function is:
`t.ppf(area to left, degrees of freedom)`

For example, for a 95% confidence interval, there is an area of 0.95 centered under the t-distribution curve, which leaves a remaining area of 0.05 for the two tails of the distribution, which implies an area of 0.025 in each tail. To find a critical value corresponding to the upper 0.025 area, note that the area to the left of this critical value will then be 0.975.

Here is an example of Python code to generate the critical value t_c for a 95% confidence interval and 15 degrees of freedom.

PYTHON CODE

```
# import function from scipy.stats library
from scipy.stats import t

# define parameters called area to left and degrees of freedom df
Area_to_left = 0.975
df = 15

# use t.ppf function to calculate critical values associated with t-
```

```
# use round function to round answer to 3 decimal places
round (t.ppf(Area_to_left, df), 3)
```

The resulting output will look like this:

```
2.131
```

Once the margin of error is calculated, the confidence interval is formed in the same way as the previous section, namely:

$$\begin{aligned}\text{Lower Bound of the Confidence Interval} &= \text{Point Estimate} - \text{Margin of Error}\\ \text{Upper Bound of the Confidence Interval} &= \text{Point Estimate} + \text{Margin of Error}\end{aligned}$$

EXAMPLE 4.4

Problem

A company's human resource administrator wants to estimate the average commuting distance for all 5,000 employees at the company. Since it is impractical to collect commuting distances from all 5,000 employees, the administrator decides to sample 16 employees and collects data on commuting distance from each employee in the sample. The sample data indicates a sample mean of 15.8 miles with a standard deviation of 3.2 miles. Calculate a 99% confidence interval and provide a conclusion regarding the confidence interval.

Solution

Since the sample size is 16 employees, the degrees of freedom is one less than the sample size, which is $df = 15$. For a 99% confidence interval, the corresponding critical value is $t_c = 2.947$.

The margin of error is calculated as follows:

$$E = t_c \frac{s}{\sqrt{n}} = 2.947 \cdot \frac{3.2}{\sqrt{16}} = 2.36$$

The 99% confidence interval is calculated as follows:

$$\begin{aligned}\text{Lower Bound of the Confidence Interval} &= \text{Point Estimate} - \text{Margin of Error}\\ &= 15.8 - 2.36 = 13.44\\ \text{Upper Bound of the Confidence Interval} &= \text{Point Estimate} + \text{Margin of Error}\\ &= 15.8 + 2.36 = 18.16\end{aligned}$$

Concluding statement:

The administrator can be 99% confident that the true population mean commuting distance for all employees is between 13.44 to 18.16 miles.

Confidence Interval for Proportions

We can also calculate a confidence interval for a population proportion based on the use of sample data. The basis for the confidence interval will be the application of a normal approximation to the binomial distribution. Recall from Discrete and Continuous Probability Distributions that a *binomial distribution* is a probability distribution for a discrete random variable where there are only two possible outcomes of an experiment. A **proportion** measures the number of *successes* in a sample. For example, if 10 survey respondents out of 50

indicate they are planning to travel internationally within the next 12 months, then the proportion is 10 out of 50, which is 0.2, or 20%. Note that the term *success* does not necessarily imply a positive outcome. For example, a researcher might be interested in the proportion of smokers among U.S. adults, and the number of smokers would be considered the number of successes.

Some terminology will be helpful:
p represents the population proportion, which is typically unknown.
$\hat{p}$ represents the sample proportion.
x represents the number of successes in the sample.
n represents the sample size.

Here are the requirements to use this procedure:

a. A random sample is selected from the population.
b. Verify that the normal approximation to the binomial distribution is appropriate by ensuring that both $n\hat{p}$ and $n(1-\hat{p})$ are both at least 5, where $\hat{p}$ represents the sample proportion.

The sample proportion $\hat{p}$ is calculated as the number of successes divided by the sample size:

$$\hat{p} = \frac{x}{n}$$

When these requirements are met, the margin of error (E) for a confidence interval for proportions is calculated according to the following formula:

$$E = z_c \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Where:

z_c is called the critical value of the normal distribution and is calculated as the z-score, which includes the area corresponding to the confidence level between $-z_c$ and $+z_c$.

$\hat{p}$ represents the sample proportion.

n is the sample size.

Once the margin of error is calculated, the confidence interval is formed in the same way as the previous section. In this case, the point estimate is the sample proportion $\hat{p}$:

$$\begin{aligned} \text{Lower Bound of the Confidence Interval} &= \text{Point Estimate} - \text{Margin of Error} \\ \text{Upper Bound of the Confidence Interval} &= \text{Point Estimate} + \text{Margin of Error} \end{aligned}$$

EXAMPLE 4.5

Problem

A medical researcher wants to know if there has been a statistically significant change in the proportion of smokers from five years ago, when the proportion of adult smokers in the United States was approximately 28%. The researcher selects a random sample of 1,500 adults, and of those, 360 respond that they are smokers. Calculate a 95% confidence interval for the true population proportion of adults who smoke. Also, determine if there has been a statistically significant change in the proportion of smokers as compared to five years ago.

Solution

For a 95% confidence interval, the corresponding critical value is $z_c = 1.960$.

Start off by calculating the sample proportion $\hat{p}$:

$$\hat{p} = \frac{x}{n} = \frac{360}{1{,}500} = 0.24$$

Verify that the normal approximation to the binomial distribution is appropriate by ensuring that both $n\hat{p}$ and $n(1-\hat{p})$ are both at least 5, where $\hat{p}$ represents the sample proportion.

For this example, $n\hat{p} = (1{,}500)(0.24) = 360$, and $n(1-\hat{p}) = (1{,}500)(1-0.24) = 1{,}140$. Both of these results are at least 5, which verifies that the normal approximation to the binomial distribution is appropriate.

The margin of error is then calculated as follows:

$$E = z_c\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 1.960\sqrt{\frac{0.24(1-0.24)}{1{,}500}} = 0.022$$

The 95% confidence interval is calculated as follows:

$$\begin{aligned}\text{Lower Bound of the Confidence Interval} &= \text{Point Estimate} - \text{Margin of Error}\\ &= 0.24 - 0.022 = 0.218\\ \text{Upper Bound of the Confidence Interval} &= \text{Point Estimate} + \text{Margin of Error}\\ &= 0.24 + 0.022 = 0.262\end{aligned}$$

Concluding statement:

The researcher can be 95% confident that the true population of adult smokers in the United States is between 0.218 and 0.262, which can also be written as 21.8% to 26.2%. Since this 95% confidence interval excludes the previous value of 28% from five years ago, the researcher can state that there has been a statistically significant decrease in the proportion of smokers as compared to five years ago.

Sample Size Determination

When collecting sample data in order to construct a confidence interval for a mean or proportion, how does the researcher determine the optimal sample size? Too small of a sample size may lead to a wide confidence interval that is not very useful. Too large of a sample size can result in wasted resources if a smaller sample size would be sufficient. Sampling is covered in some depth in Handling Large Datasets. Here, we review methods to determine minimum sample size requirements when constructing confidence intervals for means or proportions. Note that the desired margin of error plays a key role in this sample size determination.

Sample Size for Confidence Interval for the Mean

When determining a confidence interval for a mean, a researcher can use the margin of error formula for the mean; solving this formula algebraically for the sample size (n), the following minimum sample size formula can be used to determine the minimum sample size needed to achieve a certain margin of error:

Sample size formula for confidence interval for mean:

$$E = z_c\frac{\sigma}{\sqrt{n}} \rightarrow n = \left(\frac{z_c\sigma}{E}\right)^2$$

Where:
z_c is the critical value of the normal distribution.
σ is the population standard deviation.
E is the desired margin of error.

Note that for sample size calculations, sample size results are rounded up to the next higher whole number.

For example, if the formula previously shown results in a sample size of 59.2, then the sample size will be rounded to 60.

EXAMPLE 4.6

Problem

A benefits analyst is interested in a 90% confidence interval for the mean salary for chemical engineers. What sample size should the analyst use if a margin of error of $1,000 is desired? Assume a population standard deviation of $8,000.

Solution

For a 90% confidence interval, the corresponding critical value is $z_c = 1.645$. The population standard deviation is given as $8,000, and the margin of error is given as $1,000.

Using the sample size formula:

$$n = \left(\frac{z_c \sigma}{E}\right)^2 = \left(\frac{1.645 \cdot 8000}{1000}\right)^2 = 173.2$$

Round to the next higher whole number, so the desired sample size is 174.

The analyst should target a sample size of 174 chemical engineers for a salary-related survey in order to them calculate a 90% confidence interval for the mean salary of chemical engineers.

Sample Size for Confidence interval for a Proportion

In Example 4.5, how would the researcher know the optimal sample size to use to collect sample data on the proportion of adults who are smokers? The margin of error formula can be used to derive the sample size for a proportion as follows:

$$n = \hat{p}\,(1 - \hat{p})\left(\frac{z_c}{E}\right)^2$$

Where:
z_c is the critical value of the normal distribution.
$\hat{p}$ is the sample proportion.
E is the desired margin of error.

Notice in this formula, it's assumed that some prior estimate for the sample proportion $\hat{p}$ is available, perhaps from historical data or previous studies. If a prior estimate for the sample proportion is not available, then use the value of 0.5 for $\hat{p}$.

DETERMINING SAMPLE SIZE

When determining sample size needed for a confidence interval for a proportion:

If a prior estimate for the sample proportion is available, then that prior estimate should be utilized.

- If a prior estimate for the sample proportion is not available, then use $\hat{p} = 0.5$.

EXAMPLE 4.7

Problem

Political candidate Smith is planning a survey to determine a 95% confidence interval for the proportion of voters who plan to vote for Smith. How many people should be surveyed? Assume a margin of error is 3%.

a. Assume there is no prior estimate for the proportion of voters who plan to vote for candidate Smith.
b. Assume that from prior election results, approximately 42% of people previously voted for candidate Smith.

Solution

a. For a 95% confidence interval, the corresponding critical value is $z_c = 1.960$. The margin of error is specified as 0.03. Since a prior estimate for the sample proportion is unknown, use a value of 0.5 for $\hat{p}$. Using the sample size formula:

$$n = \hat{p}\,(1 - \hat{p})\left(\frac{z_c}{E}\right)^2 = 0.5(1 - 0.5)\left(\frac{1.960}{0.03}\right)^2 = 1{,}067.1$$

Round to the next higher whole number, so the desired sample size is 1,068 people to be surveyed.

b. Since a prior estimate for the sample proportion is available, use the value of 0.42 for $\hat{p}$. Using the sample size formula:

$$n = \hat{p}\,(1 - \hat{p})\left(\frac{z_c}{E}\right)^2 = 0.42\,(1 - 0.42)\left(\frac{1.960}{0.03}\right)^2 = 1039.8$$

Round to the next higher whole number, so the desired sample size is 1,040 people to be surveyed. Note that having a prior estimate of the sample proportion will result in a smaller sample size requirement. This is a general conclusion, so if a researcher has a prior estimate for the population proportion, this will result in a smaller sample size requirement as compared to the situation where no prior estimate is available.

Bootstrapping Methods

In the earlier discussion of confidence intervals, certain requirements needed to be met in order to use the estimation methods. For example, when calculating the confidence interval for a mean, this requirement was indicated:

- The sample size is at least 30, or the underlying population is known to follow a normal distribution.

When calculating the confidence interval for a proportion, this requirement was noted:

- Verify that the normal approximation to the binomial distribution is appropriate by ensuring that both $n\hat{p}$ and $n(1 - \hat{p})$ are at least 5.

What should a researcher do when these requirements are not met? Fortunately, there is another option called "bootstrapping" that can be used to find confidence intervals when the underlying distribution is unknown or if one of the conditions is not met. This bootstrapping method involves repeatedly *taking samples with replacement*.

Sampling with replacement means that when a sample is selected, the sample is replaced back in the dataset before selecting the next sample. For example, a casino dealer plans to take a sample of two cards from a standard 52-card deck. If the sampling is to be done with replacement, then after selecting the sample of the first card, this card is then returned to the deck before the second card is selected. This implies that the same data value can appear multiple times in a given sample since once a data value is selected for the sample, it is

replaced back in the dataset and is eligible to be selected again for the sample.

For example, a researcher who has collected 100 samples to use for inference may want to estimate a confidence interval for the population mean. The researcher can then resample, with replacement, from this limited set of 100 observations. They would repeatedly sample 100 and calculate the sample mean. These means would then represent the distribution of means (remember the discussion of sampling distribution). Typically, for *bootstrapping*, repeated samples are taken hundreds or thousands of times and then the sample mean is calculated for each sample.

The term "bootstrapping" comes from the old saying "pull yourself up by your bootstraps" to imply a task that was accomplished without any outside help. In the statistical sense, **bootstrapping** refers to the ability to estimate parameters based solely on one sample of data from the population without any other assumptions.

The bootstrapping method is considered a **nonparametric method** since the technique makes no assumptions about the probability distribution from which the data are sampled. (**Parametric methods**, by contrast, assume a specific form for the underlying distribution and require estimating parameters.)

Since bootstrapping requires a large number of repeated samples, software (such as Excel, Python, or R) is often used to automate the repetitive sampling procedures and construct the confidence intervals.

The bootstrapping procedure for confidence interval estimation for a mean or proportion is as follows:

1. Start out with a random sample of size n. Collect many random bootstrap samples of size n—for example, hundreds or thousands of such samples. Keep in mind that the sampling is done with replacement.
2. For each sample, calculate the sample statistic, which is the sample mean $\bar{x}$ or the sample proportion $\hat{p}$.
3. Rank order the sample statistics from smallest to largest.
4. For a 95% confidence interval, find the percentiles $P_{2.5}$ and $P_{97.5}$ in the ranked data. These values establish the 95% confidence interval.
5. For a 90% confidence interval, find the percentiles P_5 and P_{95}. These values establish the 90% confidence interval.

EXAMPLE 4.8

Problem

A college administrator is developing new marketing materials to increase enrollment at the college, and the administrator is interested in a 90% confidence interval for the mean age of students attending the college.

The administrator believes the underlying distribution of ages is skewed (does not follow a normal distribution), so a bootstrapping method will be used to construct the confidence interval. The administrator selects a random sample of 20 students and records their ages as shown in Table 4.4. Use the bootstrapping method to construct the 90% confidence interval by taking repeated samples of size 10.

Student ID	Student Age
001	22
002	24
003	21
004	34

Table 4.4 Ages of 20 Randomly

Student ID	Student Age
005	29
006	23
007	21
008	20
009	19
010	21
011	25
012	28
013	22
014	37
015	24
016	31
017	23
018	19
019	26
020	20

Table 4.4 Ages of 20 Randomly Selected Students from One College

Solution

For the bootstrapping process, we will form samples of size 10 for convenience. Note that the sampling is with replacement, so once an age is selected, the age is returned back to the dataset and then that particular age might be selected again as part of the sample.

Imagine forming hundreds or thousands of such samples, each of sample size 10. For each one of these samples, calculate the sample mean (see column labeled “Sample Means”).

In the example shown in Figure 4.4, only 20 samples are taken for space considerations since showing the results of thousands of samples is not practical for this text. However, typically a bootstrapping process involves many such samples—on the order of thousands of samples—made possible by using software (such as Excel, Python, or R).

Sample Number	Random Samples of Size 10										Sample Means	Sorted Sample Means
1	37	19	34	31	34	26	26	21	20	21	26.9	21.9
2	37	19	24	37	19	31	29	34	29	28	28.7	22.2
3	31	19	20	37	20	26	25	24	37	23	26.2	22.7
4	22	24	20	22	20	24	37	19	24	26	23.8	22.9
5	23	24	19	34	20	31	22	23	28	22	24.6	23.8
6	34	19	31	37	21	21	25	28	19	21	25.6	24.4
7	25	24	21	29	21	23	21	34	24	37	25.9	24.6
8	22	26	37	20	22	21	22	29	37	25	26.1	24.6
9	23	22	31	22	24	37	29	21	22	20	25.1	25.1
10	23	31	19	34	37	22	25	20	37	37	28.5	25.5
11	29	31	31	22	37	23	23	28	20	21	26.5	25.6
12	31	22	19	22	28	22	21	23	37	37	26.2	25.9
13	21	20	24	28	21	20	22	22	21	23	22.2	25.9
14	22	37	21	31	19	26	19	23	23	34	25.5	26.1
15	29	31	22	25	26	19	22	26	25	21	24.6	26.2
16	26	20	23	19	23	26	19	29	19	23	22.7	26.2
17	24	31	19	21	19	22	19	19	21	24	21.9	26.5
18	21	20	20	34	34	21	25	23	22	24	24.4	26.9
19	20	26	22	24	21	24	21	26	26	19	22.9	28.5
20	21	31	19	21	34	24	25	28	28	28	25.9	28.7

Figure 4.4 Bootstrap Samples and Corresponding Sample Means

Next, sort the sample means from smallest to largest (see column labeled “Sorted Sample Means”).

The last step is to calculate the 90% confidence interval based on these sorted sample means. For a 90% confidence interval, find the percentiles P_5 and P_{95}. These values establish the 90% confidence interval.

To find the 5th percentile (P_5), multiply the percentile (5%) times the number of data values, which is 20. The result is 1, so to find the 5th percentile, add the first and second data values together and divide by 2. For this example, add 21.9 to 22.2 and divide by 2. The result is 22.05.

To find the 95th percentile (P_{95}), multiply the percentile (95%) times the number of data values, which is 20. The result is 19, so to find the 95th percentile, add the 19th and 20th data values together and divide by 2. For this example, add 28.5 to 28.6 and divide by 2. The result is 28.6.

Based on the bootstrapping method, the 90% confidence interval is (22.05, 28.6).

Python provides a function called `bootstrap()` as part of the `scipy.stats` library to automate this bootstrapping process.

Within the `bootstrap()` function, the user can specify the number of resamples to be performed as part of the bootstrapping process. In this example, we will use 50,000 samples by way of the *n_resample* parameter.

PYTHON CODE



```
from scipy.stats import bootstrap
import numpy as np

#define random sample of ages
ages = [22, 23, 25, 31, 24, 21, 28, 23, 21, 20, 22, 19, 34, 19, 37, 26, 29, 21,
24, 20]

#convert ages to sequence
ages = (ages,)
```

```
#use bootstrap function for confidence interval for the mean
conf_interval = bootstrap(ages, np.mean, confidence_level=0.95,
                          random_state=1, n_resamples = 50000,
method='percentile')

#print the confidence interval
print(conf_interval.confidence_interval)
```

The resulting output will look like this:

```
ConfidenceInterval(low=22.45, high=26.7)
```

Using 50,000 bootstrapped samples, a 95% confidence interval is generated as (22.45, 26.7).

EXPLORING FURTHER

Simulating Confidence Intervals

A number of websites and resources, such as the online textbook Online Statistics Education: A Multimedia Course of Study (https://openstax.org/r/estimation) (http://onlinestatbook.com/) (project leader: David M. Lane, Rice University), allow the user to simulate confidence intervals for means or proportions using simulated sample data. It is useful to observe how many intervals contain an assumed value for the population mean or population proportion. The Stapplet (https://openstax.org/r/stapplet) website provides a similar tool.

Using Python to Calculate Confidence Intervals

When calculating confidence intervals, data scientists typically make use of technology to help streamline and automate the analysis. Python provides built-in functions for confidence interval calculations, and several examples are shown in Table 4.5. Students are encouraged to try examples themselves and experiment to use Python to assist with these calculations.

Table 4.5 provides a summary of various functions available within the `SciPy library` (https://openstax.org/r/spy) for confidence interval calculations:

Usage	Python Function Name	Syntax
Calculate confidence interval for the mean when population standard deviation is known, given sample mean, population standard deviation, and sample size (uses normal distribution).	`norm.interval()`	`norm.interval(conf_level, sample_mean,standard_dev/ sqrt (n))`
Calculate confidence interval for the mean when population standard deviation is unknown, given sample mean, sample standard deviation, and sample size (uses t-distribution).	`t.interval()`	`t.interval(conf_level, degrees_freedom, sample_mean,standard_dev/ sqrt (n))`
Calculate confidence interval for a proportion (uses normal distribution).	`proportion_confint()`	`proportion_confint(success, sample_size, 1 - confidence_level)`

Table 4.5 Python Functions for Confidence Intervals

EXAMPLE 4.9

Problem

Repeat Example 4.3 to calculate a 90% confidence interval, but use Python functions to calculate the confidence interval.

Recall from Example 4.3, the sample mean is 12.5 hours with a population standard deviation of 6.3 hours. The sample size is 50 students, and we are interested in calculating a 90% confidence interval.

Solution

Use the Python function `norm.interval()` as shown:

PYTHON CODE

```
import scipy.stats as stats
import numpy as np
import math

# Enter sample mean, population standard deviation and sample size
sample_mean = 12.5
population_standard_deviation = 6.3
sample_size = 50
```

```
# Confidence level
confidence_level = 0.90

# standard error
standard_error = population_standard_deviation / math.sqrt(sample_size)

# Calculate confidence interval using norm.interval function
stats.norm.interval(confidence_level, sample_mean, standard_error)
```

The resulting output will look like this:

```
(11.03451018636739, 13.965489813632608)
```

EXAMPLE 4.10

Problem

Repeat Example 4.4 to calculate a 99% confidence interval, but use Python functions to calculate the confidence interval.

Recall from Example 4.4, the sample mean is 15.8 miles with a standard deviation of 3.2 miles. The sample size is 26 employees, and we are interested in calculating a 99% confidence interval.

Solution

Use the Python function `t.interval()` as shown:

PYTHON CODE



```
# Enter sample mean, sample standard deviation, and sample size
sample_mean = 15.8
sample_standard_deviation = 3.2
sample_size = 26

# Degrees of freedom (sample size - 1)
degrees_of_freedom = sample_size - 1

# Confidence level
confidence_level = 0.99

# standard error
standard_error = sample_standard_deviation / math.sqrt(sample_size)
```

```
# Calculate confidence interval using t.interval function
t.interval(confidence_level, degrees_of_freedom, sample_mean, standard_error)
```

The resulting output will look like this:

```
(14.050684356083625, 17.549315643916376)
```

4.2 Hypothesis Testing

Learning Outcomes

By the end of this section, you should be able to:

- 4.2.1 Apply hypothesis testing methods to test statistical claims involving one sample.
- 4.2.2 Use Python to assist with hypothesis testing calculations.
- 4.2.3 Conduct hypothesis tests to compare two means, two proportions, and matched pairs data.

Constructing confidence intervals is one method used to estimate population parameters. Another important statistical method to test claims regarding population parameters is called *hypothesis testing*.

Hypothesis testing is an important tool for data scientists in that it is used to draw conclusions using sample data, and it is also used to quantify uncertainty associated with these conclusions.

As an example to illustrate the use of hypothesis testing in a data science applications, consider an online retailer seeking to increase revenue though a text messaging ad campaign. The retailer is interested in examining the hypothesis that a text messaging ad campaign leads to a corresponding increase in revenue. To test out the hypothesis, the retailer sends out targeted text messages to certain customers but does not send the text messages to another group of customers. The retailer then collects data for purchases made by customers who received the text message ads versus customers who did not receive the ads. Using hypothesis testing methods, the online retailer can come to a conclusion regarding the marketing campaign to transmit ads via text messaging. **Hypothesis testing** involves setting up a null hypothesis and alternative hypotheses based on a claim to be tested and then collecting sample data. A **null hypothesis** represents a statement of no effect or no change in the population; the null hypothesis provides a statement that the value of a population parameter is equal to a value. An **alternative hypothesis** is a complementary statement to the null hypothesis and makes a statement that the population parameter has a value that differs from the null hypothesis.

Hypothesis testing allows a data scientist to assess the likelihood of observing results under the assumption of the null hypothesis and make judgements about the strength of the evidence against the null hypothesis. It provides the foundation for various data science investigations and plays a role at different stages of the analyses such as data collection and validation, modeling related tasks, and determination of statistical significance.

For example, a local restaurant might claim that the average delivery time for food orders is 30 minutes or less. To test the claim, food orders can be placed and corresponding delivery times recorded. If the sample data appears to be inconsistent with the null hypothesis, then the decision is to reject the null hypothesis.

Testing Claims Based on One Sample

For our discussion, we will examine hypothesis testing methods for claims involving means or proportions. (Hypothesis testing can also be performed for standard deviation or variance, though that is beyond the scope of this text.)

The steps for hypothesis testing are as follows:

1. Set up a null and alternative hypothesis based on the claim. Identify whether the null or the alternative hypothesis represents the claim.
2. Collect relevant sample data to investigate the claim.
3. Determine the correct distribution to be used to analyze the hypothesis test (e.g., normal distribution or t-distribution).
4. Analyze the sample data to determine if the sample data is consistent with the null hypothesis. This will involve statistical calculations involving what is called a *test statistic* and a *p*-value (discussed in the next section).
5. If the sample data is inconsistent with the null hypothesis, then the decision is to "reject the null hypothesis." If the sample data is consistent with the null hypothesis, then the decision is to "fail to reject the null hypothesis."

The first step in the hypothesis testing process is to write what is a called a "null" hypothesis and an "alternative" hypothesis. These hypotheses will be statements regarding an unknown population parameter, such as a population mean or a population proportion. These two hypotheses are based on the claim under study and will be written as complementary statements to each other. When one of the hypotheses is true, the other hypothesis must be false (since they are complements of one another). Note that either the null or alternative hypothesis can represent the claim.

The null hypothesis is labeled as H_0 and is a statement of equality—for example, the null hypothesis might state that the mean time to complete an exam is 42 minutes. For this example, the null hypothesis would be written as:

$$H_0 : \mu = 42 \text{ minutes}$$

Where the notation μ refers to the population mean time to complete an exam.
The null hypothesis will always contain one of the following three symbols: $=, \leq, \text{or } \geq$.

The alternative hypothesis is labeled as H_a and will be the complement of the null hypothesis. For example, if the null hypothesis contains an equals symbol, then the alternative hypothesis will contain a not equals symbol.

If the null hypothesis contains a $\geq$ symbol, then the alternative hypothesis will contain a $<$ symbol.
If the null hypothesis contains a $\leq$ symbol, then the alternative hypothesis will contain a $>$ symbol.

Note that the alternative hypothesis will always contain one of the following three symbols: $\neq, <, \text{or } >$.

To write the null and alternative hypotheses, start off by translating the claim into a mathematical statement involving the unknown population parameter. Depending on the wording, the claim can be placed in either the null or the alternative hypothesis. Then write the complement of this statement as the other hypothesis (see Table 4.6).

When translating claims about a population mean, use the symbol μ as part of the null and alternative hypotheses. When translating claims about a population proportion, use the symbol p as part of the null and alternative hypotheses.

Table 4.6 provides the three possible setups for the null and alternative hypotheses when conducting a one-sample hypothesis test for a population mean. Notice that for each setup, the null and alternative hypotheses are complements of one another. Note that the value of "k" will be replaced by some numerical constant taken from the stated claim.

Setup A	Setup B	Setup C
$Ho : \mu = k$	$Ho : \mu \leq k$	$Ho : \mu \geq k$
$Ha : \mu \neq k$	$Ha : \mu > k$	$Ha : \mu < k$

Table 4.6 Possible Setups for the Null and Alternative Hypotheses for a Population Mean (One Sample)

Table 4.7 provides the three possible setups for the null and alternative hypotheses when conducting a one-sample hypothesis test for a population proportion. Note that the value of "k" will be replaced by some numerical constant taken from the stated claim.

Setup D	Setup E	Setup F
$Ho : p = k$	$Ho : p \leq k$	$Ho : p \geq k$
$Ha : p \neq k$	$Ha : p > k$	$Ha : p < k$

Table 4.7 Possible Setups for the Null and Alternative Hypotheses for a Population Proportion (One Sample)

Follow these steps to determine which of these setups to use for a given hypothesis test:

1. Determine if the hypothesis test involves a claim for a population mean or population proportion. If the hypothesis test involves a population mean, use either Setup A, B, or C.
2. If the hypothesis test involves a population proportion, use either Setup D, E, or F.
3. Translate a phrase from the claim into a mathematical symbol and then match this mathematical symbol with one of the setups in Table 4.6 and Table 4.7.

For example, if a claim for a hypothesis test for a mean references the phrase "at least," note that the phrase "at least" corresponds to a *greater than or equals symbol*. A greater than or equals symbol appears in Setup C in Table 4.6, so that would be the correct setup to use.

If a claim for a hypothesis test for a proportion references the phrase "different than," note that the phrase "different than" corresponds to a *not equals* symbol. A not equals symbol appears in Setup D in Table 4.6, so that would be the correct setup to use.

EXAMPLE 4.11

Problem

Write the null and alternative hypotheses for the following claim:
An auto repair facility claims the average time for an oil change is less than 20 minutes.

Solution

Note that the claim involves the phrase "less than," which translates to a less than symbol. A less than symbol must be used in the alternative hypothesis. The complement of less than is a greater than or equals symbol, which will be used in the null hypothesis. The claim refers to the population mean time for an oil change, so the symbol μ will be used. Notice that this setup will correspond to Setup C from Table 4.6. Thus the setup of the null and alternative hypothesis will be as follows:

$$H_0 : \mu \geq 20$$
$$H_a : \mu < 20$$

EXAMPLE 4.12

Problem

Write the null and alternative hypotheses for the following claim:

A medical researcher claims that the proportion of adults in the United States who are smokers is at most 25%.

Solution

Note that the claim involves the phrase "at most," which translates to a less than or equals symbol. A less than or equals symbol must be used in the null hypothesis. The complement of less than or equals is a greater than symbol, which will be used in the alternative hypothesis. The claim refers to the population proportion of smokers, so the symbol p will be used. Notice that this setup will correspond to Setup E from Table 4.7. Thus, the setup of the null and alternative hypotheses will be as follows:

$$H_0 : p \leq 0.25$$
$$H_a : p > 0.25$$

The basis of the hypothesis test procedure is to assume that the null hypothesis is true to begin with and then examine the sample data to determine if the sample data is consistent with the null hypothesis. Based on this analysis, you will decide to either reject the null hypothesis or fail to reject the null hypothesis.

It is important to note that these are the only two possible decisions for any hypothesis test, namely:

- Reject the null hypothesis, or
- Fail to reject the null hypothesis.

Keep in mind that this decision will be based on statistical analysis of sample data, so there is a small chance of coming to a wrong decision. The only way to be absolutely certain of your decision is to test the entire population, which is typically not feasible. Since the hypothesis test decision is based on a sample, there are two possible errors that can be made by the researcher:

- The researcher rejects the null hypothesis when in fact the null hypothesis is actually true, or a **Type I error**.
- A researcher fails to reject the null hypothesis when the null hypothesis is actually false, or a **Type II error**.

In hypothesis testing, the maximum allowed probability of making a Type I error is called the **level of significance**, denoted by the Greek letter alpha, α. From a practical standpoint, the level of significance is the probability value used to determine when the sample data indicates significant evidence against the null hypothesis. This level of significance is typically set to a small value, which indicates that the researcher wants the probability of rejecting a true null hypothesis to be small. Typical values of the level of significance used in hypothesis testing are as follows: $\alpha = 0.01$, $\alpha = 0.05$, or $\alpha = 0.10$.

Once the null and alternative hypotheses are written and a level of significance is selected, the next step is for the researcher to collect relevant sample data from the population under study. It's important that a representative random sample is selected, as discussed in Handling Large Datasets. After the sample data is collected, two calculations are performed: *test statistic* and *p*-value.

The **test statistic** is a numerical value used to assess the strength of evidence against a null hypothesis and is calculated from sample data that is used in hypothesis testing. As noted in Estimating Parameters with Confidence Intervals, a *sample statistic* is a numerical value calculated from a sample of observations drawn from a larger population such as a sample mean or sample proportion. The sample statistic is converted to a **standardized test statistic** such as a *z*-score or t-score based on the assumption that the null hypothesis is

true.

Once the test statistic has been determined, a probability, or *p*-value, is created. A ***p*-value** is the probability of obtaining a sample statistic with a value as extreme as (or more extreme than) the value determined by the sample data under the assumption that the null hypothesis is true. To calculate a *p*-value, we will find the area under the relevant probability distribution, such as finding the area under the normal distribution curve or the area under the t-distribution curve. *Since the p-value is a probability, any p-value must always be a numerical value between 0 and 1 inclusive.*

When calculating a *p*-value as the area under the probability distribution curve, the corresponding area will be determined using the location under the curve, which favors the rejection of the null hypothesis as follows:

1. If the alternative hypothesis contains a "less than" symbol, the hypothesis test is called a "left tailed" test and the *p*-value will be calculated as the area to the *left* of the test statistic.
2. If the alternative hypothesis contains a "greater than" symbol, the hypothesis test is called a "right tailed" test and the *p*-value will be calculated as the area to the *right* of the test statistic.
3. If the alternative hypothesis contained a "not equals" symbol, the hypothesis test is called a "two tailed" test and the *p*-value will be calculated as the sum of the area to the *left* of the negative test statistic and the area to the *right* of the positive test statistic.

The smaller the p-value, the more the sample data deviates from the null hypothesis, and this is more evidence to indicate that the null hypothesis should be rejected.

The final step in the hypothesis testing procedure is to come to a final decision regarding the null hypothesis. This is accomplished by comparing the *p*-value with the level of significance and applying the following rule:

- If the *p*-value $\leq$ level of significance, then the decision is to "reject the null hypothesis."
- If the *p*-value $>$ level of significance, then the decision is to "fail to reject the null hypothesis."

Recall that these are the only two possible decisions that a researcher can deduce when conducting a hypothesis test.

Often, we would like to translate these decisions into a conclusion that is easier to interpret for someone without a statistical background. For example, a decision to "fail to reject the null hypothesis" might be difficult to interpret for those not familiar with the terminology of hypothesis testing. These decisions can be translated into concluding statements such as those shown in Table 4.8.

	Select the decision from the hypothesis test:	Is the claim in the null or the alternative hypothesis?	Then, use this concluding statement:
Row 1	Reject the null hypothesis	Claim is the null hypothesis	*There is enough evidence to reject the claim.*
Row 2	Reject the null hypothesis	Claim is the alternative hypothesis	*There is enough evidence to support the claim.*
Row 3	Fail to reject the null hypothesis	Claim is the null hypothesis	*There is not enough evidence to reject the claim.*
Row 4	Fail to reject the null hypothesis	Claim is the alternative hypothesis	*There is not enough evidence to support the claim.*

Table 4.8 How to Establish a Conclusion from a Hypothesis Test

Testing Claims for the Mean When the Population Standard Deviation Is Known

Recall that in the discussion for confidence intervals we examined the confidence interval for the population mean when the population standard deviation is known (normal distribution is used) or when the population standard deviation is unknown (t-distribution is used). We follow the same procedure when conducting hypothesis tests.

In this section, we discuss hypothesis testing for the mean when the population standard deviation is known. Although the population standard deviation is typically unknown, in some cases a reasonable estimate for the population standard deviation can be obtained from past studies or historical data.

Here are the requirements to use this procedure:

a. A random sample is selected from the population.
b. The sample size is at least 30, or the underlying population is known to follow a normal distribution.
c. The population standard deviation (σ) is known.

When these requirements are met, the normal distribution is used as the basis for conducting the hypothesis test.

For this procedure, the test statistic will be a *z*-score and the standardized test statistic is calculated as follows:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Where:
$\bar{x}$ is the sample mean.
μ is the hypothesized value of the population mean, which comes from the null hypothesis.
σ is the population standard deviation.
n is the sample size.

Example 4.13 provides a step-by-step example to illustrate the hypothesis testing process.

EXAMPLE 4.13

Problem

A random sample of 100 college students finds that students spend an average of 14.4 hours per week on social media. Use this sample data to test the claim that college students spend at least 15 hours per week on social media. Use a level of significance of 0.05 and assume the population standard deviation is 3.25 hours.

Solution

The first step in the hypothesis testing procedure is to write the null and alternative hypotheses based on the claim. Notice that the claim involves the phrase “at least,” which translates to a greater than or equals symbol. A greater than or equals symbol must be used in the null hypothesis. The complement of greater than or equals is a less than symbol, which will be used in the alternative hypothesis. The claim refers to the population mean time that college students spend on social media, so the symbol μ will be used. The claim corresponds to the null hypothesis.

Notice that this setup will correspond to Setup C from Table 4.6. Thus the setup of the null and alternative hypotheses will be as follows:

$$H_0 : \mu \geq 15 \text{ [claim]}$$
$$H_a : \mu < 15$$

Since the population standard deviation is known in this example, the normal distribution will be used to conduct the hypothesis test.

Next, calculate the standardized test statistic:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{14.4 - 15}{3.25/\sqrt{100}} = -1.846$$

Notice that since the alternative hypothesis contains a "less than" symbol, the hypothesis test is called a "left tailed" test and the *p*-value will be calculated as the area to the *left* of the test statistic. The area under the standard normal curve to the left of a *z*-score of -1.846 is approximately 0.0324. Thus, the *p*-value for this hypothesis test is 0.0324.

The area under the standard normal curve can be found using software such as Excel, Python, or R.

For example, in Python, the area under the normal curve to the left of a *z*-score of -1.846 can be obtained using the function `norm.cdf()`.

The syntax for using this function is
`norm.cdf(x, mean, standard_deviation)`

Where:
`x` is the measurement of interest—in this case, this will be the test statistic of -1.846.
`mean` is the mean of the standard normal distribution, which is 0.
`standard_deviation` is the standard deviation of the standard normal distribution, which is 1.

Here is the Python code to calculate the *p*-value for this example:

PYTHON CODE



```
# from scipy.stats import norm
from scipy.stats import norm

# define parameters x, mean and standard_deviation:
x = -1.846
mean = 0
standard_deviation = 1
# use norm.cdf function to calculate normal probability - note this is the area
to the left of the x-value
# subtract this result from 1 to obtain area to the right of the x-value
# use round function to round answer to 4 decimal places
round (norm.cdf(x, mean, standard_deviation), 4)
```

The resulting output will look like this:

```
0.0324
```

The final step in the hypothesis testing procedure is to come to a final decision regarding the null

hypothesis. This is accomplished by comparing the *p*-value with the level of significance. In this example, the *p*-value is 0.032 and the level of significance is 0.05: since the *p*-value $\leq$ level of significance, then the decision is to "reject the null hypothesis."

Conclusion: The decision is to reject the null hypothesis; the claim corresponds to the null hypothesis. This scenario corresponds to Row 1 in Table 4.8, which means "there is enough evidence to reject the claim" that college students spend at least 15 hours per week on social media.

Testing Claims for the Mean When the Population Standard Deviation Is Unknown

In this section, we discuss one sample hypothesis testing for the mean when the population standard deviation is unknown. This is a more common application of hypothesis testing for the mean in that the population standard deviation is typically unknown; however, the sample standard deviation can be calculated based on the sample data.

Here are the requirements to use this procedure:

a. A random sample is selected from the population.
b. The sample size is at least 30, or the underlying population is known to follow a normal distribution.
c. The population standard deviation (σ) is unknown; the sample standard deviation (s) is known.

When these requirements are met, the t-distribution is used as the basis for conducting the hypothesis test.

For this procedure, the test statistic will be a t-score and the standardized test statistic is calculated as follows:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Where:
$\bar{x}$ is the sample mean.
μ is the hypothesized value of the population mean, which comes from the null hypothesis.
s is the sample standard deviation.
n is the sample size.

Example 4.14 provides a step-by-step example to illustrate this process.

EXAMPLE 4.14

Problem

A smartphone manufacturer claims that the mean battery life of its latest smartphone model is 25 hours. A consumer group decides to test this hypothesis and collects a sample of 50 smartphones and determines that the mean battery life of the sample is 24.1 hours with a sample standard deviation of 4.1 hours. Use this sample data to test the claim made by the smartphone manufacturer. Use a level of significance of 0.10 for this analysis.

Solution

The first step in the hypothesis testing procedure is to write the null and alternative hypotheses based on the claim. Notice that the claim involves the phrase "is 25 hours," where "is" will translate to an equals symbol. An equals symbol must be used in the null hypothesis. The complement of an "equals" symbol is a "not equals" symbol, which will be used in the alternative hypothesis. The claim refers to the population mean battery life of smartphones, so the symbol μ will be used. The claim corresponds to the null hypothesis. Notice that this setup will correspond to Setup A from Table 4.6. Thus the setup of the null and

alternative hypotheses will be as follows:

$$H_0 : \mu = 25 \text{ [claim]}$$
$$H_a : \mu \neq 25$$

Since the population standard deviation is unknown in this example, the t-distribution will be used to conduct the hypothesis test.

Next, calculate the standardized test statistic as follows:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{24.1 - 25}{4.1/\sqrt{50}} = -1.552$$

Notice that since the alternative hypothesis contains a "not equals" symbol, the hypothesis test is called a "two tailed" test and the *p*-value will be calculated as the area to the *left* of the negative test statistic plus the area to the right of the positive test statistic. The area under the t-distribution curve to the left of a t-score of -1.552 is approximately 0.0635 (using degrees of freedom $df = 49$). The area under the t-distribution curve to the right of a t-score of 1.552 is approximately 0.0635. Thus, the *p*-value for this hypothesis test is the sum of the areas, which is 0.127.

The area under the t-distribution curve can be found using software.

For example, in Python, the area under the t-distribution curve to the left of a t-score of -1.552 can be obtained using the function `t.cdf()`

The syntax for using this function is
`t.cdf(x, df)`

Where:
`x` is the measurement of interest—in this case, this will be the test statistic of -1.846.
`df` is the degrees of freedom.

Here is the Python code to calculate the *p*-value for this example:

PYTHON CODE



```
# from scipy.stats import t
from scipy.stats import t

# define parameters test_statistic and degrees of freedom df
x = -1.552
df = 49

# use t.cdf function to calculate area under t-distribution curve - note this is
the area to the left of the x-value
# subtract this result from 1 to obtain area to the right of the x-value
# use round function to round answer to 4 decimal places
round (t.cdf(x, df=49), 4)
```

The resulting output will look like this:

```
0.0635
```

The final step in the hypothesis testing procedure is to come to a final decision regarding the null hypothesis. This is accomplished by comparing the *p*-value with the level of significance. In this example, the *p*-value is 0.127 and the level of significance is 0.10: since the *p*-value $>$ level of significance, then the decision is to "fail to reject the null hypothesis."

Conclusion: The decision is to fail to reject the null hypothesis; the claim corresponds to the null hypothesis. This scenario corresponds to Row 3 in Table 4.8, which means "there is not enough evidence to reject the claim" that the mean battery life of its latest smartphone model is 25 hours.

Testing Claims for the Proportion

In this section, we discuss one sample hypothesis testing for proportions. Recall that a sample proportion is measuring "how many out of the total." For example, a medical researcher might be interested in testing a claim that the proportion of patients experiencing side effects from a new blood pressure–lowering medication is less than 4%.

Here are the requirements to use this procedure:

a. A random sample is selected from the population.
b. Verify that the normal approximation to the binomial distribution is appropriate by ensuring that both $n\hat{p}$ and $n(1-\hat{p})$ are both at least 5.

The sample proportion $\hat{p}$ is calculated as the number of successes divided by the sample size:

$$\hat{p} = \frac{x}{n}$$

When these requirements are met, the normal distribution is used as the basis for conducting the hypothesis test.

For this procedure, the test statistic will be a *z*-score and the standardized test statistic is calculated as follows:

$$z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}}$$

Where:
$\hat{p}$ represents the sample proportion.
p is the hypothesized value of the population proportion, which is taken from the null hypothesis.
n is the sample size.

Here is a step-by-step example to illustrate this hypothesis testing process.

EXAMPLE 4.15

Problem

A college professor claims that the proportion of students using artificial intelligence (AI) tools as part of their coursework is less than 45%. To test the claim, the professor selects a sample of 200 students and surveys the students to determine if they use AI tools as part of their coursework. The results from the survey indicate that 74 out of 200 students use AI tools. Use this sample data to test the claim made by the

college professor. Use a level of significance of 0.05 for this analysis.

Solution

The first step in the hypothesis testing procedure is to write the null and alternative hypothesis based on the claim. Notice that the claim involves the phrase "less than," which will translate to a less than symbol. A less than symbol must be used in the alternative hypothesis. The complement of a "less than" symbol is a "greater than or equal" symbol, which will be used in the null hypothesis. The claim refers to the population proportion of students using AI tools, so the symbol p will be used. Notice that this setup will correspond to Setup F from Table 4.7. Thus, the setup of the null and alternative hypotheses will be as follows:

$$H_0 : p \geq 0.45 \quad H_a : p < 0.45 \text{ [claim]}$$

The sample proportion can be calculated as follows:

$$\hat{p} = \frac{x}{n} = \frac{74}{200} = 0.37$$

Verify that the normal approximation to the binomial distribution is appropriate by ensuring that both $n\hat{p}$ and $n(1 - \hat{p})$ are both at least 5, where $\hat{p}$ represents the sample proportion.

For this example, $n\hat{p} = (200)0.37 = 74$, and $n(1 - \hat{p}) = (200)(1 - 0.37) = 126$. Both of these results are at least 5, which verifies that the normal approximation to the binomial distribution is appropriate.

Since the requirements for a hypothesis test for a proportion have been met, the normal distribution will be used to conduct the hypothesis test.

Next, calculate the standardized test statistic as follows:

$$z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} = \frac{0.37 - 0.45}{\sqrt{0.45(1-0.45)/200}} = -2.274$$

Notice that since the alternative hypothesis contains a "not equals" symbol, the hypothesis test is called a "left tailed" test and the *p*-value will be calculated as the area to the left of the test statistic. The area under the normal curve to the *left* of a *z*-score of -2.274 is approximately 0.012. Thus, the *p*-value for this hypothesis test is 0.012.

Here is the Python code to calculate the *p*-value for this example:

PYTHON CODE



```
# from scipy.stats import norm
from scipy.stats import norm

# define parameters x, mean, and standard_deviation:
x = -2.274
mean = 0
standard_deviation = 1
# use norm.cdf function to calculate normal probability - note this is the area
to the left of the x-value
```

```
# subtract this result from 1 to obtain area to the right of the x-value
# use round function to round answer to 3 decimal places
round (norm.cdf(x, mean, standard_deviation), 4)
```

The resulting output will look like this:

```
0.0115
```

The final step in the hypothesis testing procedure is to come to a final decision regarding the null hypothesis. This is accomplished by comparing the *p*-value with the level of significance. In this example, the *p*-value is 0.012 and the level of significance is 0.05: since the *p*-value $<$ level of significance, then the decision is to "reject the null hypothesis."

Conclusion: The decision is to reject the null hypothesis; the claim corresponds to the alternative hypothesis. This scenario corresponds to Row 2 in Table 4.8, which means "there is enough evidence to support the claim" that the proportion of students using artificial intelligence (AI) tools as part of their coursework is less than 45%.

Using Python to Conduct Hypothesis Tests

Python provides several functions to assist with hypothesis testing, as the `ztest()` can be used for hypothesis testing for the mean when population standard deviation is known and another Python function called `ttest_1samp()` can be used for hypothesis testing for the mean when population standard deviation is unknown. See the example:

EXAMPLE 4.16

Problem

A medical researcher wants to test a claim that the average oxygen level for females is greater than 75 mm Hg. In order to test the claim, a sample of 15 female patients is selected and the following oxygen levels are recorded:

76, 75, 74, 81, 76, 77, 71, 74, 73, 79

Use the `ttest_1samp()` function in Python to conduct a hypothesis test to test the claim. Assume a level of significance of 0.05.

Solution

The first step in the hypothesis testing procedure is to write the null and alternative hypotheses based on the claim. Notice that the claim involves the phrase "greater than 75 mm Hg," where "greater than" will translate to a greater than symbol. A greater than symbol must be used in the alternative hypothesis. The complement of a "greater than" symbol is a "less than or equals" symbol, which will be used in the null hypothesis. The claim refers to the population mean oxygen level, so the symbol μ will be used. The claim corresponds to the alternative hypothesis. Notice that this setup will correspond to Setup B from Table 4.6. Thus the setup of the null and alternative hypotheses will be as follows:

$$H_0 : \mu \leq 75$$
$$H_a : \mu > 75 \text{ [claim]}$$

Since the population standard deviation is unknown in this example, the t-distribution will be used to conduct the hypothesis test.

To use the `ttest_1samp()` function in Python, the syntax is as follows:

`Ttest_1samp(data_array, null_hypothesis_mean)`

The function will then return the value of the test statistic and the two-tailed *p*-value.

In this example we are interested in the right tailed *p*-value, so the Python program will calculate one-half of the *p*-value returned by the function.

Here is the Python code for this example:

PYTHON CODE



```
import scipy.stats as stats
import numpy as np

# Enter data array
oxygen_level = np.array([76, 75, 74, 81, 76, 77, 71, 74, 73, 79])

# Hypothesized population mean
null_hypothesis_mean = 75

# Perform one-sample t-test
test_stat, two_tailed_p_value = stats.ttest_1samp(oxygen_level,
null_hypothesis_mean)

# Obtain the one tailed p-value
one_tailed_p_value = two_tailed_p_value / 2

# print out test statistic and p-value and round to 3 decimal places
print("Test Statistic = ", test_stat)
print("p-value = ", one_tailed_p_value)
```

The resulting output will look like this:

```
Test Statistic = 0.6512171447631733
p-value = 0.2655900985964262
```

The final step in the hypothesis testing procedure is to come to a final decision regarding the null hypothesis. This is accomplished by comparing the *p*-value with the level of significance. In this example, the *p*-value is 0.266 and the level of significance is 0.05: since the *p*-value > level of significance, the decision is to "fail to reject the null hypothesis."

Conclusion: The decision is to fail to reject the null hypothesis; the claim corresponds to the alternative hypothesis. This scenario corresponds to Row 4 in Table 4.8, which means "there is not enough evidence to support the claim" that the average oxygen level for female patients is greater than 75 mm Hg.

Hypothesis Testing for Two Samples

In the previous section, we examined methods to conduct hypothesis testing for one sample. Researchers and data scientists are also often interested in conducting hypothesis tests for two samples. A researcher might want to compare two means or two proportions to determine if two groups are significantly different from one another. For example, a medical researcher might be interested in comparing the mean blood pressure between two groups of individuals: one group that is taking a blood pressure–lowering medication and another group that is taking a placebo. A government researcher might be interested in comparing the proportion of smokers for male versus female adults.

The same general hypothesis testing procedure will be used for two samples, but the setup of the null and alternative hypotheses will be structured to reflect the comparison of two means or two proportions.

As a reminder, the general hypothesis testing method introduced in the previous section includes the following steps:

1. Set up a null and an alternative hypothesis based on the claim.
2. Collect relevant sample data to investigate the claim.
3. Determine the correct distribution to be used to analyze the hypothesis test (e.g., normal distribution or t-distribution).
4. Analyze the sample data to determine if the sample data is consistent with the null hypothesis. This will involve statistical calculations involving a test statistic and a *p*-value, which are discussed next.
5. If the sample data is inconsistent with the null hypothesis, then the decision is to "reject the null hypothesis." If the sample data is consistent with the null hypothesis, then the decision is to "fail to reject the null hypothesis."

As discussed for one-sample hypothesis testing, when writing the null and alternative hypotheses, start off by translating the claim into a mathematical statement involving the unknown population parameters. The claim could correspond to either the null or the alternative hypothesis. Then write the complement of this statement in order to write the other hypothesis. The difference now as compared to the previous section is that the hypotheses will be comparing two means or two proportions instead of just one mean or one proportion.

When translating claims about a population mean, use the symbols μ_1 and μ_2 as part of the null and alternative hypotheses to represent the two unknown population means. When translating claims about a population proportions, use the symbols *p*1 and *p*2 as part of the null and alternative hypotheses to represent the two unknown population proportions. Refer to Table 4.9 and Table 4.10 for the various setups to construct

the null and alternative hypotheses for two-sample hypothesis testing.

Setup A	Setup B	Setup C
$H_0 : \mu_1 = \mu_2$	$H_0 : \mu_1 \leq \mu_2$	$H_0 : \mu_1 \geq \mu_2$
$H_a : \mu_1 \neq \mu_2$	$H_a : \mu_1 > \mu_2$	$H_a : \mu_1 < \mu_2$

Table 4.9 Three Possible Setups for the Null and Alternative Hypotheses for a Population Mean (Two Samples)

Setup D	Setup E	Setup F
$H_0 : p_1 = p_2$	$H_0 : p_1 \leq p_2$	$H_0 : p_1 \geq p_2$
$H_a : p_1 \neq p_2$	$H_a : p_1 > p_2$	$H_a : p_1 < p_2$

Table 4.10 Three Possible Setups for the Null and Alternative Hypotheses for a Population Proportion (Two Samples)

Comparing Two Means

In this section, we discuss two-sample hypothesis testing for the mean when the population standard deviations are unknown. Typically, in real-world applications, population standard deviations are unknown, so this method has considerable practical appeal.

One other consideration when conducting hypothesis testing for two samples is to determine if the two samples are dependent or independent. Two samples are considered to be **independent samples** when the sample values from one population are not related to the sample values taken from the second population. In other words, there is not an inherent paired or matched relationship between the two samples. Two samples are considered to be **dependent samples** if the samples from one population can be paired or matched to the samples taken from the second population. Dependent samples are sometimes also called *paired samples*.

For example, if a researcher were to collect data for the ages for a random sample of 15 men and a random sample of 15 women, there would be no inherent relationship between these two samples, and so these samples are considered independent samples.

On the other hand, let's say a researcher were to collect data on the ages of a random sample of 15 adults and also collect data on the ages of those adults' corresponding (15) spouses/partners; these would be considered dependent samples in that there is an inherent pairing of domestic partners.

EXAMPLE 4.17

Problem

Determine if the following samples are dependent or independent samples:

Sample #1: Scores on a statistics exam for 40 women college students

Sample #2: Scores on a statistics exam for 35 men college students

Solution

These two samples are independent because there is no inherent match-up between the women students from Sample #1 and the men students from Sample #2; it is not possible to analyze either of the samples as dependent, or paired, samples.

EXAMPLE 4.18

Problem

Determine if the following samples are dependent or independent samples:

Sample #1: Blood pressure readings on a sample of 50 patients in a clinical trial before taking any medication. Sample #2: Blood pressure readings on the same group of 50 patients after taking a blood pressure–lowering medication.

Solution

These two samples are dependent. The samples can be paired as before/after blood pressure readings for the same patient.

Here are the requirements to use this procedure:

a. The samples are random and independent.
b. For each sample, the sample sizes are both at least 30, or the underlying populations are known to follow a normal distribution.
c. The population standard deviations (σ) are unknown; the sample standard deviations (s) are known.

When these requirements are met, the t-distribution is used as the basis for conducting the hypothesis test.

In this discussion, we assume that the population variances are not equal, which leads to the following formula for the test statistic.

For this procedure, the test statistic will be a t-score and the standardized test statistic is calculated as follows:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Where:

$\bar{x}_1, \bar{x}_2$ are the sample means.

μ_1, μ_2 are the hypothesized values of the population means.

s_1, s_s are the sample standard deviations.

n_1, n_2 are the sample sizes.

Note: For the t-distribution, the degrees of freedom will be the smaller of $n_1 - 1$ and $n_2 - 1$.

Example 4.19 illustrates this hypothesis testing process.

EXAMPLE 4.19

Problem

A software engineer at a communications company claims that the average salary of women engineers is less than that of men engineers. To test the claim, a human resource administrator selects a random sample of 45 women engineers and 50 men engineers and collects statistical data as shown in Table 4.11. (Dollar amounts are in thousands of dollars.)

	Women Engineers	Men Engineers
Sample Mean	$\bar{x}_1 = 72700$	$\bar{x}_2 = 76900$
Sample Standard Deviation	$s_1 = 9800$	$s_2 = 10700$
Sample Size	$n_1 = 45$	$n_2 = 50$

Table 4.11 Sample Data for Women and Men Engineers

Use this sample data to test the claim made by the software engineer. Use a level of significance of 0.05 for this analysis.

Solution

The first step in the hypothesis testing procedure is to write the null and alternative hypotheses based on the claim. Notice that the claim mentions "average salary of women engineers is less than that of men engineers, where "less than" will translate to a less than symbol. A less than symbol must be used in the alternative hypothesis. The complement of a "less than" symbol is a "greater than or equals" symbol, which will be used in the null hypothesis. Let μ_1 represent the population mean salary for women engineers and let μ_2 represent the population mean salary for men engineers. The null hypothesis can be considered to represent that we expect there to be "no difference" between the salaries of women and men engineers. Note that this setup will correspond to Setup C from Table 4.9. The claim corresponds to the alternative hypothesis. Thus, the setup of the null and alternative hypotheses will be as follows:

$$\begin{aligned} H_0 &: \mu_1 \geq \mu_2 \\ H_a &: \mu_1 < \mu_2 \text{ [claim]} \end{aligned}$$

Since the population standard deviations are unknown in this example, the t-distribution will be used to conduct the hypothesis test.

Next, calculate the standardized test statistic as follows:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{(72700 - 76900) - (0)}{\sqrt{\frac{9800^2}{45} + \frac{10700^2}{50}}} = -1.997$$

Notice that since the alternative hypothesis contains a "less than" symbol, the hypothesis test is called a "left tailed" test and the *p*-value will be calculated as the area under the t-distribution curve to the *left* of the test statistic. The area under the t-distribution curve to the *left* of a t-score of -1.997 (using degrees of freedom $df = 44$) is approximately 0.025. Thus, the *p*-value for this hypothesis test is 0.025.

The final step in the hypothesis testing procedure is to come to a final decision regarding the null hypothesis. This is accomplished by comparing the *p*-value with the level of significance. In this example, the *p*-value is 0.025 and the level of significance is 0.04: since the *p*-value $<$ level of significance, then the decision is to "reject the null hypothesis."

Conclusion: The decision is to reject the null hypothesis; the claim corresponds to the alternative hypothesis. This scenario corresponds to Row 2 in Table 4.8, and thus "there is enough evidence to support the claim" that the average salary of women engineers is less than that of men engineers.

This problem can also be solved using the Python function called `ttest_ind_from_stats()`.

There is also a Python function to perform a two-sample t-test given raw data for the two groups, and this Python function is `ttest_ind()`.

For the function `ttest_ind_from_stats()`, the syntax is as follows:

PYTHON CODE

```
stats.ttest_ind_from_stats(sample_mean1, sample_standard_deviation1,
sample_size1, sample_mean2, sample_standard_deviation2, sample_size2)
```

Here is the Python code that can be used to solve Example 4.19. Note that this function returns the two-tailed *p*-value. Since Example 4.19 involves a left-tailed test, the desired *p*-value is one-half of the two-tailed *p*-value.

PYTHON CODE

```
import scipy.stats as stats
import numpy as np

#Example 4.19 - Two Sample t-test given sample means, sample standard deviations

sample_mean1 = 72700
sample_standard_deviation1 = 9800
sample_size1 = 45

sample_mean2 = 76900
sample_standard_deviation2 = 10700
sample_size2 = 50

test_statistic, two_tailed_p_value = stats.ttest_ind_from_stats(sample_mean1,
sample_standard_deviation1, sample_size1, sample_mean2,
sample_standard_deviation2, sample_size2)

left_tailed_p_value = two_tailed_p_value / 2

print("test statistic = ", "%.3f" % test_statistic)
print("p-value = ", "%.3f" % left_tailed_p_value)
```

The resulting output will look like this:

```
test statistic = -1.988
p-value = 0.025
```

Comparing Matched Pairs Data

Hypothesis Testing for Two Samples discussed hypothesis testing for two *independent samples*. In this section, we turn our attention to hypothesis testing for *dependent samples*, often called **matched pairs**, paired samples, or paired data.

Recall that two samples are considered to be dependent if the samples from one population can be paired, or matched, to the samples taken from the second population.

This analysis method is often used to analyze before/after data to determine if a significant change has occurred. In a clinical trial, a medical researcher may be interested in comparing before and after blood pressure readings on a patient to assess the efficacy of a blood pressure–lowering medication. Or an employer might compare scores on a safety exam for a sample of employees before and after employees have taken a course on safety procedures and practices. A parent might want to compare before/after SAT scores for students who have taken an SAT preparation course. This type of analysis can help to assess the effectiveness and value of such training.

When conducting a hypothesis test for paired data such as before/after data, a key step will be to calculate the "difference" for each pair of data values, as follows:

$$\text{Difference (d)} = \text{(before data)} - \text{(after data)}$$

Note that when calculating these differences, the result can be either a positive or a negative value.

Here are the requirements to use this procedure:

1. The samples are random and dependent (i.e., paired data).
2. The populations are normally distributed or the number of data pairs is at least 30.

When these requirements are met, the t-distribution is used as the basis for conducting the hypothesis test.

The test statistic will be the average of the differences across the data pairs (labeled as $\bar{d}$), where:

$$\bar{d} = \frac{\Sigma d}{n}$$

For this procedure, the standardized test statistic is calculated as follows:

$$t = \frac{\bar{d} - \mu_d}{s_d / \sqrt{n}}$$

Where:

$\bar{d}$ is the mean of the differences between paired data.

μ_d is the hypothesized population mean of the differences.

s_d is the standard deviation of the differences between paired data.

n is the number of data pairs.

Note: For the t-distribution, the degrees of freedom will be $n - 1$, where n is the number of data pairs.

The setup of the null and alternative hypotheses will follow one of the setups shown in Table 4.12:

Setup A	Setup B	Setup C
$H_0 : \mu_d = k$ $H_a : \mu_d \neq k$	$H_0 : \mu_d \leq k$ $H_a : \mu_d > k$	$H_0 : \mu_d \geq k$ $H_a : \mu_d < k$

Table 4.12 Possible Setups for Null and Alternative Hypothesis for Paired Data Hypothesis Testing

Example 4.20 illustrates this hypothesis testing process.

EXAMPLE 4.20

Problem

A pharmaceutical company is testing a cholesterol-lowering medication in a clinical trial. The company claims that the medication helps to lower cholesterol levels. To test the claim, a group of 10 patients is selected for a clinical trial. Before and after cholesterol measurements are taken on the same patients, where the "after" measurement is recorded after the patient has taken the medication for a six-month time period as shown in Table 4.13.

Patient	Cholesterol Level Before Taking the Medication	Cholesterol Level After Taking the Medication	Difference (Before Reading - After Reading)
1	218	210	8
2	232	241	-9
3	259	223	36
4	265	244	21
5	248	227	21
6	298	273	25
7	263	252	11
8	281	276	5
9	290	281	9
10	271	259	12
			$\bar{d} = 13.9$ $s_d = 12.414$

Table 4.13 Sample Data for Paired Data Example

Solution

The first step in the hypothesis testing procedure is to write the null and alternative hypotheses based on the claim. Notice that the company claims that the medication helps to lower cholesterol levels. If the medication did reduce cholesterol levels, then we would expect the after data to be less than the before data. In this case, the difference for "before - after" would be a positive value, which is greater than zero. A greater than symbol must be used in the alternative hypothesis. The complement of a "greater than" symbol is a "less than or equals" symbol, which will be used in the null hypothesis. This setup will correspond to Setup C from Table 4.12. The claim corresponds to the alternative hypothesis. Thus the setup

of the null and alternative hypotheses will be as follows:

$$\begin{aligned} H_0 : \mu_d &\leq 0 \\ H_a : \mu_d &> 0 \text{ [claim]} \end{aligned}$$

The mean of the differences is average of the differences column shown in Table 4.13. The mean of the differences column is 13.9, and the standard deviation of the differences column is 12.414.

Next, calculate the standardized test statistic as follows:

$$t = \frac{\bar{d} - \mu_d}{s_d/\sqrt{n}} = \frac{13.9 - 0}{12.414/\sqrt{10}} = 3.541$$

Notice that since the alternative hypothesis contains a "greater than" symbol, the hypothesis test is called a "right tailed" test and the *p*-value will be calculated as the area under the t-distribution curve to the *right* of the test statistic. The area under the t-distribution curve to the right of a t-score of 3.541 (using degrees of freedom $df = 9$) is approximately 0.003. Thus the *p*-value for this hypothesis test is 0.003.

The final step in the hypothesis testing procedure is to come to a final decision regarding the null hypothesis. This is accomplished by comparing the *p*-value with the level of significance. In this example, the *p*-value is 0.003 and the level of significance is 0.05: since the *p*-value $<$ level of significance, then the decision is to "reject the null hypothesis."

Conclusion: The decision is to reject the null hypothesis; the claim corresponds to the alternative hypothesis. This scenario corresponds to Row 2 in Table 4.8, and "there is enough evidence to support the claim that the medication helps to lower cholesterol levels."

Testing Claims for Two Proportions

Data scientists and researchers are often interested in comparing two proportions. For example, a medical researcher might be interested in comparing the percentage of people contracting a respiratory virus for those taking a vaccine versus not taking a vaccine.

In this section, we discuss two-sample hypothesis testing for proportions, where the notation p_1 refers to the population proportion for the first group and p_2 refers to the population proportion for the second group.

A researcher will take samples from the two populations and then calculate two sample proportions ($\hat{p}$) as follows:

$$\begin{aligned} \hat{p}_1 &= \frac{x_1}{n_1} \\ \hat{p}_2 &= \frac{x_2}{n_2} \end{aligned}$$

Where:
x_1 is the number of successes in the first sample.
n_1 is the sample size for the first sample.

x_2 is the number of successes in the second sample.
n_2 is the sample size for the second sample.

$\hat{p}_1$ is the sample proportion for the first sample.
$\hat{p}_2$ is the sample proportion for the second sample.

It is also useful to calculate a weighted estimate ($\bar{p}$) for $\hat{p}_1$ and $\hat{p}_2$, as follows:

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

Here are the requirements to use this procedure:

a. The samples are random and independent.
b. Sample sizes are large enough to use a normal distribution as an approximation for a binomial distribution. To confirm this, ensure that the following four quantities are all at least 5: $n_1\overline{p}$, $n_1\left(1-\overline{p}\right)$, $n_2\overline{p}$, $n_2\left(1-\overline{p}\right)$

When these requirements are met, the normal distribution is used as the basis for conducting the hypothesis test.

For this procedure, the test statistic will be a *z*-score, and the standardized test statistic is calculated as follows:

$$z=\frac{(\hat{p}_1-\hat{p}_2)-(p_1-p_2)}{\sqrt{\overline{p}\left(1-\overline{p}\right)\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$$

Here is a step-by-step example to illustrate this hypothesis testing process.

EXAMPLE 4.21

Problem

A drug is under investigation to address lung cancer. As part of a clinical trial, 250 people took the drug and 230 people took a placebo (a substance with no therapeutic effect). From the results of the study, 211 people taking the drug were cancer free, and 172 taking the placebo were cancer free. At a level of significance of 0.01, can you support the claim by the pharmaceutical company that the proportion of subjects who are cancer free after taking the actual cancer drug is greater than the proportion for those who took the placebo?

Solution

The first step in the hypothesis testing procedure is to write the null and alternative hypotheses based on the claim. Notice that the claim mentions "proportion of subjects who are cancer free after taking the actual cancer drug is greater than the proportion for those who took the placebo," where "greater than" will translate to a greater than symbol. A greater than symbol must be used in the alternative hypothesis. The complement of a "greater than" symbol is a "less than or equals" symbol, which will be used in the null hypothesis. Let p_1 represent the proportion of people who take the medication and are cancer free, and let p_2 represent the proportion of people who take the placebo and are cancer free. The null hypothesis assumes there is no difference in the two proportions such that p_1 is equal to p_2 (which can also be considered as $p_1-p_2=0$). This setup will correspond to Setup E from Table 4.10. The claim corresponds to the alternative hypothesis. Thus the setup of the null and alternative hypotheses will be as follows:

$$\begin{aligned} H_0 &: p_1 \leq p_2 \\ H_a &: p_1 > p_2 \ \text{[claim]} \end{aligned}$$

First, check that the requirements for this hypothesis test are met.

Calculate the weighted estimate ($\overline{p}$) for p_1 and p_2, as follows:

$$\overline{p}=\frac{x_1+x_2}{n_1+n_2}=\frac{211+172}{250+230}=\frac{383}{480}=0.798$$

Check that the following four quantities are all at least 5: $n_1\overline{p}$, $n_1\left(1-\overline{p}\right)$, $n_2\overline{p}$,$n_2\left(1-\overline{p}\right)$.

$$\begin{aligned} n_1\bar{p} &= (250)(0.798) = 199.5 \\ n_1(1-\bar{p}) &= (250)(1-0.798) = 50.5 \\ n_2\bar{p} &= (230)(0.798) = 183.54 \\ n_2(1-\bar{p}) &= (230)(1-0.798) = 46.46 \end{aligned}$$

Since these results are at least 5, the requirements are met.

Calculate the sample proportions as follows:

$$\begin{aligned} \hat{p}_1 &= \frac{x_1}{n_1} = \frac{211}{250} = 0.844 \\ \hat{p}_2 &= \frac{x_2}{n_2} = \frac{172}{230} = 0.748 \end{aligned}$$

Next, calculate the standardized test statistic as follows:

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\bar{p}\left(1-\bar{p}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(0.844 - 0.748) - 0}{\sqrt{\left(0.798\right)\left(1-0.798\right)\left(\frac{1}{250} + \frac{1}{230}\right)}} = 2.621$$

Notice that since the alternative hypothesis contains a "greater than" symbol, the hypothesis test is called a "right tailed" test and the *p*-value will be calculated as the area under the normal distribution curve to the right of the test statistic. The area under the normal distribution curve to the right of a *z*-score of 2.621 is approximately 0.004. Thus the *p*-value for this hypothesis test is 0.004.

The final step in the hypothesis testing procedure is to come to a final decision regarding the null hypothesis. This is accomplished by comparing the *p*-value with the level of significance. In this example, the *p*-value is 0.004 and the level of significance is 0.01: since the *p*-value $<$ level of significance, then the decision is to "reject the null hypothesis."

Conclusion: The decision is to reject the null hypothesis; the claim corresponds to the alternative hypothesis. This scenario corresponds to row #2 in Table 4.8, and "there is enough evidence to support the claim" that the proportion of subjects who are cancer free after taking the actual cancer drug is greater than the proportion for those who took the placebo. This indicates that the medication is effective in addressing lung cancer.

4.3 Correlation and Linear Regression Analysis

Learning Outcomes

By the end of this section, you should be able to:

- 4.3.1 Create scatterplots and calculate and interpret correlation coefficients.
- 4.3.2 Perform linear regression and determine the best-fit linear equation.
- 4.3.3 Use Python to calculate correlation coefficients and determine equations of linear regression models.

We briefly introduced correlation analysis at the beginning of this chapter, but now we want to dig in a little deeper. Data scientists are often interested in knowing if there are relationships, or a *correlation*, between two numeric quantities. For example, a medical researcher might be interested in investigating if there is a relationship between age and blood pressure. A college professor might be interested to know if there is a relationship between time spent on social media by students and corresponding academic grades.

Correlation analysis allows for the determination of a statistical relationship between two numeric quantities, or variables—an independent variable and a dependent variable. The **independent variable** is the variable that you can change or control, while the **dependent variable** is what you measure or observe to see how it

responds to those changes. Regression analysis takes this a step further by quantifying the relationship between the two variables, and it can be used to predict one quantity based on a second quantity, assuming there is a significant correlation between the two quantities. For example, the value of a car can be predicted based on the age of the car.

Earlier we noted many business-related applications where correlation and regression analysis are used. For instance, regression analysis can be used to establish a mathematical equation that relates a dependent variable (such as sales) to an independent variable (such as advertising expenditure). But it has many applications apart from business as well. An automotive engineer is interested in the correlation between outside temperature and battery life for an electric vehicle, for instance.

Our discussion here will focus on linear regression—analyzing the relationship between one dependent variable and one independent variable, where the relationship can be modeled using a linear equation.

Scatterplots and Correlation

In correlation analysis, we study the relationship between **bivariate data**, which is data collected on two variables where the data values are paired with one another. Correlation measures the association between two numeric variables. We may be interested in knowing if there is a correlation between bond prices and interest rates or between the age of a car and the value of the car. To investigate the correlation between two numeric quantities, the first step is to collect (x, y) data for the two numeric quantities of interest and then create a scatterplot that will graph the (x, y) ordered pairs. The independent, or explanatory, quantity is labeled the x-variable, and the dependent, or response, quantity is labeled the y-variable.

For example, let's say that a financial analyst wants to know if the price of Nike stock is correlated with the value of the S&P 500 (Standard & Poor's 500 stock market index). To investigate this, monthly data can be collected for Nike stock prices and value of the S&P 500 for a period of time, and a scatterplot can be created and examined. A **scatterplot, or scatter diagram**, is a graphical display intended to show the relationship between two variables. The setup of the scatterplot is that one variable is plotted on the horizontal axis and the other variable is plotted on the vertical axis. Each pair of data values is considered as an (x, y) point, and the various points are plotted on the diagram. A visual inspection of the plot is then made to detect any patterns or trends on the scatter diagram. Table 4.14 shows the relationship between the Nike stock price and its S&P value on a monthly basis over a one-year time period.

Date	S&P 500 (x)	Nike Stock Price (y)
Month 1	2912.43	87.18
Month 2	3044.31	98.58
Month 3	3100.29	98.05
Month 4	3271.12	97.61
Month 5	3500.31	111.89
Month 6	3363.00	125.54
Month 7	3269.96	120.08
Month 8	3621.63	134.70
Month 9	3756.07	141.47
Month 10	3714.24	133.59

Table 4.14 Nike Stock Price ($) and Value of S&P 500 over a One-Year Time Period
(source: Yahoo! Finance)

Date	S&P 500 (x)	Nike Stock Price (y)
Month 11	3811.15	134.78
Month 12	3943.34	140.45

Table 4.14 Nike Stock Price ($) and Value of S&P 500 over a One-Year Time Period
(source: Yahoo! Finance)

To assess **linear correlation**, examine the graphical trend of the data points on the scatterplot to determine if a straight-line pattern exists (see Figure 4.5). If a linear pattern exists, the correlation may indicate either a positive or a negative correlation. A positive correlation indicates that as the independent variable increases, the dependent variable tends to increase as well, or as the independent variable decreases, the dependent variable tends to decrease (the two quantities move in the same direction). A negative correlation indicates that as the independent variable increases, the dependent variable decreases, or as the independent variable decreases, the dependent variable increases (the two quantities move in opposite directions). If there is no relationship or association between the two quantities, where one quantity changing does not affect the other quantity, we conclude that there is no correlation between the two variables.

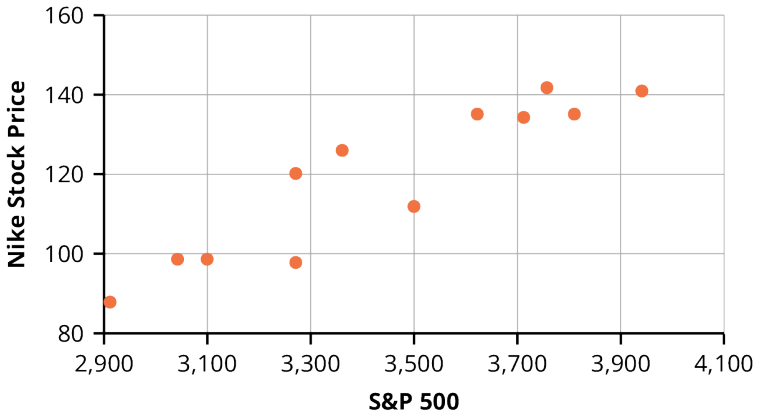


Figure 4.5 Scatterplot of Nike Stock Price ($) and Value of S&P 500
(source: Yahoo! Finance)

From the scatterplot in the Nike stock versus S&P 500 example, we note that the trend reflects a positive correlation in that as the value of the S&P 500 increases, the price of Nike stock tends to increase as well.

When inspecting a scatterplot, it may be difficult to assess a correlation based on a visual inspection of the graph alone. A more precise assessment of the correlation between the two quantities can be obtained by calculating the numeric correlation coefficient (referred to using the symbol r).

The **correlation coefficient** is a measure of the strength and direction of the correlation between the independent variable x and the dependent variable y.

The formula for r is shown; however, software is typically used to calculate the correlation coefficient.

$$r = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{n\Sigma x^2 - (\Sigma x)^2}\sqrt{n\Sigma y^2 - (\Sigma y)^2}}$$

where n refers to the number of data pairs and the symbol Σx indicates to sum the x-values.

Note that this value of "r" is sometimes referred to as the "Pearson correlation coefficient."

Table 4.15 provides a step-by-step procedure on how to calculate the correlation coefficient r.

Step	Representation in Symbols
1. Calculate the sum of the *x*-values.	Σx
2. Calculate the sum of the *y*-values.	Σy
3. Multiply each *x*-value by the corresponding *y*-value and calculate the sum of these xy products.	Σxy
4. Square each *x*-value and then calculate the sum of these squared values.	Σx^2
5. Square each *y*-value and then calculate the sum of these squared values.	Σy^2
6. Determine the value of n, which is the number of data pairs.	n
7. Use these results to then substitute into the formula for the correlation coefficient.	$r = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{n\Sigma x^2 - (\Sigma x)^2}\sqrt{n\Sigma y^2 - (\Sigma y)^2}}$

Table 4.15 Steps for Calculating the Correlation Coefficient

Note that since *r* is calculated using sample data, *r* is considered a *sample statistic* and is used to measure the strength of the correlation for the two population variables. Recall that sample data indicates data based on a subset of the entire population.

As mentioned, given the complexity of this calculation, software is typically used to calculate the correlation coefficient. There are several options for calculating the correlation coefficient in Python. The example shown uses the `scipy.stats` library in Python which includes the function `pearsonr()` for calculating the Pearson correlation coefficient (*r*).

We will create two numerical arrays in Python using the `np.array()` function and then use the `pearsonr()` function to calculate the Pearson correlation coefficient for the (x, y) data shown in Table 4.15.

The following Python code provides the correlation coefficient for the Nike Stock Price dataset as 0.923.

PYTHON CODE

```
# import libraries
import numpy as np
from scipy.stats  import pearsonr

# establish x and y data, x-data is S&P500 value, y-data is Nike Stock Price

SP500 = np.array([2912.43, 3044.31, 3100.29, 3271.12, 3500.31, 3363.00, 3269.96,
3621.63, 3756.07, 3714.24, 3811.15, 3943.34])

Nike = np.array([87.18, 98.58, 98.05, 97.61, 111.89, 125.54, 120.08, 134.70,
141.47, 133.59, 134.78, 140.45])

#pearson r returns both value of r and corresponding p-value
```

```
r, p = pearsonr(SP500, Nike)

#print value of r , rounded to 3 decimal places

print("Correlation coefficient: ", round(r, 3))
```

The resulting output will look like this:

```
Correlation coefficient: 0.923
```

Interpret a Correlation Coefficient

Once the value of *r* is calculated, this measurement provides two indicators for the correlation:

a. the strength of the correlation based on the *value* of *r*
b. the direction of the correlation based on the *sign* of *r*

The *value* of *r* gives us this information:

- The value of *r* is always between -1 and $+1$: $-1 \leq r \leq 1$.
- The size of the correlation *r* indicates the strength of the linear relationship between the two variables. Values of *r* close to -1 or to $+1$ indicate a stronger linear relationship.
- If $r = 0$, there is no linear relationship between the two variables (*no linear correlation*).
- If $r = 1$, there is perfect positive correlation. If $r = -1$, there is perfect negative correlation. In both of these cases, all the original data points lie on a straight line.

The sign of *r* gives us this information:

- A positive value of *r* means that when x increases, y tends to increase, and when x decreases, y tends to decrease (*positive correlation*).
- A negative value of *r* means that when x increases, y tends to decrease, and when x decreases, y tends to increase (*negative correlation*).

From the statistical results shown in the Python output, the correlation coefficient *r* is 0.923, which indicates that the relationship between Nike stock and the value of the S&P 500 over this time period represents a strong, positive correlation.

Test a Correlation Coefficient for Significance

The correlation coefficient, *r*, tells us about the strength and direction of the linear relationship between x and y. The sample data are used to compute *r*, the correlation coefficient for the sample. If we had data for the entire population (that is, all measurements of interest), we could find the population correlation coefficient, which is labeled as the Greek letter ρ (pronounced "rho"). But because we have only sample data, we cannot calculate the population correlation coefficient. The sample correlation coefficient, *r*, is our estimate of the unknown population correlation coefficient.

- $\rho =$ population correlation coefficient (unknown)
- $r =$ sample correlation coefficient (known; calculated from sample data)

An important step in the correlation analysis is to determine if the correlation is significant. By this, we are asking if the correlation is strong enough to allow meaningful **predictions** for y based on values of x. One method to test the significance of the correlation is to employ a hypothesis test. The hypothesis test lets us decide whether the value of the population correlation coefficient ρ is close to zero or significantly different from zero. We decide this based on the sample correlation coefficient *r* and the sample size n.

If the test concludes that the correlation coefficient is significantly different from zero, we say that the correlation coefficient is significant.

- Conclusion: There is sufficient evidence to conclude that there is a significant linear relationship between the two variables because the correlation coefficient is significantly different from zero.
- What the conclusion means: There is a significant linear relationship between the two variables. If the test concludes that the correlation coefficient is not significantly different from zero (it is close to zero), we say that correlation coefficient is not significant.

A hypothesis test can be performed to test if the correlation is significant. A hypothesis test is a statistical method that uses sample data to test a claim regarding the value of a population parameter. In this case, the hypothesis test will be used to test the claim that the population correlation coefficient ρ is equal to zero.

Use these hypotheses when performing the hypothesis test:

- Null hypothesis: $H_0 : \rho = 0$
- Alternate hypothesis: $H_a : \rho \neq 0$

The hypotheses can be stated in words as follows:

- Null hypothesis H_0: The population correlation coefficient *is not* significantly different from zero. There *is not* a significant linear relationship (correlation) between x and y in the population.
- Alternate hypothesis H_a: The population correlation coefficient is significantly different from zero. There *is* a significant linear relationship (correlation) between x and y in the population.

A quick shorthand way to test correlations is the relationship between the sample size and the correlation. If $|r| \geq \frac{2}{\sqrt{n}}$, then this implies that the correlation between the two variables demonstrates that a linear relationship exists and is statistically significant at approximately the 0.05 level of significance. As the formula indicates, there is an inverse relationship between the sample size and the required correlation for significance of a linear relationship. With only 10 observations, the required correlation for significance is 0.6325; for 30 observations, the required correlation for significance decreases to 0.3651; and at 100 observations, the required level is only 0.2000.

INTERPRETING THE SIGNIFICANCE OF THE SAMPLE CORRELATION COEFFICIENT

- If *r* is significant and the scatterplot shows a linear trend, the line can be used to predict the value of y for values of x that are within the domain of observed *x*-values.
- If *r* is not significant OR if the scatterplot does not show a linear trend, the line should *not* be used for prediction.
- If *r* is significant and the scatterplot shows a linear trend, the line may *not* be appropriate or reliable for prediction *outside* the domain of observed *x*-values in the data.

EXAMPLE 4.22

Problem

Suppose that the chief financial officer (CFO) of a corporation is investigating the correlation between stock prices and unemployment rate over a period of 10 years and finds the correlation coefficient to be -0.68. There are 10 (x, y) data points in the dataset. Should the CFO conclude that the correlation is significant for the relationship between stock prices and unemployment rate based on a level of significance of 0.05?

Solution

When using a level of significance of 0.05, if $|r| \geq \frac{2}{\sqrt{n}}$, then this implies that the correlation between the two variables demonstrates a linear relationship that is statistically significant at approximately the 0.05 level of significance. In this example, we check if $|-0.68|$ is greater than or equal to $\frac{2}{\sqrt{n}}$ where $n = 10$. Since $\frac{2}{\sqrt{10}}$ is approximately 0.632, this indicates that the *r*-value of -0.68 is greater than $\frac{2}{\sqrt{n}}$, and thus the correlation is deemed significant.

Linear Regression

A regression model is typically developed once the correlation coefficient has been calculated and a determination has been made that the correlation is significant. The regression model can then be used to predict one quantity (the dependent variable) based on the other (the independent variable). For example, a business may want to establish a correlation between the amount the company spent on advertising (the independent variable) versus its recorded sales (the dependent variable). If a strong enough correlation is established, then the business manager can predict sales based on the amount spent on advertising for a given period. In this discussion we will focus on linear regression, where a straight line is used to model the relationship between the two variables. Once a straight-line model is developed, this model can then be used to predict the value of the dependent variable for a specific value of the independent variable.

Method of Least Squares and Residuals

To create a linear model that best fits the (x, y) points on a scatterplot, researchers generally use the **least squares method** based on a mathematical algorithm designed to minimize the squared distances from the (x, y) data points to the fitted line.

Recall from algebra that the equation of a straight line is given by:

$$y = mx + b$$

where m is the slope of the line and b is the *y*-intercept of the line.

The slope measures the steepness of the line, and the *y*-intercept is that point on the y-axis where the graph crosses, or intercepts, the y-axis.

In linear regression analysis, the equation of the straight line is written in a slightly different way using the model

$$\hat{y} = a + bx$$

In this format, b is the slope of the line, and a is the *y*-intercept of the line. The notation $\hat{y}$ is called y-hat and is used to indicate a predicted value of the dependent variable y for a certain value of the independent variable x.

If a line extends uphill from left to right, the slope is a positive value (see Figure 4.6; if the line extends downhill from left to right, the slope is a negative value).

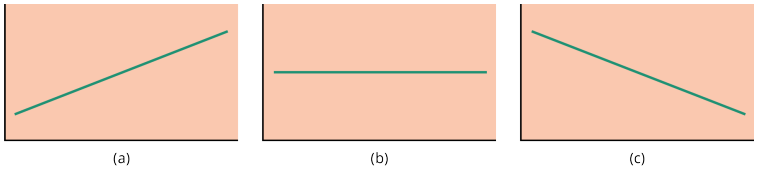


Figure 4.6 Three examples of graphs of $\hat{y} = a + bx$. (a) If $b > 0$, the line slopes upward to the right. (b) If $b = 0$, the line is horizontal. (c) If $b < 0$, the line slopes downward to the right.

When generating the equation of a line in algebra using $y = mx + b$, two (x, y) points were required to generate the equation. However, in regression analysis, all the (x, y) points in the dataset will be utilized to develop the linear regression model.

The first step in any regression analysis is to create the scatterplot. Then proceed to calculate the correlation coefficient *r* and check this value for significance. If we think that the points show a linear relationship, we would like to draw a line on the scatterplot. This line can be calculated through a process called linear regression. However, we only calculate a regression line if one of the variables helps to explain or predict the other variable. If x is the independent variable and y the dependent variable, then we can use a regression line to predict y for a given value of x.

As an example of a regression equation, assume that a correlation exists between the monthly amount spent on advertising and the monthly revenue for a Fortune 500 company. After collecting (x, y) data for a certain time period, the company determines the regression equation is of the form

$$\hat{y} = 9376.7 + 61.8x$$

where x represents the monthly amount spent on advertising (in thousands of dollars), $\hat{y}$ represents the monthly revenues for the company (in thousands of dollars), the slope is 61.8, and the *y*-intercept is 9376.7.

A scatterplot of the (x, y) data is shown in Figure 4.7.

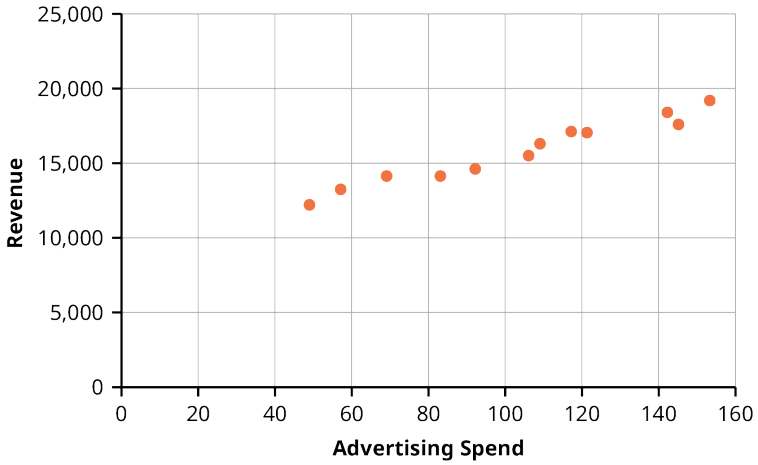


Figure 4.7 Scatterplot of Revenue versus Advertising for a Fortune 500 Company ($000s)

The company would like to predict the monthly revenue if its executives decide to spend $150,000 on advertising next month. To determine the estimate of monthly revenue, let $x = 150$ in the regression equation and calculate a corresponding value for $\hat{y}$:

$$\begin{aligned} \hat{y} &= 9376.7 + 61.8x \\ \hat{y} &= 9376.7 + 61.8(150) \\ \hat{y} &= 18646.7 \end{aligned}$$

This predicted value of y indicates that the anticipated revenue would be $18,646,700, given the advertising spend of $150,000.

Notice that from past data, there may have been a month where the company actually did spend $150,000 on advertising, and thus the company may have an actual result for the monthly revenue. This actual, or observed, amount can be compared to the prediction from the linear regression model to calculate what is called a residual.

A **residual** is the difference between an observed *y*-value and the predicted *y*-value obtained from the linear regression equation. As an example, assume that in a previous month, the actual monthly revenue for an advertising spend of $150,000 was $19,200,000, and thus $y = 19{,}200$. The residual for this data point can be

calculated as follows:

$$\begin{aligned}\text{Residual} &= (\text{observed } y\text{-value}) - (\text{predicted } y\text{-value})\\ \text{Residual} &= y - \hat{y}\\ \text{Residual} &= 19200 - 18646.7 = 553.3\end{aligned}$$

Notice that residuals can be positive, negative, or zero. If the observed *y*-value exactly matches the predicted *y*-value, then the residual will be zero. If the observed *y*-value is greater than the predicted *y*-value, then the residual will be a positive value. If the observed *y*-value is less than the predicted *y*-value, then the residual will be a negative value.

When formulating the best-fit linear regression line to the points on the scatterplot, the mathematical analysis generates a linear equation where the sum of the squared residuals is minimized. This analysis is referred to as the **method of least squares**. The result is that the analysis generates a linear equation that is the "best fit" to the points on the scatterplot, in the sense that the line minimizes the differences between the predicted values and observed values for y; this is generally referred to as the **best-fit linear equation**.

EXAMPLE 4.23

Problem

Suppose that the chief financial officer of a corporation has created a linear model for the relationship between the company stock and interest rates. When interest rates are at 5%, the company stock has a value of $94. Using the linear model, when interest rates are at 5%, the model predicts the value of the company stock to be $99. Calculate the residual for this data point.

Solution

A residual is the difference between an observed *y*-value and the predicted *y*-value obtained from the linear regression equation:

$$\begin{aligned}\text{Residual} &= (\text{observed } y\text{-value}) - (\text{predicted } y\text{-value})\\ \text{Residual} &= y - \hat{y}\\ \text{Residual} &= 94 - 99 = -5\end{aligned}$$

The goal in regression analysis is to determine the coefficients a and b in the following regression equation:

$$\hat{y} = a + bx$$

Once the (x, y) has been collected, the slope (b) and *y*-intercept (a) can be calculated using the following formulas:

$$\begin{aligned}b &= \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{n\Sigma x^2 - (\Sigma x)^2}\\ a &= \frac{\Sigma y}{n} - b\frac{\Sigma x}{n}\end{aligned}$$

where n refers to the number of data pairs and Σx indicates sum of the *x*-values.

Notice that the formula for the *y*-intercept requires the use of the slope result (b), and thus the slope should be calculated first, and the *y*-intercept should be calculated second.

When making predictions for y, it is always important to plot a scatter diagram first. If the scatterplot indicates that there is a linear relationship between the variables, then it is reasonable to use a best-fit line to make predictions for y, given x is within the domain of *x*-values in the sample data, *but not necessarily for x-values outside that domain*.

USING TECHNOLOGY FOR LINEAR REGRESSION

Typically, technology is used to calculate the best-fit linear model as well as calculate correlation coefficients and scatterplot. Details of using Python for these calculations are provided in Using Python for Correlation and Linear Regression.

Assumptions for Linear Regression

Testing the significance of the correlation coefficient requires that certain assumptions about the data are satisfied. The premise of this test is that the data are a sample of observed points taken from a larger population. We have not examined the entire population because it is not possible or feasible to do so. We are examining the sample to draw a conclusion about whether the linear relationship that we see between x and y data in the sample data provides strong enough evidence that we can conclude that there actually is a linear relationship between x and y data in the population.

The regression line equation that we calculate from the sample data gives the best-fit line for our particular sample. We want to use this best-fit line for the sample as an estimate of the best-fit line for the population (Figure 4.8). Examining the scatterplot and testing the significance of the correlation coefficient helps us determine if it is appropriate to do this.

These are the assumptions underlying the test of significance:

1. There is a linear relationship in the population that models the average value of y for varying values of x. In other words, the expected value of y for each particular value lies on a straight line in the population. (We do not know the equation for the line for the population. Our regression line from the sample is our best estimate of this line in the population.)
2. The *y*-values for any particular *x*-value are normally distributed about the line. This implies that there are more *y*-values scattered closer to the line than are scattered farther away. Assumption (1) implies that these normal distributions are centered on the line: the means of these normal distributions of *y*-values lie on the line.
3. The standard deviations of the population *y*-values about the line are equal for each value of x. In other words, each of these normal distributions of *y*-values has the same shape and spread about the line.
4. The residual errors are mutually independent (no pattern).
5. The data are produced from a well-designed random sample or randomized experiment.

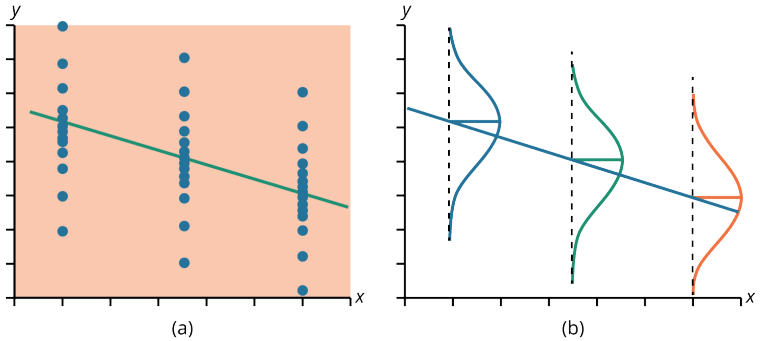


Figure 4.8 Best-Fit Line. Note that the *y*-values for each *x*-value are normally distributed about the line with the same standard deviation. For each *x*-value, the mean of the *y*-values lies on the regression line. More *y*-values lie near the line than are scattered farther away from the line

Using Python for Correlation and Linear Regression

Once a correlation has been deemed significant, a linear regression model is developed. The goal in the regression analysis is to determine the coefficients a and b in the following regression equation:

$$\hat{y} = a + bx$$

These formulas can be quite cumbersome, especially for a significant number of data pairs, and thus software is often used (such as Excel, Python, or R).

Develop a Linear Regression Model

In the following example, Python will be used for the following analysis based on a given dataset of (x, y) data.

- Create a scatterplot.
- Calculate the correlation coefficient.
- Construct the best-fit linear equation.
- Predict values of the dependent variable for given values of the independent variable.

There are various ways to accomplish these tasks in Python, but we will use several functions available within the `scipy` library, such as:

`Pearsonr()` for correlation coefficient

`linregress()` for linear regression model

`plt.scatter()` for scatterplot generation

EXAMPLE 4.24

Problem

A marketing manager is interested in studying the correlation between the amount spent on advertising and revenue at a certain company. Twelve months of data were collected as shown in Table 4.16. (Dollar amounts are in thousands of dollars.)

Use Python to perform the following analysis:

a. Generate a scatterplot.
b. Calculate the correlation coefficient r.
c. Determine if the correlation is significant (use level of significance of 0.05).
d. Construct a linear regression model.
e. Use the model to predict the revenue for a certain month where the amount spent on advertising is $100.

Month	Advertising Spend	Revenue
Jan	49	12210
Feb	145	17590
Mar	57	13215
Apr	153	19200
May	92	14600
Jun	83	14100
Jul	117	17100
Aug	142	18400

Table 4.16 Revenue versus Advertising for a Company (in $000s)

Month	Advertising Spend	Revenue
Sep	69	14100
Oct	106	15500
Nov	109	16300
Dec	121	17020

Table 4.16 Revenue versus Advertising for a Company (in $000s)

Solution

a. The first step in the analysis is to generate a scatterplot. (Recall that scatterplots were also discussed in Scatterplots and Correlation.)
The Python function `plt.scatter()` can be used to generate the scatterplot for the (x, y) data. Note that we consider advertising to be the independent variable (x-data) and revenue to be the dependent variable (y-data) since revenue depends on the amount of advertising.

Here is the Python code to generate the scatterplot:

PYTHON CODE

```
# import the matplotlib library
import matplotlib.pyplot as plt

# define the x-data, which is amount spent on advertising
x = [49, 145, 57, 153, 92, 83, 117, 142, 69, 106, 109, 121]

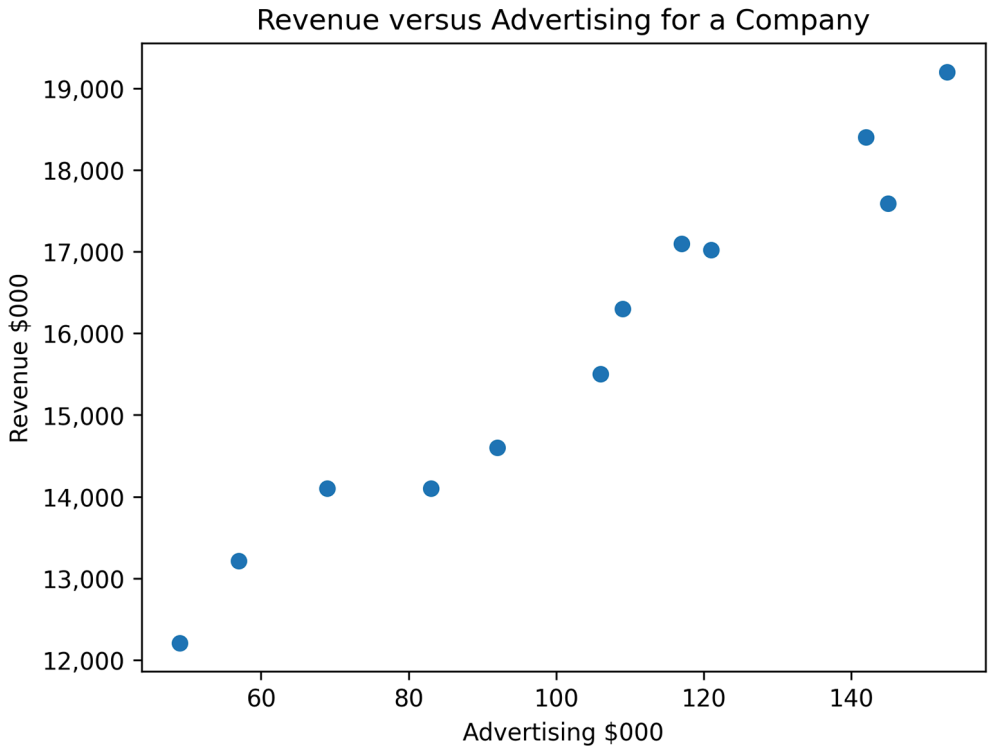
# define the y-data, which is revenue
y = [12210, 17590, 13215, 19200, 14600, 14100, 17100, 18400, 14100, 15500, 16300,
17020]

# use the scatter function to generate a time series graph
plt.scatter(x, y)

# Add a title using the title function
plt.title("Revenue versus Advertising for a Company")

# Add labels to the x and y axes by using xlabel and ylabel functions
plt.xlabel("Advertising $000")
plt.ylabel ("Revenue $000")
```

The resulting output will look like this:



b. The Python function `pearsonr()` can be used to generate the correlation coefficient, *r*.

Here is the Python code to calculate the correlation coefficient:

PYTHON CODE

```
# import libraries
import numpy as np
from scipy.stats  import pearsonr

# define the x-data, which is amount spent on advertising
x = [49, 145, 57, 153, 92, 83, 117, 142, 69, 106, 109, 121]

# define the y-data, which is revenue
y = [12210, 17590, 13215, 19200, 14600, 14100, 17100, 18400, 14100, 15500, 16300,
17020]

#pearson r returns both value of r and corresponding p-value
r, p = pearsonr(x, y)

#print value of r, rounded to 3 decimal places
```

```
print("Correlation coefficient: ", round(r, 3))
```

The resulting output will look like this:

```
Correlation coefficient: 0.981
This value of r = 0.981 indicates a strong, positive correlation between
advertising spend and revenue.
```

c. When using a level of significance of 0.05, if $|r| \geq \frac{2}{\sqrt{n}}$, then this implies that the correlation between the two variables demonstrates a linear relationship that is statistically significant at approximately the 0.05 level of significance. In this example, we check if $|0.981|$ is greater than or equal to $\frac{2}{\sqrt{n}}$ where $n = 12$. Since $\frac{2}{\sqrt{12}}$ is approximately 0.577, this indicates that the *r*-value of 0.981 is greater than $\frac{2}{\sqrt{n}}$, and thus the correlation is deemed significant.

d. The linear regression model will be of the form:

$$\hat{y} = a + bx$$

To determine the slope and *y*-intercept of the regression model, the Python function `linregress()` can be used.
This function takes the x and y data as inputs and provides the slope and intercept as well as other regression-related output.

Here is the Python code to generate the slope and *y*-intercept of the best-fit line:

PYTHON CODE

```
# import libraries
import numpy as np
from scipy.stats  import linregress

# define the x-data, which is amount spent on advertising
x = [49, 145, 57, 153, 92, 83, 117, 142, 69, 106, 109, 121]

# define the y-data, which is revenue
y = [12210, 17590, 13215, 19200, 14600, 14100, 17100, 18400, 14100, 15500, 16300,
17020]

#pearson r returns both value of r and corresponding p-value
slope, intercept, r_value, p_value, std_err  = linregress(x, y)

print ("slope  =", slope)
print("intercept = ", intercept)
```

The resulting output will look like this:

```
slope = 61.79775818816664
intercept = 9376.698881009072
```

Based on this Python equation, the regression equation can be written as

$$\begin{aligned}\hat{y} &= a + bx \\ \hat{y} &= 9376.7 + 61.8x\end{aligned}$$

where x represents the amount spent on advertising (in thousands of dollars) and y represents the amount of revenue (in thousands of dollars).

It is also useful to plot this best-fit line superimposed on the scatterplot to show the strong correlation between advertising spend and revenue, as follows (see Figure 4.9):

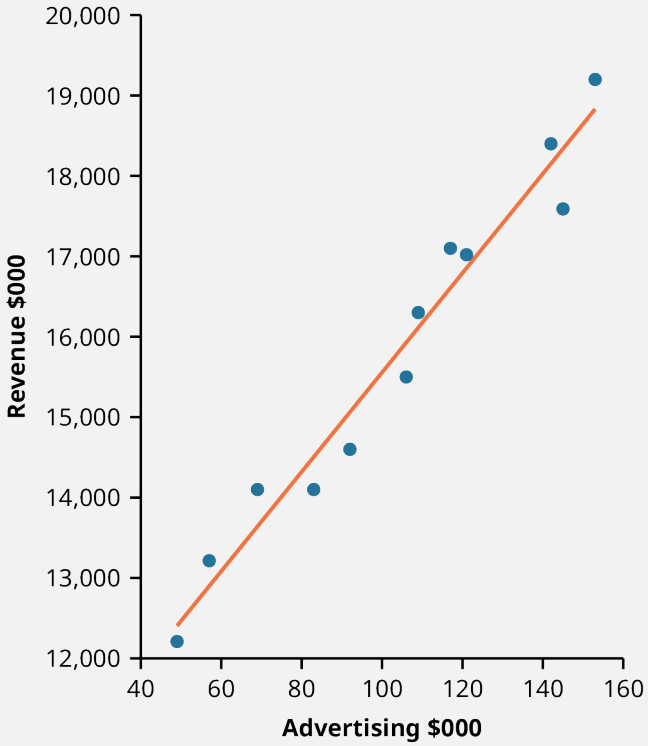


Figure 4.9 Best-Fit Line Superimposed on the Scatterplot

e. The regression equation is: $\hat{y} = 9{,}376.7 + 61.8x$.
The given amount of advertising spend of $100 corresponds to a given *x*-value, so replace "x" with 100 in this regression equation to generate a predicted revenue:

$$\begin{aligned}\hat{y} &= 9376.7 + 61.8x \\ \hat{y} &= 9376.7 + 61.8(100) \\ \hat{y} &= 15556.7\end{aligned}$$

Since both advertising and revenue are expressed in thousands of dollars, the conclusion is that if advertising spend is $100,000, the predicted revenue for the company will be $15,556,700.

Interpret and Apply the Slope and *y*-Intercept

The slope of the line, b, describes how changes in the variables are related. It is important to interpret the slope of the line in the context of the situation represented by the data. You should be able to write a sentence interpreting the slope in plain English.

Interpretation of the Slope

The slope of the best-fit line tells us how the dependent variable (y) changes for every one-unit increase in the independent (x) variable, on average.

In the previous example, the linear regression model for the monthly amount spent on advertising and the monthly revenue for a company for 12 months was generated as follows:

$$\begin{aligned} \hat{y} &= a + bx \\ \hat{y} &= 9376.7 + 61.8x \end{aligned}$$

Since the slope was determined to be 61.8, the company can interpret this to mean that for every $1,000 spent on advertising, on average, this will result in an increase in revenues of $61,800.

The intercept of the regression equation is the corresponding *y*-value when $x = 0$.

Interpretation of the Intercept

The intercept of the best-fit line tells us the expected mean value of y in the case where the x-variable is equal to zero.

However, in many scenarios it may not make sense to have the x-variable equal zero, and in these cases, the intercept does not have any meaning in the context of the problem. In other examples, the *x*-value of zero is outside the range of the x-data that was collected. In this case, we should not assign any interpretation to the *y*-intercept.

In the previous example, the range of data collected for the x-variable was from $49,000 to $153,000 spent per month on advertising. Since this interval does not include an *x*-value of zero, we would not provide any interpretation for the intercept.

4.4 Analysis of Variance (ANOVA)

Learning Outcomes

By the end of this section, you should be able to:

- 4.4.1 Set up and apply one-way analysis of variance hypothesis test.
- 4.4.2 Use Python to conduct one-way analysis of variance.

In Testing Claims Based on One Sample, we reviewed methods to conduct hypothesis testing for two means; however, there are scenarios where a researcher might want to compare three or more means to determine if any of the means are different from one another. For example, a medical researcher might want to compare three different pain relief medications to compare the average time to provide relief from a migraine headache. **Analysis of variance (ANOVA)** is a statistical method that allows a researcher to compare three or more means and determine if the means are all statistically the same or if at least one mean is different from the others. The **one-way ANOVA** focuses on one independent variable. For two independent variables, a method called "two-way ANOVA" is applicable, but this method is beyond the scope of this text. We explore the one-way ANOVA next.

One-Way ANOVA

To compare three or more means, the sample sizes for each group can be different sample sizes (samples sizes do not need to be identical). For one-way ANOVA hypothesis testing, we will follow the same general

outline of steps for hypothesis testing that was discussed in Hypothesis Testing.

The first step will be to write the null and alternative hypotheses. For one-way ANOVA, the null and alternative hypotheses are always stated as the following:

H_0: All population means are equal: $\mu_1 = \mu_2 = \mu_3 = \ldots$

H_a: At least one population mean is different from the others.

Here are the requirements to use this procedure:

a. The samples are random and selected from approximately normal distributions.
b. The samples are independent of one another.
c. The population variances are approximately equal.

In this discussion, we assume that the population variances are not equal.

When these requirements are met, the *F*-distribution is used as the basis for conducting the hypothesis test.

The ***F*-distribution** is a skewed distribution (skewed to the right), and the shape of the distribution depends on two different degrees of freedom, referred to as degrees of freedom for the numerator and degrees of freedom for the denominator.

Figure 4.10 shows the typical shape of the F-distribution:

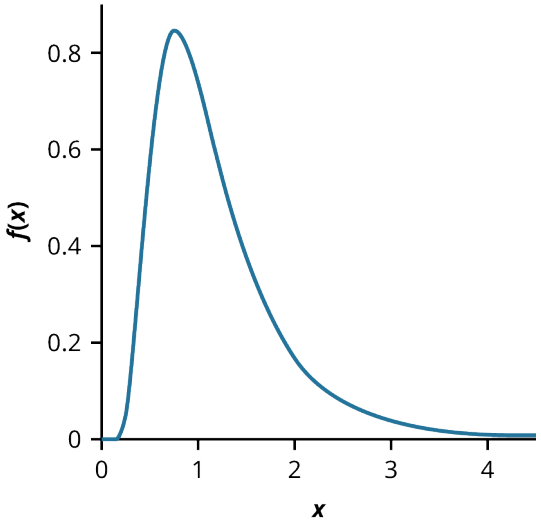


Figure 4.10 Shape of the *F*-Distribution

The test statistic for this hypothesis test will be the ratio of two variances, namely the ratio of the variance between samples to the ratio of the variances within the samples.

$$\text{Test Statistic} = \frac{\text{Variance between samples}}{\text{Variance within samples}}$$

The numerator of the test statistic (variance between samples) is sometimes referred to as variation due to treatment or explained variation.

The denominator of the test statistic (variance within samples) is sometimes referred to as variation due to error, or unexplained variation.

The details for a manual calculation of the test statistic are provided next; however, it is very common to use software for ANOVA analysis.

For the purposes of this discussion, we will assume we are comparing three means, but keep in mind the ANOVA method can be applied to three or more means.

1. Find the mean for each of the three samples. Label these means as $\bar{x}_1, \bar{x}_2, \bar{x}_3$. Find the variance for each of the three samples. Label these variances as s_1^2, s_2^2, s_3^2.
2. Find the grand mean (label this as $\bar{\bar{x}}$). This is the sum of all the data values from the three samples, divided by the overall sample size.
3. Calculate the quantity called "sum of squares between (SSB)" according to the following formula:
$$SSB = n_1\left(\bar{x}_1 - \bar{\bar{x}}\right)^2 + n_2\left(\bar{x}_2 - \bar{\bar{x}}\right)^2 + n_3\left(\bar{x}_3 - \bar{\bar{x}}\right)^2$$
4. Calculate the quantity called "sum of squares within (SSW)" according to the following formula:
$$SSW = \left(n_1 - 1\right)s_1^2 + \left(n_2 - 1\right)s_2^2 + \left(n_3 - 1\right)s_3^2$$
5. Calculate the degrees of freedom for the numerator (df_n) and degrees of freedom for the denominator (df_d).
   For degrees of freedom for the numerator, this is calculated as the number of groups minus 1. For example, if a medical researcher is comparing three different pain relief medications to compare the average time to provide relief from a migraine headache, there are three groups, and thus the degrees of freedom for the numerator would be $3 - 1 = 2$.
   For degrees of freedom for the denominator, this is calculated as the total of the sample sizes minus the number of groups.
6. Calculate the quantity called "mean square between (MSB)" according to the following formula:
$$MSB = \frac{SSB}{df_n}$$
7. Calculate the quantity called "mean square within (MSW)" according to the following formula:
$$MSW = \frac{SSW}{df_d}$$
8. Calculate the test statistic (F), according to the following formula:
$$F = \frac{MSB}{MSW}$$

Once the test statistic is obtained, the *p*-value is calculated as the area to the right of the test statistic under the F-distribution curve (note that the ANOVA hypothesis test is always considered a "right-tail" test).

Usually, the results of these computations are organized in an ANOVA summary table like Table 4.17.

Variation	Sum of Squares	Degrees of Freedom	Mean Squares	F Test Statistic	*p*-value
Between	SSB	df_n	MSB	$\frac{MSB}{MSW}$	Area to the right of the F test statistic
Within	SSW	df_d	MSW		N/A

Table 4.17 ANOVA Summary Table

Using Python for One-Way ANOVA

As mentioned earlier, due to the complexity of these calculations, technology is typically used to calculate the test statistic and *p*-value.

Using Python, the `f_oneway()` function is provided as part of the `scipy` library, and this function provides both the test statistic and *p*-value for a one-way ANOVA hypothesis test.

The syntax is to call the function with the data arrays as arguments, as in:

```
f_oneway(Array1, Array2, Array3, …)
```

The function returns both the F test statistic and the *p*-value for the one-way ANOVA hypothesis test.

The details of this Python function are shown in Example 4.25.

Here is a step-by-step example to illustrate this ANOVA hypothesis testing process.

EXAMPLE 4.25

Problem

An airline manager claims that the average arrival delays for flights from Boston to Dallas are the same for three airlines: Airlines A, B, and C. The following arrival delay data (in minutes) is collected on samples of flights for the three airlines (Table 4.18):

Airline A	Airline B	Airline C
8	13	11
12	0	15
7	2	9
9	7	11
11	6	12
4	5	N/A

Table 4.18 Arrival Delay Data (in Minutes) for Three Airlines

Use this sample data to test the claim that the average arrival delays are the same for three airlines: Airlines A, B, and C. Use a level of significance of 0.05 for this analysis. Assume the samples are independent and taken from approximately normal distributions and each population has approximately the same variance.

Solution

The first step in the hypothesis testing procedure is to write the null and alternative hypothesis based on the claim.

For an ANOVA hypothesis test, the setup of the null and alternative hypotheses is shown here. Notice that the claim mentions "the average arrival delays are the same for three airlines," which corresponds to the null hypothesis.

H_0: All three population means are equal: $\mu_A = \mu_B = \mu_C$ [Claim]

H_a: At least one population mean is different from the others.

The following Python program generates the corresponding F test statistic and *p*-value for this ANOVA analysis:

PYTHON CODE

```
# import libraries
import numpy as np
from scipy.stats  import f_oneway

# enter airline arrival data

AirlineA = np.array([8, 12, 7, 9, 11, 4])
AirlineB = np.array([13, 0, 2, 7, 6, 5])
AirlineC = np.array([11, 15, 9, 11, 12])

#f_oneway function returns both the test statistic and p-value
f_oneway(AirlineA, AirlineB, AirlineC)
```

The resulting output will look like this:

```
F_onewayResult(statistic=4.388264306955828, pvalue=0.03314999857295516)
```

From this Python output, the test statistic is $F = 4.388$ and the *p*-value is 0.033.

The final step in the hypothesis testing procedure is to come to a final decision regarding the null hypothesis. This is accomplished by comparing the *p*-value with the level of significance. In this example, the *p*-value is 0.033 and the level of significance is 0.05: since the *p*-value $<$ level of significance, the decision is to "reject the null hypothesis."

Conclusion: The decision is to reject the null hypothesis. The claim corresponds to the null hypothesis. This scenario corresponds to the first row in Table 4.18 and thus the conclusion is: There is enough evidence to reject the claim that the average arrival delays are the same for three airlines. This implies that at least one of the means is different from the others.

Note: The ANOVA test method does not specify which mean is different from the others. Further statistical analysis is needed to make this determination.

Key Terms

alternative hypothesis a complementary statement regarding an unknown population parameter used in hypothesis testing

analysis of variance (ANOVA) statistical method to compare three or more means and determine if the means are all statistically the same or if at least one mean is different from the others

best-fit linear equation an equation of the form $\hat{y} = a + bx$ that provides the best-fit straight line to the (x, y) data points

bivariate data data collected on two variables where the data values are paired with one another

bootstrapping a method to construct a confidence interval that is based on repeated sampling and does not rely on any assumptions regarding the underlying distribution

central limit theorem describes the relationship between the sample distribution of sample means and the underlying population

confidence interval an interval where sample data is used to provide an estimate for a population parameter

confidence level the probability that the interval estimate will contain the population parameter, given that the estimation process on the parameter is repeated over and over

correlation a measure of association between two numeric variables

correlation analysis a statistical method used to evaluate and quantify the strength and direction of the linear relationship between two quantitative variables

correlation coefficient a measure of the strength and direction of the linear relationship between two variables

critical value *z*-score that cuts off an area under the normal curve corresponding to a specified confidence level

dependent samples samples from one population that can be paired or matched to the samples taken from the second population

dependent variable in correlation analysis, the variable being studied or measured; the dependent variable is the outcome that is measured or observed to determine the impact of changes in the independent variable

F distribution a skewed probability distribution that arises in statistical hypothesis testing, such as ANOVA analysis

hypothesis testing a statistical method to test claims regarding population parameters using sample data

independent samples the sample from one population that is not related to the sample taken from the second population

independent variable in correlation analysis, the variable that is manipulated or changed in an experiment or study; the value of the independent variable is controlled or chosen by the experimenter to observe its effect on the dependent variable

inferential statistics statistical methods that allow researchers to infer or generalize observations from samples to the larger population from which they were selected

least squares method a method used in linear regression that generates a straight line fit to the data values such that the sum of the squares of the residual is the least sum possible

level of significance (a) the maximum allowed probability of making a Type I error; the level of significance is the probability value used to determine when the sample data indicates significant evidence against the null hypothesis

linear correlation a measure of the association between two variables that exhibit an approximate straight-line fit when plotted on a scatterplot

margin of error an indication of the maximum error of the estimate

matched pairs samples from one population that can be paired or matched to the samples taken from the second population

method of least squares a mathematical method to generate a linear equation that is the “best fit” to the

points on the scatterplot in the sense that the line minimizes the differences between the predicted values and observed values for y

modeling the process of creating a mathematical representation that describes the relationship between different variables in a dataset; the model is then used to understand, explain, and predict the behavior of the data

nonparametric methods statistical methods that do not rely on any assumptions regarding the underlying distribution

null hypothesis statement of no effect or no change in the population

***p*-value** the probability of obtaining a sample statistic with a value as extreme as (or more extreme than) the value determined by the sample data under the assumption that the null hypothesis is true

parametric methods statistical methods that assume a specific form for the underlying distribution

point estimate a sample statistic used to estimate a population parameter

prediction a forecast for the dependent variable based on a specific value of the independent variable generated using the linear model

proportion a measure that expresses the relationship between a part and the whole; a proportion represents the fraction or percentage of a dataset that exhibits a particular characteristic or falls into a specific category

regression analysis a statistical technique used to model the relationship between a dependent variable and one or more independent variables

residual the difference between an observed *y*-value and the predicted *y*-value obtained from the linear regression equation

sample mean a point estimate for the unknown population mean chosen as the most unbiased estimate of the population

sample proportion chosen as the most unbiased estimate of the population, calculated as the number of successes divided by the sample size: $p = \frac{x}{n}$

sample statistic a numerical summary or measure that describes a characteristic of a sample, such as a sample mean or sample proportion

sampling distribution a probability distribution of a sample statistic based on all possible random samples of a certain size from a population or the distribution of a statistic (such as the mean) that would result from taking random samples from the same population repeatedly and calculating the statistic for each sample

scatterplot (or scatter diagram) graphical display that shows values of the independent variable plotted on the x-axis and values of the dependent variable plotted on the y-axis

standard error of the mean the standard deviation of the sample mean, calculated as the population standard deviation divided by the square root of the sample size

standardized test statistic a numerical measure that describes how many standard deviations a particular value is from the mean of a distribution; a standardized test statistic is typically used to assess whether an observed sample statistic is significantly different from what would be expected under a null hypothesis

t-distribution a bell-shaped, symmetric distribution similar to the normal distribution, though the t-distribution has “thicker tails” as compared to the normal distribution

test statistic a numerical value used to assess the strength of evidence against a null hypothesis, calculated from sample data that is used in hypothesis testing

Type I error an error made in hypothesis testing where a researcher rejects the null hypothesis when in fact the null hypothesis is actually true

Type II error an error made in hypothesis testing where a researcher fails to reject the null hypothesis when the null hypothesis is actually false

unbiased estimator a statistic that provides a valid estimate for the corresponding population parameter without overestimating or underestimating the parameter

variable a characteristic or attribute that can be measured or observed.

Group Project

Project A: Confidence Intervals

Conduct a survey of students in your class or at your school to collect data on the number of hours worked per week by one or two predefined sociodemographic characteristics, such as gender, age, geographic neighborhood of residence, first-generation college student, etc. As a group, create 90% confidence intervals for the average number of hours worked for each category of students. Compare the confidence intervals and come to any relevant conclusions for the average number of hours worked by each category of students.

Project B: Hypothesis Testing

As a group, develop a hypothesis for the proportion of white cars in your city. To collect sample data, go to your school parking lot or a nearby parking lot of some kind and determine the sample proportion of white cars. Then, use this data to conduct a hypothesis test for your original hypothesis.

Project C: Correlation and Regression Analysis

For each student, collect data on shoe length and person's height. Create a scatterplot of this (x, y) data and calculate the correlation coefficient. Test the correlation coefficient for significance. If the correlation is significant, develop a linear regression model to predict a person's height based on shoe length.

As a second assignment on the same subject, students should research a certain model car, such as Honda Accord or Toyota Camry. Search websites for used car prices for the selected model and record (x, y) data where x is the age of the car and y is the price of the car. Create a scatterplot of this (x, y) data and calculate the correlation coefficient. Test the correlation coefficient for significance. If the correlation is significant, develop a linear regression model to predict a car's price based on age of the vehicle.

Quantitative Problems

1. An automotive designer wants to create a 90% confidence interval for the mean length of a body panel on a new model electric vehicle. From a sample of 35 cars, the sample mean length is 37 cm and the sample standard deviation is 2.8 cm.
   a. Create a 90% confidence interval for the population mean length of the body panel using Python.
   b. Create a 90% confidence interval for the population mean length of the body panel using Excel.
   c. Create a 95% confidence interval for the population mean length of the body panel using Python.
   d. Create a 95% confidence interval for the population mean length of the body panel using Excel.
   e. Which confidence interval is narrower, the 90% or 95% confidence interval?

2. A political candidate wants to conduct a survey to generate a 95% confidence interval for the proportion of voters planning to vote for this candidate. Use a margin of error of 3% and assume a prior estimate for the proportion of voters planning to vote for this candidate is 58%. What sample size should be used for this survey?

3. A medical researcher wants to test the claim that the mean time for ibuprofen to relieve body pain is 12 minutes. A sample of 25 participants are put in a medical study, and data is collected on the time it takes for ibuprofen to relieve body pain. The sample mean time is 13.2 minutes with a sample standard deviation of 3.7 minutes.
   a. Calculate the *p*-value for this hypothesis test using Python.
   b. Calculate the *p*-value for this hypothesis test using Excel.
   c. Test the researcher's claim at a 5% level of significance.

4. A consumer group is studying the percentage of drivers who wear seat belts. A local police agency states that the percentage of drivers wearing seat belts is at least 70%. To test the claim, the consumer group selects a random sample of 150 drivers and finds that 117 are wearing seat belts.

a. Calculate the *p*-value for this hypothesis test using Python.
b. Calculate the *p*-value for this hypothesis test using Excel.
c. Use a level of significance of 0.10 to test the claim made by the local police agency.

5. A local newspaper is comparing the costs of a meal at two local fast-food restaurants. The newspaper claims that there is no difference in the average price of a meal at the two restaurants. A sample of diners at each of the two restaurants is taken with the following results:

Restaurant A: Sample mean price of a meal $= \$17.55$, sample standard deviation $= \$7.45$, sample size $= 15$
Restaurant B: Sample mean price of a meal $= \$22.75$, sample standard deviation $= \$8.10$, sample size $= 18$

a. Calculate the *p*-value for this hypothesis test using Python.
b. Calculate the *p*-value for this hypothesis test using Excel.
c. Use a level of significance of 0.10 to test the claim made by the newspaper. Assume population variances are not equal.

6. A health official claims the proportion of men smokers is greater than the proportion of women smokers. To test the claim, the following data was collected:

Men: Sample size $= 150$, Number of smokers $= 41$
Women: Sample size $= 140$, Number of smokers $= 31$

a. Calculate the *p*-value for this hypothesis test using Python
b. Calculate the *p*-value for this hypothesis test using Excel.
c. Use a level of significance of 0.05 to test the claim made by the health official.

7. A software company has been tracking revenues versus cash flow in recent years, and the data is shown in the table below. Consider cash flow to be the dependent variable.

Revenues ($000s)	Cash Flow ($000s)
212	81
239	84
218	81
297	99
287	91

Table 4.19 Software Co. Revenue vs. Cash Flow

a. Calculate the correlation coefficient for this data (all dollar amounts are in thousands) using Python.
b. Calculate the correlation coefficient using Excel.
c. Determine if the correlation is significant using a significance level of 0.05.
d. Construct the best-fit linear equation for this dataset.
e. Predict the cash flow when revenue is 250.
f. Calculate the residual for the revenue value of 239.

8. A human resources administrator calculates the correlation coefficient for salary versus years of experience as 0.68. The dataset contains 10 (x, y) data points. Determine if the correlation is significant or not significant at the 0.05 level of significance.

9. A biologist is comparing three treatments of light intensity on plant growth and is interested to know if the average plant growth is different for any of the three treatments.
Sample data is collected on plant growth (in centimeters) for the three treatments, and data is summarized in the table below.
Use an ANOVA hypothesis test method with 0.05 level of significance.

Treatment A	Treatment B	Treatment C
4.3	3.9	2.8
3.7	4.1	2.9
3.1	3.6	3.5
3.4	3.5	3.9
4.1	4.3	4.4
4.0	4.4	4.3
N/A	n/a	4.1

Table 4.20

a. Calculate the *p*-value for this hypothesis test using Python.
b. Determine if the average plant growth is different for any of the three treatments.



5

Time Series and Forecasting

Figure 5.1 Time series analysis involves the examination of data points collected at specific time intervals; it is crucial for identifying trends and patterns that inform predictions and strategic decisions. (credit: modification of work "Bitcoin steigend" by Tim Reckmann/Flickr, CC BY 2.0)

Chapter Outline

5.1 Introduction to Time Series Analysis
5.2 Components of Time Series Analysis
5.3 Time Series Forecasting Methods
5.4 Forecast Evaluation Methods

Introduction

The future can be very unpredictable. Wouldn't it be wonderful to know whether a given stock price will rise or fall in value over the next year, which baseball team will win the next World Series, or if it might rain on the particular day that you happen to be planning to go to the beach? A manufacturer may want to know how the demand for their products changes over time. Politicians would like to know how high their favorability with voters will be on election day. While we cannot see the future, there may be patterns that point to certain outcomes as being more likely than others. In this chapter we explore the concept of time series and develop the necessary tools to discover trends and repeating patterns hidden within time series data to forecast future values of the series and quantify the uncertainty of those predictions.

5.1 Introduction to Time Series Analysis

Learning Outcomes

By the end of this section, you should be able to:

- 5.1.1 Define time series data.
- 5.1.2 Identify examples of time series data in real-world applications.

Data can change over time, and it is important to be able to identify the ways that the data may change in order to make reasonable forecasts of future data. Does the data seem to be steadily rising or falling overall? Are there ups and downs that occur at regular intervals? Maybe there are predictable fluctuations in the data

that are tied closely to the seasons of the year or multiyear business cycles.

Time series analysis allows for the examination of data points collected or recorded at specific time intervals, enabling the identification of trends, patterns, and seasonal variations crucial for making informed predictions and decisions across a variety of industries. It is widely used in fields such as business, finance, economics, environmental and weather science, information science, and many other domains in which data is collected over time.

Some examples of the areas in which time series analysis is used to solve real-world problems include the following:

- **Forecasting and prediction.** Making predictions about future (unknown) data based on current and historical data may be the most important use of time series analysis. Businesses use such methods to drive strategy decisions and increase future revenues. Policymakers rely on forecasting to predict demographic changes that might influence elections and the need for public services.
- **Risk management.** Time series analysis can help to quantify risk. For example, insurance companies may forecast trends that affect the number and total dollar amounts of claims homeowners may make in the future, which directly affects insurance premiums.
- **Anomaly detection.** If data seems to deviate wildly from predictions, then this could point to a major change in some feature of the data or uncover a hidden factor that will need to be analyzed more carefully. For example, if a family's spending patterns increase significantly beyond predicted levels, this could indicate either a sudden increase in household earnings or potential fraud.
- **Health care and epidemiology.** Time series analysis is often used to monitor patient's vitals, study how cancer progresses in individuals from various populations, track disease outbreaks, and predict future need for health care resources such as vaccines or additional beds in hospitals, among other things.

Time series analysis involves a number of statistical and computational techniques to identify patterns, make predictions, or gain insights into the data. The basic steps in time series analysis closely follow the data science cycle as described in What Are Data and Data Science? and are as follows:

1. **Problem Identification/Definition.** What is it that you need to know? Do you want to predict sales of a product over the next few years? Are you interested in studying the patterns of sunspots over time? Is your investment in a stock likely to lead to profit or loss?
2. **Data Gathering.** Once you have a question in mind, now you need data to analyze. This step can be very time-consuming if a suitable dataset does not already exist. In the best-case scenario, others have already collected data that you might be able to use. In the worst-case scenario, you would need to arrange for data collection by surveys, experiments, observations, or other means. Note that data collection is covered in some detail in Overview of Data Collection Methods.
3. **Data Cleaning.** Real-world data rarely comes in a form that is immediately able to be analyzed. Most often, the data must be cleaned, which includes dealing with missing data in some way and formatting the data appropriately so that a computer can understand it. This topic is covered in Web Scraping and Social Media Data Collection.
4. **Model Selection.** The next step is to fit the data to a model. Indeed, you may find that you must create numerous models and variations on those models before you are satisfied with the result.
5. **Model Evaluation.** This is perhaps the most important step. Any model that you create will be limited in its usefulness if it is not accurate enough. The model should include some indication of how confident you can be in its predictions. Moreover, when new observations become available, the model's forecasts should be checked against them and adjusted as necessary to better fit the data.

What Is a Time Series?

Any set of data that consists of numerical measurements of the same variable collected and organized according to regular time intervals may be regarded as **time series** data. For example, see Table 5.1 on the

S&P 500, an aggregate index of the top 500 publicly traded companies on the stock market, for the past several years. Here, the variable is the S&P index value at the last trading day of the year (*point-in-time* data).

Year	S&P Index at Year-End
2013	1848.36
2014	2058.90
2015	2043.94
2016	2238.83
2017	2673.61
2018	2506.85
2019	3230.78
2020	3756.07
2021	4766.18
2022	3839.50
2023	4769.83

Table 5.1 S&P Index Values from 2013 through 2023 (source: https://www.nasdaq.com/market-activity/index/spx/historical)

While the table is informative, it is not particularly easy to use to find trends or make predictions. A visualization would be better. Figure 5.2 displays the time series data using a time series chart (essentially a line chart). Notice that while the general trend is increasing, there are also dips in the years 2018 and 2022.

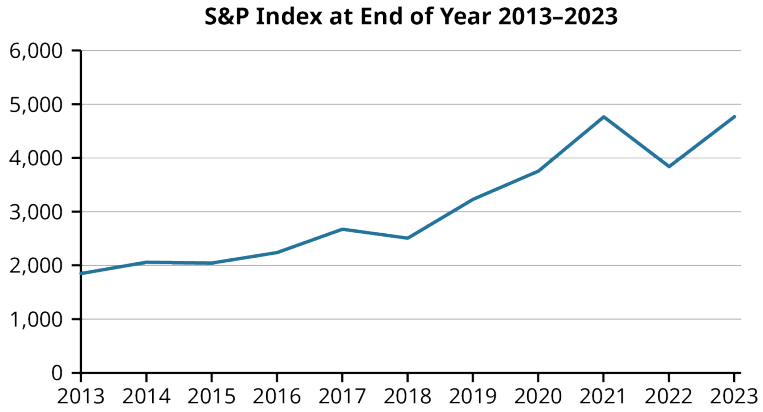


Figure 5.2 Line Chart (or Time Series Chart) of the S&P 500 Index. The chart makes it easier to spot the general upward trend over the past decade. (data source: adapted from S&P 500 [SPX] Historical Data)

The example shown in Figure 5.3 is a simple time series with only a single measure (S&P 500 index value) tracked over time. As long as we keep in mind that this data represents yearly values of the S&P 500 index starting in the year 2013, it is more efficient to consider the values alone as an ordered list.

1848.36, 2058.9, 2043.94, 2238.83, 2673.61, 2506.85, 3230.78, 3756.07, 4766.18, 3839.5, 4769.83

In mathematics, an ordered list of numbers is called a **sequence**. The individual values of the sequence are called **terms**. An abstract sequence may be denoted by (x_n) or $(x_n)_{1 \leq n \leq N}$. In both notations, n represents the index of each value of the sequence, while the latter notation also specifies the range of index values for the sequence (n takes on all index values from 1 to N). That is,

$$(x_n)_{1\le n\le N} = (x_1, x_2, x_3, x_4, \ldots, x_N)$$

We will use the standard term *time series* to refer to a sequence of time-labeled data and use the term *sequence* when discussing the terms of the time series as an ordered list of values.

Not every sequence of data points is a time series, though. The set of current populations of the countries of the world is not a time series because the data are not measured at different times. However, if we focused on a single country and tracked its population year by year, then that would be a time series. What about a list of most popular baby names by year? While there is a time component, the data is not numerical, and so this would not fall under the category of time series. On the other hand, information about how many babies are named "Jennifer" each year would constitute time series data.

Typically, we assume that time series data are taken at equal time intervals and that there are no missing values. The terms of a time series tend to depend on previous terms in some way; otherwise, it may be impossible to make any predictions about future terms.

Time Series Models

A **time series model** is a function, algorithm, or method for finding, approximating, or predicting the values of a given time series. The basic idea behind time series models is that previous values should provide some indication as to how future values behave. In other words, there is some kind of function that takes the previous values of the time series as input and produces the next value as output:

$$x_{n+1} = f(x_n, x_{n-1}, x_{n-2}, \ldots, x_1)$$

However, in all but the most ideal situations, a function that predicts the next value of the time series with perfect accuracy does not exist. Random variation and other factors not accounted for in the model will produce **error**, which is defined as the extent to which predictions differ from actual observations. Thus, we should always incorporate an error term (often denoted by the Greek letter ε, called "epsilon") into the model. Moreover, instead of expecting the model to produce the next term exactly, the model generates predicted values. Often, the predicted values of a times series are denoted by $(\hat{x}_n)$ to distinguish them from the actual values (x_n). Thus,

$$\hat{x}_{n+1} = f(x_n, x_{n-1}, x_{n-2}, \ldots, x_1, \varepsilon)$$

In Time Series Forecasting Methods and Forecast Evaluation Methods, we will delve into the details of building time series models and accounting for error.

Time Series Forecasting

Typically, the goal of time series analysis is to make predictions, or *extrapolations*, about future values of the time series, a process known as **forecasting**. As a general rule, the accuracy of a forecast *decreases* as predictions are made further into the future. When future predictions become no more accurate than tossing a coin or rolling a die, then forecasting at the point or beyond becomes ineffective. In practice, time series models are updated regularly to accommodate new data.

Depending on the situation and the nature of the data, there are many different ways to forecast future data. The simplest of all methods, which is known as the **naïve** or **flat forecasting method**, is to use the most recent value as the best guess for the next value. For example, since the S&P 500 had a value of 3,839.5 at the end of 2022, it is reasonable to assume that the value will be relatively close to 3,839.5 at the end of 2023. Note, this would correspond to the time series model, $\hat{x}_{n+1} = x_n$. The naïve method has only limited use in practice.

Instead of using only the last observed value to predict the next one, a better approach might be to take into consideration a number of values, $x_n, x_{n-1}, x_{n-2}, \ldots$, to find the estimate $\hat{x}_{n+1}$. One might average the last T values together (for some predefined value of T). This is known as a *simple moving average*, which will be

defined explicitly in Components of Time Series Analysis. For now, let's illustrate the idea intuitively. Suppose we used the average of the most recent $T = 3$ terms to estimate the next term. The time series model would be:

$$\hat{x}_{n+1} = \frac{x_n + x_{n-1} + x_{n-2}}{3}$$

Based on the data in Table 5.1, the prediction for the S&P index value at the end of 2023 would work out as follows.

$$\frac{4{,}769.83 + 3{,}839.5 + 4{,}766.18}{3} = 4{,}458.5$$

Another simple method of forecasting is to fit the data to a linear regression model and then use the regression model to predict future values. (See Linear Regression for details on linear regression.) Linear regression does a better job at capturing the overall direction of the data compared to using only the last data point. On the other hand, linear regression will not be able to model more subtle structures in the data such as cyclic patterns. Moreover, there is a hidden assumption that the data rises or falls more or less uniformly throughout the period we would like to predict, which is often not a valid assumption to make.

EXAMPLE 5.1

Problem

Find a linear regression model for the data from Table 5.1 and use it to forecast the value of the S&P at the end of the years 2024 and 2025.

Solution

Using standard statistical software, the linear regression is found to be $\hat{y} = 304.437t - 611{,}288$, where t is the year. Alternatively, in Excel, the regression line can be added to the line graph using the "Trendline" feature, with option "Linear." A graph of the regression line is shown in Figure 5.3.

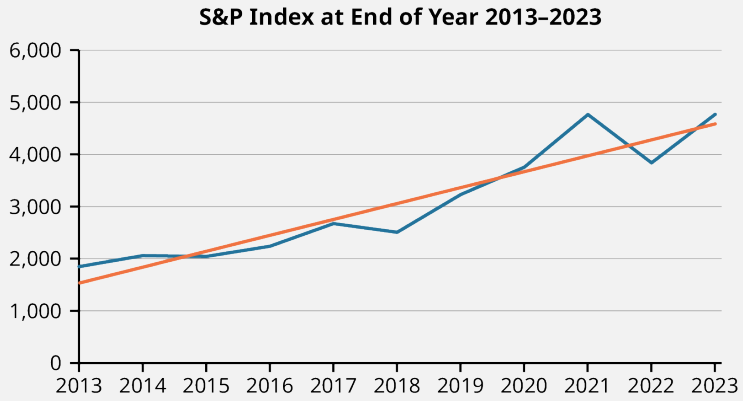


Figure 5.3 Line Chart of the S&P 500 Index with the Linear Regression (data source: adapted from S&P 500 [SPX] Historical Data)

The forecast for the end of year 2024 is $\hat{y} = 304.437(2024) - 611{,}288 = 4{,}892.5$.

Similarly, the prediction for 2025 is $\hat{y} = 304.437(2025) - 611{,}288 = 5{,}196.9$.

In this chapter, we will develop more sophisticated tools that can detect patterns that averages or linear models simply cannot find, including autoregressive models, moving averages, and the autoregressive integrated moving average (ARIMA) model. However, creating a fancy model for time series data is only part of the process. An essential part of the process is testing and evaluating your model. Thus, we will also explore measures of error and uncertainty that can determine how good the model is and how accurate its forecasts

might be.

Examples of Time Series Data

Time series data is typically measured or observed at regular time intervals, such as daily, weekly, monthly, or annually. While we do hope to find some structure, pattern, or overall trend in the data, there is no requirement that time series data follows a simple, predictable pattern. Thus, we should always be careful when using any statistical or predictive models to analyze time series data. Regular validation and testing of the chosen models against new data are essential to ensure their reliability and effectiveness, which are concepts that we will explore in later sections.

Time series data could represent short-term, frequent observations, such as minute-by-minute monitoring of stock prices. It makes sense to measure stock prices at very short time intervals because they tend to be so volatile. Data is considered to be *volatile* or displaying high **volatility** if the individual data points show significant fluctuations from the mean, typically due to factors that are difficult to analyze or predict. Volatility can be measured using statistics such as variance and standard deviation; however, it is important to realize that even the level of volatility can change over time. Measuring data at more frequent intervals does help to reduce or control volatility, but it is still difficult to forecast volatile data to make future predictions.

Other sets of data represent more long-term measurements. For example, meteorologists may collect weather data, including temperature, atmospheric pressure, rainfall, wind speed and direction, in a particular location on a daily basis. Over multiple years, seasonal patterns emerge. Time series analysis may be used to pick up on the seasonal variations and then, after taking those variations into account, may find that the overall trend for temperatures is rising or falling in the long term.

Sales data may be measured weekly or monthly. This kind of data is useful in identifying peak sales periods and slumps when customers are not buying as much. Important business decisions are often driven by analysis of time series data.

Finally, very long-term time series may be useful in the sciences. Monitoring levels of atmospheric carbon over time and comparing these measurements with the levels found by extracting ice cores from Antarctica can demonstrate a statistically significant change to long-term cycles and patterns.

Working with and Visualizing Time Series Data in Excel

You may recall from What Are Data and Data Science? that spreadsheet applications are very well suited for basic analysis and visualization of time series data. Simply create two columns, one for the time periods and the other for the data. Highlight both columns (together with their headings) and insert a line chart using the "Recommended Charts" feature in Excel[1], resulting in a chart such as the one shown in Figure 5.4.

1 Microsoft Excel for Microsoft 365

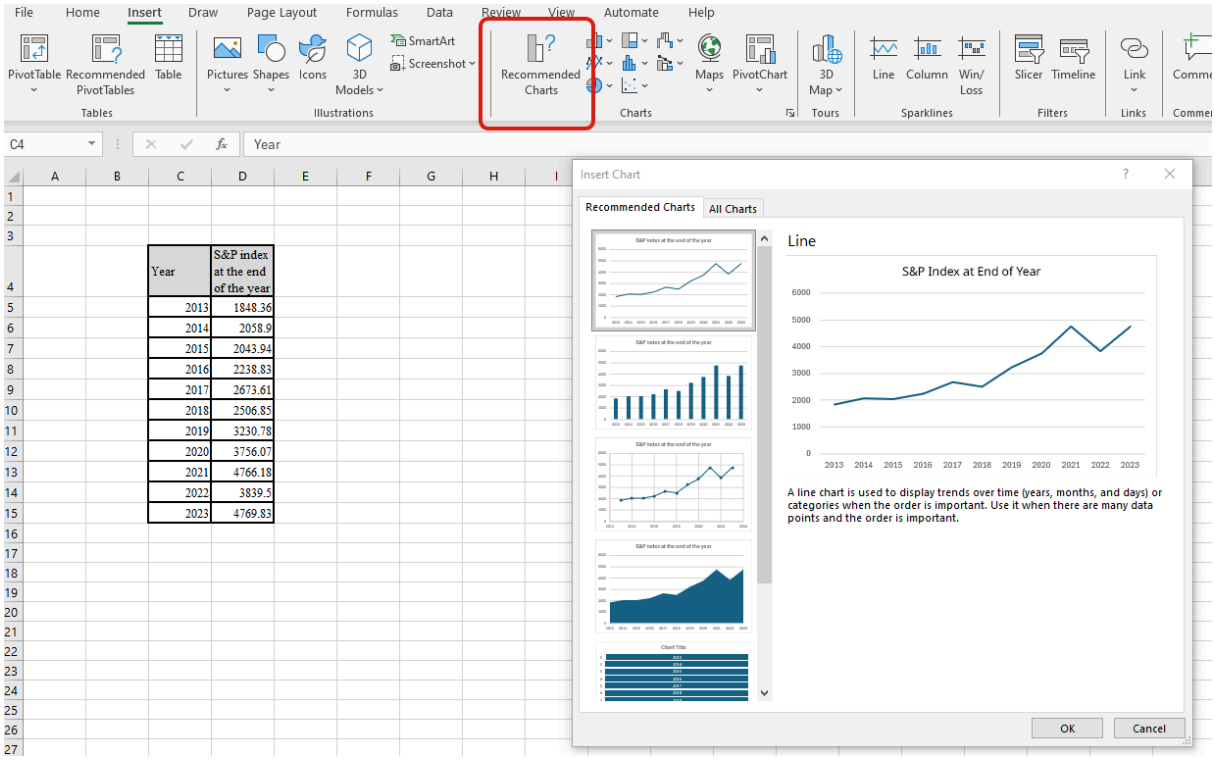


Figure 5.4 Using Excel to Create a Line Chart. Instead of using "Recommended Charts," you could also choose line chart from the panel of choices located on the same tab. (Used with permission from Microsoft)

Working with and Visualizing Time Series Data in Python

In Python, the pandas library (https://openstax.org/r/pydata) is incredibly useful for reading in CSV files and managing them as data structures called *DataFrames*. It also includes a simple command for visualizing the time series, `plot()`. Here is how to create a line chart of the daily values of the S & P 500 Index from 2014 through 2024 from the dataset SP500.csv (https://openstax.org/r/SP500).

PYTHON CODE

```
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as ticker

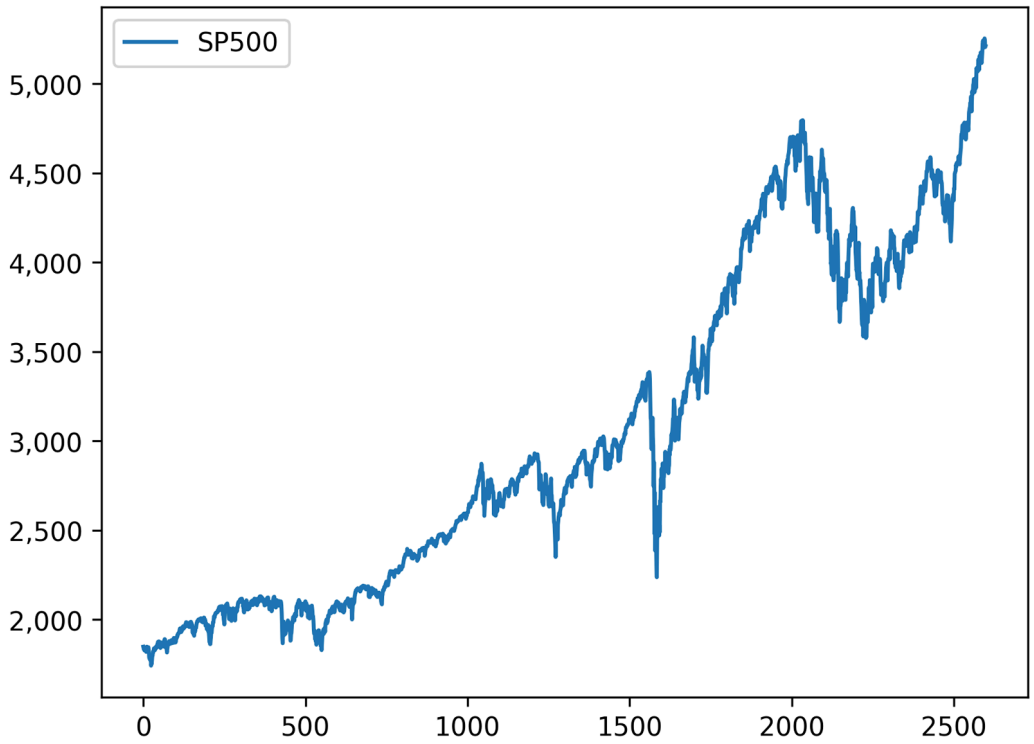
# Read the CSV file into a Pandas DataFrame
df = pd.read_csv('SP500.csv')

# Create a plot
fig, ax = plt.subplots()
df.plot(ax=ax)

# Use FuncFormatter to add commas to the y-axis
ax.yaxis.set_major_formatter(ticker.FuncFormatter(lambda x, pos:
'{:,}'.format(int(x))))
```

```
# Display the plot
plt.show()
```

The resulting output will look like this:



Notice that the simple plot feature did not label the x or y axes. Moreover, the x-axis does not show actual dates, but instead the number of time steps. For simple visualizations that you do not intend to share with a wider audience, this may be ok, but if you want to create better-looking and more informative graphs, use a library such as matplotlib.pyplot (https://openstax.org/r/matplotlib). In the next set of Python code, you will see the same dataset visualized using `matplotlib`. (Note that there is a line of code that uses the `to_datatime()` command. Dates and times are notoriously inconsistent from dataset to dataset, and software packages may not interpret them correctly without a little help.)

PYTHON CODE



```
# Import libraries
import pandas as pd ## for dataset management
import matplotlib.pyplot as plt ## for visualizations
from matplotlib.ticker import FuncFormatter ## formatting Y-axis values

# Read the CSV file into a Pandas DataFrame
df = pd.read_csv('SP500.csv')

# Convert the 'DATE' column to datetime format
```

```
df['Date'] = pd.to_datetime(df['Date'], errors='coerce')

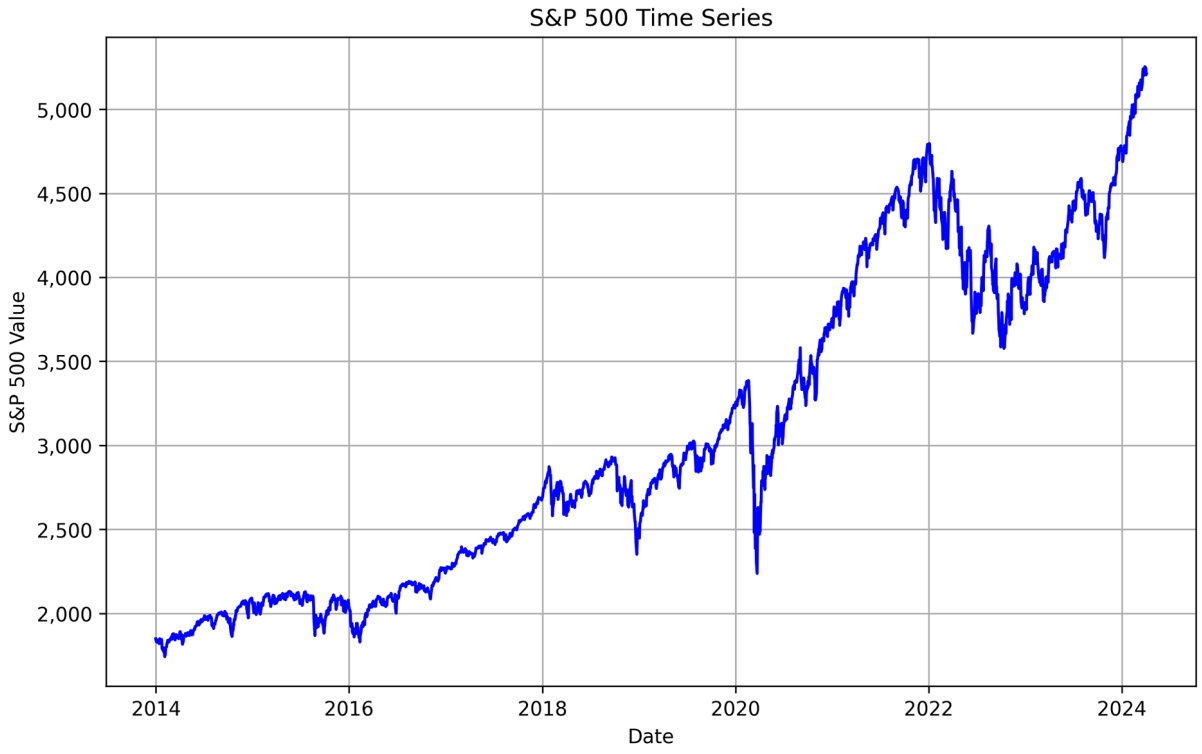
# Function to format the Y-axis values
def y_format(value, tick_number):
  return f'{value:,.0f}'

# Plotting the time series data using Matplotlib
plt.figure(figsize=(10, 6))
plt.plot(df['Date'], df['SP500'], marker='', linestyle='-', color='b')
plt.title('S&P 500 Time Series')
plt.xlabel('Date')
plt.ylabel('S&P 500 Value')
plt.grid(True)

# Apply the formatter to the Y-axis
plt.gca().yaxis.set_major_formatter(FuncFormatter(y_format))

plt.show()
```

The resulting output will look like this:



5.2 Components of Time Series Analysis

Learning Outcomes

By the end of this section, you should be able to:

- 5.2.1 Define the trend curve of a time series.
- 5.2.2 Define and identify seasonal and cyclic variations in time series data.
- 5.2.3 Discuss how noise and error affect a time series.

Time series data may be *decomposed*, or broken up into several key components. There may be an overall direction (up or down) that the data seems to be following, also known as the *trend*. Cycles and seasonal patterns may be present. Finally, noise or random variation can often be separated from the rest of the data. Time series analysis begins with identifying and extracting the components that contribute to the structure of the data.

Summarizing, the four main components of a time series are defined as follows:

1. **Trend**. The long-term direction of time series data in the absence of any other variation.
2. **Cyclic component**. Large variations in the data that recur over longer time periods than seasonal fluctuations but not with a fixed frequency.
3. **Seasonal component** (often referred to as **seasonality**). Variation in the data that occurs at fixed time periods, such as every year in the same months or every week on the same days.
4. **Noise/Random Error**. Random, unpredictable variations that occur in the time series that cannot be attributed to any of the other components previously listed. (Note, the term "random" is not the same as *statistically independent*.)

In practice, the cyclic component is often included with the trend, which is known as the **trend-cycle component**, since the cyclic component cannot be predicted in the same way that a seasonal component might be. Thus, time series data may be considered to be made up of three components: the trend (or trend-cycle) component, the seasonal component (or components), and noise or random error. Indeed, you might consider that trend, seasonal, and noise components are three separate time series that all contribute to the time series data. Although we still want to think of each component as a sequence of values, it is customary to drop the parentheses in the notation. Thus, the time series will be denoted simply by x_n rather than (x_n), and the three components will be denoted in a similar way.

Now the ways in which the components are combined in a time series analysis varies. There are additive decompositions, multiplicative decompositions, and hybrid decompositions.

In a simple **additive decomposition**, the time series is decomposed into the sum of its three components, t_n for the trend, s_n for the seasonal variation, and ε_n for the error term, as shown:

$$x_n = t_n + s_n + \varepsilon_n$$

In a strictly **multiplicative decomposition**, the time series is decomposed into the product of its components (which will look quite different from their additive counterparts, and so capital letters are used to distinguish them).

$$x_n = T_n \cdot S_n \cdot E_n$$

Finally, a hybrid decomposition in which the trend-cycle and seasonal components are multiplied while the error term remains as an additive component is often useful. In this case, we might write:

$$x_n = T_n \cdot S_n + \varepsilon_n$$

Multiplicative and hybrid decompositions can be more useful if the variation above and below the trend curve seems to be proportional to the level of the trend itself. In other words, when the trend is higher, the variation around the trend is also larger. We will focus only on additive decomposition in this text. In fact, there is a

simple transformation using properties of logarithms that changes a strictly multiplicative decomposition into an additive one as long as all of the components consist of positive values.

$$\log(x_n) = \log(T_n \cdot S_n \cdot E_n) = \log(T_n) + \log(S_n) + \log(E_n)$$

Note that regression methods may be used in conjunction with decomposition. This section will cover the fundamentals of decomposition and regression for time series analysis.

The Trend Curve and Trend-Cycle Component

As previously defined, the *trend* is the direction that the data follows when variations such as seasonal ups and downs and random noise and error are ignored. The **trend curve** (or **trendline**) models the trend by a line or curve, representing something like a *mean value* for the time series in any given time frame. By this definition, the linear regression shown in Example 5.1 would be a trend curve for the S&P 500 Index values. If there is no significant rise or fall in the long-term (that is, only short-term fluctuations or patterns exist), then the trend may be estimated by a constant called the **level**, which is the mean of all the time series data.

Visualizing a Trend Curve

There is no single agreed-upon approach to finding the trend of a time series. The mechanics of isolating a trend curve or trend-cycle component are covered in detail in Time Series Forecasting Methods. For now, we shall see by example what a trend curve might look like in a dataset.

Figure 5.5 shows the monthly consumption of coal for generating electricity in the United States from 2016 through 2022, as provided in the dataset MonthlyCoalConsumption.xlsx (https://openstax.org/r/spreadsheetsd1j) (source: https://www.statista.com/statistics/1170053/monthly-coal-energy-consumption-in-the-us/ (https://openstax.org/r/statistics). Note: The data was cleaned in the following ways: *Month* converted to a standard date format from string format; *Value* converted to numerical data from string format.)

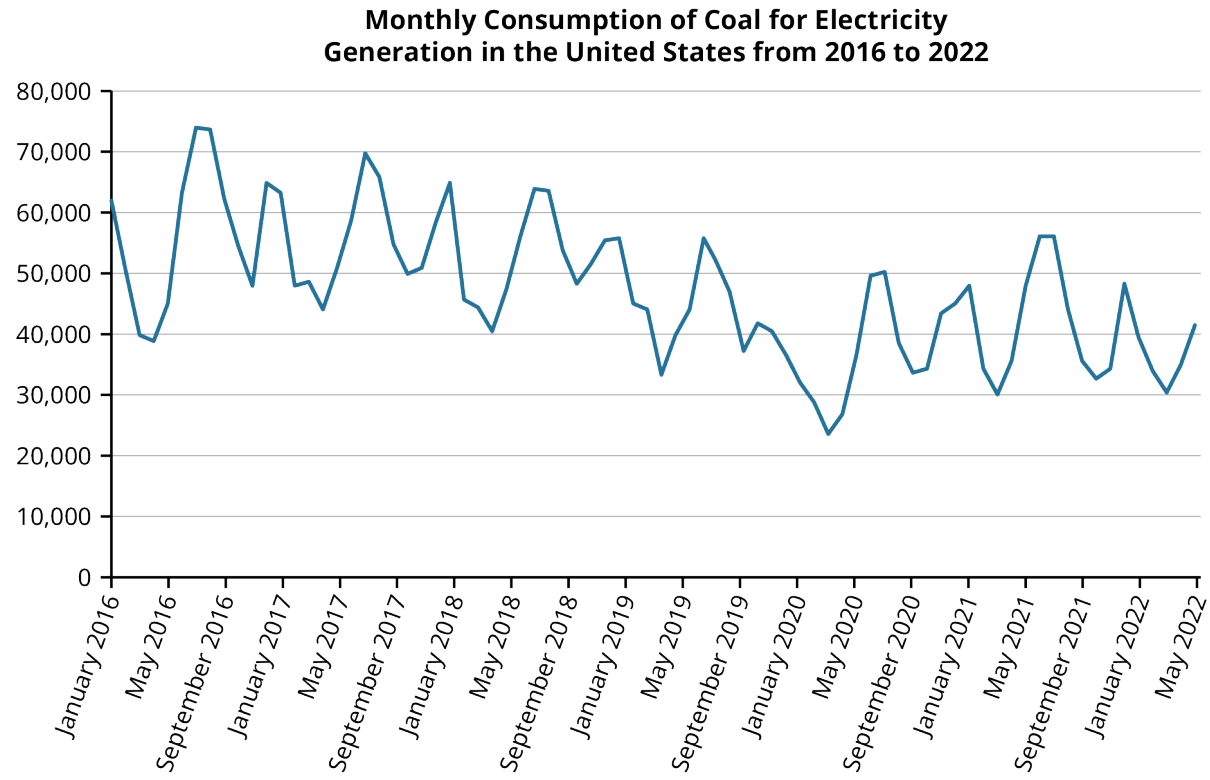


Figure 5.5 Monthly Consumption of Coal for Electricity Generation in the United States from 2016 to 2022, in 1000s of Tons (data source: https://www.statista.com/statistics/1170053/monthly-coal-energy-consumption-in-the-us/)

Although there are numerous ups and downs, there is clearly an overall trajectory of the data. If you were given the task to draw a curve that best captures the underlying trend, you might draw something like the

curve shown in Figure 5.6. (This is what we refer to as a *trendline* or *trend curve*.)

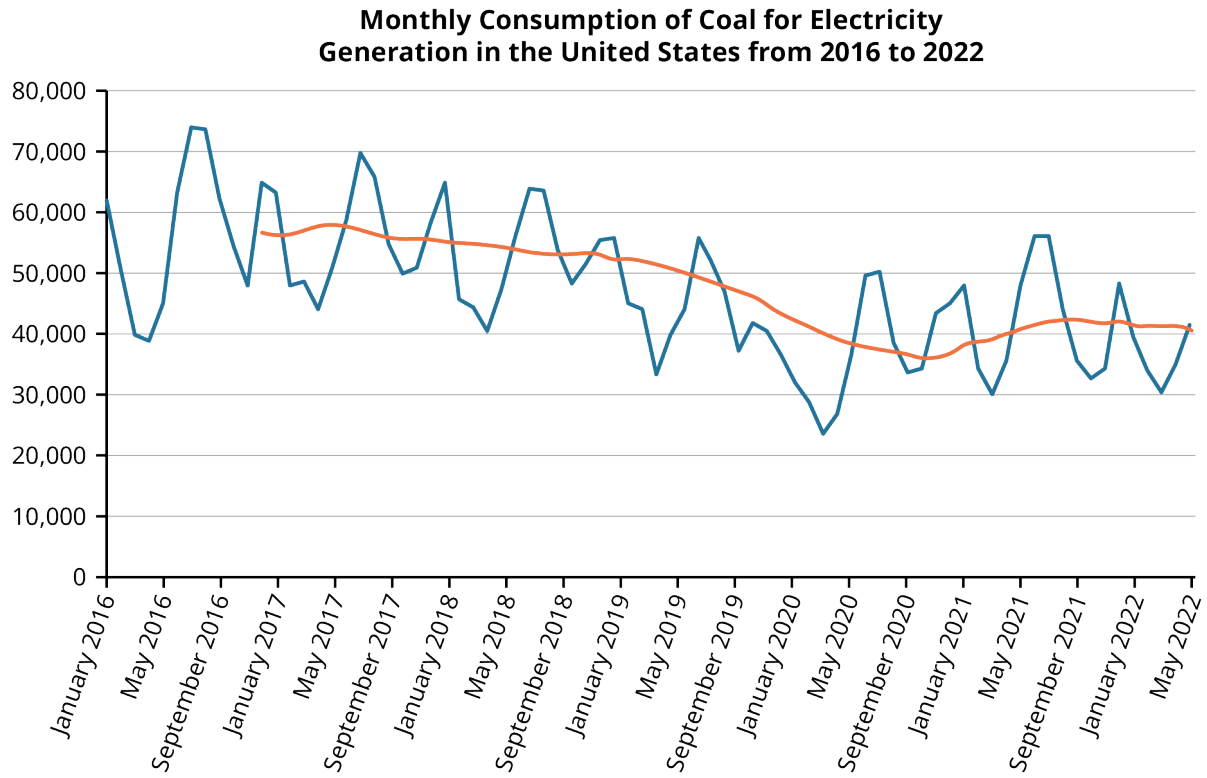


Figure 5.6 Monthly Consumption of Coal for Electricity Generation in the United States from 2016 to 2022, in 1000s of Tons with Trend Curve Shown (data source: https://www.statista.com/statistics/1170053/monthly-coal-energy-consumption-in-the-us/)

Trendline in Excel

Excel has some built-in features that make it easy to include trend curves directly onto the graph of a time series. First, make a line graph of the data in the same way as shown in Introduction to Time Series Analysis. (Note: If "Recommended Charts" does not display the type of chart that you want, you can click on the "All Charts" tab and choose another. For Figure 5.7, we chose the line chart rather than the default column chart.) Next, you can click on "Add Chart Element" in the "Chart Design" area of Excel. One of the options is "Trendline" (see Figure 5.7). To get a trendline like the one in Figure 5.6, select "More Options" and adjust the period of the trendline to 12. After adding the trendline, you can change its color and design if you wish.

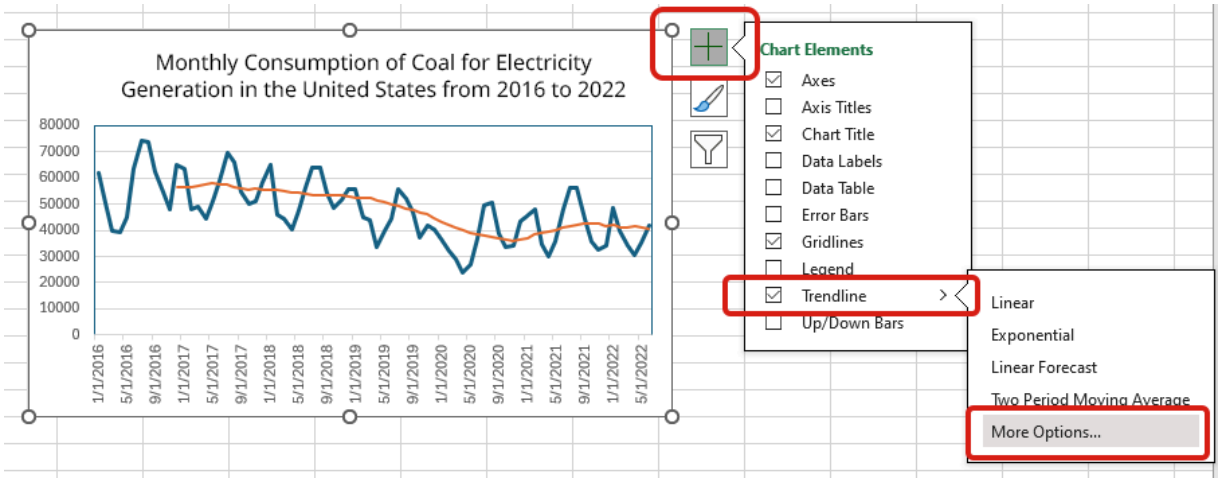


Figure 5.7 Adding a Trendline to a Line Chart in Excel (Used with permission from Microsoft)

Seasonal Variations

We've seen that *seasonality*, or seasonal variations, are predictable changes in the data occurring at regular intervals. The smallest interval of time at which the pattern seems to repeat (at least approximately so) is called its **period**.

For example, suppose you live in a temperate climate in the northern hemisphere and you record the temperature every day for a period of multiple years. You will find that temperatures are warmer in the summer months and colder in the winter months. There is a natural and predicatable pattern of rise and fall on a time scale that repeats every year. This is a classic example of seasonal variation with period 1 year.

However, the period of seasonal variation may be much longer or much shorter than a year. Suppose that you record temperatures near your home every hour for a week or so. Then you will notice a clear repeating cycle of rising and falling temperatures each day—warmer in the daytime and colder each night. This daily cycle of rising and falling temperatures is also a seasonal variation in the time series of period 24 hours. Indeed, any given time series may have multiple seasonal variation patterns at different periods.

EXAMPLE 5.2

Problem

Walt's Water Adventures is a local company that rents kayaks and canoes for river excursions. Due to mild winters in the area, WWA is able to rent these items year-round. Quarterly revenues from rentals were recorded for three consecutive years and also shown in Table 5.2. Is there a clear seasonal variation in the dataset, and if so, what is the period of the seasonal variation? When do the high points and low points generally occur within each seasonal pattern?

Quarter	Revenues ($)
2021 Spring	1799
2021 Summer	3304
2021 Fall	1279
2021 Winter	999
2022 Spring	2284
2022 Summer	3459
2022 Fall	1964
2022 Winter	1594
2023 Spring	2704
2023 Summer	3849
2023 Fall	2131
2023 Winter	1939

Table 5.2 Quarterly Revenues for Kayak and Canoe Rentals

Solution

The period is 1 year, or 4 quarters, with high values occurring in summer of each year and low values occurring in winter of each year as shown in Figure 5.8.

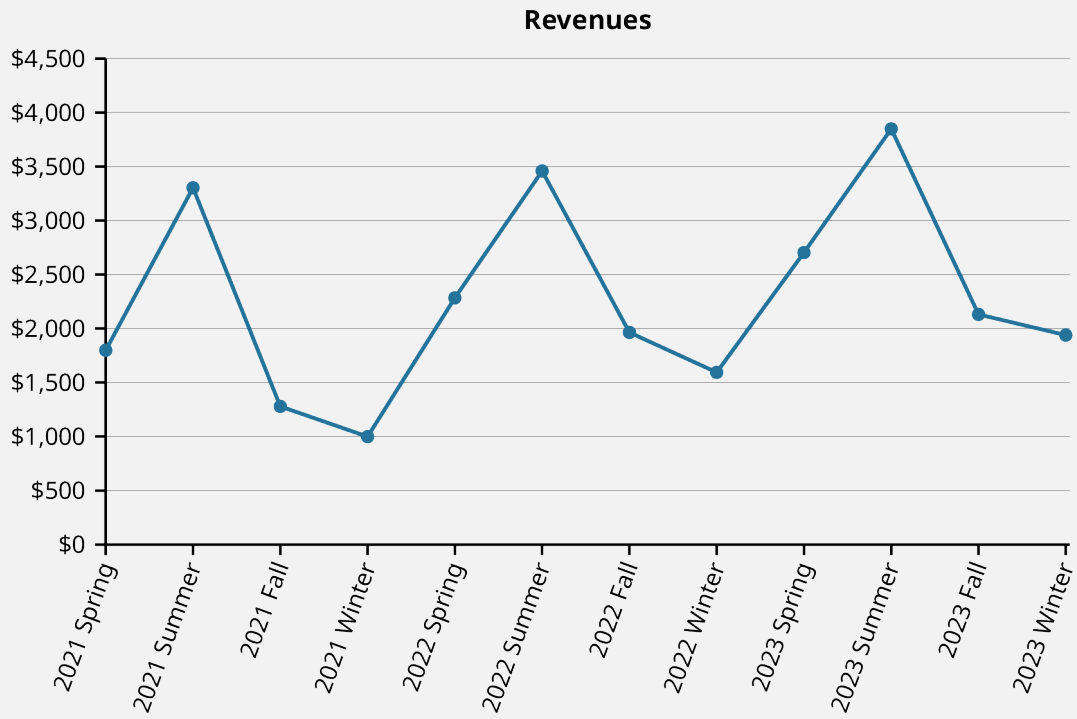


Figure 5.8 Quarterly Revenues for Kayak and Canoe Rentals

Noise and Random Variation

All data is noisy. There are often complex factors that affect data in ways that are difficult or impossible to predict. Even if all factors could be accounted for, there will inevitably be some error in the data due to mistakes, imprecise measurements, or even how the values are rounded.

Noise shows up in time series data as small random ups and downs without an identifiable pattern. When visualizing the time series, noise is the component of the data that makes the line graph look jagged. In the absence of noise or random variation, the plot of a time series would look fairly smooth.

In a time series, once the trend, cyclic variation, and seasonal variation are all quantified, whatever is "left over" in the data will generally be noise or random fluctuations. In other words, once the trend (or trend-cycle component) t_n and seasonal variation s_n are accounted for, then the noise component is just the remaining component:

$$\varepsilon_n = x_n - t_n - s_n$$

In the real world, we rarely have access to the actual t_n or s_n series directly. Instead, almost all time series forecast methods produce a model $(\hat{x}_n)$ that tries to capture the trend-cycle curve and seasonal variation. The **residuals**, which are the time series of differences between the given time series (x_n) and the model, provide a measure of the error of the model with respect to the original series. If the model is good, in the sense that it captures as much of the predictable aspects of the time series as possible, then the residuals should give an estimate of the noise component of the time series. Therefore, it is customary to use the same notation for residuals as for the noise term. Thus,

$$\varepsilon_n = x_n - \hat{x}_n$$

The residuals should have the characteristics of a truly random time series, known as **white noise**, which has the following key characteristics:

1. **Constant zero mean.** The mean should be close to zero anywhere in the series.

2. **Constant variance.** The variance should be approximately the same anywhere in the series. (Note: *Variance* is defined in Measures of Variation.)
3. **Uncorrelated.** The sequence of values at one point in the series should have a correlation close to zero when compared to any sequence of values of the same size anywhere else in the series. (Note: *Correlation* is covered in Correlation and Linear Regression Analysis.)

In the definitions given, the phrase *anywhere in the series* refers to a sample of consecutive terms taken from the series $(x_k, x_{k+1}, x_{k+2}, x_{k+3}, \ldots, x_{k+r})$, which is also called a **window**. How many terms are included in a window? That depends on the time series, but just keep in mind that the window should be large enough to accurately reflect the mean and variance of nearby points while not being so large that it fails to detect actual ups and downs present in the data. Statistical software can be used to check whether a time series has the characteristics of white noise, and we will discuss the topic further in the next section. For now, check out Figure 5.9, which displays a time series of white noise.

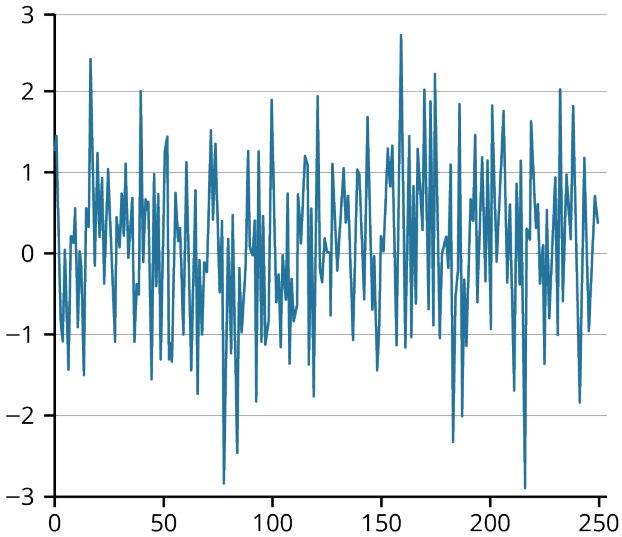


Figure 5.9 White Noise. White noise is a type of random noise. The mean is constantly near zero. Variance is pretty constant everywhere in the series, close to a value of 1. There are no observable patterns that repeat.

5.3 Time Series Forecasting Methods

Learning Outcomes

By the end of this section, you should be able to:

- 5.3.1 Analyze a time series by decomposing it into its components, including trend, seasonal and cyclic variation, and residual noise.
- 5.3.2 Compute various kinds of moving averages, weighted moving averages, and related smoothing techniques.
- 5.3.3 Describe the ARIMA procedure and how autocorrelation functions can be used to find appropriate parameters.
- 5.3.4 Use ARIMA to make forecasts, with the aid of Python.

This section focuses on analysis of a time series into its components for the purpose of building a robust model for forecasting. Although much of the analysis can be done by hand, numerous computer software applications have been developed to handle this task. Ultimately, we will see how Python can be used to decompose a time series and create a general-purpose model called the autoregressive integrated moving average model (commonly known as ARIMA) to make forecasts.

Detrending the Data and Smoothing Techniques

Detrending a time series refers to the process of separating the underlying trend component t_n from the time

series as a whole through one of two complementary operations: *isolating the trend* and then removing it or *smoothing out the trend* leaving only the seasonal and error components. Methods that reduce the error term and flatten out seasonal variations result in a trend-cycle component t_n that still closely follows the long-term ups and downs in the time series. In this case, the other components of the time series can be found by removing the trend component:

$$x_n - t_n = s_n + \varepsilon_n$$

On the other hand, there are detrending methods that transform the data into a new time series with a trendline closer to a constant line at the level of 0. In other words, after applying such a transformation, the new time series, z_n, would consist of only the seasonal and noise components:

$$z_n = s_n + \varepsilon_n$$

Common methods for isolating or removing the trend or trend-cycle component include the following:

1. **Simple moving average (SMA).** The **simple moving average (SMA)** is found by taking the average (arithmetic mean) of a fixed number of consecutive data points. This has the effect of smoothing out short-term variations in the data and isolating the trend. If the data has seasonal variation, then an SMA of the same length can effectively remove the seasonal component completely—there is more about this in Time Series Forecasting Methods.
2. **Weighted moving average (WMA).** There are several variations on the moving average idea, in which previous terms of the series are given different weights, thus termed **weighted moving average (WMA)**. One of the most common weighted moving averages is the **exponential moving average (EMA)**, in which the weights follow exponential decay, giving more recent data points much more weight than those further away in time. These techniques are more often used in forecasting as discussed later in this section, but they may also be employed as a smoothing method to isolate a trend in the data.
3. **Differencing.** The (first-order) difference of a time series is found by **differencing**, or taking differences of consecutive terms—that is, the nth term of the difference would be $x_{n+1} - x_n$. If the original time series had a linear trend, then the difference would display a relatively constant trend with only noise and seasonal variation present. Similarly, the difference of the difference, or second-order difference, can transform a quadratic trend into a constant trend. Higher-order polynomial trends can be identified using higher-order differences. Thus, we can use differencing to check for the overall trend of a certain type (linear, quadratic, etc.) while at the same time removing the original trend from the time series.
4. **Regression.** As mentioned in Example 5.1, linear regression may be applied to a time series that seems to display a uniform upward or downward trend. Using differencing to identify whether there is a trend of some particular order, then linear, quadratic, or higher-order polynomial regression could be used to find that trend.
5. **Seasonal-Trend decomposition using LOESS (STL). Seasonal-Trend decomposition using LOESS (STL)** is a powerful tool that decomposes a time series into the trend-cycle, seasonal, and noise components. Later in this section we will find out how to perform STL using Python. (Note: LOESS is a regression technique the fits a low-order polynomial—typically linear or quadratic—to small portions of the data at a time. A full understanding of LOESS is not required for using STL.)

There are, of course, more complex methods for analyzing and isolating the trends of time series, but in this text we will stick to the basics and make use of technology when appropriate.

Simple Moving Averages (SMA)

Recall that we use the term *window* to refer to consecutive terms of a time series. In what follows, assume that the size of the window is a constant, T. Given a time series with values (x_n), that is, a sequence $(x_1, x_2, x_3, x_4, \ldots, x_N)$, the simple moving average with window size T is defined as a new series (y_n) with values given by the formula

$$y_n = \frac{x_n + x_{n-1} + x_{n-2} + \ldots + x_{n-T+1}}{T} = \frac{1}{T}\sum_{k=1}^{T} x_{n-T+k}$$

This formula is not as difficult as it looks. Just think of it as the average of the last T terms. For example, given the sequence (5, 4, 2, 6, 2, 3, 1, 7), the SMA of length 4 would be found by averaging 4 consecutive terms at a time. Note that the first term of the SMA is y_4 because at least 4 terms are needed in the average.

Term $k = 4$: $\left(\mathbf{5}, \mathbf{4}, \mathbf{2}, \mathbf{6}, 2, 3, 1, 7\right) \rightarrow y_4 = \frac{5+4+2+6}{4} = \frac{17}{4} = 4.25$

Term $k = 5$: $\left(5, \mathbf{4}, \mathbf{2}, \mathbf{6}, \mathbf{2}, 3, 1, 7\right) \rightarrow y_5 = \frac{4+2+6+2}{4} = \frac{14}{4} = 3.5$

Term $k = 6$: $\left(5, 4, \mathbf{2}, \mathbf{6}, \mathbf{2}, \mathbf{3}, 1, 7\right) \rightarrow y_6 = \frac{2+6+2+3}{4} = \frac{13}{4} = 3.25$

Term $k = 7$: $\left(5, 4, 2, \mathbf{6}, \mathbf{2}, \mathbf{3}, \mathbf{1}, 7\right) \rightarrow y_7 = \frac{6+2+3+1}{4} = \frac{12}{4} = 3$

Term $k = 8$: $\left(5, 4, 2, 6, \mathbf{2}, \mathbf{3}, \mathbf{1}, \mathbf{7}\right) \rightarrow y_8 = \frac{2+3+1+7}{4} = \frac{13}{4} = 3.25$

Thus, the SMA of length 4 is the sequence (4.25, 3.5, 3.25, 3, 3.25). Note how the values of the SMA do not jump around as much as the values of the original sequence.

The number of terms to use in a simple moving average depends on the dataset and desired outcome. If the intent is to reduce noise, then a small number of terms (say, 3 to 5) may be sufficient. Sometimes inconsistencies in data collection can be mitigated by a judicious use of SMA. For example, suppose data is collected on the number of walk-in customers at the department of motor vehicles, which only operates Monday through Friday. No data is collected on the weekends, while the number of walk-ins on Mondays may be disproportionately high because people have had to wait until Monday to use their services. Here, an SMA of 7 days would smooth out the data, spreading the weekly walk-in data more evenly over all the days of the week.

EXAMPLE 5.3

Problem

Find the simple moving average of window size 3 for the data from Table 5.1 and compare with the original line graph.

Solution

The SMA will begin at index $k = 3$.

$$y_3 = \frac{1{,}848.36 + 2{,}058.9 + 2{,}043.94}{3} = 1{,}983.73$$

$$y_4 = \frac{2{,}058.9 + 2{,}043.94 + 2{,}238.83}{3} = 2{,}113.89$$

$$y_5 = \frac{2{,}043.94 + 2{,}238.83 + 2{,}673.61}{3} = 2{,}318.79$$

$$y_6 = \frac{2{,}238.83 + 2{,}673.61 + 2{,}506.85}{3} = 2{,}473.1$$

$$y_7 = \frac{2{,}673.61 + 2{,}506.85 + 3{,}230.78}{3} = 2{,}803.75$$

$$y_8 = \frac{2{,}506.85 + 3{,}230.78 + 3{,}756.07}{3} = 3{,}164.57$$

$$y_9 = \frac{3{,}230.78 + 3{,}756.07 + 4{,}766.18}{3} = 3{,}917.68$$

$$y_{10} = \frac{3{,}756.07 + 4{,}766.18 + 3{,}839.5}{3} = 4{,}120.58$$

$$y_{11} = \frac{4{,}766.18 + 3{,}839.5 + 4{,}769.83}{3} = 4{,}458.5$$

The SMA is graphed together with the original time series in Figure 5.10.

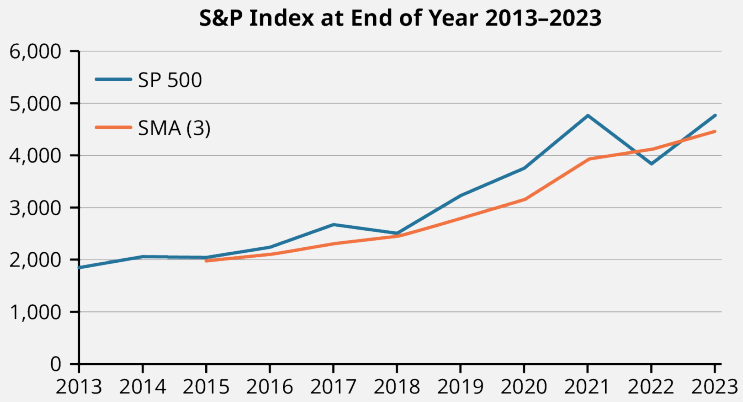


Figure 5.10 Line Chart of the S&P 500 Index Together with the Simple Moving Average of Length 3 (data source: adapted from S&P 500 [SPX] Historical Data)

Simple moving averages provide an easy way to identify a trend-cycle curve in the time series. In effect, the SMA can serve as an estimate of the t_n component of the time series. In practice, when using SMA to isolate the trend-cycle component, the terms of the SMA are **centered**, which means that when computing the terms of the SMA, y_k, the x_k term shows up right in the middle of the window, with the same number of terms to the left and to the right.

For example, the SMA found in Example 5.3 may be shifted so that $y_k = \frac{x_{k-1}+x_k+x_{k+1}}{3}$. This has the effect of shifting the index of the SMA terms back 1 so that the SMA begins at index $k = 2$.

$$y_2 = \frac{x_1 + x_2 + x_3}{3} = \frac{1{,}848.36 + 2{,}058.9 + 2{,}043.94}{3} = 1{,}983.73$$

This procedure works well when the window size is odd, but when it is even, a common practice is to compute two SMAs in which x_k is very nearly in the middle of the sequence and then average their results. For example, using the same data as in Example 5.3, if we wanted to compute a centered SMA of length 4, then the first term would be y_3, and we would find two averages.

$$\frac{x_1 + x_2 + x_3 + x_4}{4} = \frac{1{,}848.36 + 2{,}058.9 + 2{,}043.94 + 2{,}238.83}{4} = 2{,}047.51$$

$$\frac{x_2 + x_3 + x_4 + x_5}{4} = \frac{2{,}058.9 + 2{,}043.94 + 2{,}238.83 + 2{,}673.61}{4} = 2{,}253.82$$

$$y_3 = \frac{2{,}047.51 + 2{,}253.82}{2} = 2{,}150.67$$

The average of the two averages is equivalent to the following more efficient formula. First consider the previous example.

$$y_3 = \frac{1}{2}\left(\frac{x_1 + x_2 + x_3 + x_4}{4} + \frac{x_2 + x_3 + x_4 + x_5}{4}\right) = \frac{x_1}{8} + \frac{x_2}{4} + \frac{x_3}{4} + \frac{x_4}{4} + \frac{x_5}{8}$$

This suggests a general pattern for even window sizes. In summary, we have two closely related formulas for centered SMA, depending on whether T is even or odd.

$$y_k = \frac{1}{T}\left(x_{k-(T-1)/2} + \ldots + x_{k+(T-1)/2}\right), \text{ for } T \text{ odd}$$

$$y_k = \frac{x_{k-T/2}}{2T} + \frac{1}{T-2}\left(x_{k-T/2+1} + \ldots + x_{k+T/2-1}\right) + \frac{x_{k+T/2}}{2T}, \text{ for } T \text{ even}$$

Of course, there are functions in statistical software that can work out the centered SMA automatically for you.

Differencing

In mathematics, the first-order difference of a sequence $(x_n) = (x_1, x_2, x_3, x_4, \ldots, x_N)$ is another sequence, denoted $\Delta(x_k)$, consisting of the differences of consecutive terms. That is,

$$\begin{aligned} \Delta(x_n) &= \left(x_{n+1} - x_n\right) \\ &= \left(x_2 - x_1, x_3 - x_2, x_4 - x_3, \ldots, x_N - x_{N-1}\right) \end{aligned}$$

Note that there is one fewer term in the difference $\Delta(x_n)$ compared to the original time series (x_n). Differencing has the effect of leveling out a uniform upward or downward trend without affecting other factors such as seasonal variation and noise.

Moreover, the average (arithmetic mean) of the terms of difference series measures of the rate of change of the original time series, much as the slope of the linear regression does. (Indeed, if you have seen some calculus, you might notice that there is a strong connection between differencing and computing the *derivative* of a continuous function.)

EXAMPLE 5.4

Problem

Find the first-order difference time series for the data from Table 5.1 and compare with the original line graph. Find the average of the difference series and compare the value to the slope of the linear regression found in Example 5.1.

Solution

Since the original time series from Table 5.1 has $N = 10$ terms, the difference will have only 9 terms, as shown in Table 5.3.

Term n	S&P Index Term x_n	n^{th} Term of $\Delta(x_n)$
1	1848.36	N/A
2	2058.9	$2058.9 - 1848.36 = 210.54$
3	2043.94	$2043.94 - 2058.9 = -14.96$
4	2238.83	$2238.83 - 2043.94 = 194.89$
5	2673.61	$2673.61 - 2238.84 = 434.78$
6	2506.85	$2506.85 - 2673.61 = -166.76$
7	3230.78	$3230.78 - 2506.85 = 723.93$
8	3756.07	$3756.07 - 3230.78 = 525.29$
9	4766.18	$4766.18 - 3756.07 = 1010.11$

Table 5.3 First-Order Difference Time Series for the Data in Table 5.1 (source:

Term n	S&P Index Term x_n	n^{th} Term of $\Delta(x_n)$
10	3839.5	$3839.5 - 4766.18 = -926.68$
11	4769.83	$4769.83 - 3839.5 = 930.33$

Table 5.3 First-Order Difference Time Series for the Data in Table 5.1 (source: adapted from https://www.nasdaq.com/market-activity/index/spx/historical)

Figure 5.11 shows the graph of the original data together with its difference series.

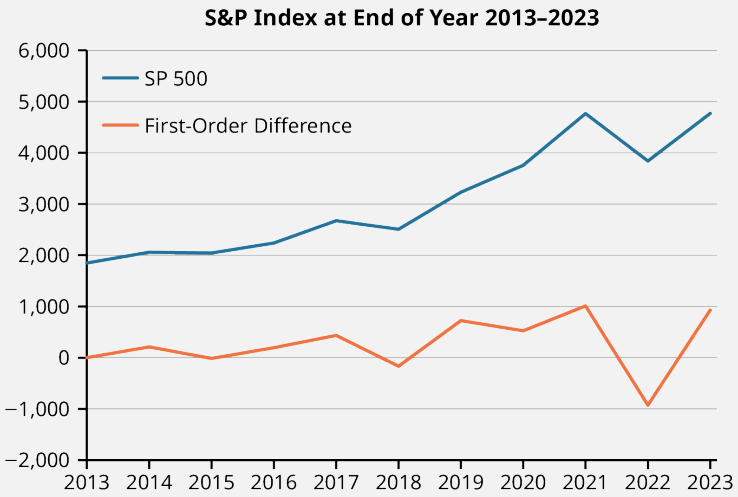


Figure 5.11 Comparison of S&P Index Values and Their Difference Series (data source: adapted from S&P 500 [SPX] Historical Data)

The average value of $\Delta(x_n)$ is:

$$\frac{1}{10}(210.54 + (-14.96) + 194.89 + 434.78 + (-166.76) + 723.93 + 525.29 + 1{,}010.11 + (-926.68) + 930.33) = 292.15$$

Comparing with the slope of the linear regression, which is about 304 (recall Example 5.1), we find the two values are relatively close to each other.

Differencing can be used to check whether a time series trend curve is essentially linear, quadratic, or polynomial of higher order. First, we need to recall a definition from algebra that is typically applied to polynomial functions but can also be applied to sequences. If a sequence (x_n) is given by a polynomial formula of the form

$$x_n = f(n) = a_r n^r + a_{r-1} n^{r-1} + \ldots + a_1 n + a_0$$

where r is a whole number, each of the coefficients a_k (for $k = 1, 2, 3, \ldots, n$) is a constant, and the leading term is nonzero ($a_r \neq 0$), then the sequence has *order* or *degree* r. (**Order** or **degree** refers to the number of terms or lags used in a model to describe the time series.)

For example, any sequence of the form $x_n = an + b$ has order 1 (and may be called *linear*). A sequence of the form $x_n = an^2 + bn + c$ has order 2 (and is called *quadratic*). Quadratic data fits well to a parabola, which would indicate a curving trend in the data.

If a sequence (x_n) has order $r > 0$, then the difference sequence $\Delta(x_n)$ has order $r - 1$. If the original sequence already is of order 0, which means the sequence is a constant sequence (all terms being the same constant), then the difference sequence simply produces another constant sequence (order 0) in which every

term is equal to 0. This important result can be proved mathematically, but we will not present the proof in this text.

As an example, consider the sequence $(x_n) = (2, 5, 8, 11, 14, 17, 20)$, which has order 1, since you can find the values using the formula $x_n = 3n - 1$. The difference sequence is $\Delta(x_n) = (3, 3, 3, 3, 3, 3)$, which is a constant sequence, so it has order 0. Notice that taking the difference one more time yields the constant 0 sequence, $\Delta(\Delta(x_n)) = (0, 0, 0, 0, 0)$, and taking further differences would only reduce the number of terms, but each term would remain at the value 0. (For simplicity, we will assume that there are always enough terms in a sequence so that the difference is also a non-empty sequence.)

The *second-order* difference, $\Delta^2(x_n) = \Delta\big(\Delta(x_n)\big)$, or difference of the difference, would reduce the order of an r-order sequence twice to $r - 2$ if $r \geq 2$ or to the constant zero sequence if $r \leq 1$. In general, a d-order difference, defined by $\Delta^d(x_n) = \Delta\big(\Delta^{d-1}(x_n)\big)$, reduces the order of an r-order sequence to $r - d$ if $r \geq d$ or to order 0 otherwise. With a little thought, you may realize that we now have a simple procedure to determine the order of a sequence that does not require us to have the formula for x_n in hand.

If a sequence (x_n) has a well-defined order (in other words, there is a polynomial formula that is a good model for the terms of the sequence, even if we do not know what that formula is yet), then the order r is the smallest whole number such that the r-order difference, $\Delta^{r+1}(x_n)$, is equal to either a constant 0 sequence or white noise.

EXAMPLE 5.5

Problem

Determine the order of the sequence (10, 9, 6, 7, 18, 45, 94, 171) using differencing.

Solution

Let (x_n) stand for the given sequence.

First-order difference: $\Delta(x_n) = (-1, -3, 1, 11, 27, 49, 77)$

Second-order difference: $\Delta^2(x_n) = (-2, 4, 10, 16, 22, 28)$

Third-order difference: $\Delta^3(x_n) = (6, 6, 6, 6, 6)$

Fourth-order difference: $\Delta^4(x_n) = (0, 0, 0, 0)$

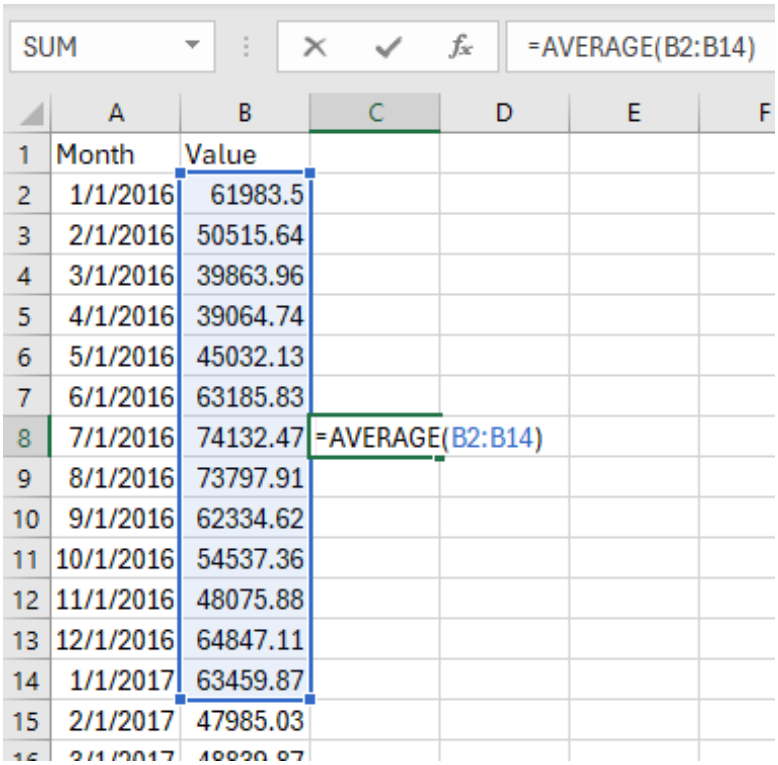
The given sequence has order 3 since the $(3 + 1)$-order difference resulted in all zeros.

We cannot hope to encounter real-world time series data this is exactly linear, quadratic, or any given degree. Every time series will have some noise in addition to possible seasonal patterns. However, the trend-cycle curve may still be well approximated by a polynomial model of some order.

SMA and Differencing in Excel

Excel can easily do simple moving averages and differencing. The ability to write one equation in a cell and then drag it down so that it works on all the rows of data is a very powerful and convenient feature. By way of example, let's explore the dataset MonthlyCoalConsumption.xlsx (https://openstax.org/r/spreadsheetsd1j). First, you will need to save the file as an Excel worksheet, as CSV files do not support equations.

Let's include a centered 13-term window SMA (see Figure 5.12). In cell C8, type "=AVERAGE(B2:B14)" as shown in Figure 5.12. Note: There should be exactly 6 rows above and 6 rows below the center row at which the formula is entered.



	A	B	C	D	E	F
1	Month	Value				
2	1/1/2016	61983.5				
3	2/1/2016	50515.64				
4	3/1/2016	39863.96				
5	4/1/2016	39064.74				
6	5/1/2016	45032.13				
7	6/1/2016	63185.83				
8	7/1/2016	74132.47	=AVERAGE(B2:B14)			
9	8/1/2016	73797.91				
10	9/1/2016	62334.62				
11	10/1/2016	54537.36				
12	11/1/2016	48075.88				
13	12/1/2016	64847.11				
14	1/1/2017	63459.87				
15	2/1/2017	47985.03				

Figure 5.12 Screenshot of a 13-term Excel Window SMA (Used with permission from Microsoft)

Next, click on the square in the lower right corner of cell C8 and drag it all the way down to cell C173, which is 6 rows before the final data value. Column C now contains the SMA time series for your data, as shown in Figure 5.13.

	A	B	C
1	Month	Value	SMA
2	1/1/2016	61983.5	
3	2/1/2016	50515.64	
4	3/1/2016	39863.96	
5	4/1/2016	39064.74	
6	5/1/2016	45032.13	
7	6/1/2016	63185.83	
8	7/1/2016	74132.47	56987
9	8/1/2016	73797.91	55910.2
10	9/1/2016	62334.62	55781.29
11	10/1/2016	54537.36	56120.9
12	11/1/2016	48075.88	57031.12
13	12/1/2016	64847.11	58094.19
14	1/1/2017	63459.87	58600.56
15	2/1/2017	47985.03	57956.64
16	3/1/2017	48839.87	56488.58
17	4/1/2017	44278.92	55540.93
18	5/1/2017	50897.5	55259.73
19	6/1/2017	58852.12	56058.3

Figure 5.13 Screenshot Showing the SMA Time Series in Column C (Used with permission from Microsoft)

You can generate the graph of the time series and the SMA together, which is shown in Figure 5.14. Compare your computed SMA with the trendline created by Excel in Figure 5.7.

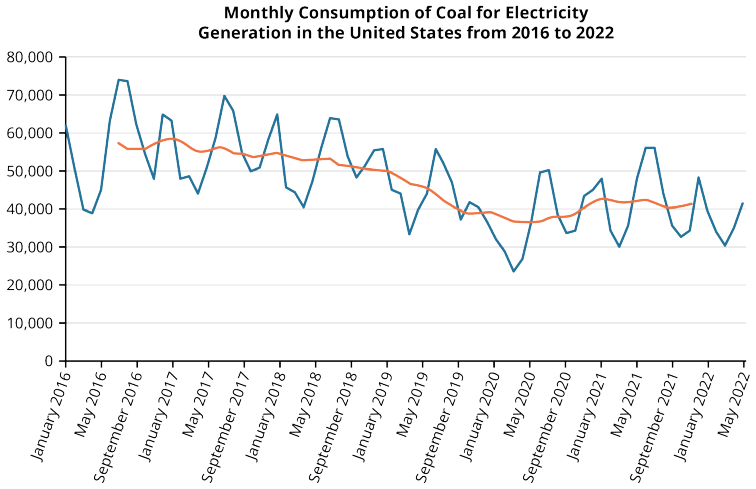


Figure 5.14 Monthly Consumption of Coal for Electricity Generation in the United States from 2016 to 2022. The SMA of window length 13 has been added to the graph in orange. (data source: https://www.statista.com/statistics/1170053/monthly-coal-energy-consumption-in-the-us/)

Let's now work out the first-order difference. The idea is very similar. Put the formula for the difference, which is nothing more than "=B3-B2," into cell D3, as in Figure 5.15.

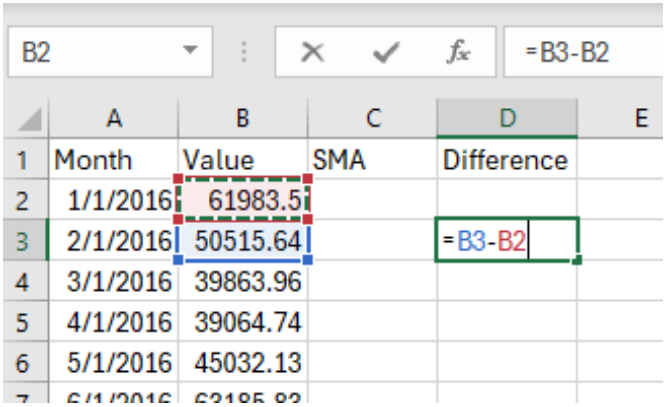


Figure 5.15 Screenshot of Excel Showing Formula for Difference in Column D (Used with permission from Microsoft)

Then drag the formula down to the end of the column. The result is shown in Figure 5.16.

	A	B	C	D
1	Month	Value	SMA	Difference
2	1/1/2016	61983.5		
3	2/1/2016	50515.64		-11467.86
4	3/1/2016	39863.96		-10651.68
5	4/1/2016	39064.74		-799.22
6	5/1/2016	45032.13		5967.39
7	6/1/2016	63185.83		18153.7
8	7/1/2016	74132.47	56987	10946.64
9	8/1/2016	73797.91	55910.2	-334.56
10	9/1/2016	62334.62	55781.29	-11463.29
11	10/1/2016	54537.36	56120.9	-7797.26
12	11/1/2016	48075.88	57031.12	-6461.48
13	12/1/2016	64847.11	58094.19	16771.23
14	1/1/2017	63459.87	58600.56	-1387.24
15	2/1/2017	47985.03	57956.64	-15474.84
16	3/1/2017	48839.87	56488.58	854.84
17	4/1/2017	44278.92	55540.93	-4560.95
18	5/1/2017	50897.5	55259.73	6618.58
19	6/1/2017	58852.12	56058.3	7954.62
20	7/1/2017	69768.68	56058.13	10916.56

Figure 5.16 Screenshot of Excel Showing Difference in Column D (Used with permission from Microsoft)

If you need to compute the second-order differences, just do the same procedure to find the difference of column D. Higher-order differences can easily be computed in the analogous way.

SMA and Differencing in Python

Computing an SMA in Python takes only a single line of code. Here is how it works on the Coal Consumption dataset (found in MonthlyCoalConsumption.xlsx (https://openstax.org/r/spreadsheetsd1j)), using a window size of 12. (Before, we used a window of 13 because odd-length windows are easier to work with than even-length ones. However, Python easily handles all cases automatically.) The key here is the function `rolling()` that creates a DataFrame of rolling windows of specified length. Then the `mean()` command computes the average of each window of data. Computing the first-order difference is just as easy, using the `diff()` command.

PYTHON CODE

```
# Import libraries
import pandas as pd # for dataset management
import matplotlib.pyplot as plt # for visualizations
from matplotlib.ticker import MaxNLocator, FuncFormatter # for graph formatting and
y-axis formatting
import matplotlib.dates as mdates # for date formatting
```

```
# Read the Excel file into a Pandas DataFrame
df = pd.read_excel('MonthlyCoalConsumption.xlsx')

# Convert 'Month' column to datetime format
df['Month'] = pd.to_datetime(df['Month'], format='%m/%d/%Y')

# Calculate a simple moving average with a window size of 12
df['SMA'] = df['Value'].rolling(window=12).mean()

# Calculate the first-order difference
df['Diff'] = df['Value'].diff()

# Function to format the Y-axis values
def y_format(value, tick_number):
  return f'{int(round(value)):,}'

# Create the plot
plt.figure(figsize=(12, 6))

# Plot the time series, SMA, and difference
plt.plot(df['Month'], df['Value'], marker='', linestyle='-', color='b',
label='Value')
plt.plot(df['Month'], df['SMA'], marker='', linestyle='-', color='r', label='SMA')
plt.plot(df['Month'], df['Diff'], marker='', linestyle='-', color='g',
label='Diff')

# Set the x-axis limits to start from the first date and end at the maximum date in
the data
plt.xlim(pd.to_datetime('01/01/2016', format='%m/%d/%Y'), df['Month'].max())

# Format the x-axis to show dates as MM/DD/YYYY and set the locator for a fixed
number of ticks
ax = plt.gca()
ax.xaxis.set_major_formatter(mdates.DateFormatter('%m/%d/%Y'))
ax.xaxis.set_major_locator(MaxNLocator(nbins=12)) # Set the number of ticks to a
reasonable value

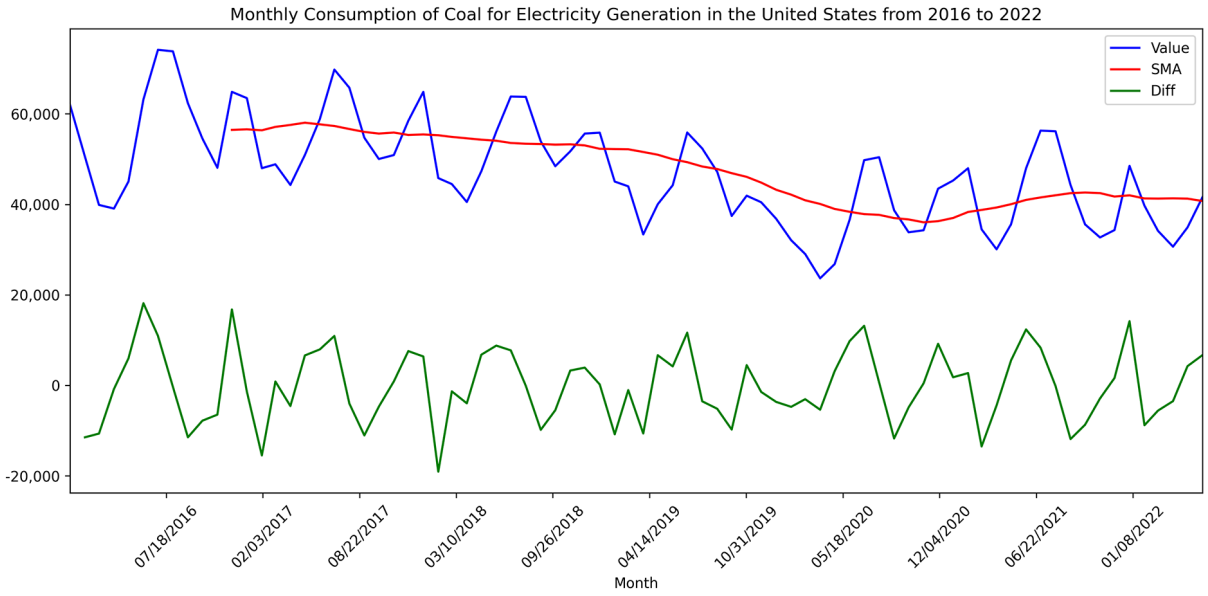
# Manually set the rotation of date labels to 45 degrees
plt.xticks(rotation=45)

# Title and labels
plt.title('Monthly Consumption of Coal for Electricity Generation in the United
States from 2016 to 2022')
plt.xlabel('Month')
plt.legend()

# Apply the formatter to the Y-axis
ax.yaxis.set_major_formatter(FuncFormatter(y_format))
```

```
plt.tight_layout() # Adjust layout to ensure labels fit well
plt.show()
```

The resulting output will look like this:



Analyzing Seasonality

Once the time series has been detrended using one of the methods already discussed (or other, more sophisticated methods), what is left over should represent only the **seasonality**, or *seasonal variation*, s_n, and error term, ε_n.

$$z_n = s_n + \varepsilon_n$$

How could we analyze and model s_n? Suppose there were no error term at all ($\varepsilon_n = 0$) and so we are looking at a time series with pure seasonal variation and no other factors. In this case, the component s_n would look like a repeating pattern, such as the pattern shown in Figure 5.17. As mentioned in Components of Time Series Analysis, the time interval that it takes before the pattern completes once and begins to repeat is called the period. In a purely seasonal component, we would expect $s_{n+P} = s_n$, where P is the period, for all values of n.

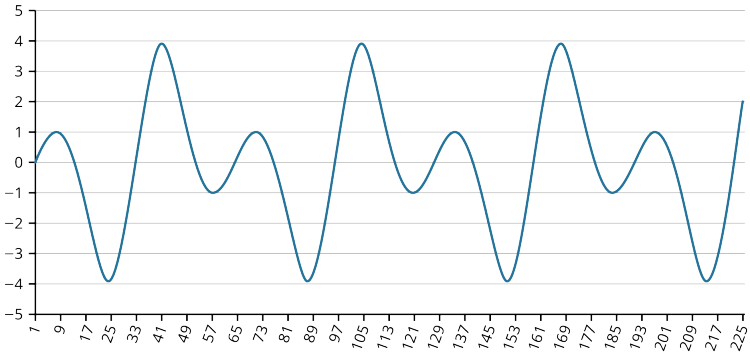


Figure 5.17 Pure Seasonal Variation. The same pattern repeats after 64 time steps, and so the period is 64.

While it is easy to spot the repeating pattern in a time series that does not have noise, the same cannot be said for the general case. There are several tools available for isolating the seasonal component of a detrended time series.

Identifying the Period

First we need to identify the period of the seasonal data. The simplest method is to use prior knowledge about the time series data, along with a visual inspection. Often, a seasonal pattern is exactly that—a pattern that repeats over the course of one year. Thus, the period would be $P = 1$ year (or 12 months, 52 weeks, 365 days, etc., depending on the time steps of your time series). For example, even without detrending the data, it was easy to identify a period of 4 quarters (1 year) in the data in Example 5.2.

If repetition of a seasonal pattern is not apparent from the data, then it may be discovered using standard statistical measures such as autocorrelation. **Autocorrelation** is the measure of correlation between a time series and the same time series shifted by some number of time steps. (Correlation and Linear Regression Analysis discussed correlation between two sets of data. Autocorrelation is the same idea applied to the same dataset with a shifted copy of itself.) The size of the shift is called the **lag**.

The **autocorrelation function (ACF)** measures the correlation at each lag starting with 0 or 1. Note, lag 0 always produces an autocorrelation of 1 since the time series is being compared to an exact copy of itself. Thus, we are only interested in the correlation coefficient at positive lags. However, small lags generally show high correlation with the original time series as well, simply because data measured within shorter time periods are generally similar to each other. So, even if the ACF shows relatively large correlation at lags of 1, 2, 3, etc., there may not be a true seasonal component present. The positive lag corresponding to the first significant **peak**, or autocorrelation value that is much larger than the previous values, may indicate a seasonal component with that period.

Later, we will see how to compute ACF of a time series using Python, but first let's explore further topics regarding seasonal components.

Seasonally Adjusted Data

Once you have determined the period of the seasonal component, you can then extract s_n from the time series. As usual, there are a number of different ways to proceed.

Let's assume that you have detrended the time series so that what remains consists of a seasonal component and noise term:

$$z_n = x_n - t_n = s_n + \varepsilon_n$$

Moreover, suppose that the period of s_n is P. Of course, with the noise term still present in the time series, we cannot expect that $s_{n+P} = s_n$ holds, but it should be the case that $s_{n+P} \approx s_n$ for every n. In other words, the data should look approximately the same when shifted by P time steps. Our job now is to isolate one complete period of data that models the seasonal variation throughout the time series. In other words, we want to produce an estimate $\hat{s}_n$ that is exactly periodic ($\hat{s}_{n+P} = \hat{s}_n$, for all n) and models the underlying seasonal component as closely as possible.

For any given index $n < P$, we expect the data points $\left(z_n, z_{n+P}, z_{n+2P}, z_{n+3P}, \ldots\right)$ to hover around the same value or, in statistical language, to have rather low variance (much lower than the time series as a whole, that is). So it seems reasonable to use the mean of this slice of data to define the value of $\hat{s}_n$. For simplicity, assume that the time series has exactly $M \cdot P$ data points (if not, then the data can be padded in an appropriate way). Then the estimate for the seasonal component would be defined for $1 \leq n < P$ by the formula

$$\hat{s}_n = \frac{1}{M}\left(z_n + z_{n+P} + z_{n+2P} + z_{n+3P} + \ldots + z_{n+(M-1)P}\right)$$

As usual, the formula looks more intimidating than the concept that it's based upon. As previously mentioned, this represents nothing more than the average of all the values of the time series that occur every P time steps.

Once $\hat{s}_n$ is computed in the range $1 \leq n < P$, extend it by repeating the same values in each period until you

reach the last index of the original series ($n = MP$). (Note: The noise term may now be estimated as the series of residuals, $\epsilon_n \approx s_n - \hat{s}_n$, which will occupy our interest in Forecast Evaluation Methods.)

Finally, the estimate $\hat{s}_n$ can then be used in conjunction with the original time series (x_n) to smooth out the seasonal variation. Note that $\hat{s}_n$ essentially measures the fluctuations of observed data x_n above or below the trend. We expect the mean of $\hat{s}_n$ to be close to zero in any window of time. Define the **seasonally adjusted** data as the time series

$$x_n - \hat{s}_n$$

A seasonally adjusted time series is closely related to the underlying trend-cycle curve, t_n, though generally it contains the noise and other random factors that t_n does not generally include.

EXAMPLE 5.6

Problem

The revenue data from Walt's Water Adventures (see Table 5.2) seems to display seasonality of period 4. Using the level as the trend, detrend the data, compute the seasonal component $\hat{s}_n$, and find the seasonally adjusted time series.

Solution

First, we need to detrend the data. The level is the average of all 12 observations. Using a calculator or software, the level is found to be $L = 2{,}275.42$. Subtract $x_n - L$ to detrend the data (see Table 5.4). For example, the first term of the detrended data would be $z_1 = x_1 - L = 1{,}799 - 2{,}275.42 = -476.42$.

There are a total of 12 observations, so there will be $M = 12/4 = 3$ periods of size $P = 4$. Using the values of z_n in the formula for $\hat{s}_n$, we find:

$$\begin{aligned}
\hat{s}_1 &= \tfrac{1}{3}(z_1 + z_5 + z_9) = \tfrac{1}{3}((-476.42) + 8.58 + 428.58) = -13.08 \\
\hat{s}_2 &= \tfrac{1}{3}(z_2 + z_6 + z_{10}) = \tfrac{1}{3}(1{,}028.58 + 1{,}183.58 + 1{,}573.58) = 1{,}261.92 \\
\hat{s}_3 &= \tfrac{1}{3}(z_3 + z_7 + z_{11}) = \tfrac{1}{3}((-996.42) + (-311.42) + (-144.42)) = -484.08 \\
\hat{s}_4 &= \tfrac{1}{3}(z_4 + z_8 + z_{12}) = \tfrac{1}{3}((-1{,}276.42) + (-681.42) + (-336.42)) = -764.75
\end{aligned}$$

Then repeat this block of 4 values as needed to obtain the entire seasonal component. Finally, the seasonally adjusted time series is the difference between the observed data and seasonal component (see Table 5.4).

k	Quarter	Revenues	Detrended z_k	Seas. Comp. $\hat{s}_k$	Seasonally Adj. $x_n - \hat{s}_n$
1	2021 Spring	1799	−476.42	−13.08	1812.08
2	2021 Summer	3304	1028.58	1261.92	2042.08
3	2021 Fall	1279	−996.42	−484.08	1763.08
4	2021 Winter	999	−1276.42	−764.75	1763.75
5	2022 Spring	2284	8.58	−13.08	2297.08
6	2022 Summer	3459	1183.58	1261.92	2197.08

Table 5.4 Quarterly Revenues for Kayak and Canoe Rentals (all columns in $). Columns for detrended data, seasonal

k	Quarter	Revenues	Detrended z_k	Seas. Comp. $\hat{s}_k$	Seasonally Adj. $x_n - \hat{s}_n$
7	2022 Fall	1964	−311.42	−484.08	2448.08
8	2022 Winter	1594	−681.42	−764.75	2358.75
9	2023 Spring	2704	428.58	−13.08	2717.08
10	2023 Summer	3849	1573.58	1261.92	2587.08
11	2023 Fall	2131	−144.42	−484.08	2615.08
12	2023 Winter	1939	−336.42	−764.75	2703.75

Table 5.4 Quarterly Revenues for Kayak and Canoe Rentals (all columns in $). Columns for detrended data, seasonal component estimate, and seasonally adjusted data are included.

Figure 5.18 shows the data along with the seasonal component and seasonally adjusted data.

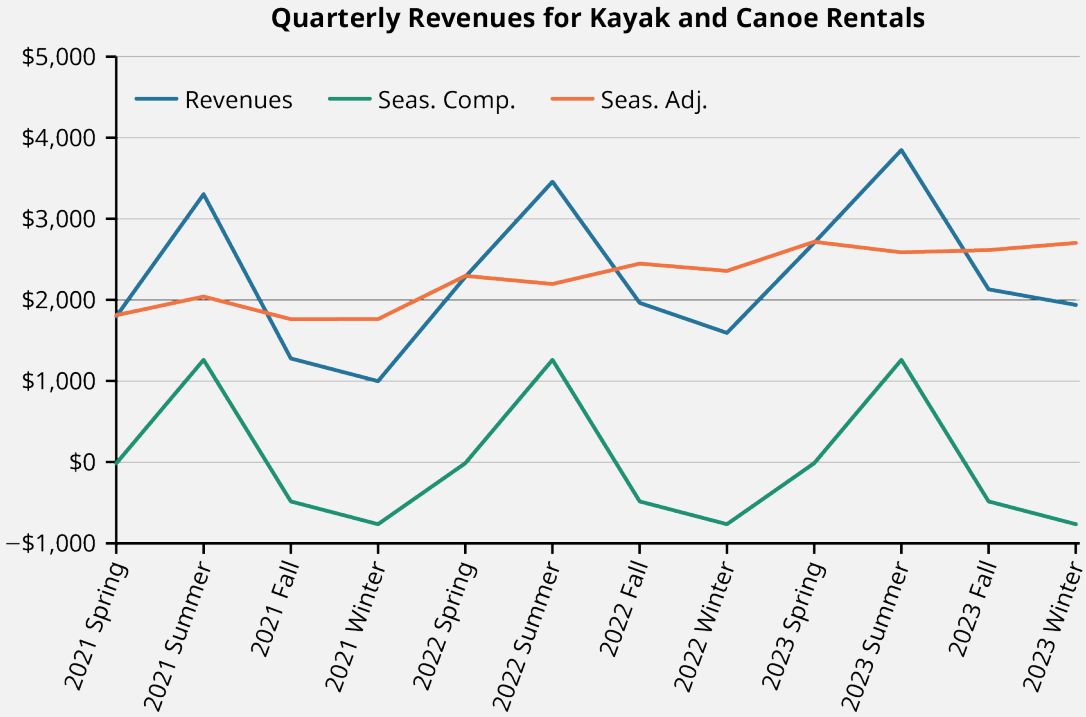


Figure 5.18 Quarterly Revenues for Kayak and Canoe Rentals. Seasonal component and seasonally adjusted time series are included on the graph.

Decomposing a Time Series into Components Using Python

There is a powerful Python library known as statsmodels (https://openstax.org/r/statsmodels) that collects many useful statistical functions together. We will use the `STL` module to decompose the time series MonthlyCoalConsumption.xlsx (https://openstax.org/r/spreadsheetsd1j) into its components. First, we plot an ACF (autocorrelation function) to determine the best value for periodicity.

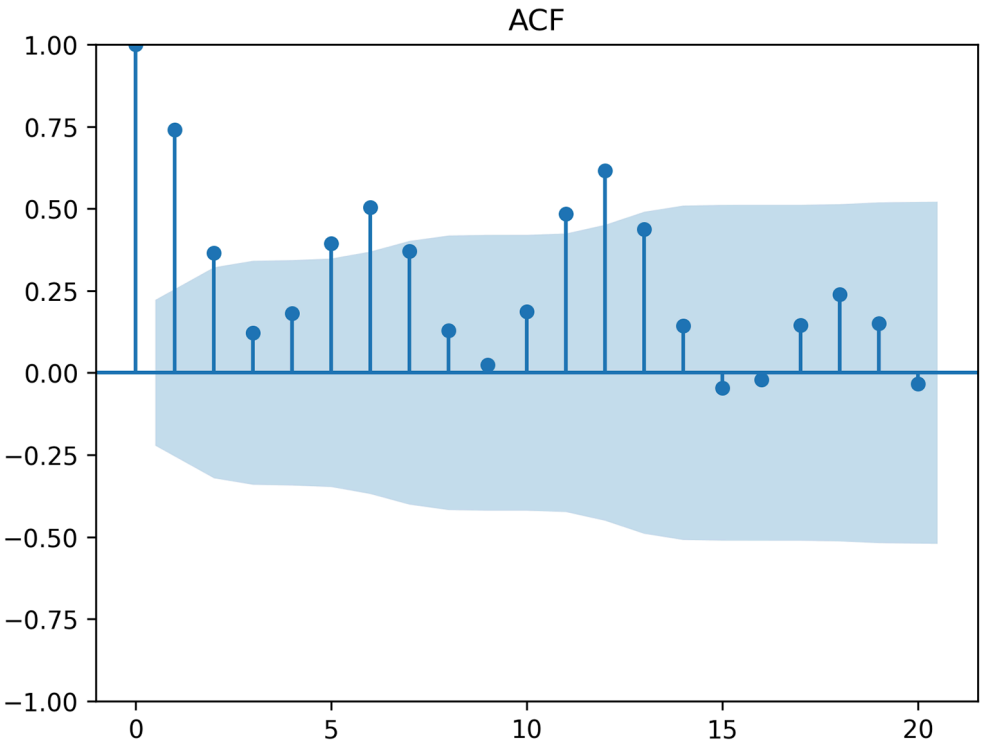
PYTHON CODE

```
# Import libraries
from statsmodels.graphics.tsaplots import plot_acf

# Read the Excel file into a Pandas Dataframe
df = pd.read_excel('MonthlyCoalConsumption.xlsx')

# Plot the ACF
plot_acf(df['Value'], lags=20, title='ACF');
```

The resulting output will look like this:



Points that are within the blue shaded area correspond to lags at which the autocorrelation is statistically insignificant. The lags that show the most significance are at 0, 1, 6, and 12. $\text{Lag} = 0$ always gives 100% autocorrelation because we are comparing the unshifted series to itself. At $\text{lag} = 1$, a high autocorrelation corresponds to the fact that consecutive data points are relatively close to each other (which is also to be expected in most datasets). Of special note are lags at 6 and 12, corresponding to a possible seasonal period of 6 or 12 in the data. We will use period 12 in our STL decomposition.

PYTHON CODE

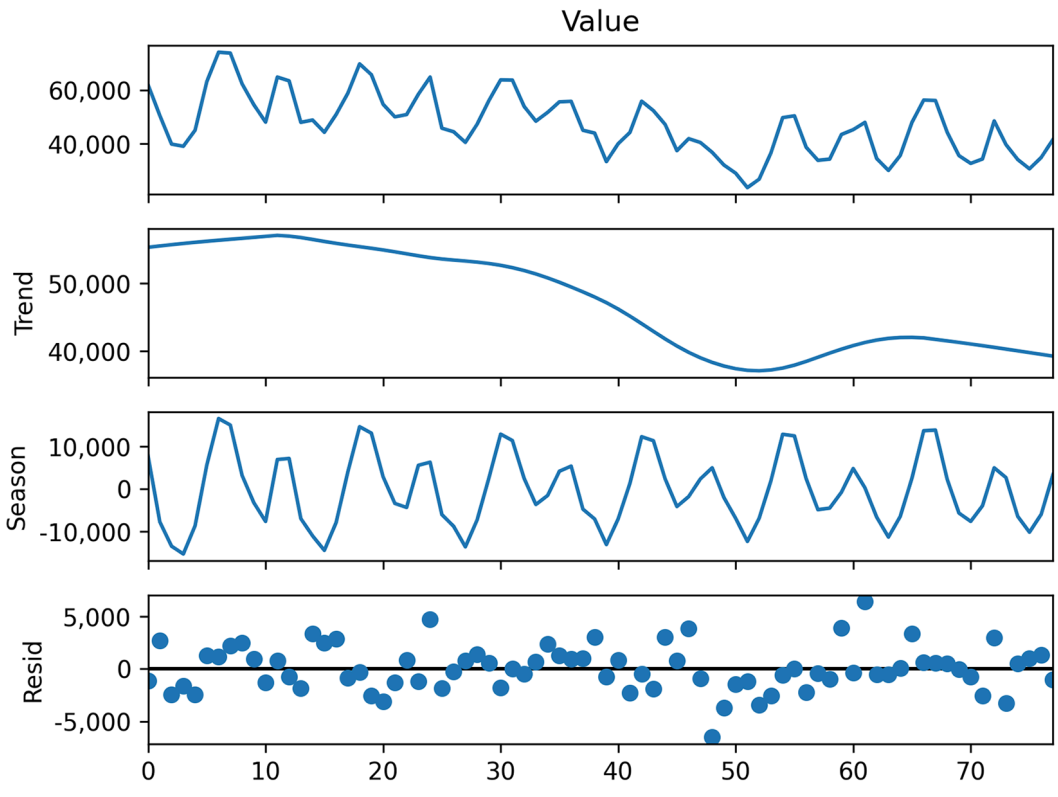
```
# Import libraries
from statsmodels.tsa.seasonal import STL
import matplotlib.pyplot as plt ## for data visualization

# Perform seasonal decomposition using STL
stl = STL(df['Value'], period=12)
result = stl.fit()

# Plot components and format the y-axis labels
fig = result.plot()
for ax in fig.axes:
  ax.yaxis.set_major_formatter(FuncFormatter(y_format))

plt.show()
```

The resulting output will look like this:



The trend, seasonal variation, and residual (noise) components are all graphed in separate plots. To see them all on the same set of axes, we can use matplotlib (noise omitted).

PYTHON CODE

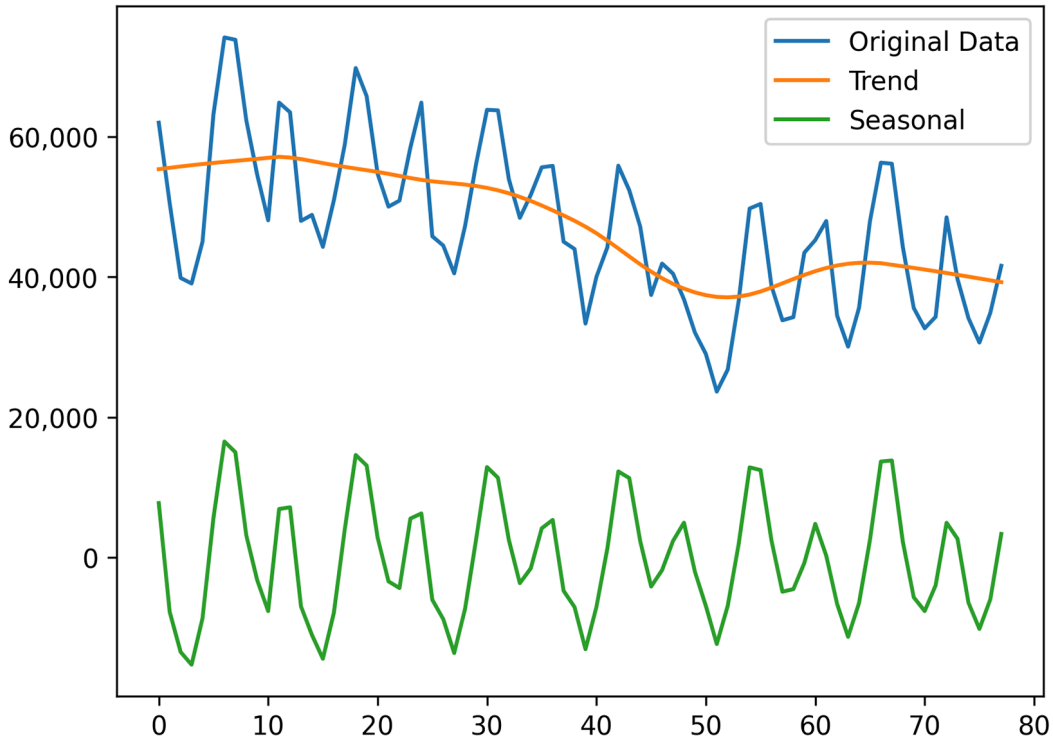
```
import matplotlib.pyplot as plt ## for data visualization

# Extract the trend and seasonal components
trend = result.trend
seasonal = result.seasonal

# Plot the original data, trend, and seasonally adjusted data
plt.plot(df['Value'], label='Original Data')
plt.plot(trend, label='Trend')
plt.plot(seasonal, label='Seasonal')
plt.legend()

# Apply the formatter to the Y-axis
plt.gca().yaxis.set_major_formatter(FuncFormatter(y_format))
plt.show()
```

The resulting output will look like this:



Weighted Moving Averages and Exponential Moving Averages

Simple moving averages were discussed in detail at the beginning of this section. Now we expand the discussion to include more sophisticated weighted moving averages and their uses in smoothing data and forecasting. Recall from our discussion in Time Series Forecasting Methods that **weighted moving averages** are moving averages in which terms are given *weights* according to some formula or rule.

Suppose that a time series (x_n) has no significant seasonal component (or the seasonal component has been removed so that remaining series is seasonally adjusted). Thus, the series has only trend-cycle and noise components:

$$x_n = t_n + \varepsilon_n$$

As already discussed, a simple moving average (SMA) has the effect of smoothing out the data and providing an easy method for predicting the next term of the sequence. For reference, here is the SMA prediction model, with a window of T terms:

$$\hat{x}_{n+1} = \frac{x_n + x_{n-1} + x_{n-2} + \ldots + x_{n-T+1}}{T}$$

This model may be regarded as a weighted moving average model (WMA) in which the weights of the most recent T terms are all equal to $\frac{1}{T}$, while all other weights are set to 0. For example, in an SMA model with $T = 3$, the weights would be $\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0, 0, 0, 0, \ldots$

$$\hat{x}_{n+1} = \frac{x_n + x_{n-1} + x_{n-2}}{3} = \frac{1}{3}x_n + \frac{1}{3}x_{n-1} + \frac{1}{3}x_{n-2} + 0x_{n-3} + 0x_{n-4} + \ldots$$

Even the naïve model, $\hat{x}_{n+1} = x_n$, counts as a WMA, with weights 1, 0, 0, ...

In principle, a weighted moving average model may be built using any coefficients, giving a general form for any WMA:

$$\hat{x}_{n+1} = a_n x_n + a_{n-1} x_{n-1} + a_{n-2} x_{n-2} + a_{n-3} x_{n-3} + \ldots$$

Of course, if we want the model to do a good job in forecasting new values of the time series, then the coefficients must be chosen wisely. Typically, the weights must be **normalized**, which means that they sum to 1.

$$a_n + a_{n-1} + a_{n-2} + a_{n-3} + \ldots = 1$$

However, there is no hard-and-fast requirement that the sum must be equal to 1.

Rather than giving the same weight to some number of previous terms as the SMA model does, you might choose to give greater weight to more recent terms. For example, one might use powers of $\frac{1}{2}$ as weights.

$$\hat{x}_{n+1} = \frac{1}{2}x_n + \frac{1}{4}x_{n-1} + \frac{1}{8}x_{n-2} + \frac{1}{16}x_{n-3} + \ldots$$

This method allows the most recent term to dominate the sum, and each successive term contributes half as much in turn. (Note: The sequence $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \ldots$ *would* be a normalized sequence if infinitely many terms were included, but even a small finite number of terms, like 5 or so, adds up to a sum that is pretty close to 1.)

This is an example of an **exponential moving average**, or **EMA**. The most general formula for the EMA model is shown. (Note that the previous example corresponds to an EMA model with $\alpha = \frac{1}{2}$).

$$\hat{x}_{n+1} = \alpha x_n + \alpha\big(1-\alpha\big)x_{n-1} + \alpha(1-\alpha)^2 x_{n-2} + \alpha(1-\alpha)^3 x_{n-3} + \ldots$$

EXAMPLE 5.7

Problem

Use EMA models with $\alpha = \frac{1}{2}$ (equivalently, 0.5) and $\alpha = 0.7$ for the S&P Index data from Table 5.1 to estimate the value of the S&P Index at the end of year 2023.

Solution

First, for $\alpha = \frac{1}{2}$, the model is:

$$\hat{x}_{n+1} = \frac{1}{2}x_n + \frac{1}{4}x_{n-1} + \frac{1}{8}x_{n-2} + \frac{1}{16}x_{n-3} + \ldots$$

Now plug in the data (remember to use the most recent data point first and work your way backward). Eight

terms were used, as the terms beyond that point are rather close to 0. Table 5.2 records the results.

For $\alpha = 0.7$, use the formula for weights,

$\alpha = 0.7$, $\alpha(1-\alpha) = 0.7(1-0.7) = 0.7(0.3) = 0.21$, $\alpha(1-\alpha)^2 = 0.7(0.3)^2 = 0.063$, etc. This time, the coefficients decrease quicker than in the case $\alpha = \frac{1}{2}$, so only 5 terms are significant. Table 5.5 shows the results.

Year	S&P Index at Year-End	Exponentially Weighted $\alpha = \frac{1}{2}$	Exponentially Weighted $\alpha = 0.7$
2013	1848.36	≈ 0	≈ 0
2014	2058.9	≈ 0	≈ 0
2015	2043.94	$\frac{1}{512} \cdot 2043.94 = 3.99$	≈ 0
2016	2238.83	$\frac{1}{256} \cdot 2238.83 = 8.75$	≈ 0
2017	2673.61	$\frac{1}{128} \cdot 2673.61 = 20.89$	≈ 0
2018	2506.85	$\frac{1}{64} \cdot 2506.85 = 39.17$	$0.00567 \cdot 2506.9 = 4.26$
2019	3230.78	$\frac{1}{32} \cdot 3230.78 = 100.96$	$0.00567 \cdot 3230.78 = 18.32$
2020	3756.07	$\frac{1}{16} \cdot 3756.07 = 234.75$	$0.0189 \cdot 3756.07 = 70.99$
2021	4766.18	$\frac{1}{8} \cdot 4766.18 = 595.77$	$0.063 \cdot 4766.18 = 300.27$
2022	3839.5	$\frac{1}{4} \cdot 3839.5 = 959.88$	$0.21 \cdot 3839.5 = 806.30$
2023	4769.83	$\frac{1}{2} \cdot 4769.83 = 2384.92$	$0.7 \cdot 4769.83 = 3338.88$
2024		SUM $= 4349$	SUM $= 4541$

Table 5.5 EMA Models (source: adapted from https://www.nasdaq.com/market-activity/index/spx/historical)

Therefore, we find estimates of 4,349 and 4,541 for the S&P value at the end of 2024, by EMA models with $\alpha = \frac{1}{2}$ and $\alpha = 0.7$, respectively. Note that larger values of α emphasize the most recent data, while smaller values allow more of the past data points to influence the average. If the time series trend experiences more variability (due to external factors that we have no control over), then an EMA with a relatively large α -value would be more appropriate than one with a small α-value.

EMA models may also serve as smoothing techniques, isolating the trend-cycle component. In this context, it's better to use an equivalent form of the EMA model:

$$\hat{x}_{n+1} = \alpha x_n + (1-\alpha)\hat{x}_n$$

In this form, we build a series of estimates $(\hat{x}_n)$ recursively by incorporating the terms of the original time series one at a time. The idea is that the next estimate, $\hat{x}_{n+1}$, is found by weighted average of the current data point, x_n, and the current estimate, $\hat{x}_n$. Note, we may set $\hat{x}_1 = x_1$ since we need to begin the process somewhere. It can be shown mathematically that this procedure is essentially the same as working out the sum of exponentially weight terms directly and is much more efficient.

EXAMPLE 5.8

Problem

Use the recursive EMA formula with $\alpha = 0.75$ to smooth the S&P Index time series (Table 5.1), and then use the EMA model to estimate the value of the S&P Index at the end of the 2024. Plot the EMA together with the original time series.

Solution

With $\alpha = 0.75$, the recursive EMA formula becomes: $\hat{x}_{n+1} = 0.75x_n + 0.25\hat{x}_n$. Table 5.6 shows the results of the computation.

Year	S&P Index at Year-End	EMA Estimate
2013	$x_1 = 1848.36$	$\hat{x}_1 = x_1 = 1848.36$
2014	$x_2 = 2058.9$	$\hat{x}_2 = 0.75x_1 + 0.25\hat{x}_1 = 0.75(1848.36) + 0.25(1848.36) = 1848.36$
2015	$x_3 = 2043.94$	$\hat{x}_3 = 0.75x_2 + 0.25\hat{x}_2 = 0.75(2058.9) + 0.25(1848.36) = 2006.27$
2016	$x_4 = 2238.83$	$\hat{x}_4 = 0.75x_3 + 0.25\hat{x}_3 = 0.75(2043.94) + 0.25(2006.27) = 2034.52$
2017	$x_5 = 2673.61$	$\hat{x}_5 = 0.75x_4 + 0.25\hat{x}_4 = 0.75(2238.83) + 0.25(2034.52) = 2187.75$
2018	$x_6 = 2506.85$	$\hat{x}_6 = 0.75x_5 + 0.25\hat{x}_5 = 0.75(2673.61) + 0.25(2187.75) = 2552.15$
2019	$x_7 = 3230.78$	$\hat{x}_7 = 0.75x_6 + 0.25\hat{x}_6 = 0.75(2506.85) + 0.25(2552.15) = 2518.17$
2020	$x_8 = 3756.07$	$\hat{x}_8 = 0.75x_7 + 0.25\hat{x}_7 = 0.75(3230.78) + 0.25(2518.17) = 3052.63$
2021	$x_9 = 4766.18$	$\hat{x}_9 = 0.75x_8 + 0.25\hat{x}_8 = 0.75(3756.07) + 0.25(3052.63) = 3580.21$
2022	$x_{10} = 3839.5$	$\hat{x}_{10} = 0.75x_9 + 0.25\hat{x}_9 = 0.75(4766.18) + 0.25(3580.21) = 4469.69$
2023	$x_{11} = 4769.83$	$\hat{x}_{11} = 0.75x_{10} + 0.25\hat{x}_{10} = 0.75(3839.5) + 0.25(4469.69) = 3997.05$
2024	N/A	$\hat{x}_{12} = 0.75x_{11} + 0.25\hat{x}_{11} = 0.75(4769.83) + 0.25)3997.05) = 4576.63$

Table 5.6 EMA Smoothing for the S&P Index Time Series (source: adapted from https://www.nasdaq.com/market-activity/index/spx/historical)

According to this model, the S&P Index will be about 4577 at the end of 2024. Figure 5.19 shows the time series together with the exponential smoothing model.

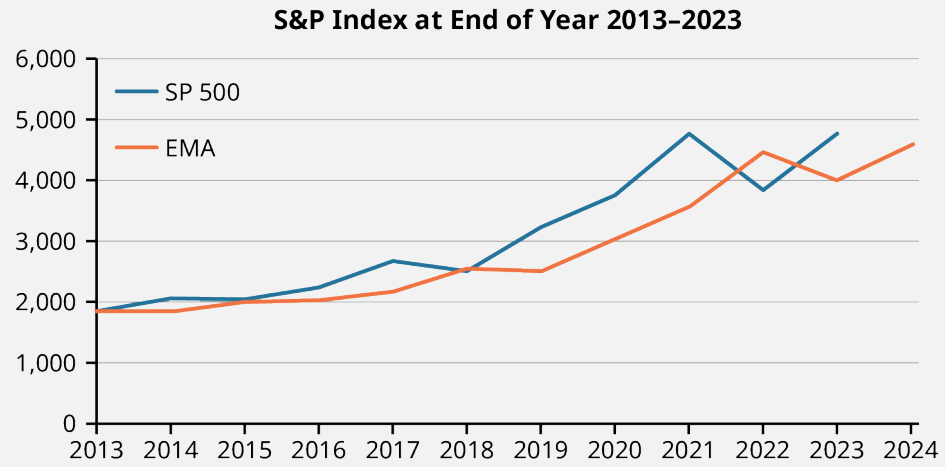


Figure 5.19 Exponential Smoothing Model for the S&P Index (data source: adapted from S&P 500 [SPX] Historical Data)

Note, EMA smoothing can be done very easily in Excel. Simply type in the formulas =B2 into cell C2 and =0.75*B2+0.25*C2 into cell C3. Then drag cell C3 down to the end of the data to compute the EMA. You can also do EMA smoothing in Python using the `ewm()` function applied to a dataset. Note, the "span" value should be set to $\frac{2}{\alpha} - 1$, as shown in this code.

PYTHON CODE

```
# Import libraries
import pandas as pd ## for dataset management

# Creating a dictionary with the given data
data = {
  'Year': [2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023],
  'SP500': [1848.36, 2058.9, 2043.94, 2238.83, 2673.61, 2506.85, 3230.78,
3756.07, 4766.18, 3839.5, 4769.83]
}
# Creating DataFrame
df = pd.DataFrame(data)

alpha = 0.75
df['ema']=df['SP500'].ewm(span=2/alpha-1, adjust=False).mean()

print(df['ema'])
```

The resulting output will look like this:

```
0   1848.360000
1   2006.265000
2   2034.521250
3   2187.752812
4   2552.145703
5   2518.173926
6   3052.628481
7   3580.209620
8   4469.687405
9   3997.046851
10  4576.634213
Name: ema, dtype: float64
```

Autoregressive Integrated Moving Average (ARIMA)

In this section, we develop a very powerful tool that combines multiple models together and is able to capture trend and seasonality. The method is called **autoregressive integrated moving average**, or **ARIMA** for short, and it consists of three main components:

1. **Autoregressive (AR) model**. This component captures patterns in the time series that are repeated,

including seasonal components as well as correlation with recent values.

2. **Integrative (I) component**. This component represents the differencing operation discussed in the section on detrending a time series.
3. **Moving average (MA) model.** Despite the name, this component is not the same as the moving average discussed previously, although there is a mathematical connection. Instead of focusing on averages of existing data points, MA models incorporate the influence of past forecast errors to predict the next value of the time series.

Before getting into the specifics of each component of ARIMA, we need to talk about the term **stationary**, which refers to a time series in which the variance is relatively constant over time, an overall upward or downward trend cannot be found, and no seasonal patterns exist. White noise is an example of a stationary time series; however, the term stationary is a bit more general, as it includes the possibility of a cyclic component in the data. Recall, a cyclic component (despite its name) is a longer-term rise or fall in the values of a time series that may occur from time to time, but the occurrences are unpredictable.

The AR and MA models work best on stationary time series. The integrative component is the tool that can detrend the data, transforming a nonstationary time series into a stationary one so that AR and/or MA may be used.

Autoregressive Models

The autoregressive models take into account a fixed number of previous values of the time series in order to predict the next value. In this way, AR models are much like WMA models previously discussed, in the sense that the next term of the time series depends on a sum of weighted terms of previous values; however, in the AR model, the terms used are the terms of the model itself.

Suppose we want to model a time series, (x_n), using an AR model. First, we should pick a value p, called the order of the model, which specifies how many previous terms to include in the model. Then we may call the model an $\mathrm{AR}\,(p)$ model. If there is a strong seasonal component of period$p$, then it makes sense to choose that value as the order.

An $\mathrm{AR}\,(p)$ model is a recursive formula of the form

$$\mathrm{AR}\,(p)_n = w_1 y_{n-1} + w_2 y_{n-2} + w_3 y_{n-3} + \ldots + w_p y_{n-p} + \varepsilon_n$$

Here, the parameters or weights, $w_1, w_2, w_3, \ldots w_p$, are constants that must be chosen so that the values of the time series (y_n) fit the known data as closely as possible (i.e., $y_n \approx x_n$ for all known observations). There are also certain restrictions on the possible values of each weight c, but we do not need to delve into those specifics in this text as we will rely on software to do the work of finding the parameters for us.

Integrative Component

The "I" in ARIMA stands for *integrative*, which is closely related to the word *integral* in calculus. Now if you have seen some calculus, you might recall that the integral is just the opposite of a *derivative*. The derivative, in turn, is closely related to the idea of differencing in a time series. Of course, you don't have to know calculus to understand the idea of an integrative model.

Rather than producing another component of the time series, the integrative component of ARIMA represents the procedure of taking differences, $\Delta^d\,(x_n)$ (up to some specific order d), until the time series becomes stationary. Often, only the first-order difference is needed, and you can check for stationarity visually. A more sophisticated method, called the **Augmented Dickey-Fuller (ADF) test**, sets up a test statistic that measures the presence of non-stationarity in a time series. Using ADF, a p-value is computed. If the p-value is below a set threshold (such as $p < 0.01$), then the null hypothesis of non-stationarity can be rejected. Next, we will show how to run the ADF test on a time series and its differences $\Delta^d\,(x_n)$ to find out the best value for d in ARIMA.

Then the AR and MA models can be produced on the resulting stationary time series. Integrating (which is the

opposite of differencing) builds the model back up so that it can ultimately be used to predict values of the original time series (x_n).

Moving Average Models

An MA model of order q, or $\text{MA}\,(q)$, takes the form

$$\text{MA}\,(q)_n = L + \varepsilon_n + b_1\varepsilon_{n-1} + b_2\varepsilon_{n-2} + b_3\varepsilon_{n-3} + \ldots + b_q\varepsilon_{n-q}$$

Here, the ε_n terms are residuals (forecast errors), and L is the level of the time series (mean value of all the terms). Again, there are a multitude of parameters that need to be chosen in order for the model to make sense. We will rely on software to handle this in general.

ARIMA Model in Python

The full ARIMA model, $\text{ARIMA}\,(p, d, q)$, incorporates all three components—autoregressive of order p, integrative with d-order differencing, and moving average of order q. For the time series MonthlyCoalConsumption.xlsx (https://openstax.org/r/spreadsheetsd1j), it turns out that the second-order difference is stationary, as shown by a series of ADF tests.

PYTHON CODE



```
# Import libraries
import pandas as pd ## for dataset management
# Import adfuller from the statsmodels library
from statsmodels.tsa.stattools import adfuller

# Read data
df = pd.read_excel('MonthlyCoalConsumption.xlsx')

print('d = 0')
result = adfuller(df['Value'])
print('ADF Statistic:', result[0])
print('p-value:', result[1])

## Get the first difference
print('\n d = 1')
df1 = df['Value'].diff().dropna()
result = adfuller(df1)
print('ADF Statistic:', result[0])
print('p-value:', result[1])

## Get the second difference
print('\n d = 2')
df2 = df1.diff().dropna()
result = adfuller(df2)
print('ADF Statistic:', result[0])
print('p-value:', result[1])
```

The resulting output will look like this:

```
d = 0
ADF Statistic: -1.174289795062662
p-value: 0.6845566772896323

d = 1
ADF Statistic: -1.7838215415905252
p-value: 0.38852291349101137

d = 2
ADF Statistic: -8.578323839498081
p-value: 7.852009937900099e-14
```

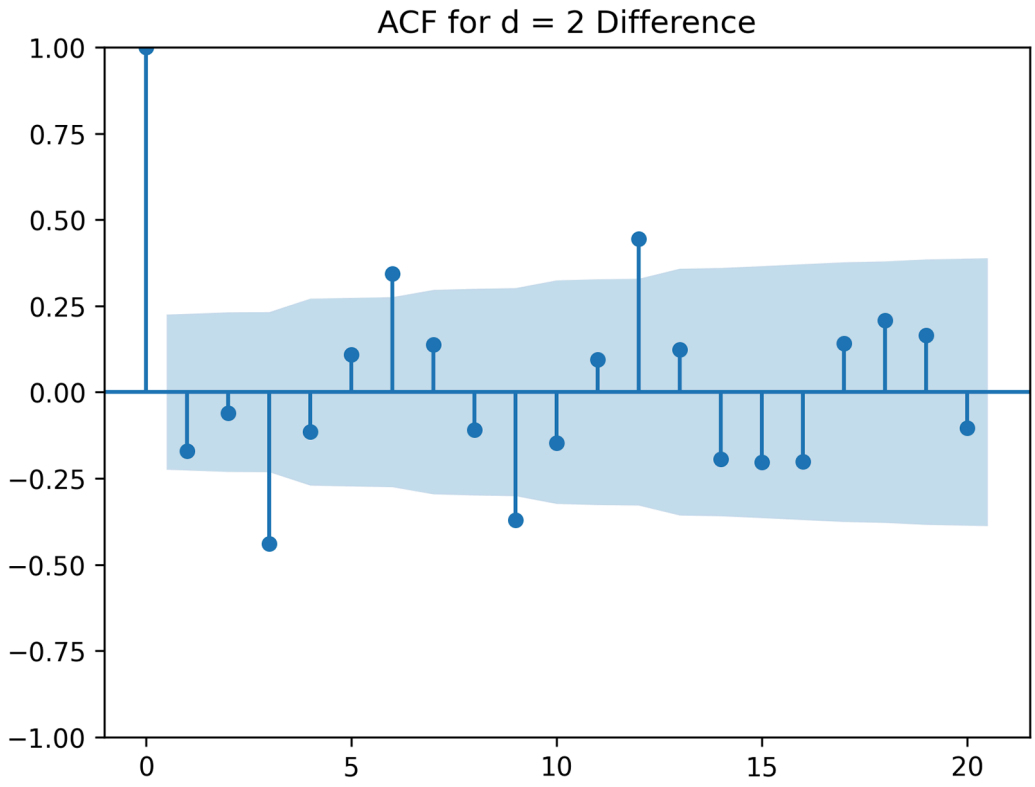
Here, a p-value of 7.852×10^{-14} indicates that the second-order difference is stationary, so we will set $d = 2$ in the ARIMA model. An ACF plot of the second difference can be used to determine p.

PYTHON CODE



```
# Import libraries
from statsmodels.graphics.tsaplots import plot_acf

# Plot the ACF of the second difference
plot_acf(df2, lags=20, title='ACF for d = 2 Difference');
```



We see a peak at lag 12, corresponding to the expected yearly seasonal variation, so we will set $p = 12$ in the ARIMA model. For this time series, positive values of q did not change the accuracy of the model at all, so we set $q = 0$ (MA model component unnecessary). The model was used to forecast 24 time steps (2 years) into the future. We are using the Python library statsmodels.tsa.arima.model (https://openstax.org/r/statsmodel) to create the ARIMA model.

PYTHON CODE

```
# Import ARIMA model from statsmodels.tsa library
from statsmodels.tsa.arima.model import ARIMA
import matplotlib.pyplot as plt ## for plotting graphs
from matplotlib.ticker import MaxNLocator ## for graph formatting
from matplotlib.ticker import FuncFormatter ## for formatting y-axis

# Fit an ARIMA model
p = 12; d = 2; q = 0 # ARIMA component orders
model = ARIMA(df['Value'], order=(p, d, q))
results = model.fit()

# Make forecasts
fc = results.forecast(steps=24)

# Plot the results
plt.figure(figsize=(10, 6))

# Plot original time series
plt.plot(df['Value'], label='Original Time Series')

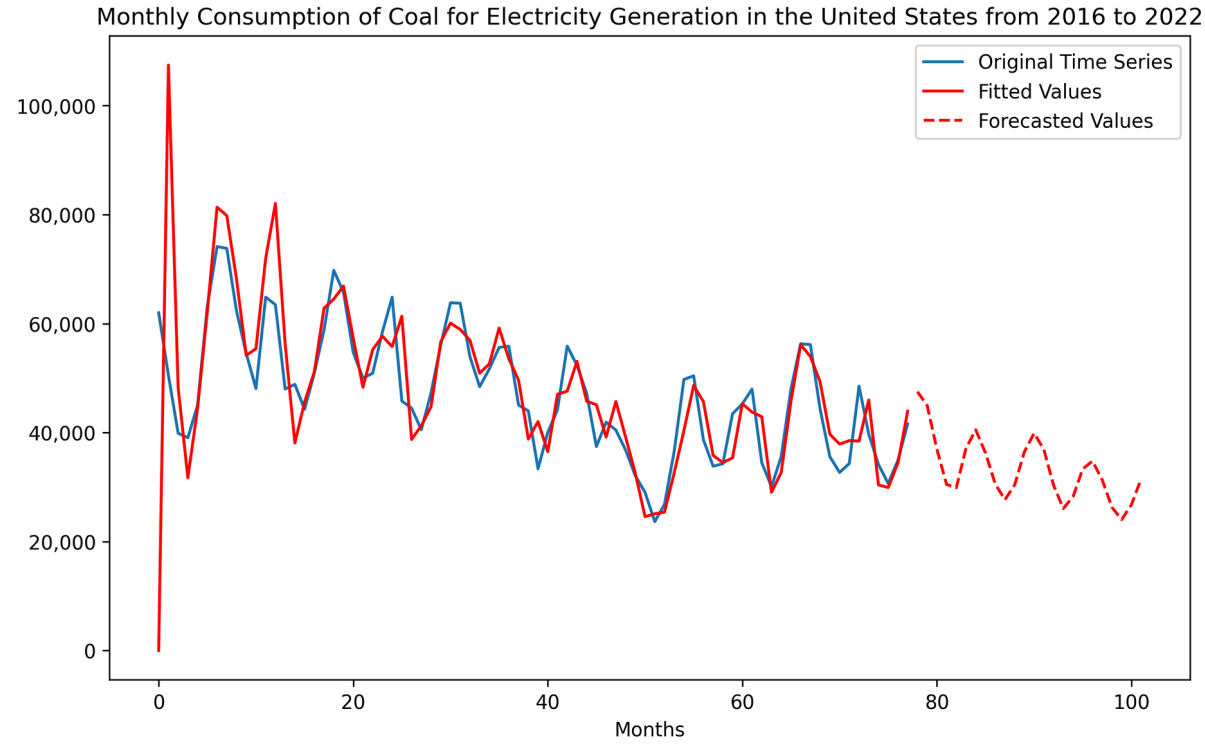
# Plot fitted and forecasted values
plt.plot(results.fittedvalues, color='red', label='Fitted Values')
plt.plot(fc, color='red', linestyle='dashed', label='Forecasted Values') ##
Forecasted values

# Set labels and legend
plt.xlabel('Months')
plt.title('Monthly Consumption of Coal for Electricity Generation in the United
States from 2016 to 2022')
plt.legend()

# Function to format the Y-axis values
def y_format(value, tick_number):
  return f'{value:,.0f}'

# Apply the formatter to the Y-axis
plt.gca().yaxis.set_major_formatter(FuncFormatter(y_format))
plt.show()
```

The resulting output will look like this:



Here, the blue curve is the original time series, and the red curve represents the ARIMA model, with forecasted values indicated by the dashed part of the curve. Although there is considerable error near the beginning of the time period, the model seems to fit the known values of the data fairly well, and so we expect that the future forecasts are reasonable.

5.4 Forecast Evaluation Methods

Learning Outcomes

By the end of this section, you should be able to:

- 5.4.1 Explain the nature of error in forecasting a time series.
- 5.4.2 Compute common error measures for time series models.
- 5.4.3 Produce prediction intervals in a forecasting example.

A time series forecast essentially predicts the *middle of the road* values for future terms of the series. For the purposes of prediction, the model considers only the trend and any cyclical or seasonal variation that could be detected within the known data. Although noise and random variation certainly do influence future values of the time series, their precise effects cannot be predicted (by their very nature). Thus, a forecasted value is the best guess of the model in the absence of an error term. On the other hand, error does affect how certain we can be of the model's forecasts. In this section, we focus on quantifying the error in forecasting using statistical tools such as prediction intervals.

Forecasting Error

Suppose that you have a model $(\hat{x}_n)$ for a given time series (x_n). Recall from Components of Time Series Analysis that the residuals quantify the error between the time series the model and serve as an estimate of the noise or random variation.

$$\varepsilon_n = x_n - \hat{x}_n$$

The better the fit of the model to the observed data, the smaller the values of ε_n will be. However, even very good models can turn out to be poor predictors of future values of the time series. Ironically, the better a model does at fitting to known data, the worse it may be at predicting future data. There is a danger of *overfitting*. (This term is defined in detail in Decision-Making Using Machine Learning Basics, but for now we do not need to go deeply into this topic.) A common technique to avoid overfitting is to build the model using only a portion of the known data, and then the model's accuracy can be tested on the remaining data that was held out.

To discuss the accuracy of a model, we should ask the complementary question: *How far away are the predictions from the true values?* In other words, we should try to quantify the total error. There are several **measures of error**, or **measures of fit**, the metrics used to assess how well a model's predictions align with the observed data. We will only discuss a handful of them here that are most useful for time series. In each formula, x_i refers to the ith term of the time series, and ε_i is the error between the actual i^{th} term and the predicted i^{th} term of the series.

a. **Mean absolute error (MAE)**: $\frac{1}{n}\sum_{i=1}^{n}|\varepsilon_i|$. A measure of the average magnitude of errors.
b. **Root mean squared error (RMSE)**: $\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\varepsilon_i{}^2\right)}$. A measure of the standard deviation of errors, penalizing larger errors more heavily than MAE does.
c. **Mean absolute percentage error (MAPE)**: $\frac{1}{n}\sum_{i=1}^{n}\left|\frac{\varepsilon_i}{x_i}\right|$. A measure of the average relative errors—that is, a percentage between the predicted values and the actual values (on average).
d. **Symmetric mean absolute percentage error (sMAPE)**: $\frac{1}{n}\sum_{i=1}^{n}\frac{2|\varepsilon_i|}{|x_i|+|\hat{x}_i|}$. Similar to MAPE, a measure of the average relative errors, but scaled so that errors are measured in relation to both actual values and predicted values.

Of these error measures, MAE and RMSE are **scale-dependent**, meaning that the error is in direct proportion to the data itself. In other words, if all terms of the time series and the model were scaled by a factor of k, then the MAE and RMSE would both be multiplied by k as well. On the other hand, MAPE and sMAPE are not scale-dependent. These measures of errors are often expressed as percentages (by multiplying the result of the formula by 100%). However, neither MAPE nor sMAPE should be used for data that is measured on a scale containing 0 and negative numbers. For example, it would not be wise to use MAPE or sMAPE as a measure of error for a time series model of Celsius temperature readings. For all of these measures of error, lower values indicate less error and hence more accuracy of the model. This is useful when comparing two or more models.

EXAMPLE 5.9

Problem

Compute the MAE, RMSE, MAPE, and sMAPE for the EMA smoothing model for the S&P Index time series, shown in Table 5.3.

Solution

The first step is to find the residuals, ε_n, and take their absolute values. For RMSE, we will need the squared residuals, included in Table 5.7. We also include the terms $\left|\frac{\varepsilon_i}{x_i}\right|$ and $\frac{2|\varepsilon_i|}{|x_i|+|\hat{x}_i|}$, which are used for computing MAPE and sMAPE, respectively. Notice all formulas are an average of some kind, but RMSE requires an additional step of taking a square root.

Year	S&P Index at Year-End	EMA Estimate	Abs. Residuals $\lvert\varepsilon_n\rvert = \lvert x_n - \widehat{x}_n\rvert$	$(\varepsilon_n)^2$	$\lvert\frac{\varepsilon_i}{x_i}\rvert$	$\frac{2\lvert\varepsilon_i\rvert}{\lvert x_i\rvert+\lvert\widehat{x}_i\rvert}$
2013	1848.36	1848.36	0	0	0	0
2014	2058.9	1848.36	210.54	44327.09	0.102	0.108
2015	2043.94	2006.27	37.67	1419.03	0.018	0.019
2016	2238.83	2034.52	204.31	41742.58	0.091	0.096
2017	2673.61	2187.75	485.86	236059.94	0.182	0.2
2018	2506.85	2552.15	45.3	2052.09	0.018	0.018
2019	3230.78	2518.17	712.61	507813.01	0.221	0.248
2020	3756.07	3052.63	703.44	494827.83	0.187	0.207
2021	4766.18	3580.21	1185.97	1406524.84	0.249	0.284
2022	3839.5	4469.69	630.19	397139.44	0.164	0.152
2023	4769.83	3997.05	772.78	597188.93	0.162	0.176
N/A	N/A	**Average:**	**453.52**	**339008.62**	**0.127**	**0.137**

Table 5.7 Data Needed to Compute the MAE, RMSE, MAPE, and SMAPE for EMA Smoothing Model

$\text{MAE} = 453.52$. (Predicted values are [on average] about 450 units away from true values.)

$\text{RMSE} = \sqrt{339008.62} = 582.24$. (The errors between predicted and average values have a standard deviation of about 580 from the ideal error of 0.)

$\text{MAPE} = 0.127, \text{or } 12.7\%$. (Predicted values are [on average] within about 13% of their true values.)

$\text{sMAPE} = 0.137, \text{or } 13.7\%$. (When looking at both predicted values and true values, their differences are on average about 14% from one another.)

These results may be used to compare the EMA estimate to some other estimation/prediction method to find out which one is more accurate. The lower the errors, the more accurate the method.

Prediction Intervals

A forecast is often accompanied by a **prediction interval** giving a range of values the variable could take with some level of probability. For example, if the prediction interval of a forecast is indicated to be 80%, then the prediction interval contains a range of values that should include the actual future value with a probability of 0.8. Prediction intervals were introduced in Analysis of Variance (ANOVA) in the context of linear regression, so we won't get into the details of the mathematics here. The key point is that we want a measure of *margin of error*, E_n (depending on n), such that future observations of the data will be within the interval $\widehat{x}_n \pm E_n$ with probability α, where α is a chosen level of confidence.

Here, we will demonstrate how to use Python to obtain prediction intervals.

The Python library `statsmodels.tsa.arima.model` (https://openstax.org/r/statsmodel) contains functions for finding confidence intervals as well. Note that the command `get_forecast()` is used here rather than `forecast()`, as the former contains more functionality.

PYTHON CODE

```
### Please run all code from previous section before running this ###

# Set alpha to 0.2 for 80% confidence interval
forecast_steps = 24
forecast_results = results.get_forecast(steps=forecast_steps, alpha=0.2)

# Extract forecast values and confidence intervals
forecast_values = forecast_results.predicted_mean
confidence_intervals = forecast_results.conf_int()

# Plot the results
plt.figure(figsize=(10, 6))

# Plot original time series
plt.plot(df['Value'], label='Original Time Series')

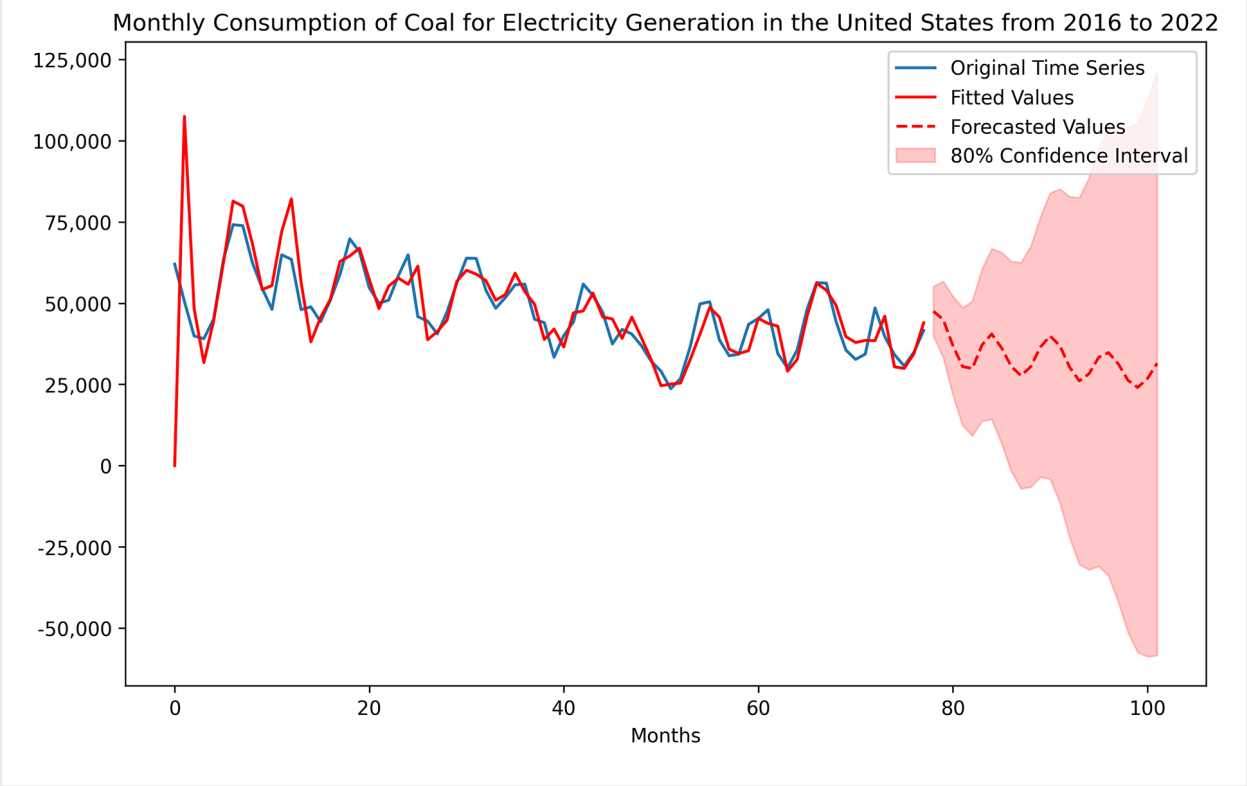
# Plot fitted values
plt.plot(results.fittedvalues, color='red', label='Fitted Values')

# Plot forecasted values with confidence intervals
plt.plot(forecast_values, color='red', linestyle='dashed', label='Forecasted
Values')
plt.fill_between(
  range(len(df), len(df) + forecast_steps),
  confidence_intervals.iloc[:, 0],
  confidence_intervals.iloc[:, 1],
  color='red', alpha=0.2,
  label='80% Confidence Interval'
)

# Set labels and legend
plt.xlabel('Months')
plt.title('Monthly Consumption of Coal for Electricity Generation in the United
States from 2016 to 2022')
plt.legend()

# Apply the formatter to the Y-axis
plt.gca().yaxis.set_major_formatter(FuncFormatter(y_format))
plt.show()
```

The resulting output will look like this:



The forecast data (dashed curve) is now surrounded by a shaded region. With 80% probability, all future observations should fit into the shaded region. Of course, the further into the future we try to go, the more uncertain our forecasts will be, which is indicated by the wider and wider confidence interval region.

Datasets

Note: The primary datasets referenced in the chapter code may also be [downloaded here (https://openstax.org/r/drive1)](https://openstax.org/r/drive1).

Key Terms

additive decomposition time series decomposition into the sum of its components
augmented Dickey-Fuller (ADF) test statistical test for stationarity of a time series
autocorrelation measure of correlation between a time series and a shifted copy of itself
autoregressive (AR) model model or component of a model that captures how the time series depends on its own previous values
cyclic component large variations in the data that recur over longer time periods than seasonal fluctuations, having no fixed frequency
detrending one of two complementary operations that separate the trend component from a time series
differencing found by taking differences of consecutive terms, that is, $x_{n+1} - x_n$
error extent to which predictions differ from actual observations
exponential moving average (EMA) type of weighted moving average in which the most recent values are given larger weights and the weights decay exponentially the further back in time the values are in the series
forecasting making predictions about future, unknown values of a time series
integrative (I) component component of the ARIMA model that represents the differencing operation
lag number of time steps that a time series is shifted for the purpose of computing autocorrelation
level the mean of all the time series data
mean absolute error (MAE) measure of error; $\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|\varepsilon_i|$
mean absolute percentage error (MAPE) measure of relative error; $\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}|\frac{\varepsilon_i}{x_i}|$
measures of error (or measures of fit) the metrics used to assess how well a model's predictions align with observed data
multiplicative decomposition time series decomposition into the product of its components
naïve (or flat) forecasting method using only the last observed value of a time series to predicting the next term
noise (or random error) random, unpredictable variations that occur in the time series that cannot be attributed to the trend, cycles, or seasonal components
normalized (weights) weights or coefficients that add up to a total of 1
order (or degree) the number of terms or lags used in a model to describe the time series; for a sequence that is modeled by a polynomial formula, the value of the exponent on the term of highest order of the polynomial formula that models the sequence
peak when computing ACF, a correlation value that is significantly larger than the previous values
period smallest time interval over which a seasonal variation pattern repeats
prediction interval range of values the variable could take with some level of probability in a forecast
residuals the difference between the given time series and the model $(x_n - \hat{x}_n)$ quantifying the error of the model, or noise (random variation) of the time series
root mean squared error (RMSE) measure of error; $\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\varepsilon_i{}^2\right)}$
scale-dependent regarding a measure or error that is in direct proportion with the data
seasonal component (seasonality) variation in the data that occurs at fixed time periods
Seasonal-Trend decomposition using LOESS (STL) a powerful tool that decomposes a time series into the trend-cycle, seasonal, and noise components
sequence ordered list of numbers, measurements, or other data
simple moving average (SMA) average (arithmetic mean) of a fixed number of consecutive data points in a time series
stationary characterizing a time series in which the variance is relatively constant over time, an overall upward or downward trend cannot be found, and no seasonal patterns exist
symmetric mean absolute percentage error (sMAPE) measure of relative error;

$$\text{sMAPE} = \frac{1}{n}\sum_{i=1}^{n}\frac{2|\epsilon_i|}{|x_i|+|\hat{x}_i|}$$

terms in mathematics, the individual values of a sequence
time series data that has a time component, or an ordered sequence of data points
time series analysis the examination of data points collected at specific time intervals, enabling the identification of trends, patterns, and seasonal variations crucial for making informed predictions and decisions
time series model function, algorithm, or method for finding, approximating, or predicting the values of a given time series
trend long-term direction of time series data in the absence of any other variation
trend curve (or trendline) line or curve that models the trend of a time series
trend-cycle component component of a time series that combines the trend and long-term cycles
volatility displaying significant fluctuations from the mean, typically due to factors that are difficult to analyze or predict
weighted moving average (WMA) moving average in which terms are given weights according to some formula or rule
white noise time series data that has constant mean value close to zero, constant variance, and no correlation from one part of the series to another
window term used to describe a fixed number of consecutive terms of a time series



Group Project

Project A – Smoothing out the Stock Market

Predicting the ups and downs of the stock market is a notoriously difficult task. On the other hand, if anyone could obtain accurate forecasts of future stock prices, then they could make a lot of money by buying up stocks at low prices and selling them later at higher prices. It is often useful just to predict whether the trend of a stock is upward or downward in the near future. In this project, you will use standard time series smoothing techniques to estimate trends in the S&P 500 Index.

Using the data set SP500.csv (https://openstax.org/r/spreadsheetsd1j), perform multiple techniques to smooth the time series including simple moving averages with various window lengths and exponential moving averages.

a. Produce SMA trendlines with window sizes 10, 30, 60, and 120. Graph the trendlines on the same set of axes as the original graph and describe the results. What happens to the trendline as the window size increases?
b. Produce EMA trendlines with $\alpha = 0.25, 0.5$ and 0.75. Graph the trendlines on the same set of axes as the original graph and describe the results. What happens to the trendline as the value of α increases?
c. Now compare your results from part a and part b. As a group, discuss the advantages and disadvantages of using SMA vs. EMA. What characteristics of the data are emphasized (or de-emphasized) by each method as the parameters (window size and α, respectively) change.

Project B – ARIMA Analysis of Coal Consumption

As a group, explore the time series in the data set MonthlyCoalConsumption.xlsx (https://openstax.org/r/spreadsheetsd1j).

a. Find the smallest order of differencing so that the time series becomes stationary.
b. Then run an ACF diagram and look for any spikes that might indicate periodicity of a seasonal component.
c. Run ARIMA analysis of the time series using the parameters for p and d established by your findings about periodicity and stationarity, trying different values of q. Evaluate the model's accuracy by computing the AIC score for each combination of parameters.
d. As a group, decide which parameters (p, d, q) you think will give the most accurate forecast. Then forecast

up to 30 time units into the future and produce confidence intervals for your forecasts. Interpret your results by indicating how likely your forecasts are to be within a range of values.

Critical Thinking

1.

a. What characteristics define a time series?

b. Which of the following are examples of time series data?

i. The vital statistics of a cancer patient taken right before a major surgery

ii. The monthly expenses of a small company recorded over a period of five years

iii. Daily temperature, rainfall, humidity, and wind speeds measured at a particular location over a few months

iv. Student final grades in all sections of a course at a university

2. Consider the graph in the figure as shown below.

a. Is the trend curve generally increasing, decreasing, or staying level over the entire time period?

b. The data set USATemps1961-2023.csv (https://openstax.org/r/spreadsheetsd1j) contains the time series from which the graph was created. Find the moving average trendline with period 10 for the data.

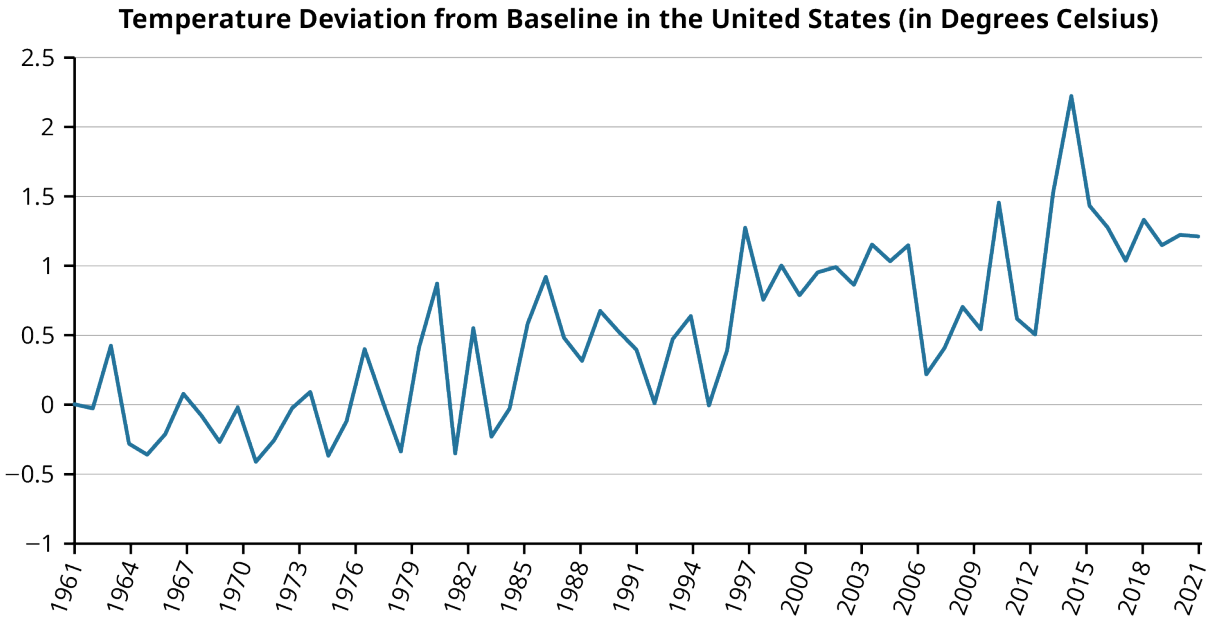
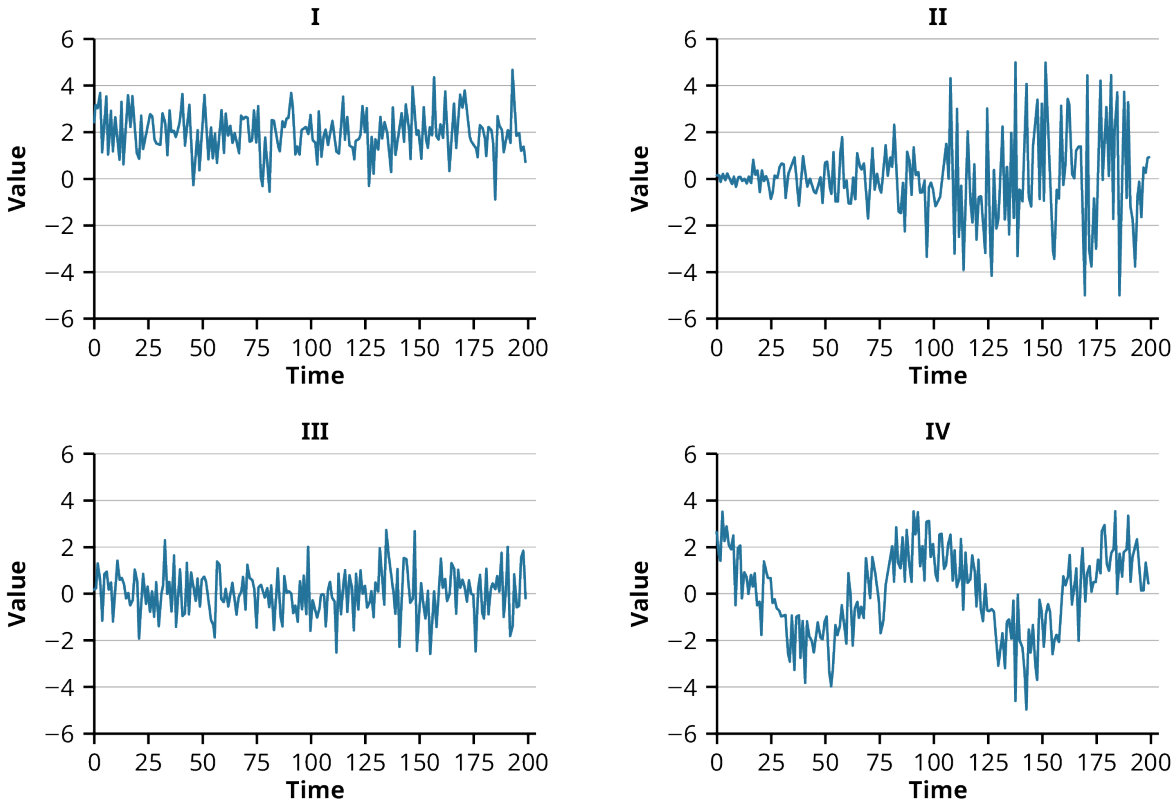


Figure 5.20

3.

a. What are the characteristics of white noise? Why is it important that the residuals of a time series model be white noise?

b. Determine which of the following graphs most likely represents white noise.



4. Four time series models were produced to model a given dataset. Various measures of error were run on each of the models, the results of which are shown in the table.

	MAE	RMSE	MAPE
Model 1	103.1	131.2	0.097
Model 2	251.7	207.9	0.136
Model 3	110.7	129.0	0.115
Model 4	89.8	125.3	0.191

Table 5.8 Measures of Error

a. Identify the most accurate model based only on:
i. MAE
ii. RMSE
iii. MAPE

b. Which model is likely to produce the least accurate forecasts overall, and why?
c. Which model is likely to produce the most accurate forecasts overall, and why?

Quantitative Problems

1. A U.S. company located in New Hampshire records weekly sales of its main product for a period of 3 years. If a seasonal variation in sales exists, what would be the most likely period of the seasonal variation?

2. Consider the time series given in the table as shown.

Month	Value
1	880.7
2	727.2
3	798.5
4	504.1
5	888.4
6	725.8
7	793.4
8	499.0
9	891.7
10	722.0
11	789.1
12	501.6

Table 5.9 Time Series

a. Identify the period of the seasonal component.
b. Using a centered simple moving average (SMA) of window size equal to the period you found in part a, identify a trend-cycle component, t_n, in the data.
c. Detrend the data by subtracting the SMA you found in part b.
d. Identify the seasonal component, $\hat{s}_n$, using the method of averaging.
e. Find the seasonally adjusted time series.

3. In **Collecting and Preparing Data**, the following data was given, as provided in the table, showing daily new cases of COVID-19. Since new cases were not reported on Saturdays and Sundays, those weekend cases were added to the Monday cases. One way to deal with the missing data is to use a centered simple moving average to smooth the time series.

Date	Weekday	New Case
10/18/2021	Monday	3115
10/19/2021	Tuesday	4849

Table 5.10 Sample of COVID-19 Data Cases within 23 Days (source: https://data.cdc.gov/Case-Surveillance)

10/20/2021	Wednesday	3940
10/21/2021	Thursday	4821
10/22/2021	Friday	4357
10/23/2021	Saturday	0
10/24/2021	Sunday	0
10/25/2021	Monday	8572
10/26/2021	Tuesday	4463
10/27/2021	Wednesday	5323
10/28/2021	Thursday	5012
10/29/2021	Friday	4710
10/30/2021	Saturday	0
10/31/2021	Sunday	0
11/1/2021	Monday	10415
11/2/2021	Tuesday	5096
11/3/2021	Wednesday	6882
11/4/2021	Thursday	5400
11/5/2021	Friday	6759
11/6/2021	Saturday	0
11/7/2021	Sunday	0
11/8/2021	Monday	10069
11/9/2021	Tuesday	5297

Table 5.10 Sample of COVID-19 Data Cases within 23 Days (source: https://data.cdc.gov/Case-Surveillance)

a. What is the most appropriate window size to use for centered SMA to address the issue of missing data in their analysis of COVID-19 data from the CDC?

b. Perform the SMA and illustrate the results by a graph.

4. Consider the data set USATemps1961-2023.csv (https://openstax.org/r/spreadsheetsd1j).
   a. Use an ACF plot to determine the period of a potential seasonal component.
   b. Use STL to decompose the time series into trend-cycle, seasonal, and noise components.
5. Use the recursive EMA formula with $\alpha = 0.6$ to smooth the time series found in USATemps1961-2023.csv (https://openstax.org/r/spreadsheetsd1j), and then use the EMA model to forecast the next value of the series.
6. A very small time series is given in the table, along with a model.

x_n	$\hat{x}_n$
122	117
108	135
172	152
190	169
167	186

Table 5.12 Time Series

   a. Find error measures MAE, RMSE, MAPE, and sMAPE for the model.
   b. Which of the error measures are scale-dependent? What would this imply about a similar time series and model in which all values are multiplied by 10?



6

Decision-Making Using Machine Learning Basics

Figure 6.1 Machine learning may be used to classify all kinds of data. Facial recognition technology relies heavily on machine learning algorithms, such as random forest models. (credit: modification of work "Humanoid Robot Uprising" by Steve Jurvetson/ Flickr, CC BY 2.0)

Chapter Outline

6.1 What Is Machine Learning?
6.2 Classification Using Machine Learning
6.3 Machine Learning in Regression Analysis
6.4 Decision Trees
6.5 Other Machine Learning Techniques

Introduction

Imagine a world in which your face is your passport. No need for keys, cards, or passwords; your identity is simply your unique appearance. In this world, machine learning plays the role of gatekeeper, using the visible features of your face to identify you and grant you access. Machine learning has a broad range of applications that go beyond facial recognition, transforming the way we interact with technology. As we'll see in this chapter and Deep Learning and AI Basics, machine learning is also used in medical image analysis, satellite image interpretation, and augmented reality (e.g., in gaming, education/training, and architecture/design). It is used to power recommendation systems and for predictive analytics. In this chapter, we explore the fundamentals of classification, clustering, and regression algorithms that allow machines to learn how to label data accurately and make useful decisions that can be put to use in these many applications.

6.1 What Is Machine Learning?

Learning Outcomes

By the end of this section, you should be able to:

- 6.1.1 Summarize the differences between supervised and unsupervised learning in machine learning.
- 6.1.2 Describe the roles of training and testing datasets.
- 6.1.3 Use common measures of accuracy and error to determine the fitness of a model.
- 6.1.4 Explain the concepts of overfitting and underfitting and how these problems can adversely affect the model.

A **machine learning (ML) model** is a mathematical and computational model that attempts to find a relationship between input variables and output (response) variables of a dataset. The way that the model "learns" is by adjusting internal parameters until the model meets a certain level of **accuracy** (defined as a measure of how correct the model is when used to predict. We will have more to say on accuracy later). What sets apart a machine learning model from a typical mathematical model is that the user does not have full control over all the parameters used by the model. Moreover, there may be hundreds, thousands, or even millions of parameters that are used and adjusted within a given ML model—way more than a human could reasonably track.

The part of that dataset that is used for initial learning is the **training set (or data)**. There is often a **testing set (or data)** as well, which (as the name implies) can be used to determine whether the model is accurate enough. As a general rule of thumb, about 60–80% of available data will be used for training and the remaining data used for testing. After a machine learning model is trained, it can be used to make predictions as new input data is fed in and the responses produced. There may be multiple rounds of training and testing before the ML model is ready to make predictions or decisions.

The machine learning life cycle typically follows these steps:

1. **Problem formulation/identification:** Clearly state the problem you want the model to solve.
2. **Data collection and preparation:** Collect relevant data and clean it for analysis.
3. **Feature selection/engineering:** Choose important aspects of the data for the model.
4. **Model/algorithm selection:** Select a suitable machine learning algorithm for the task.
5. **Model/algorithm training:** Feed in the training data to teach the model patterns in the data.
6. **Model validation:** Test how well the model predicts outcomes.
7. **Model implementation:** Put the model to work in real-world scenarios.
8. **Performance monitoring and enhancement:** Keep an eye on the model's accuracy and update as needed.
9. **Continuous improvement and refinement:** Repeat and refine steps based on feedback and changes.

Throughout the process, be sure to keep ethical considerations in mind to ensure fairness, privacy, and transparency. For example, were the data collected with the proper permissions? Does the training data accurately reflect the same diversity of data that the algorithm will ultimately be used on? Currently there are serious questions as to the ethical implications of using facial recognition algorithms in law enforcement, as training images may have been gathered from people without their explicit consent, and the models created often reflect biases that can result in discriminatory outcomes, disproportionately affecting certain demographic groups, such as people of color or women.

Bias—error introduced by overall simplistic or overly rigid models that do not capture important features of the data—can be hard to identify! One example that is famous in data science circles (though it may be apocryphal) is the "Russian tank problem" (Emspak, 2016). As the story goes, the United States military trained an early ML model to distinguish between Russian and US tanks. The training set consisted of photos of many

different kinds of tanks used by both countries. Although the ML model produced very accurate results on the training and test sets, it performed terribly out in the field. Apparently, most of the pictures of Russian tanks were low quality and blurry, while those of the US tanks were high quality. They had inadvertently trained their ML model to distinguish between low- and high-quality photos regardless which county's tanks were depicted in them!

Supervised vs. Unsupervised Learning

Data may be labeled or unlabeled. **Labeled data** has specific labels, categories, or classes associated with each data point. For example, a dataset containing the height, weight, cholesterol levels, and various other health-related measures for patients at a clinic, along with an indication as to whether each is susceptible to heart disease or not, would be labeled data. The label is the value (yes or no) of the variable "susceptible to heart disease." **Unlabeled data** does not have a specific label, category, or class associated with its data points. For example, in product marketing, we might collect data representing customer attributes such as age, income, and purchase history, and we may want to identify distinct groups of customers with similar characteristics without knowing ahead of time what those groups (labels) will be.

Supervised learning uses labeled data in its training and testing sets. Parameters are adjusted in the model based on the correctness of the results as compared with the known labels. **Unsupervised learning** trains on unlabeled data. Often, both methods will be used in the same project. Unsupervised learning might be employed as an initial step to find patterns in the dataset, allowing the data to be labeled in some way. Then supervised learning techniques may be applied to the labeled data as a training step. The model created in this way may then be used to determine the best labels for future (unlabeled) data.

Supervised Learning

Supervised learning is analogous to a student learning from an instructor. At first, the student may answer many questions incorrectly, but the instructor is there to correct the student each time. When the student gives (mostly) correct responses, there is evidence that the student has sufficiently learned the material and would be able to answer similar questions correctly out in the real world where it really matters.

The goal of supervised learning is to produce a model that gives a mapping from the set of inputs or features to a set of output values or labels (see Figure 6.2). The model is function $y = f(X)$, where X is the input array and y is the output value or label. The way that it does this is through training and testing. When training, the algorithm creates a correspondence between input features and the output value or label, often going through many iterations of trial and error until a desired level of accuracy is achieved. The model is then tested on additional data, and if the accuracy remains high enough, then the model may be deployed for use in real-world applications. Otherwise, the model may need to be adjusted substantially or abandoned altogether.

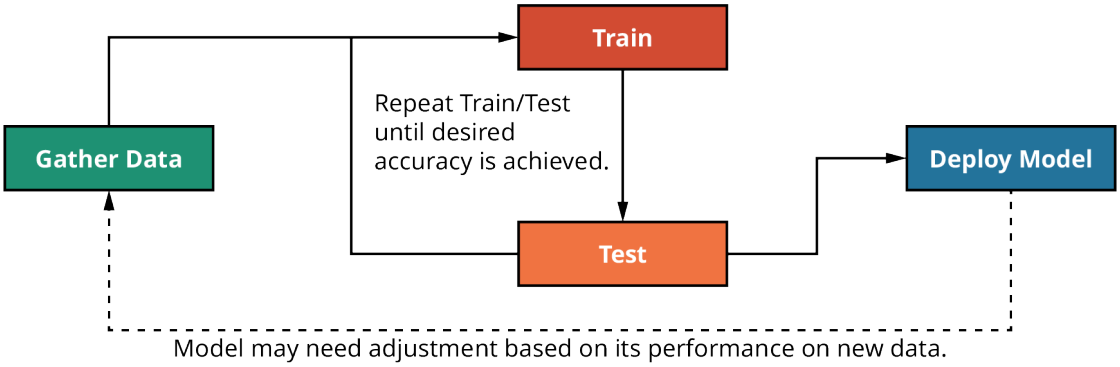


Figure 6.2 The Supervised Learning Cycle. The supervised learning cycle consists of gathering data, training on some part of the data, testing on another part of the data, and deployment to solve problems about new data.

Examples of supervised learning models include linear regression (discussed in Correlation and Linear Regression Analysis), logistic regression, naïve Bayes classification, decision trees, random forests, and many

kinds of neural networks. Linear regression may not seem like a machine learning algorithm because there are no *correction* steps. A formula simply produces the line of best fit. However, the regression formula itself represents an optimization that reduces the error in the values predicted by the regression line versus the actual data. If the linear regression model is not accurate enough, additional data could be collected to improve the accuracy of the model, and once the regression line is found, it may then be used to predict values that were not in the initial dataset. Linear regression has been developed in previous chapters, and you will see logistic regression, Bayes methods, decision trees, and random forests later in this chapter.

Unsupervised Learning

Unsupervised learning is like building a puzzle without having the benefit of a picture to compare with. It is not impossible to build a puzzle without the picture, but it may be more difficult and time-consuming.

The main advantage of unsupervised learning is that the training data does not need to be labeled. All the data is considered input, and the output would typically be information about the inherent structure of the data. Common tasks are to determine clusters, patterns, and shapes within the dataset. For example, if the current locations in latitude and longitude of every person in the world were somehow known and collected into a dataset, then an algorithm may discover locations of high densities of people—hence *learn* of the existence of cities!

Unsupervised learning algorithms include k-means clustering, DBScan, and some kinds of neural networks. More generally, most clustering algorithms and dimension-reduction methods such as principal component analysis (PCA) and topological data analysis (TDA) fall into the category of unsupervised learning. You will encounter the k-means and DBScan algorithms in this chapter. The more advanced topics of PCA and TDA are outside the scope of this text.

Variations and Hybrid Models

Although we will only cover supervised and unsupervised models, it is important to note that there are some additional models of machine learning.

When some of the data in a dataset has labels and some does not, a *semi-supervised learning algorithm* may be appropriate. The labeled data can be run through a supervised learning algorithm to generate a predictive model that can then be used to label the unlabeled data, called *pseudo-data*. From this point, the dataset can be regarded as completely labeled data, which can then be analyzed using supervised learning.

Reinforcement learning introduces rewards for accuracy while also penalizing incorrectness. This model most closely follows the analogy of a student-teacher relationship. Many algorithms in artificial intelligence use reinforcement learning. While reinforcement generally improves accuracy, there can be issues in locking the model into a rigid state. For example, have you ever noticed that when you search online for a type of product, you begin to see advertisements for those kinds of products everywhere—even after you've purchased it? It's as if the algorithm has decided that the only product you ever wish to purchase is what you have just searched for!

Training and Testing a Model

How do we know how good a model is? The key is to compare its output against known output. In supervised learning models, there is always a metric that measures efficacy of the model. During the training phase, this metric helps to determine how to adjust the model to make it more accurate. During the testing phase, the metric will be used to evaluate the model and determine if it is good enough to use on other datasets to make predictions. In this section, we will explore the supervised learning cycle in more detail.

Model Building

Suppose we are given a dataset with inputs or features X and labels or values y. Our goal is to produce a function that maps $y = f(X)$. In practice, we do not expect that our model can capture the function f with

100% accuracy. Instead, we should find a model $\widehat{f}$ such that predicted values $\hat{y} = \widehat{f}(X)$ are *close enough to* y. In most cases, the metric we use to determine the fitness of our model $\widehat{f}$ is some measure of error between $\hat{y}$ and y (e.g., see the calculation of residuals of a linear regression in Correlation and Linear Regression Analysis). The way that error is measured is specific to the type of model that we are building and the nature of the outputs. When the outputs are labels, we may use a count or percentage of misclassifications. When the outputs are numerical, we may find the total distances (absolute, squared, etc.) between predicted and actual output values.

First, the data are separated into a training set and testing set. Usually about 60–80% of the data is used for training with the remaining amount used for testing. While training, parameters may be adjusted to reduce error. The result of the training phase will be the model $\widehat{f}$. Then, the model is tested on the testing set and error computed between predicted values and known outputs. If the model meets some predefined threshold of accuracy (low enough error), then the model can be used with some confidence to make predictions about new data.

Measures of Accuracy

As mentioned previously, it is important to measure how good our machine learning models are. The choice of a specific measure of accuracy depends on the nature of the problem. Here are some common accuracy measures for different machine learning tasks:

1. Classification
   - Accuracy is the ratio of correct predictions to the total number of predictions.
   - **Precision** (p) is the ratio of true positive predictions (TP) to the total number of positive (true positive plus false positive) predictions $(TP + FP)$. This is useful when the response variable is true/false and we want to minimize false positives. $p = \frac{TP}{TP+FP}$
   - **Recall** (r) is the ratio of true positive predictions (TP) to the total number of actual positives (true positive plus false negative $(TP + FN)$ predictions. This is useful when the response variable is true/false and we want to minimize false negatives. $r = \frac{TP}{TP+FN}$
   - **F1 Score** is a combination of precision (p) and recall (r). $F1 = \frac{2(p)(r)}{p+r} = \frac{2TP}{2TP+FP+FN}$
2. Regression (comparison of the predicted values, $\hat{y}_i$, and the actual values, y_i)
   - **Mean absolute error (MAE)**: $\frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$
   - **Mean absolute percentage error (MAPE)**: $\frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i-\hat{y}_i}{y_i}\right|$
   - **Mean squared error (MSE)**: $\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2$
   - **Root mean squared error (RMSE)**: $\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}$
   - *R-squared* is the proportion of the variance in the response variable that is predictable from the input variable(s).

Note that MAE and RMSE, also discussed in Forecast Evaluation Methods, have the same units as the data itself. MSE and RMSE can be very sensitive to outliers because the differences are squared. MAPE (also introduced in Forecast Evaluation Methods) is often represented in percent form, which you can easily find by multiplying the raw MAPE score by 100%. The R-squared measure, which is discussed in Inferential Statistics and Regression Analysis in the context of linear regression, can be computed by statistical software packages, though we will not make significant use of it in this chapter. In all cases, the smaller the value of the error term, the more accurate the modeling would be.

Due to the tedious calculations involved in these measures, statistical software packages are typically used to compute them in practice. But it is important for every data scientist to understand how the formulas work so that results can be evaluated and interpreted correctly. Next, we will work out a small example by hand so you can have experience with the formulas.

EXAMPLE 6.1

Problem

A test for COVID-19 was applied to 1,000 patients. Two hundred thirty-five tested positive for COVID-19, while the remaining 765 tested negative. Of those that tested positive, 198 turned out to carry the COVID-19 virus, and 37 did not have the virus. Of those that tested negative, 63 patients did in fact have the virus, while the remaining 702 did not. Compute the accuracy, precision, recall, and F1 scores for this experiment.

Solution

First, the total number of correct predictions were: 198 true positive predictions + 702 true negative predictions, for a total of $TP + TN = 900$. So the accuracy is $\frac{900}{1{,}000} = 0.9(90\%)$.

For precision, we use $TP = 198$ and FP (false positives) $= 37$, and we have: $p = \frac{TP}{TP+FP} = \frac{198}{198+37} = 0.843$ (84.3%).

For recall, we use $TP = 198$ and FN (false negatives) $= 63$, and we have: $r = \frac{TP}{TP+FN} = \frac{198}{198+63} = 0.759$ (75.9%).

Finally, $F1 = \frac{2(p)(r)}{p+r} = \frac{2(0.843)(0.759)}{0.843+0.759} = 0.799$ (79.9%).

EXAMPLE 6.2

Problem

The score on an exam is positively correlated with the time spent studying for the exam. You have collected the following data (pairs of study time in hours and exam scores):

(9.8, 83.0), (8.2, 87.7), (5.2, 61.6), (8.0, 77.8), (2.1, 42.2), (6.8, 62.1), (2.3, 30.9), (9.5, 94.4), (6.6, 76.2), (9.5, 93.1)

Use 80% of the dataset to create a linear model and discuss the accuracy of the model on the remaining testing set.

Solution

Since there are 10 data points, we use 8 of them for training, as 8 is 80% of 10.

Training data: (9.8, 83.0), (8.2, 87.7), (5.2, 61.6), (8.0, 77.8), (2.1, 42.2), (6.8, 62.1), (2.3, 30.9), (9.5, 94.4)

The linear regression model that best fits the training data is $y = 21.7 + 7.1x$. (See Correlation and Linear Regression Analysis for details on linear regression.) We will compute MAE, MAPE, MSE, and RMSE for the remaining two data points (test set) as in Table 6.1.

x	Prediction	Actual Value	Difference (Error)
2.3	$21.7 + 7.1\,(2.3) = 38.03$	30.9	-7.13
9.5	$21.7 + 7.1(9.5) = 89.15$	94.4	5.25

Table 6.1 Test Set

- $\text{MAE} = \frac{1}{2}(|-7.13| + |5.25|) = 6.19$

- MAPE $= \frac{1}{2}\left(\left|\frac{-7.13}{30.9}\right| + \left|\frac{5.25}{94.4}\right|\right) = 0.143$, or 14.3%
- MSE for cubic model: $\frac{1}{2}\left((-7.13)^2 + 5.25^2\right) = 39.2$
- RMSE for cubic model: $\sqrt{39.2} = 6.26$

The MAE and RMSE both show that the model's predictions are off by an average of about 6.2 points from actual values. The MAPE suggests that the model has an error of about 14.3% on average.

Overfitting and Underfitting

Can 100% accuracy be attained from a machine learning algorithm? In all but the most trivial exercises, the answer is a resounding ***no***. Real-world data is noisy, often incomplete, and rarely amenable to a simple model. (Recall from Collecting and Preparing Data that we defined noisy data as data that contains errors, inconsistencies, or random fluctuations that can negatively impact the accuracy and reliability of data analysis and interpretation.) Two of the most common issues with machine learning algorithms are overfitting and underfitting.

Overfitting happens when the model is adjusted to fit the training data too closely, resulting in a complex model that is too specific to the training data. When such a model is given data outside of the training set, it may perform worse, a phenomenon known as high **variance**.

Underfitting occurs when the model is not robust enough to capture essential features of the data. This could be because of too small a dataset to train on, choosing a machine learning algorithm that is too rigid, or a combination of both. A model that suffers from underfitting may be said to have *high bias*.

Often, a model that has low variance will have high bias, and a model that has low bias will have high variance. This is known as the *bias-variance trade-off*.

Overfitting

It might at first seem desirable to produce a model that makes no mistakes on the training data; however, such "perfect" models are usually terrible at making predictions about data not in the training set. It is best to explain this idea by example.

Suppose we want to create a model for the relationship between *x* and *y* based on the given dataset,

$$S = (x, y) = (2, 10), (4, 9), (6, 6), (8, 7), (10, 4), (12, 5)$$

How would we go about creating the model? We might first split S into a training set, $S_{\text{train}} = \{(2, 10), (4, 9), (6, 6), (8, 7)\}$, and testing set, $S_{\text{test}} = \{(10, 4), (12, 5)\}$. It can be shown that the cubic equation, $\hat{y} = \frac{1}{8}x^3 - \frac{7}{4}x^2 + \frac{13}{2}x + 3$, fits the training data perfectly. However, there is no reason to expect that the relationship between *x* and *y* is truly cubic. Indeed, the predictions of the cubic model on S_{test} are terrible! See Table 6.2.

x	Prediction, Cubic Model	Actual Value, *y*
10	18	4
12	45	5

Table 6.2 Cubic Model Predictions

The linear model (linear regression) that represents a best fit for the first four data points is $\hat{y} = -0.6x + 11$. Figure 6.3 shows the data points along with the two models.

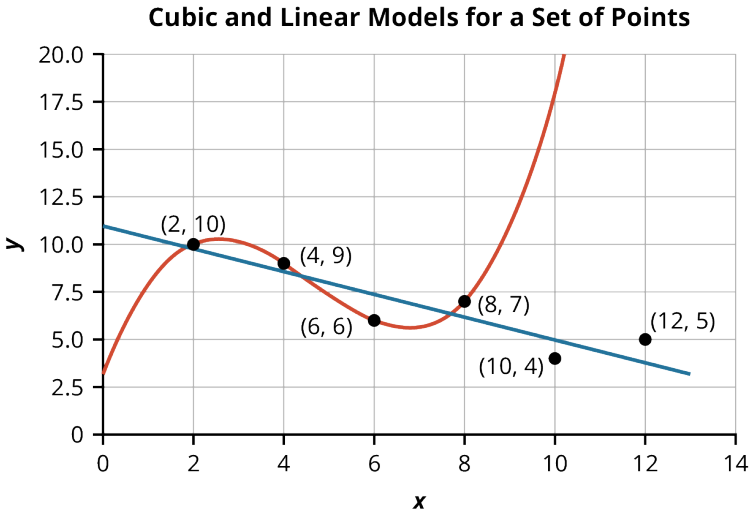


Figure 6.3 Data Points along a Cubic Model and a Linear Model. The four data points in S_{train} can be fitted onto a cubic curve (0% error), but it does poorly on additional data from the testing data, S_{test}. The cubic curve overfits the training data. The regression line does not fit the training data exactly; however, it does a better job predicting additional data.

To get a good sense of how much better the linear model is compared to the cubic model in this example, we will compute the MAPE and RMSE for each model, as shown in Table 6.3.

x_i	Actual Value, y_i	Prediction, y_i, Cubic Model	$\lvert y_i - \hat{y}_i \rvert$, Cubic Model	Prediction, $\hat{y}_i$, Linear Model	$\lvert y_i - \hat{y}_i \rvert$, Linear Model
2	**10**	10	0	9.8	0.2
4	**9**	9	0	8.6	0.4
6	**6**	6	0	7.4	1.4
8	**7**	7	0	6.2	0.8
10	**4**	18	14	5	1
12	**5**	45	40	3.8	1.2

Table 6.3 MAPE and RMSE Calculated for Each Model

- MAPE for cubic model: $\frac{1}{6}(0/10 + 0/9 + 0/6 + 0/7 + 14/4 + 40/5) = 1.917$, or 191.7%
- RMSE for cubic model: $\sqrt{\frac{1}{6}\left(0^2 + 0^2 + 0^2 + 0^2 + 14^2 + 40^2\right)} = 17.3$
- MAPE for linear model: $\frac{1}{6}(0.2/10 + 0.4/9 + 1.4/6 + 0.8/7 + 1/4 + 1.2/5) = 0.15$, or 15%
- RMSE for linear model: $\sqrt{\frac{1}{6}\left(0.2^2 + 0.4^2 + 1.4^2 + 0.8^2 + 1^2 + 1.2^2\right)} = 0.93$

The MAPE and RMSE values for the linear model are much lower than their respective values for the cubic model even though the latter predicts four values with 100% accuracy.

Underfitting

A model that is underfit may perform well on some input data and poorly on other data. This is because an underfit model fails to detect some important feature(s) of the dataset.

EXAMPLE 6.3

Problem

A student recorded their time spent on each homework set in their statistics course. Each HW set was roughly similar in length, but the student noticed that they spent less and less time completing them as the semester went along. Find the linear regression model for the data in Table 6.4 and discuss its effectiveness in making further predictions.

HW Set	1	2	3	4	5	6
Time (hr)	9.8	5.3	3.4	2.6	1.9	1.7

Table 6.4 Homework Set Completion Times

Solution

There is a definitive downward trend in the data. A simple linear regression model for this data is:

$$\hat{y} = -1.47143x + 9.26667$$

However, the line of best fit does not seem to fit the data all that well. There is no way that a linear function can *bend* the way the data points seem to do in this example (see Figure 6.4).

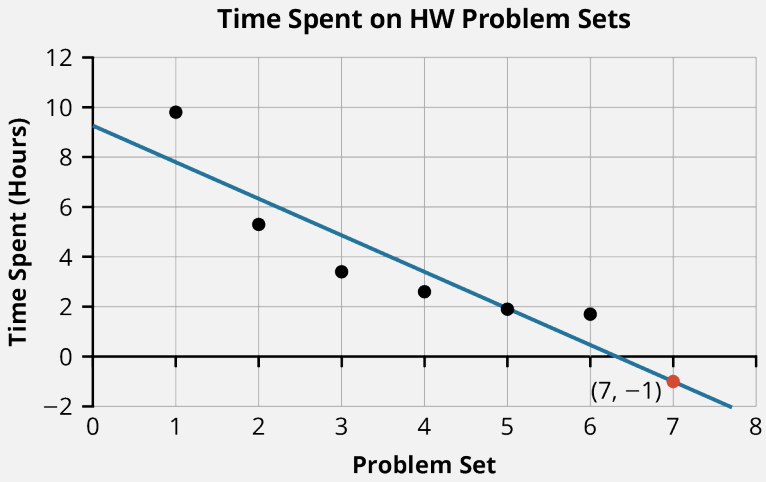


Figure 6.4 Linear Model That Best Fits the Data in Example 6.3. This graph contains a scatterplot of the six data points together with the linear model that best fits the data. This is a clear case of underfitting.

Another issue with using the linear model for the data in Table 6.4 is that eventually the values of $\hat{y}$ become negative! According to this model, it will take about -1 hours to complete the 7th homework set, which is absurd. Perhaps a decreasing exponential model, shown in Figure 6.5, would work better here. Proper model selection is the most effective way of dealing with underfitting.

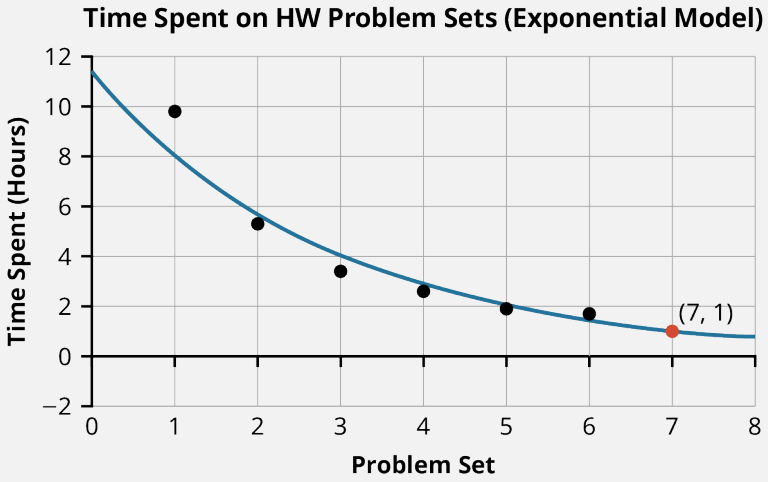


Figure 6.5 Exponential Model That Best Fits the Data in Example 6.3. This graph contains a scatterplot of the six data points together with the exponential model that best fits the data. Under this model, the prediction of $\hat{y} = 1$ when $x = 7$ seems much more reasonable.

6.2 Classification Using Machine Learning

Learning Outcomes

By the end of this section, you should be able to:

- 6.2.1 Perform logistic regression on datasets and interpret the results.
- 6.2.2 Perform k-means clustering on datasets.
- 6.2.3 Define the concept of density-based clustering and use DBScan on datasets.
- 6.2.4 Interpret the confusion matrix in clustering or classifying data.

Classification problems come in many flavors. For example, suppose you are tasked with creating an algorithm that diagnoses heart disease. The input features may include biological sex, age, blood pressure data, and cholesterol levels. The output would be either: yes (diagnose heart disease) or no (do not diagnose heart disease). That is, your algorithm should classify patients as "yes" or "no" based on an array of features, or *symptoms* in medical terminology. Logistic regression is one tool for classification when there are only two possible outputs. This is often called a **binary (binomial) classification** problem. If the goal is to classify data into more than two classes or categories, then the problem is referred to as **multiclass (multinomial) classification**.

Other kinds of classification problems involve finding two or more clusters in the data. A **cluster** is a collection of data points that are closer to one another than to other data points, according to some definition of *closeness*. For example, certain characteristics of music, such as tempo, instrumentation, overall length, and types of chord patterns used, can be used as features to classify the music into various genres, like rock, pop, country, and hip-hop. A supervised machine learning algorithm such as a decision tree (see Decision Trees) or random forest (see Other Machine Learning Techniques) may be trained and used to classify music into the various predefined genres. Alternatively, an unsupervised model such as k-means clustering could be used to group similar-sounding songs together without necessarily adhering to a concept of *genre*.

This section will focus on logistic regression techniques and clustering algorithms such as k-means and DBScan. We'll use measures of accuracy including the confusion matrix.

Logistic Regression

A **logistic regression** model $L(X)$ takes input vectors X (features) and produces an output of "yes" or "no." Much like linear regression, the logistic regression model first fits a continuous function based on known,

labeled data. However, instead of fitting a line to the data, a sigmoid function is used. A **sigmoid function** is a specific type of function that maps any real-valued number to a value between 0 and 1. The term *sigmoid* comes from the fact that the graph has a characteristic "S" shape (as shown in Figure 6.6), and *sigma* (σ) is the Greek letter that corresponds to our letter *S*. We often use the notation $\sigma(x)$ to denote a sigmoid function. The basic formula for the sigmoid is:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

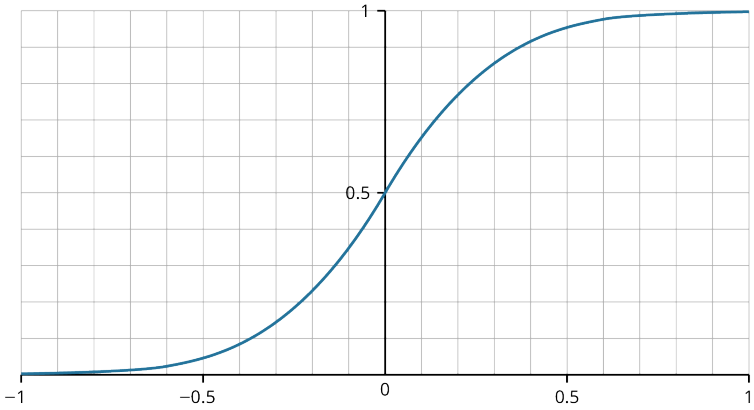


Figure 6.6 Graph of a Typical Sigmoid Function

One advantage of the sigmoid over a linear function is that the values of the sigmoid stay close to 0 when x is negative and close to 1 when x is positive. This makes it ideal for classifying arbitrary inputs x into one of two classes, "yes" (1) or "no" (0). In fact, we can interpret the *y*-axis as probability that an event occurs. The sigmoid function can be shifted right or left and compressed or expanded horizontally as necessary to fit the data. In the following formula, the parameters a and b control the position and shape of the sigmoid function, respectively.

$$\sigma(a + bx) = \frac{1}{1 + e^{-(a+bx)}}$$

As complicated as this formula may seem at first, it is built up from simple elements. The presence of $a + bx$ suggests that there may be a linear regression $y = a + bx$ lurking in the background. Indeed, if we solve to find the inverse function of $\sigma(x)$, we can use it to isolate $a + bx$, which in turns allows us to use standard linear regression techniques to build the logistic regression. Recall that the natural logarithm function, ln n, is the inverse function for the exponential, e^x. In what follows, let p stand for $\sigma(x)$.

$$p = \sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\begin{aligned}
\frac{1}{p} &= 1 + e^{-x} \\
\frac{1}{p} - 1 &= e^{-x} \\
e^{-x} &= \frac{1}{p} - 1 = \frac{1-p}{p} \\
-x &= \ln\left(\frac{1-p}{p}\right) \\
x &= -\ln\left(\frac{1-p}{p}\right) = \ln\left[\left(\frac{1-p}{p}\right)^{-1}\right] = \ln\left(\frac{p}{1-p}\right)
\end{aligned}$$

In the final line, we used the power property of logarithms, $\ln A^n = n \ln A$, with power $n = -1$. The function we've just obtained is called the logit function.

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right)$$